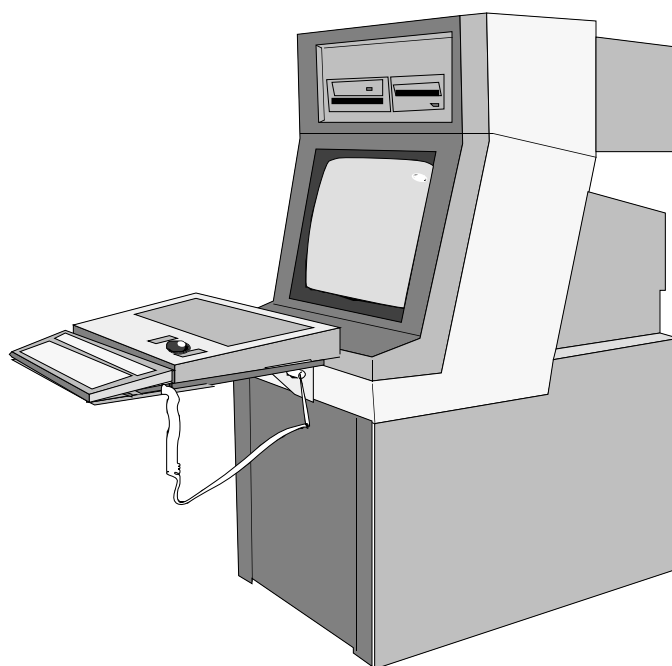


Universal StationX Service

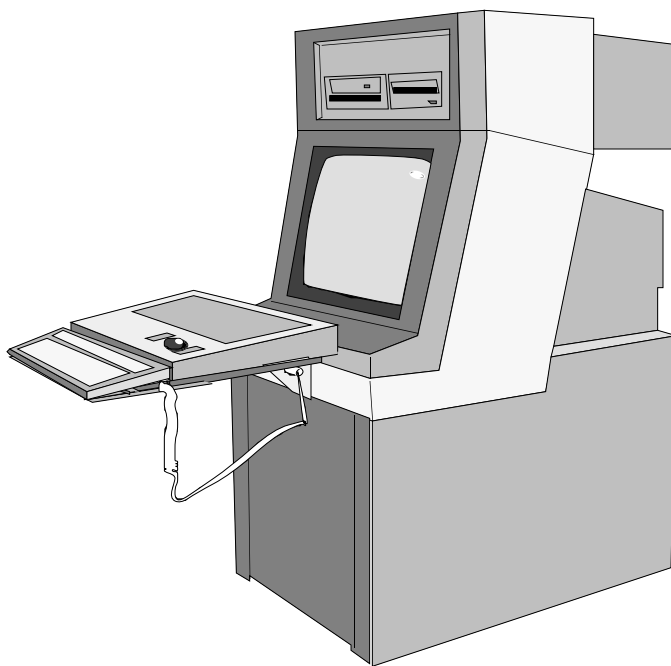
UX13-510



Universal Station^X

Universal Station^X Service

**UX13-510
Release 521
CE Compliant
4 / 97**



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Revision 03 – April 11, 1997

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About This Publication

This manual provides instructions for maintenance and repair of the Universal Station^X (Universal Station with X-Windows, or U^XS). The periodic replaceable items (printer ribbons, etc.) that are used with the U^XS are listed. Options are also discussed. Each of the Optimum Replaceable Units (ORUs) are addressed regarding cleaning, adjustments, test/troubleshooting, removal/replacement, spare part numbers, and startup.

This publication supports **TotalPlant** Solution (TPS) Network Release 521 and earlier software releases. TPS is the evolution of TDC 3000^X.

This publication supports CE Compliant equipment. Any equipment designated as “CE Compliant” complies with the European Union EMC and Health and Safety Directives. All equipment shipping into European Union countries after January 1, 1996 requires this type of compliance—denoted by the “CE Mark.”

Change bars are used to indicate paragraphs, tables, or illustrations containing changes that have been made to this manual. Pages revised only to correct minor typographical errors contain no change bars.

Standard Symbols

Scope

The following defines standard symbols used in this publication.

ATTENTION

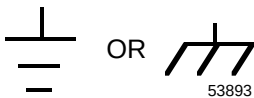
Notes inform the reader about information that is required, but not immediately evident.

CAUTION

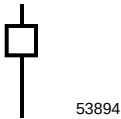
Cautions tell the user that damage may occur to equipment if proper care is not exercised.

WARNING

Warnings tell the reader that potential personal harm or serious economic loss may occur if instructions are not followed.



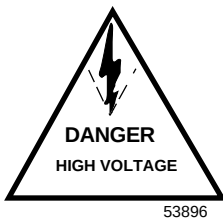
Ground connection to building safety ground.



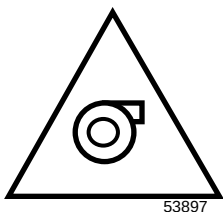
Ground stake for building safety ground.



Electrical Shock Hazard—can be lethal.



Electrical Shock Hazard—can be lethal.



Rotating Fan—can cause personal injury.

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Acronyms

Coprocessor.....	HP9000-382 Processor
CRT	Cathode Ray Tube
DAT	Digital Audio Tape
HVTS.....	Hardware Verification Test System
K2LCN.....	Processor/Memory/LCN Interface
K4LCN.....	Processor/Memory/LCN Interface
LAN	Local Area Network
LCN	Local Control Network
LED	Light Emitting Diode
MB.....	Megabyte
ORU.....	Optimum Replaceable Unit
PDGI.....	Peripheral Display Generator Interface
RAM	Random Access Memory
RGB.....	Red/Green Blue (refers to three lead coax)
ROM	Read Only Memory
SCSI.....	Small Computer Communications Interface
TPDG I/O.....	Turbo Peripheral Display Generator Input/Output
TPDGC	Turbo Peripheral Display Generator (Input/Output)
TPDGX	Turbo Peripheral Display Generator (R200)
WSI I/O.....	Workstation Interface Input /Output
WSI	Workstation Interface
WSI2 I/O.....	Workstation Interface Input /Output
WSI2	Workstation Interface
WSI2R I/O	Workstation Interface Input /Output

Glossary of Terms

Coprocessor.....	HP9000-382 Processor
Pixel.....	Smallest Graphic Element

References

Publication Title	Publication Number	Binder Title	Binder Number
<i>TPS Universal Station^X Specification and Technical Data</i>	UX03-502	System Summary - 2	TPS 3010-2
<i>TPS System Site Planning</i>	SW02-550	System Site Planning	TPS 3020-1
<i>Local Control Network Planning</i>	SW02-501	System Site Planning	TPS 3020-1
<i>LCN Guidelines - Implementation, Troubleshooting, and Service</i>	LC09-510	LCN Installation	TPS 3025
<i>LCN System Installation</i>	SW20-500	LCN Installation	TPS 3025
<i>LCN System Checkout</i>	SW20-510	LCN Installation	TPS 3025
<i>Five/Ten-Slot Module Service</i>	LC13-500	LCN Service-2	TPS 3060-2
<i>Test System Executive</i>	SW13-510	LCN Service-3	TPS 3060-3
<i>Hardware Verification Test System</i>	SW13-511	LCN Service-3	TPS 3060-3
<i>Core Module Test System</i>	SW13-512	LCN Service-3	TPS 3060-3
<i>LCNI Network Communications Test</i>	SW13-513	LCN Service-3	TPS 3060-3
<i>Universal Station^X System Administration Manual (Release 200)</i>	UX11-402	Universal Station ^X	TPS 3095
<i>Keyboards</i>	SW09-508	Implementation/ Engineering Operations - 1	TPS 3032-1

NOTE: Further reference to OEM publications are listed in Appendix A.

Section 1 – Universal Station^X Introduction

1.1 Hardware Overview

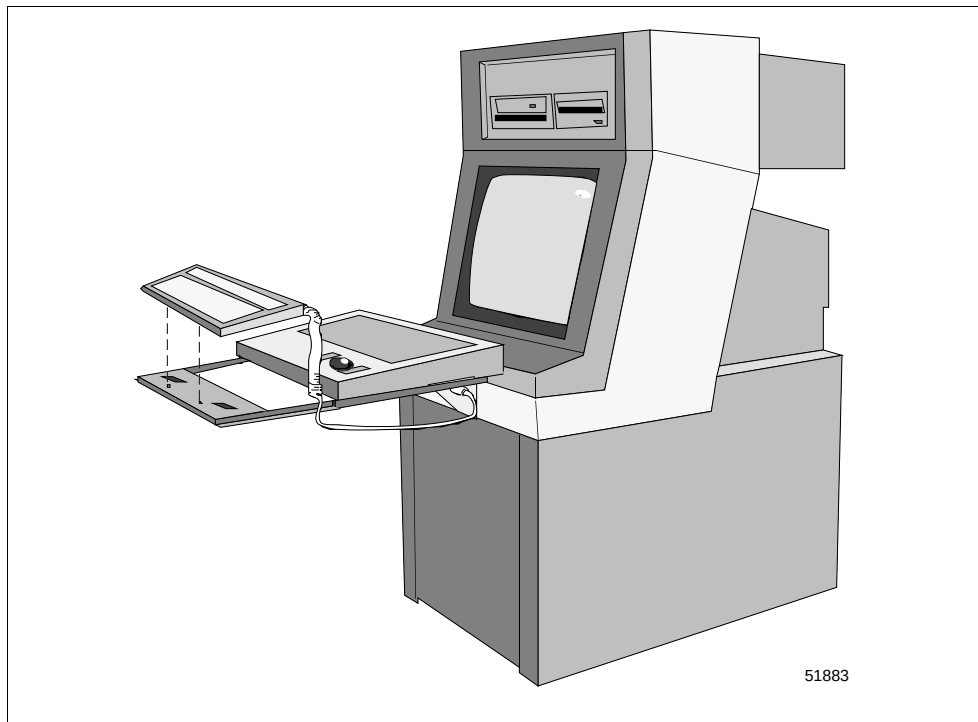
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1.3	Universal Station ^X Shutdown	14
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1.5	Universal Station ^X Operation	18
1.6	Universal Station ^X Node Troubleshooting	20

General introduction

The Universal Station^X provides the man-machine interface to the TPS system integrated with a UNIX Workstation to provide an XWindow open-systems environment.

Figure 1-1 Universal Station^X



1.1 Hardware Overview, Continued

Universal Station^X components

The Universal Station^X contains the following components

- unique board (electronics) complement
 - TPDGX, Turbo Display Generator and TPDG I/O
 - Hard Disk Drive Tray (HDDT) and HDDT I/O
 - WSI2, Work Station Interface and WSI2 I/O (or WSI2R I/O)
 - K2LCN-8/K4LCN-8, processor/LCN Interface/8 megabyte memory board and CLCN A/B (replaces LCN I/O)
- high-resolution color monitor (1 or 2)—1280 x 1024 pixels
- 525 MB/1 GB Hard Drive mounted on HDDT
- DAT (Digital Audio Tape) drive (optional)
- cursor Control Function
 - Touchscreen capability on the monitor (optional)
 - Trackball built into keyboard tray (optional)
 - Mouse (optional)
- QWERTY Operator keyboard
- U^XS Auxiliary Keyboard (Engineer's Keyboard)
- 150 MB Cartridge Drive (optional)
- CD-ROM drive (provides access to UNIX documentation)
- Media Access Unit, provides access to an 802.3 LAN (Ethernet)
- coprocessor Laser Printer (optional—connected directly to PIN)
- LCN Printer (optional)
- alarm Annunciator

The CD-ROM drive can be mounted in place of either the DAT or cartridge drive.

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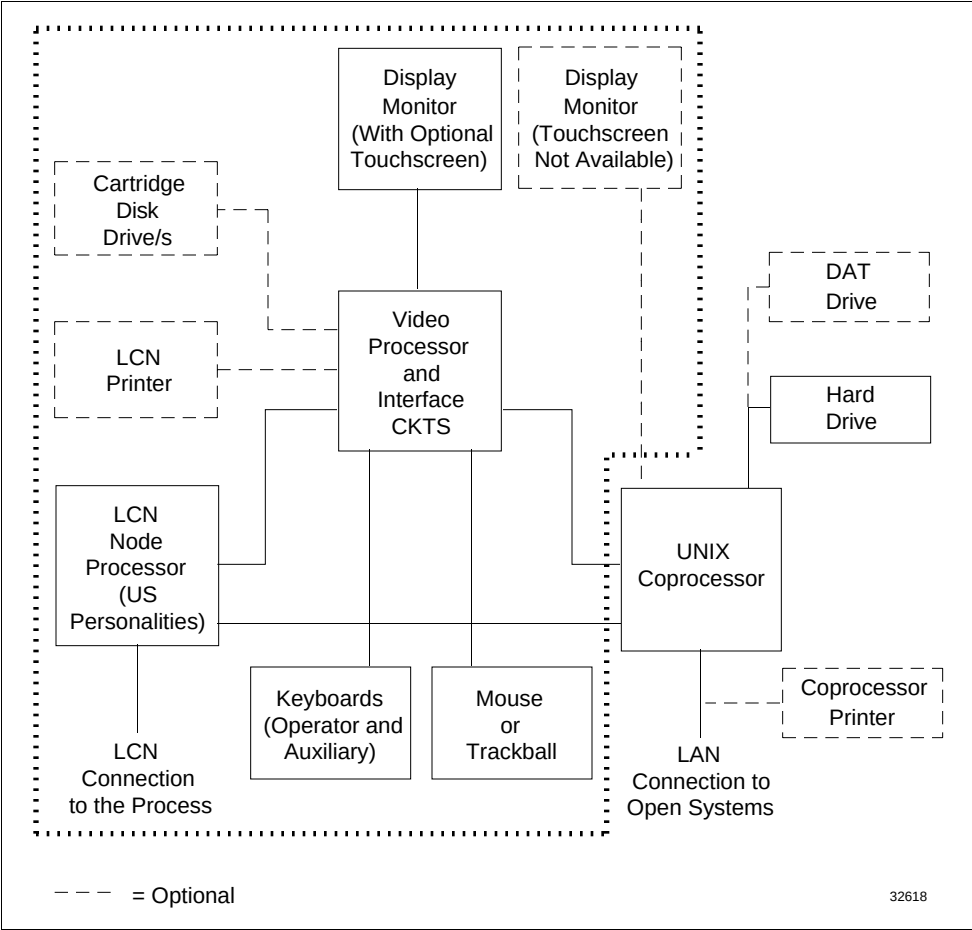
1.1 Hardware Overview, Continued

Functional description

The Universal Station^X provides a simultaneous view of the process and LAN (Ethernet environments.) An embedded PA-RISC UNIX coprocessor (HP 743) shares the monitor display and keyboard; however, functions are executed in their native processor. An WSI2 with optional graphics is required to support a second monitor.

Illustration

Figure 1-2 Block Diagram—LCN Perspective



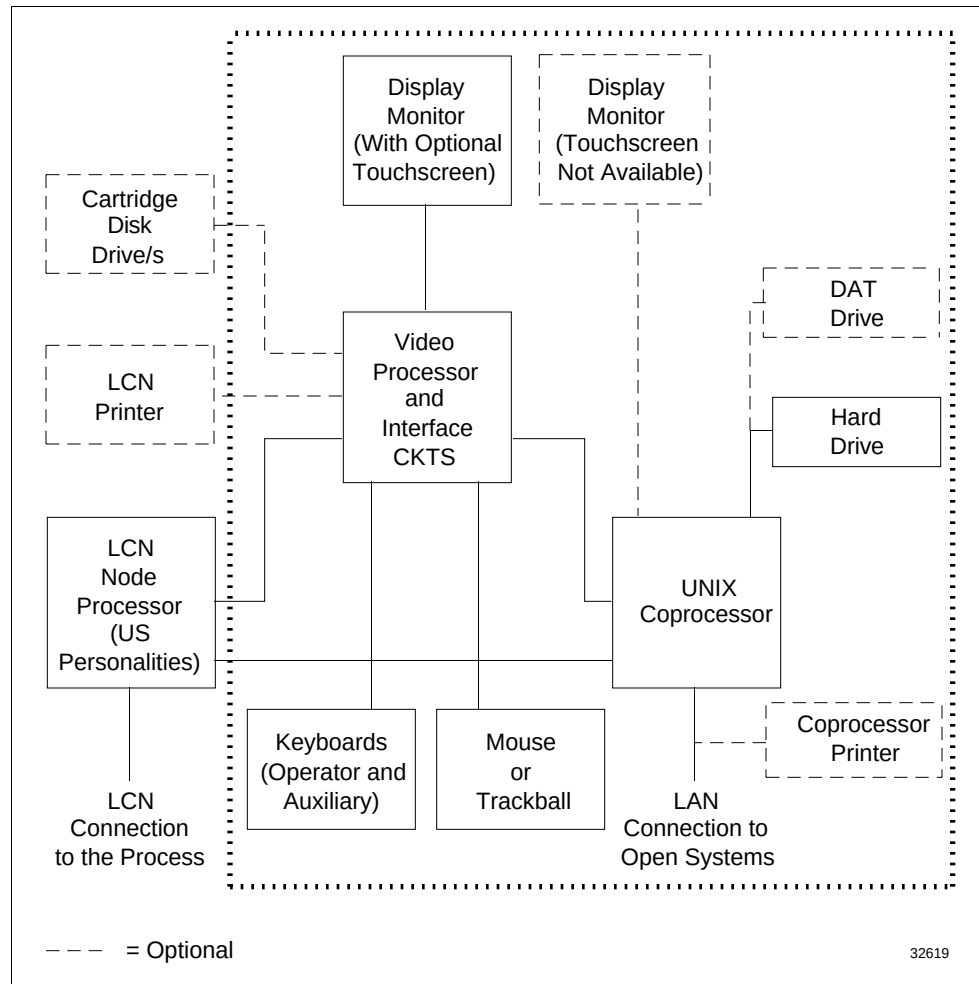
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1.1 Hardware Overview, Continued

Illustration

Universal Station^X provides for a window to the external world. The window to the world is accomplished by the coprocessor system on the WSI board, its hard drive, DAT drive, printer, and LAN connection.

Figure 1-3 Block Diagram—Coprocessor Perspective



Continued on next page

1.1 Hardware Overview, Continued

ATTENTION

ATTENTION—Different assemblies were required for CE Compliancy. Some have new board names (New assembly numbers) and some still have the same name (but a new tab number on the part number). Boards with new names are discussed separately below.

Monitor

The 19" pillow-tube color monitor provides 1280 X 1024 pixel resolution. A second monitor is optional. When present, a second monitor acts as if it is an extension of the first (lower) monitor.

TPDGX board

The TPDGX (Turbo Peripheral Display Generator) is a high-speed dual ported graphics board supporting X Windows graphics from the WSI board and TPS graphics.

The TPDGX contains an MC68020 main processor, 16 MB RAM, and a TMS 34020 display controller, which provides a display resolution of 1280 x 1024 pixels. It is implemented as a superset of all EPDG features including 640 x 448 graphics resolution and peripheral device control.

The TPDGX board is located in Slot 5 of the Five-Slot Module Chassis.

TPDG I/O board

This I/O module board provides the input/output connections for the TPDG board (other than the free-edge ribbon cables connecting the TPDG and the WSI boards):

The TPDG I/O board is located in Slot 5 of the rear I/O Board Chassis.

TPDGC I/O board

This CE Compliant I/O module board provides the input/output connections for the TPDG board (other than the free-edge ribbon cables connecting the TPDG and the WSI boards):

The TPDGC board is located in Slot 5 of the rear I/O Board Chassis.

HDDT board

Hard Disk Drive Tray can support one or two 525 MB or 1 GB hard drives.
Connects to WSI2 board by a ribbon cable on the front edge of the boards.

The HDDT is located in slots 3 and 4 due to the height of the board.

HDDT I/O board

The HDDT I/O board provides a SCSI interface to the DAT drive and CD-ROM drive. A terminator is required if the cable drive is not present.

Continued on next page

1.1 Hardware Overview, Continued

WSI2 board	A module board that provides connections to the HP743 coprocessor and LAN daughter boards. The WSI2 board holds 32 MB, 64 MB or 128 MB of memory including shared memory accessible to both the TPS and coprocessors. The WSI2 board is located in Slot 2 of the Five-Slot Module Chassis.
WSI2 I/O board	An I/O module board that provides the input/output connections for the WSI2 board (other than the free-edge ribbon cables connecting the TPDG and the WSI2 boards). This board is used in factory built Release 200 units. The WSI2 I/O board is located in Slot 2 of the rear I/O Board Chassis.
WSI2R I/O board	An I/O module board that provides the input/output connections for the WSI2 board (other than the free-edge ribbon cables connecting the TPDG and the WSI2 boards). This board is used in Release 100 units that were field upgraded to Release 200 status. This board has a SCSI connector to connect to the 400 MB hard drive. The WSI2 I/O board is located in Slot 2 of the rear I/O Board Chassis.
Coprocessor daughter board	This UNIX coprocessor is a daughter board inserted into the WSI2 board. The board is a PA-RISC processor with RAM memory • 32 MB, with one 32 MB daughter board or two 16 MB daughter boards • 64 MB, with one 64 MB daughter board or two 32 MB daughter boards • 128 MB, with two 64 MB daughter boards The coprocessor executes HP-UX version 9.0 (or later) operating system.
LAN daughter board	This LAN interface is a daughter board mounted on the WSI2 board. It connects to the Ethernet transceiver (MAU) through the WSI2 I/O board and a two meter LAN interface cable.
K2LCN-8 board	A standard TPS module board. This board provides the processor (68020), memory (8 megabytes) and LCN interface to execute the TPS software personality. The K2LCN-8 must have revision B (or later) firmware. The K2LCN-8 board is located in Slot 1 of the Five-Slot Module Chassis.

Continued on next page

1.1 Hardware Overview, Continued

K4LCN-8 board	A standard TPS module board. This board provides the processor (68040), memory (8 megabytes), and LCN interface to execute the TPS software personality. The K4LCN-8 board is located in Slot 1 of the Five-Slot Module Chassis.
LCN I/O board	This board provides the K2LCN/K4LCN board with the I/O connection to the LCN and is located in Slot 1 of the rear I/O chassis.
CLCN A/B board	This board provides the K2LCN/K4LCN board with the I/O connection to the LCN and is located in Slot 1 of the rear I/O chassis. This board is the CE Compliant replacement for the LCN I/O board.
Removable media drives	Drive(s) for the following types of removable media are located above the monitor • cartridge drive, single or dual. The second cartridge drive, if installed, replaces the DAT drive. • Digital Audio Tape (DAT) drive, single with optional single cartridge drive. • CD-ROM—UNIX bookset access
Coprocessor storage	The coprocessor can have two types of storage media • one or two 525 MB or 1 GB Winchester drive(s) (hard drive) • an optional 2 GB Digital Audio Tape (DAT) used for backup of the hard drive and loading of coprocessor software.
Operator Keyboard	The Operator Keyboard provides access to the TPS system functionality, controlled by the level of access allowed by the keyswitch setting • operator functions • supervisor functions • engineer functions Refer to the <i>Process Operations Manual</i>, <i>Keyboards</i>, and engineering functions manuals for a more detailed description of this functionality.
UXS Auxiliary Keyboard	The UXS Auxiliary Keyboard provides additional engineering functionality not available on the Operator Keyboard. This keyboard can be made to emulate many devices connected to the Ethernet. Refer to the <i>Keyboards</i> manual. The keyboard can be placed on the extended shelf that is part of the keyboard tray.

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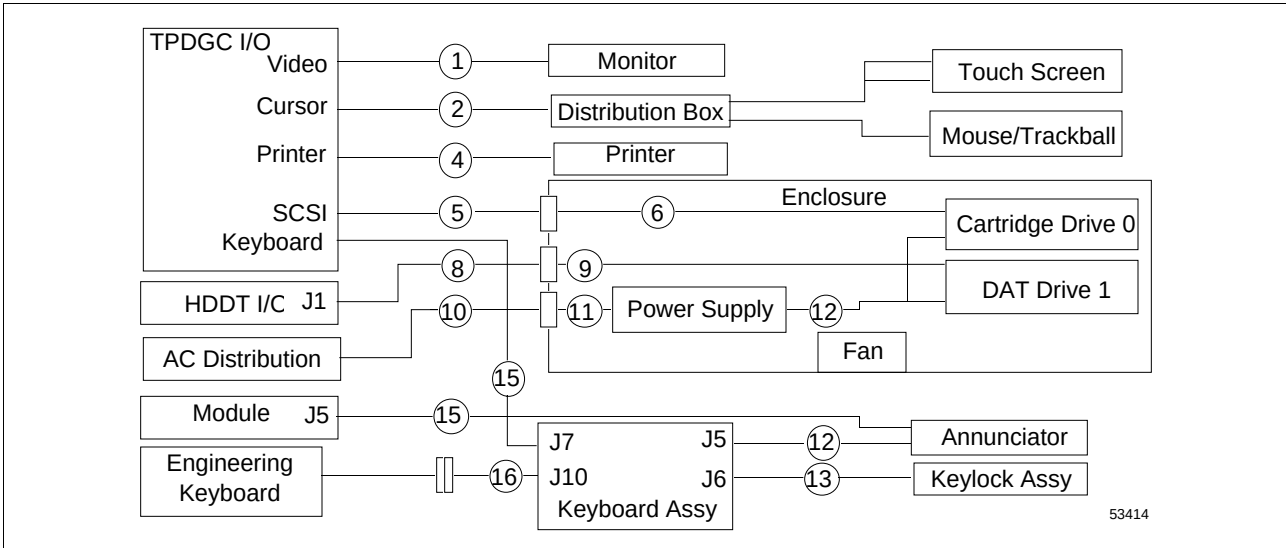
1.1 Hardware Overview, Continued

Installation procedures	Follow standard TPS Universal Station installation procedures covered in <i>LCN System Installation</i> manual.
Universal Station ^X LAN components	There are two Universal Station^X LAN components: • LAN Interface Cable from the WSI I/O board to the MAU • MAU (Media Access Unit) Ethernet Transceiver
LAN interface cable	The LAN interface cable must be installed during U ^X S installation. This cable connects the WSI2 I/O board LAN port (J7) to the MAU.
Ethernet transceiver	The provided MAU is connected to the existing Ethernet network. Consult the local LAN administrator before attempting the Ethernet connection.
ATTENTION	ATTENTION—The MAU has two DIP switches; both must be in position 0.

1.2 Universal Station^X Cabinet Cabling

CE Compliant cable connections Cable referencing for a CE Compliant Universal Station^X in classic furniture is shown in Figure 1-4. Refer to Table 1-1 for cable identification.

Figure 1-4 U^XS Cabling—Classic Furniture with Touchscreen and Mouse/Trackball



Continued on next page

1.2 Universal Station^X Cabinet Cabling, Continued

CE Compliant cable reference table

Table 1-1 is a list of cable references for Figure 1-4.

Table 1-1 Universal Station^X Cabling—Classic Furniture

Reference Number	Part Number	Application Notes
1	51191784-300	Cable, CRT Signal
2	51308119-100	TPDGC I/O Distribution Box Cable
3	51308123-100	CE Compliant Smartscreen Cable
4	51308103-003	Printer Cable – 3 meters/9.8 ft.
	51308103-008	Printer Cable – 8 meters/26.3 ft.
	51308103-015	Printer Cable — 15 meters/49.2 ft.
5	51192054-102	Cable,SCSI Shielded, 50-Pin
6	51305070-202	Cable, SCSI
7		Not Applicable
8	51192054-102	Cable, SCSI, Shielded, 50-Pin
9	51305070-103	Cable, SCSI
10	51308094-600	Classic Module AC Power Cable
11	51305342-100	Power Supply Enclosure AC Cable (Power Input to Skynet Power Supply)
12	51305341-100	Power Supply Enclosure DC Cable
13		Not Applicable
14	51303420-586	Module status cable
15	51308126-100	Keyboard Cable
16	51304496-026	Engineering Keyboard Interface Cable
17	51304047-652	Annunciator Power cable
18	51304029-500	Keylock Switch Cable

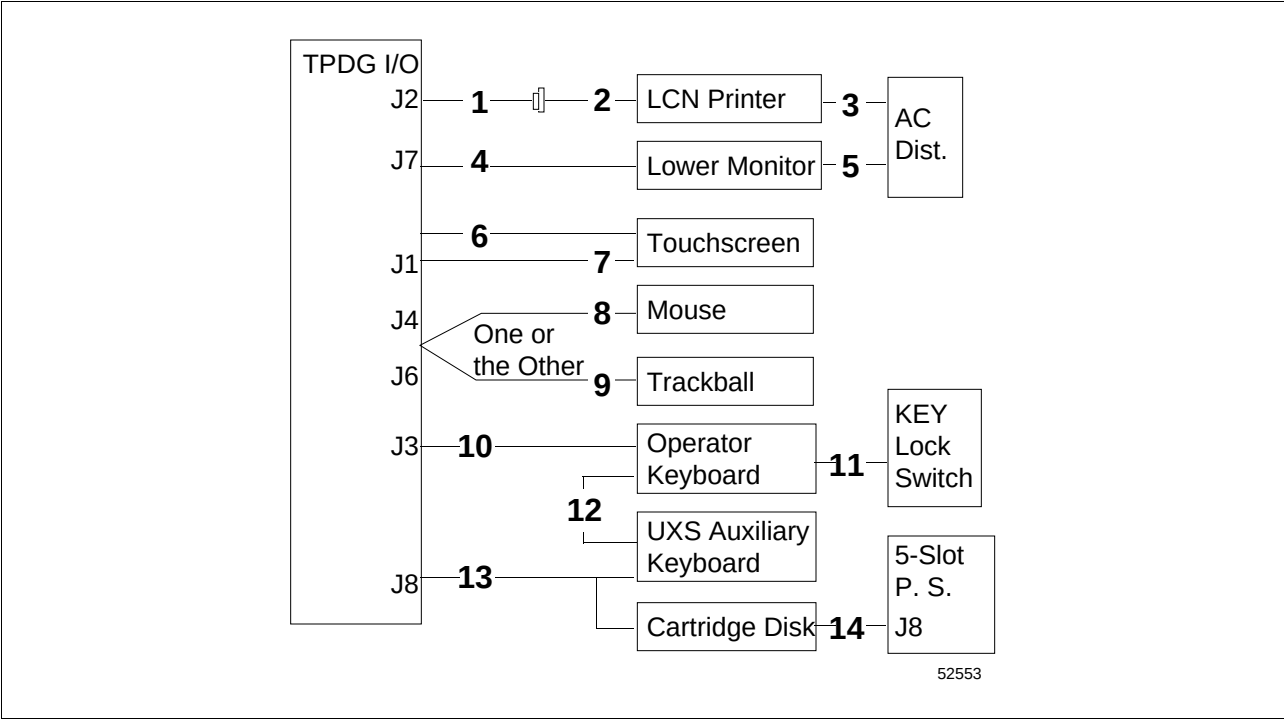
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1.2 Universal Station^X Cabinet Cabling, Continued

Pre-CE Compliant cable connections

Figure 1-5 illustrates TPDG I/O cabling for stations **manufactured before CE Compliance requirements**. Refer to Table 1-2 for cable identification.

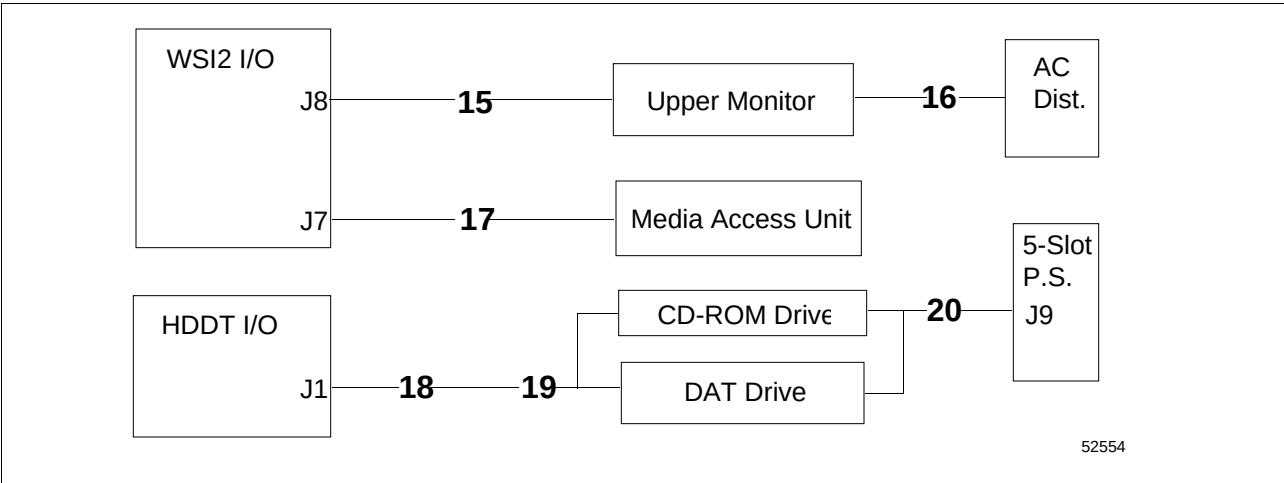
Figure 1-5 Pre-CE Compliant Universal Station^X TPDG I/O Cabling



Pre-CE Compliant cable connections

Figure 1-6 illustrates WSI2 I/O and HDDT I/O cabling for stations **manufactured before CE Compliance**. Refer to Table 1-2 for cable identification.

Figure 1-6 Pre-CE Compliant Universal Station^X WSI2 I/O and HDDT I/O Cabling



Continued on next page

1.2 Universal Station^X Cabinet Cabling, Continued

Pre-CE Compliant
cable reference table

Table 1-2 Cabling Key List

Reference Number	Part Number	Identification
1	51304887-100	Printer Signal Cable (3 meter)
2	51196218-300	Printer Ferrite Cable
3	—	LCN Printer Power cable provided with printer
4	51191784-100	Monitor Signal Cable
5	51304374-100	Monitor Power Cable, 110 Vac
	51304374-200	Monitor Power Cable, 220 Vac
6	51304030-150	Touchscreen Interface Cable
7	51304031-150	Touchscreen Power Cable
8	51304841-100	Mouse Cable
9	51108313-200	Trackball Cable
10	51303409-122	Operator Keyboard Signal Cable
11	51304029-100	Keyboard Switch Cable
12	51304496-026	Roving Engineer Keyboard Cable
13	51304842-100	LCN Cartridge Disk Interface Cable
14	51304769-300	LCN Cartridge Disk Power Cable
15	51191784-100	Monitor Signal Cable
16	51304374-100	Monitor Power Cable, 110 Vac
	51304374-200	Monitor Power Cable, 220 Vac
17	51191783-100	MAU Signal Cable
18	51192054-103	HDDT to Bracket SCSI Cable, Upper Pod 3.0 m
	51192054-104	HDDT to Bracket SCSI Cable, Table Top 1.5 m
19	51305070-200	Bracket to DAT/CD-ROM SCSI Cable
20	51304769-300	DAT/Cartridge/CD-ROM Power Cable
	51304585-200	Fan Power Cable—No Cartridge or DAT Drive
	51304585-300	Fan Power Cable—Cartridge or DAT Drive

Continued on next page

1.2 Universal Station^X Cabinet Cabling, Continued

Cable connections Figures 1-7 and 1-8 illustrate cabling for the stations that have been upgraded. Refer to Table 1-3 for cable identification.

Figure 1-7 Universal Station^X TPDG I/O Cabling—Upgraded Units

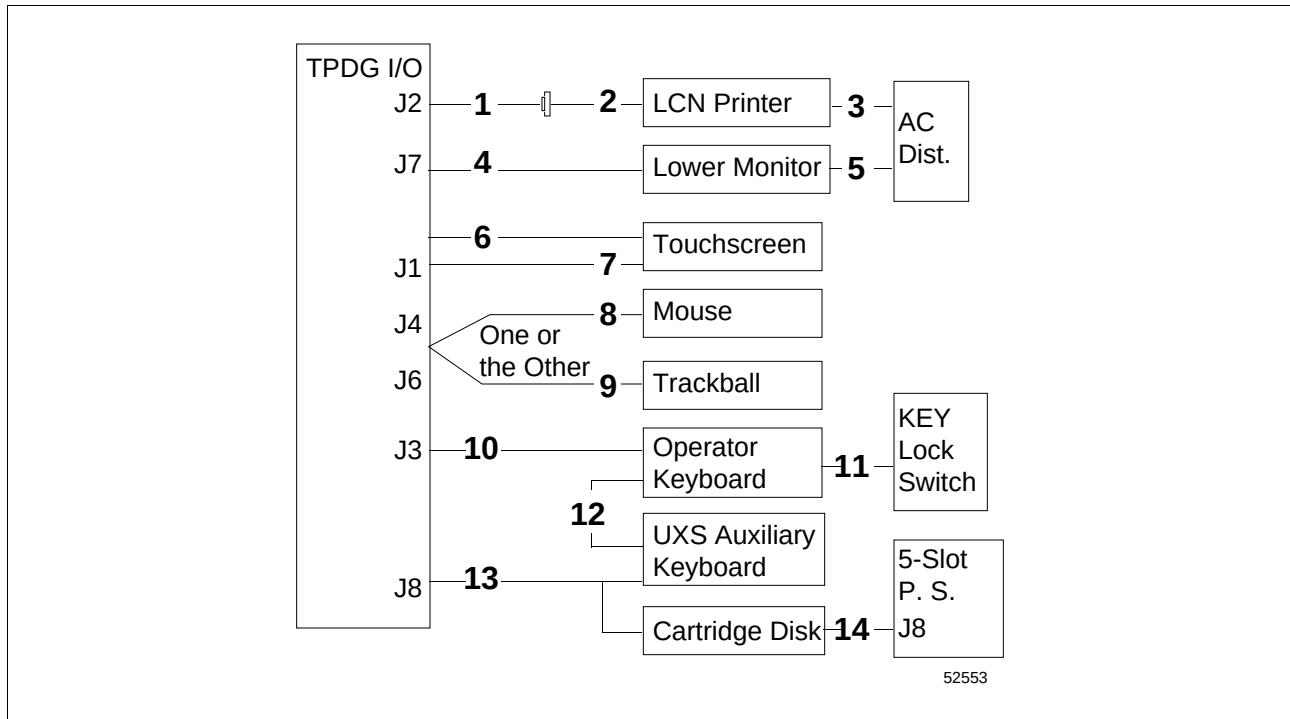
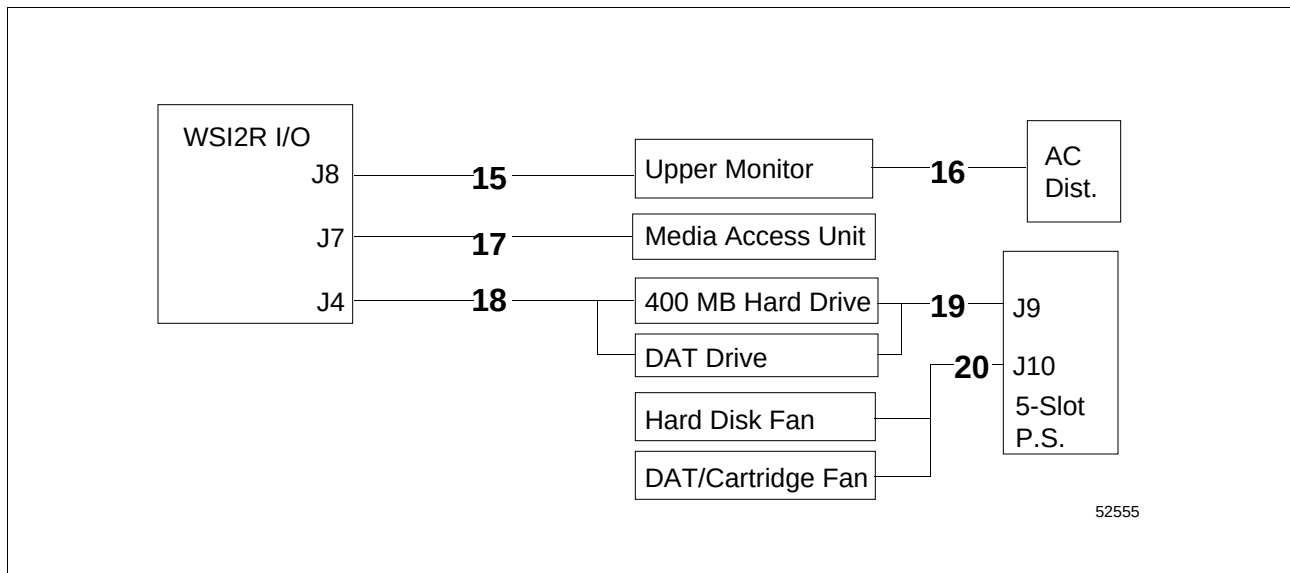


Figure 1-8 Universal Station^X WSI2R I/O Cabling—Upgraded Units



Continued on next page

1.2 Universal Station^X Cabinet Cabling, Continued

Cable part numbers

Table 1-3 references cable item numbers for Figures 1-7 and 1-8.

Table 1-3 Cabling Key List

Reference Number	Part Number	Identification
1	51304887-100	Printer Signal Cable (3 meter)
2	51196218-300	Printer Ferrite Cable
3	—	LCN Printer Power Cable Provided with Printer
4	51191784-100	Monitor Signal Cable
5	51304374-100	Monitor Power Cable, 110 Vac
	51304374-200	Monitor Power Cable, 220 Vac
6	51304030-150	Touchscreen Interface Cable
7	51304031-150	Touchscreen Power Cable
8	51304841-100	Mouse Cable
9	51108313-200	Trackball Cable
10	51303409-122	Operator Keyboard Signal Cable
11	51304029-100	Keyboard Switch Cable
12	51304496-026	Roving Engineer Keyboard Cable
13	51304842-100	LCN Cartridge Disk Interface Cable
14	51304769-300	LCN Cartridge Disk Power Cable
15	51191784-100	Monitor Signal Cable
16	51304374-100	Monitor Power Cable, 110 Vac
	51304374-200	Monitor Power Cable, 220 Vac
17	51191783-100	MAU Signal Cable
18	51304837-100	Hard Disk/DAT Drive SCSI Interface Cable w DAT
	51304839-100	Hard Disk Drive SCSI Interface Cable without DAT
19	51304836-100	Hard Disk Power Cable with DAT Drive
	51304838-100	Hard Disk Power Cable without DAT Drive
20	51304585-200	Drive Fan Power Cable without Cartridge or DAT
	51304585-300	Drive Fan Power Cable with Cartridge or DAT

1.3 Universal Station^X Shutdown

CAUTION

CAUTION—It is imperative that both the LCN node processor and UNIX coprocessor operations be terminated prior to powering off the Universal Station^X.

Failure to do so can cause damage to the coprocessor file system. It also causes a disruption to the orderly token passing scheme on the LCN. (Node goes isolated.)

Following the shutdown procedure ensures the integrity of the process environment and the coprocessor file system.

ATTENTION

ATTENTION—The user logout steps are not needed if there is no user logged onto the coprocessor at the time. This situation can be recognized by observing if a login/shutdown prompt is present across the bottom of the display.

Procedure Overview

The shutdown procedures include these operations:

- User logout
- Coprocessor shutdown
- LCN node processor shutdown
- Power off

Follow the steps in Table 1-4 to properly power down the station.

User logout

Steps 1-5 are required if a user is currently logged onto the coprocessor. Begin with step 6 if the user has already logged out.

Table 1-4 Shutdown Procedure—Coprocessor Logout

Step	Action	Remarks
1	Move the display cursor to a blank portion of the display screen using the mouse (or trackball).	The Work Space menu can only be accessed from a blank portion of the screen.
2	Press the left mouse button (or select button on trackball assembly).	This displays the Work Space menu.
3	Select the X-MENU	X-MENU appears.
4	Select the LOGOUT target.	Quit Mwm ? prompt appears.
5	Select the OK target.	Logout process commences. The login/shutdown window appears after several seconds.

Continued on next page

1.3 Universal Station^X Shutdown, Continued

ATTENTION

ATTENTION—The user must be authorized to shutdown the coprocessor—see IWSM in the *Universal Station^X Administration Manual*.

Table 1-4 Shutdown Procedure—Coprocessor Logout, Continued

Step	Action	Remarks
6	Enter your user name and password.	Do not press the carriage return or enter key after entering the password.
7	Select the SHUTDOWN WSI target.	This will initiate the coprocessor shutdown operation. A message indicating the coprocessor is in the idle state appears after several seconds. Note: An optional method of coprocessor shutdown can be forced by using the shutdown switch on the WSI2 board free edge (see Figure 6-3) and waiting for the yellow indicator to turn off. This performs a proper shutdown of the coprocessor file system, but it does not perform an orderly departure from the LAN. Other LAN nodes will see it as a failure of this node. This technique should be avoided. If you do use this procedure return the switch to the Normal position.

Continued on next page

1.3 Universal Station^X Shutdown, Continued

LCN node processor shutdown

Table 1-4 Shutdown Procedure—Coprocessor Logout, Continued

Step	Action	Remarks
8	Press the CONSOLE STATUS key on the operator keyboard.	The Console Status display appears.
9	Select your Universal Station ^X .	The selected node number turns white.
10	Select the SHUTDOWN target.	“Press enter to execute” prompt appears at the top of the display.
11	Press ENTER.	The shutdown of the LCN node processor in the Universal Station ^X commences. The Console Status display disappears from the screen.
12	Option: Do a second SHUTDOWN from another US.	If you do a second SHUTDOWN sequence, the node will go to FAILED instead of ISOLATED and you avoid several error messages in the journal.

Continued on next page

1.3 Universal Station^X Shutdown, Continued

Power off

Table 1-4 Shutdown Procedure—Coprocessor Logout, Continued

Step	Action	Remarks
13	Turn OFF the power switch on the 5-slot chassis power supply.	The front of the power supply is accessible through the rear door of the station furniture. This only removes power from the five-slot chassis.
14	Turn OFF the circuit breaker located on the AC power distribution panel.	It is located inside the door under the operator keyboard. (top right) This step removes power from all components in the Universal Station ^X furniture. Free-standing printers can be separately powered.

1.4 Universal Station^X Startup

Reference

Refer to *Universal Station^X System Administration Manual* and *Universal Station^X User's Guide* for boot and startup procedures.

1.5 Universal Station^X Operation

Power switches

There are two power ON/power OFF devices in a Universal Station^X.

- 15 amp, AC breaker located in side the front door, under the keyboard.
 - AC power switch located on the Five-Slot Module, that controls the module power and all power to all the peripherals except the monitor.
-

Keyboards

Refer to the *Keyboards* manual.

Cartridge operation

Table 1-5 Cartridge Operating Procedure

Removable Media Drive	Function	Operation
Cartridge Drive	Load	Insert the cartridge in the drive—the LED will flash until the drive has spun up to speed.
	Unload	Press the unload button—the LED will flash until the drive has unloaded. Remove the cartridge by pulling the cartridge out of the drive.
	Access	LED will light when the cartridge is accessed.

DAT operation

Table 1-6 DAT Operating Procedure

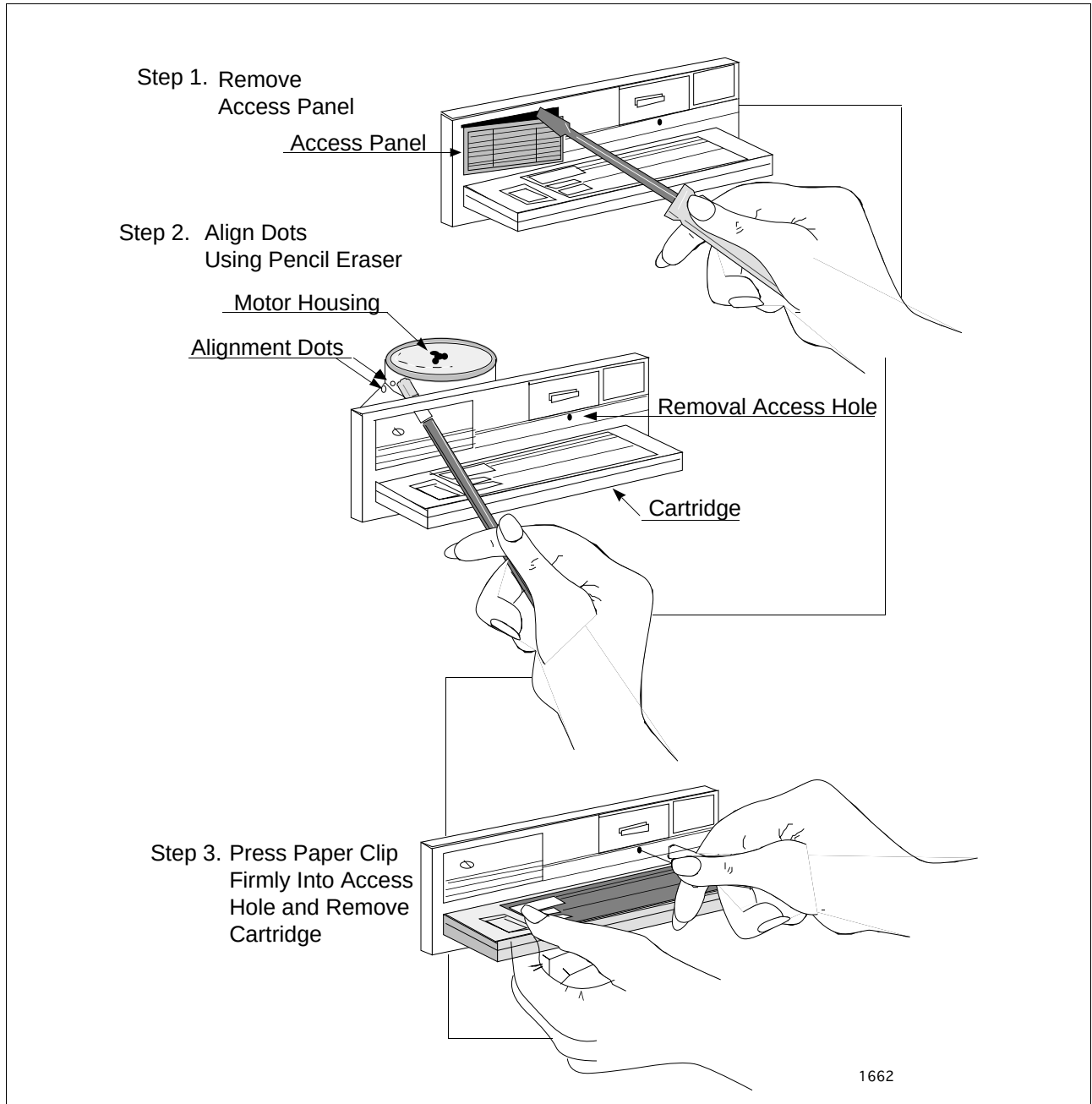
Removable Media Drive	Function	Operation
Digital Audio Tape (DAT) Drive	Load	Insert the cassette with the arrow on top of the cassette pointing toward the drive. The LEDs will flash until the cassette is loaded.
	Unload	Press button to Unload. LED will flash until cassette is unloaded.
	Errors	Refer to DAT section in this manual.

Continued on next page

1.5 Universal Station^X Operation, Continued

Procedure

Figure 1-9 Powered Down Cartridge Removal



1.6 Universal Station^X Node Troubleshooting

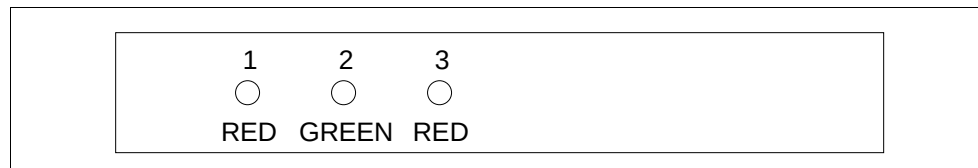
Overview

Use standard hardware practices common to the Five-Slot Module for troubleshooting/service procedures. The added UNIX system components have status indicators similar to those found in existing U^XSs.

TPDGX status

The TPDGX board has three status indicator LEDs. They are found on the left side along the front edge of the board.

Figure 1-10 TPDGX Indicators



Use the table below to interpret TPDGX LED indicators:

Table 1-7 TPDGX Indicators

LED	Indication	Action
1	Board fails self-test	Replace module board
2	Board passes self-test	Good indication—hardware OK
3	Bus transaction error	Investigate further—possible backpanel problems

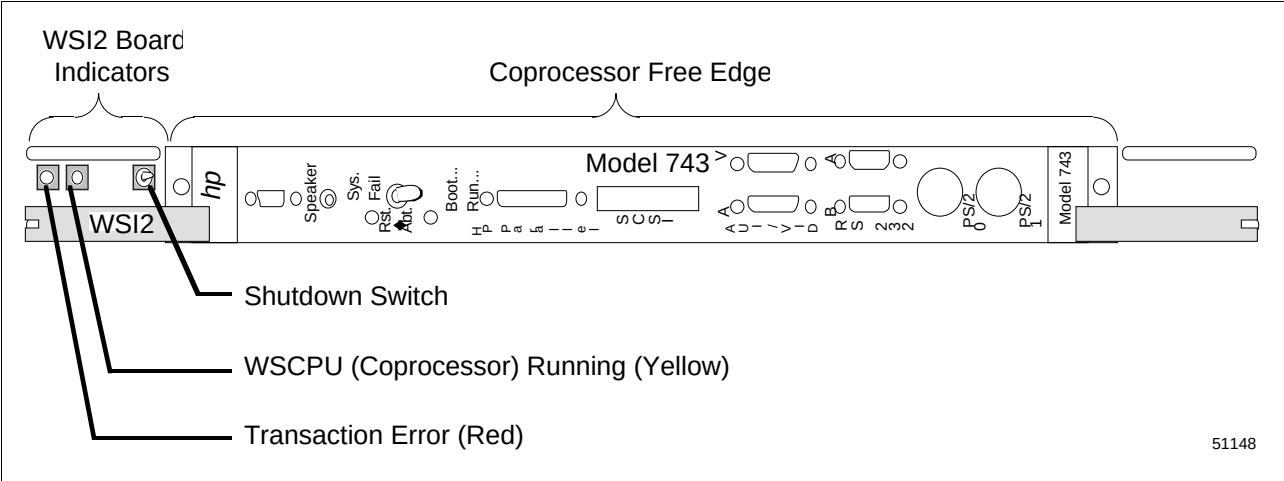
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1.6 Universal Station^X Node Troubleshooting,

Continued

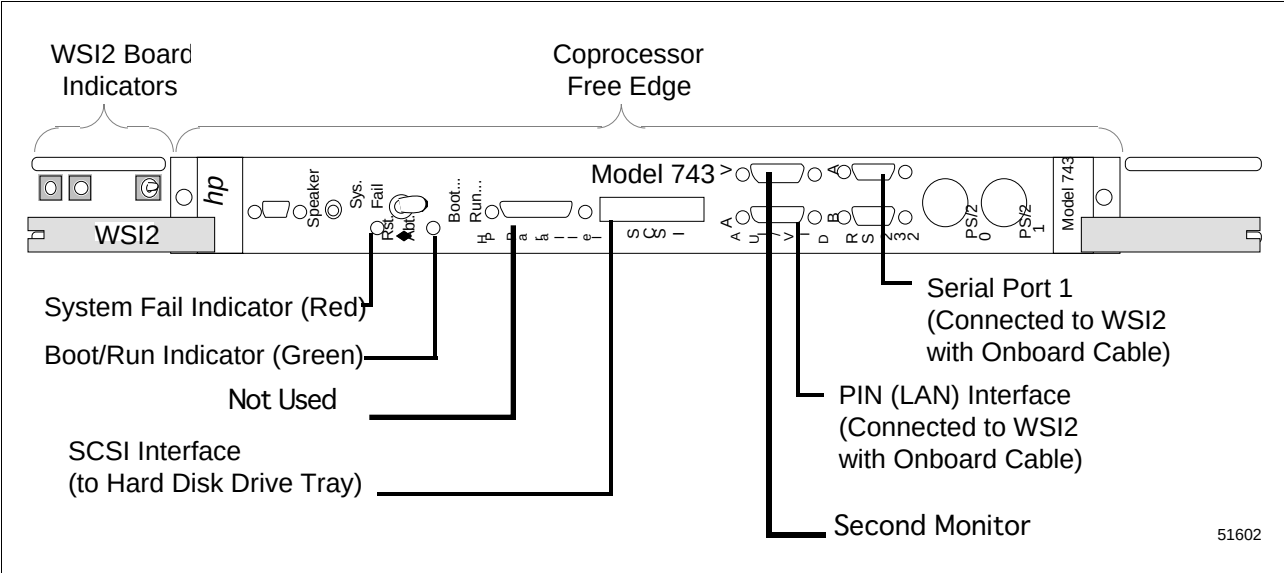
WSI status The WSI2 board has four status indicator LEDs. They are found on the left side along the front edge of the board (as viewed from power supply side of 5-slot module).

Figure 1-11 WSI Indicators



Use the figure below to interpret PA-RISC LED indicators:

Figure 1-12 Coprocessor Indicators



Continued on next page

1.6 Universal Station^X Troubleshooting, Continued

Coprocessor does not boot

Table 1-8 Coprocessor Troubleshooting

Step	Action
1	DO NOT TURN POWER OFF! Booting the coprocessor can take up to 5 minutes. If, after 6 minutes the banner does not appear at the bottom of the Monitor screen, proceed to Step 2; otherwise go the Step 3.
2	Ensure the boards (TPDG, TPDGC (or TPDG I/O), WSI, WSI I/O, K2LCN/K4LCN, CLCN A/B (or LCN I/O) are properly seated in the correct slot positions with all cabling correctly installed and secure.
3	At Power ON: • The HP 382 searches its SCSI (pronounced “scuzzy”) bus for a bootable device. • Finding the internal hard drive, it loads the Unix file system. • The coprocessor performs FSCK, which automatically repairs inconsistencies in the file system. ATTENTION This repair adds time to the coprocessor boot process (less than 5 minutes total boot time for 400 MB hard drives. DO NOT GET IMPATIENT AND CYCLE POWER DURING FSCK REPAIR. Unit will recover from an improper shutdown, but may fail if the FSCK operation is interrupted during repairs. This may leave the file system in a nonrepairable state and require the drive to be reformatted and the Unix/Honeywell software to be reloaded. However, if the coprocessor completes the FSCK process, the Honeywell X-Window software is initialized and the coprocessor LOGIN banner is displayed.
4	If the LOGIN banner does not appear within 10 minutes: • Place the Shutdown switch on the WSI board to the shutdown position. This will disable the coprocessor. • Wait 1 minute before turning the Power switch to the OFF position. • Return the Shutdown switch to the Normal position. • Contact your local service representative or call the Honeywell Technical Assistance Center to resolve the problem.

ATTENTION

ATTENTION—If the coprocessor displays the LOGIN banner, refer to the *Universal Station^X User's Guide* for complete booting procedures.

Continued on next page

1.6 Universal Station^X Troubleshooting, Continued

Using Terminal 0
(TERM0)

Table 1-9 Coprocessor Troubleshooting Using a VGA

Step	Action
1	To recover from a failure which prevents the coprocessor from booting, it is necessary to connect a Dec VT-100 compatible monitor to J3 of the WSI I/O board, and an HP compatible keyboard to J1 of the WSI I/O board.

Section 2 – Universal Station^X Service

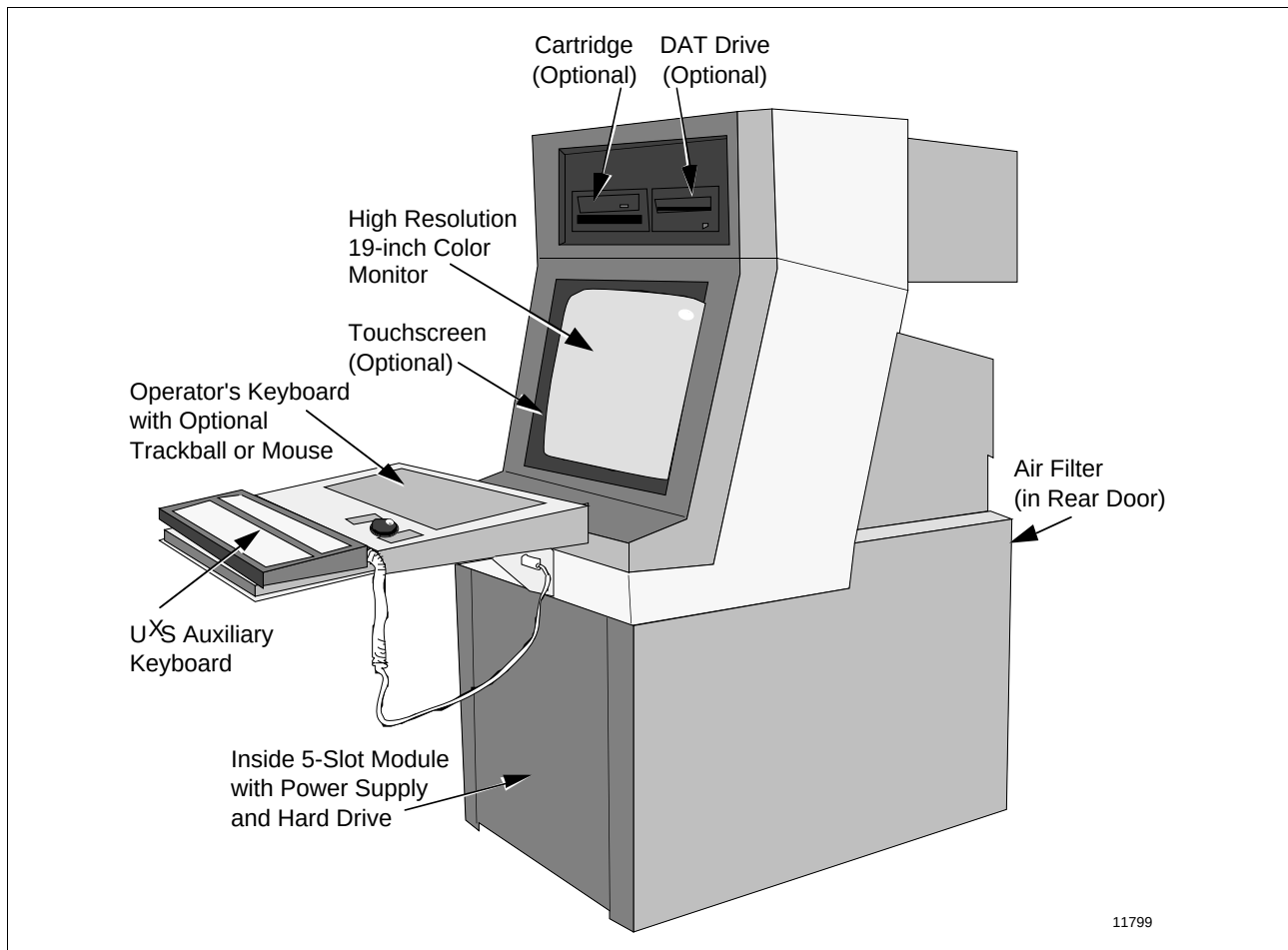
2.1 Universal Station^X Components

Section contents

	Topic	See Page
	SECTION 2 – UNIVERSAL STATION^X SERVICE.....	25
2.1	Universal Station ^X Components.....	25
2.2	Periodic Replacement Items	26

General introduction Initially, each Universal Station^X is delivered in an existing Universal Station furniture configuration. The various components that comprise the Universal Station^X are shown in Figure 2-1.

Figure 2-1 Universal Station^X ORU Locations



2.2

Periodic Replacement Items

Items requiring periodic replacement

Table 2-1 lists the items replaced periodically. They are in alphabetical order with the applicable Honeywell part numbers. Each device within the Universal Station has its own section in this manual, except for the devices common to all USs, such as the Cartridge drives, printers, mouse, and trackball.

Periodic replacement parts

Table 2-1 Periodic Replacement Parts

Item Name	Part Number*
Air Filter, Cabinet – 74 x 17 cm (29 x 6 inches)	51201201-100
Air Filter, Console – 42 x 35 cm (16 x 13 inches)	51201201-200
Battery, WSI	51196250-200
Cartridge Disk (20 MB)	51195169-100
Cartridge Disk Device Cleaning Kit (20 MB)	51195169-101
35 MB Multidisk 150 MB compatible 5.25" cartridge	51196485-100
150 MB Multidisk 150 MB compatible 5.25" cartridge	51196485-400
Cartridge Drive Cleaning Kit—includes • cleaning cartridge • three 0.25 oz bottles of cleaning solution • two extra cleaning pads	51196485-500
Cartridge Drive Cleaning Solution—refill kit • three 0.25 oz bottles of cleaning solution	51196485-600
Wiper Arm Pads (quantity of 2)	51196485-700
DAT Blank Tape	51191814-100
DAT Head Cleaning Cassette Kit (Useful Life = 25 uses)	51196257-200 (HP 92283K)
Keyboard Blank Insert	51400759-100
Laser Printer Toner Cartridge	
Ribbon, Black – ASPI-41	51195091-700

Section 3 – Monitor Service

3.1 Overview

Section contents

These are the topics covered in this section:

Topic		See Page
SECTION 3 – MONITOR SERVICE		27
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3.4	Cleaning	29
3.5	Monitor Cabling	29
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3.7	Removing the Monitor	30
3.8	Installing the Monitor	32
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3.10	Monitor Color Bias Adjustments	35
3.11	Monitor Focus Adjustment	36
3.12	Vertical Linearity Adjustment	37
3.13	Horizontal Linearity Adjustment	38
3.14	Side Pin Cushion Control	39
3.15	Further Adjustments	39
3.16	Fuse Replacement	40
3.17	Spare Parts	40

Optional monitor

There can be a second monitor in the station. The second monitor connects to the WSI2 I/O board.

3.2 Monitor Description

Specifications

Table 3-1 Monitor Specifications

Type	19 Inch
Resolution	1280 x 1024 pixels
Frequency	76-80 kHz
Interface Video Sync Impedance Video	Analog RGB Composite on Green 75 ohm
AC Input Voltage Frequency Power	90 to 132 198 to 264 47 - 63 Hz 165 watts
Weight	25.4 kgs (56 lb)

3.3 Termination Switch Configuration

Switch setting

The termination switch identified in Figure 3-2 should always be in the 75-ohm position.

3.4 Cleaning the Monitor CRT

Monitor Screen

An anti-reflection coating is provided on the CRT screen to reduce eye strain. Soil and fingerprints reduce legibility and the effectiveness of the anti-reflection treatment. To remove soil or fingerprints from the screen, do the following:

- Use a cleaner (such as window cleaning solution) to clean the screen face. Be careful not to scratch or mar the anti-reflective treatment. Before using an untested cleaner, spot test an area at the edge of the screen to determine if the results are satisfactory.
- Remove the cleaner with a damp cloth and wipe the surface with a soft, dry cloth.

Monitor interior

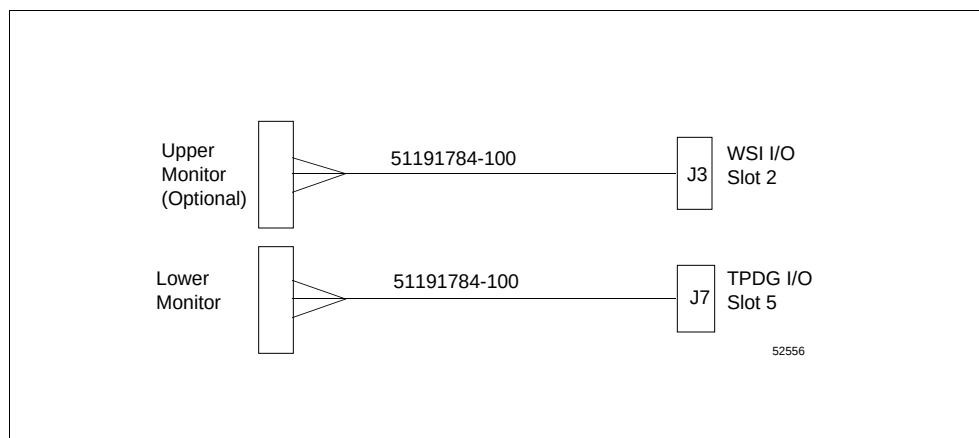
High static electric fields exist inside the monitor during normal operation, attracting dust through the console ventilation slots. Since dust can collect on charged surfaces, it is recommended that you occasionally clean the inside of the monitor, especially around the area of the CRT anode cap. The environmental conditions determine how often the area around the anode cap should be cleaned.

- To remove accumulated dust, use a soft dry, lint-free cloth, a soft brush, or a small vacuum hose to clean the monitor's interior.
- Make sure not to disturb or damage components or wiring connections.
- Check for improperly fitted connectors, sharply bent or unintentionally exposed wiring, and broken plugs or plug insulators.

3.5 Monitor Cabling

Illustration

Figure 3-1 Monitor Cabling



3.6 Monitor Access

Procedure

To access the monitor, remove the rear cover:

Table 3-2 Accessing the Monitor

Step	Action
1	Remove the two hex-head screws on the bottom rear (one on each side) from the cover.
2	Slide the cover off to the rear.

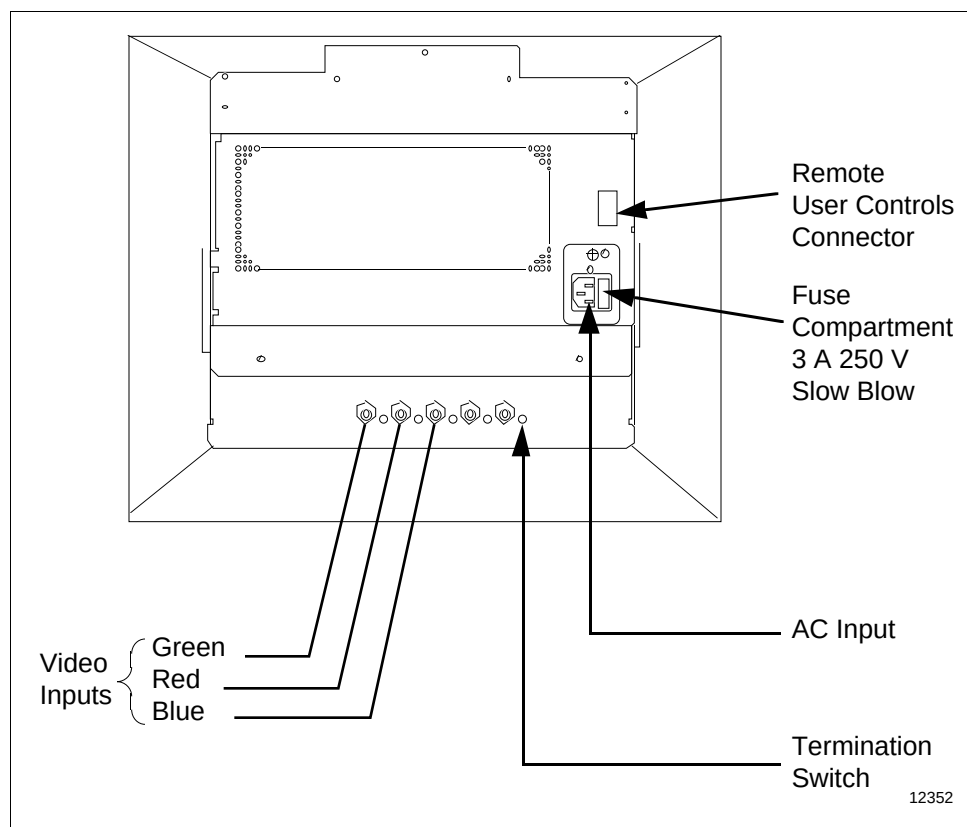
3.7 Removing the Monitor

WARNING

WARNING—Execute the normal LCN and coprocessor shutdown procedure before removing power from the station.

Illustration

Figure 3-2 Monitor Cable Connections



Continued on next page

3.7 Removing the Monitor, Continued

Procedure

To remove the monitor from the Universal Station^X, follow the procedure in Table 3-3.

Table 3-3 Removing the Monitor

Step	Action
1	Open cabinet door below keyboards and set the main power circuit breaker, located in the upper-right corner, to the down (OFF) position.
2	Open rear door below CRT chassis and set the module power switch to OFF.
3	Disconnect the ac power cable from the back of the monitor.
4	Disconnect the three coax cables from the rear of the monitor.
5	Disconnect the remote cable assembly from the remote connector at the rear of the monitor.
6	Ensure that all cables are clear of the monitor.
7	Loosen the retaining blocks that hold the monitor in place.
8	Rotate the retaining blocks out of the way to allow the slide tray to pass.
8	Slide the monitor out of the cabinet as far as it will go.
9	Press the detent button on each slide rail.
10	ATTENTION Because the monitor assembly weighs more than 80 pounds, use two people to slide the monitor out of the slide rail.
11	ATTENTION Use eye protection when handling a monitor.

3.8 Installing the Monitor

Procedure

To install (replace) the monitor from the Universal Station^X, follow the procedure in Table 3-4.

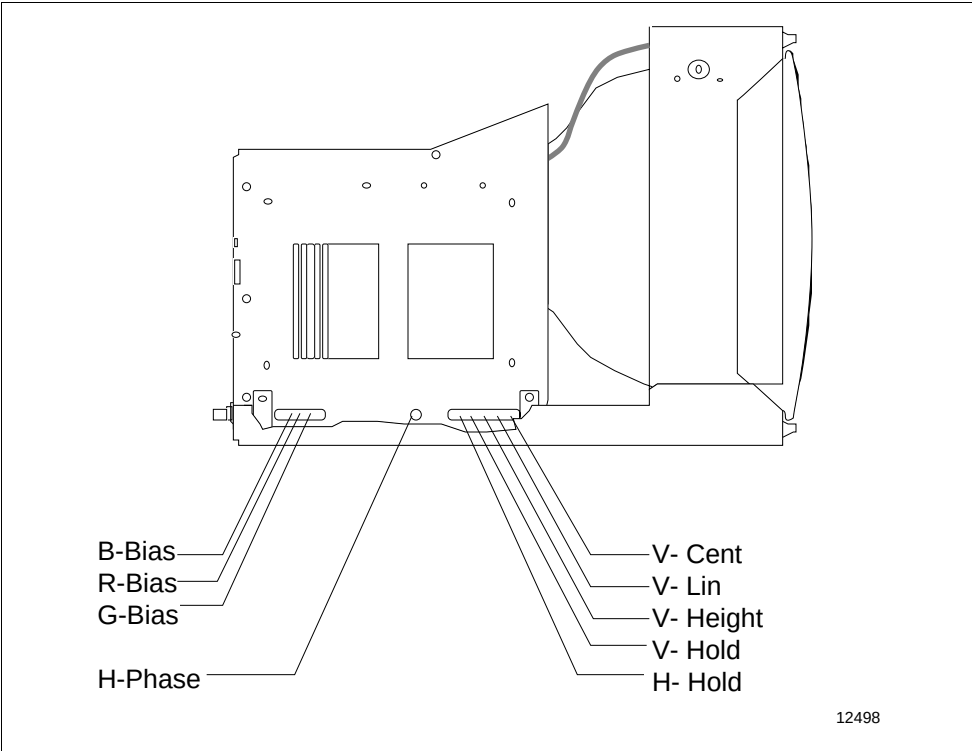
Table 3-4 Replacing the Monitor

Step	Action
1	Ensure that the termination switch is in the 75-ohm position.
2	ATTENTION Because the monitor assembly weighs more than 80 pounds, use 2 people to lift the monitor and slide it into the slide rails, pushing it all the way to the bezel. The slide rails will lock into position automatically—listen for the click.
3	Reverse steps 1 through 8 in Table 3-3.
4	Connect the 3 coax cables to the back of the monitor. Ensure that the color-coding of the cables matches the printed labels on the monitor.
5	Reconnect the monitor control harness to the rear of the monitor.
6	Connect the ac power cord to the monitor.

3.9 Monitor Adjustments

Monitor adjustments High-resolution monitor adjustments are as shown in Figure 3-3 and Figure 3-4.

Figure 3-3 Right Side Monitor Adjustments

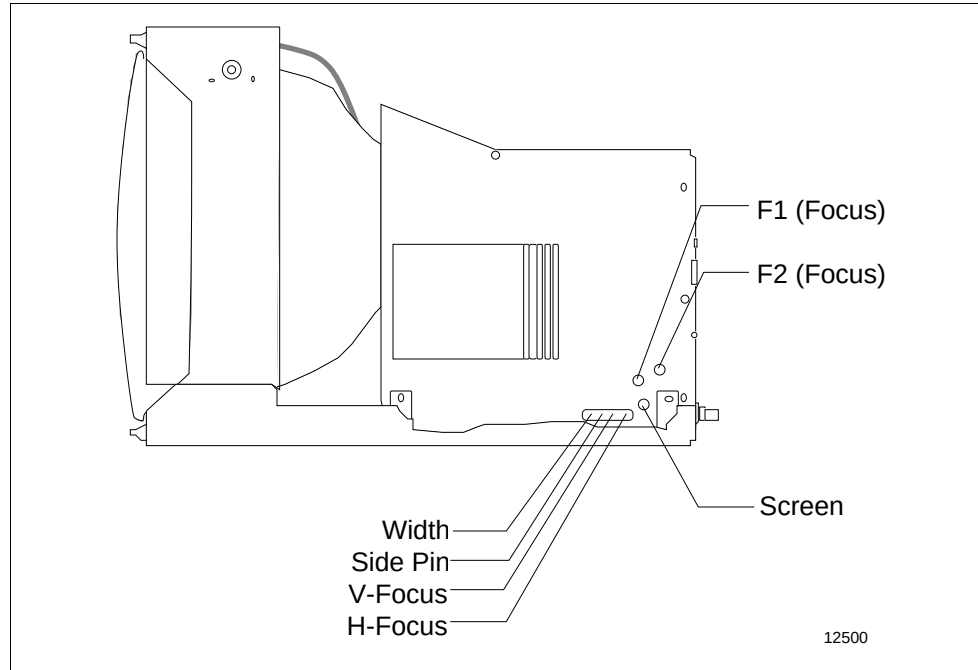


Continued on next page

3.9 Monitor Adjustments, Continued

Illustration

Figure 3-4 Left Side Monitor Adjustments



3.10 Monitor Color Bias Adjustments

CRT Bias

Use Native Window, full screen, to make CRT adjustments. The CRT Bias adjustment controls are located on the input video circuits, which reside on the main board. If required, make the adjustment before attempting to balance the color.

ATTENTION

ATTENTION—Before adjusting color level, be sure to degauss the monitor.

Procedure

Use the procedures in Table 3-4 to adjust the CRT Bias.

Table 3-5 Adjusting CRT Bias

Step	Action
1	Ensure that the Brightness is at its minimum position
2	Align the adjustment slot by setting each cathode bias control: • Red Bias (R Bias) • Green Bias (G Bias) • Blue Bias (B Bias) to the 12 o'clock position
3	Turn the screen control (Figure 3-3) clockwise until a dim raster appears on the screen. Then turn counterclockwise until the background raster begins to disappear.
4	Observe the screen (in a darkened area) with a cupped hand or an eyepiece and adjust each cathode bias control to a balanced level. • Turn the R BIAS control clockwise until a dim, red background appears. • Then turn counterclockwise until the background begins to disappear. • Repeat these steps for both G BIAS and B BIAS controls.
5	Turn the Brightness control to its maximum position until a dim, gray background appears. If the gray shifts in hue towards any color, repeat Step 4.
6	Turn the Brightness control to its normal position.

3.11 Monitor Focus Adjustment

Scope

The following adjustment procedure is designed to be done using a video test generator. However, in the customer's environment, this could be done on a screen filled with small text. As you adjust the focus be sure to observe the four corners, the midpoint of each edge, as well as the center of the screen.

Procedure

Table 3-6 Adjusting the Monitor Focus

Step	Action
1	Set the video test generator to display a white grid test pattern.
2	Turn the Contrast Control for the maximum display luminance.
3	Adjust the Brightness control until the background is no longer visible.
4	Use the F2 focus control to adjust the focus pot until the vertical axis line at the center of the test pattern and is as thin as possible, at both the center and edge of the screen.
5	Use the F1 control to adjust the focus pot until the horizontal axis line is at the center of the test pattern is as thin as possible, at both the center and edge of the screen.
6	Repeat Steps 4 and 5 at least three times until sharpest overall image is achieved.

3.12 Vertical Linearity Adjustment

Scope

Vertical linearity is the proportionate size of the top and bottom of the display, with reference to the center. If the top and bottom of the display are not linear, use the Vertical Linearity (V-LIN) control shown in Figure 3-3 to make the adjustment. HVTS PDGI Test 18 (pattern of squares) is available for this purpose—measure the size of the squares at the top, middle, and bottom of the screen.

Procedure

Table 3-7 Adjusting Vertical Linearity

Step	Action
1	ATTENTION This requires that the station be taken out of service. Set up the pattern using Test 18 of the HVTS PDGI.
2	Turn the V-LIN control clockwise to increase the size of the top grid boxes and to decrease the size of the bottom grid boxes (see Figure 3-3). Turn the control counterclockwise for the opposite effect.
3	Adjust the V-LIN control until the top boxes are equal in height to the bottom boxes. In the optimum condition, the top and bottom boxes are equal to each other and the box in the center of the display. If the V-LIN control is significantly adjusted, use the Vertical Centering (V-Cent) control to readjust the vertical centering.

3.13 Horizontal Linearity Adjustment

Scope

Horizontal linearity refers to the proportionate size of the left and right sides of the display. If the sides of the display are not linear, with reference to the center, use the Horizontal Linearity (H-LIN) control (L802) to adjust the left and right sides of the display.

Procedure

Table 3-8 Adjusting Horizontal Linearity

Step	Action
1	WARNING Make sure you do not touch exposed terminals. Set up the pattern using Test 18 of the HVTS PDGI.
2	Rotate the thumbwheel on the coil (L802), stretching or compressing the width of the grid blocks, to change the left side of the display.
3	Rotate the thumbwheel until the width of the left grid block is equal to the right grid block.
	ATTENTION If the H-LIN control is adjusted significantly, readjust the Horizontal centering control (H-Cent), located in the center of the mother board to adjust the horizontal centering.

3.14 Side Pin Cushion Control

Scope

The side pin cushion (SIDE PIN) control adjusts the straightness of the left and right sides of the display. It also corrects pin cushion distortion that occurs naturally with the CRT. Other distortions, such as trapezoidal and parallelogram, are functions of tube performance and cannot be corrected with the SIDE PIN control.

Procedure

Table 3-9 Adjusting Side Pincushion

Step	Action
1	Set up the pattern using Test 18 of the HVTS PDGI.
2	Turn the SIDE PIN control (see Figure 3-4) until the left and right display are as straight as possible.

3.15 Further Adjustments

WARNING

WARNING—Adjustment of controls not mentioned in this procedure must not be attempted. The adjustment controls have been sealed at the factory and an attempt to adjust them will **RESULT IN DAMAGE** to the board and will void the warranty.

3.16 Fuse Replacement

Procedure

The spare fuse is located in the right side of the AC power connector housing and is accessed by extracting the small compartment holding the spare fuse.

3.17 Spare Parts

Spare parts list

Table 3-10 Monitor Spare Parts

Part Number	Description
*51197002-200	High Resolution Monitor, 20" pillow tube, including CE Compliant control harness
51190577-232	Fuse, time-delay, 3.0 amp, 250 volts, .2 in. diameter by .8 in. long
*51191784-100	Cable, Signal

* ORU = Optimum Replaceable Unit

Section 4 – Touchscreen Service

4.1 Overview

Section contents These are the topics covered in this section:

Topic		See Page
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4.3	Touchscreen Alignment Verification	42
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4.5	Removing the Touchscreen Assembly	47
4.6	Replacing the Touchscreen Assembly.....	48
4.7	Spare Parts.....	49

Scope The touchscreen consists of LEDs that are directed across the monitor face in a matrix pattern. Each LED has a detector circuit located opposite the LED. By placing a finger on the surface of the monitor face, a beam is broken in two directions providing X and Y coordinates of the location.

Description The touchscreen assembly is mounted inside the monitor bezel and is accessible from the rear of the station. See Figure 4-2.

4.2 Cleaning the Touchscreen Window

Procedure

Table 4-1 Cleaning the Touchscreen

Step	Action
1	Gently wipe the four inside edges of the touchscreen window bezel with a slightly damp, soft cloth.
2	Use the acid brush provided with the station to brush out the dust from the slot (if your station has a slotted window) and to brush the LEDs mounted on the touchscreen frame. ATTENTION To do this, the monitor must be backed out of the station.

4.3 Touchscreen Alignment Verification

WARNING

WARNING—It is imperative that both the LCN node processor and Unix coprocessor operations be terminated before turning the power OFF on the Universal Station^X.

Self-test

When a Universal Station^X node is first powered up or is reset, the firmware executes a self-test. The results of the Touchscreen self-test can be displayed on the screen by pressing the CONS STATS button on the Operator Keyboard.

Procedure

Table 4-2 Verifying the Touchscreen Alignment

Step	Action
1	Turn the power switch to the US node to the ON position or, if power is already on, after momentarily moving the Shutdown switch on the WSI free edge to the shutdown position, press the RESET button on the Operator's Keyboard.
2	After the cursor symbols appear in the upper left-hand side of the display, press the CONS STATS button. Now analyze the display.

Interpretation

What follows is an excerpt of the explanation of the data fields found in the *Customer Resource Manual*, Tab 15, Page B05. The BEAM ANALYSIS field and the SIGNAL VALUE field are the two fields useful to service personnel. The rest of the description is for Engineering debug.

ATTENTION

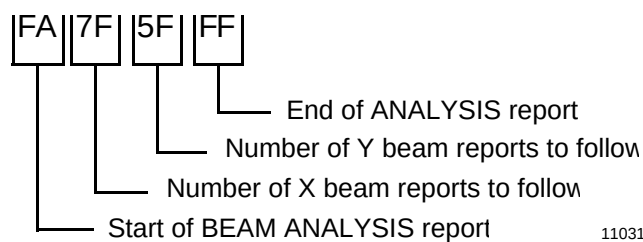
ATTENTION—There are two different report formats; one reports a numeric value of beam strength, the other reports a go/no go value. The go/no go value report is in the same format as the other value except that the value is 00 for good and 77 for bad.

This value is reported by the firmware in newer touchscreen units. The change was made because units with a low value reported were being replaced when they were still functional, but dusty.

Continued on next page

4.3 Touchscreen Alignment Verification, Continued

BEAM analysis



11031

- Each failed beam report will be two bytes long. The first byte (of the two center bytes) is the byte address and the second byte is the fault found with the beam.
- Addressing is from top down and left to right.

The fault codes are:

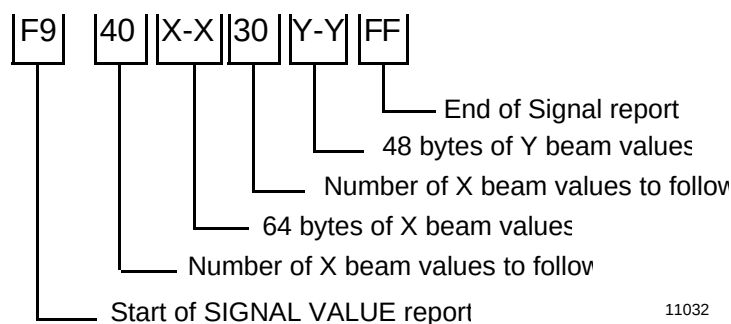
- 00 Firmware could not determine the problem
- 01 Failed photo-transistor
- 02 Failed LED
- 03 Failed photo-transistor and LED
- 04 Individual components appear to work separately, but not together; failure cannot be isolated.

EXAMPLE: FA 01 03 02 00 FF shows 1 failed X beam (01), a failed address of 3 (03), and is a failed LED. No failed Y beams (00).

ATTENTION

ATTENTION—A failed beam could be a beam blocked by dust, etc.

SIGNAL VALUE



11032

Continued on next page

4.3 Touchscreen Alignment Verification, Continued

Interpretation

The Signal Value is the differential value between the ambient reading and the reading with the beam on.

- Any reading above 00 should work, but the factory doesn't ship any unit that has a value less than 05.
- A reading of 01 or 02 may be reported as failed.
- A reading of 00 is failed.
- The screen will continue to work as beams fail, but the resolution will suffer.
- If any values are found to be below 05 then the screen should be cleaned and then tested again.
- The screen doesn't need to be replaced until the BEAM ANALYSIS reports one or more failed beams.

EXAMPLE:

SIGNAL VALUE F9 40 [49 51 0D 4B 38 47 28 40 44 20 10 18 2E 4D
37

1B 27 3B 10 23 32 11 0D 17 35 44 4C 33 10 18 2E 28 58 58 11 31 35 4B
1B 1C 3E 56 48 11 44 21 57 4F 15 3B 4D 56 53 42 55 52 59 21 4E 02 59
59 50 42] 30 [3A 31 04 25 3B 45 2B 25 40 2B 27 4C 30 2E 28 37 44 26
1A 47 1D 2D 1E 2C 40 2F 25 15 28 33 14 2B 1B 0A 37 17 30 16 0A 1B
2A 32 27 2F 19 23 22 2B] FF

4.4 Touchscreen Alignment Procedure

Touchscreen to CRT alignment

This procedure determines if the touchscreen is in alignment with targets on the monitor. Calibration is done by adjusting the width, height, and centers of the test image on the CRT to match the position of the touchscreen. You will be using four places on the face of the CRT as calibration points. They are the middle right and left edges, and the middle vertical top and bottom edges. Figure 4-1 shows the pattern.

Table 4-3 Aligning the Touchscreen

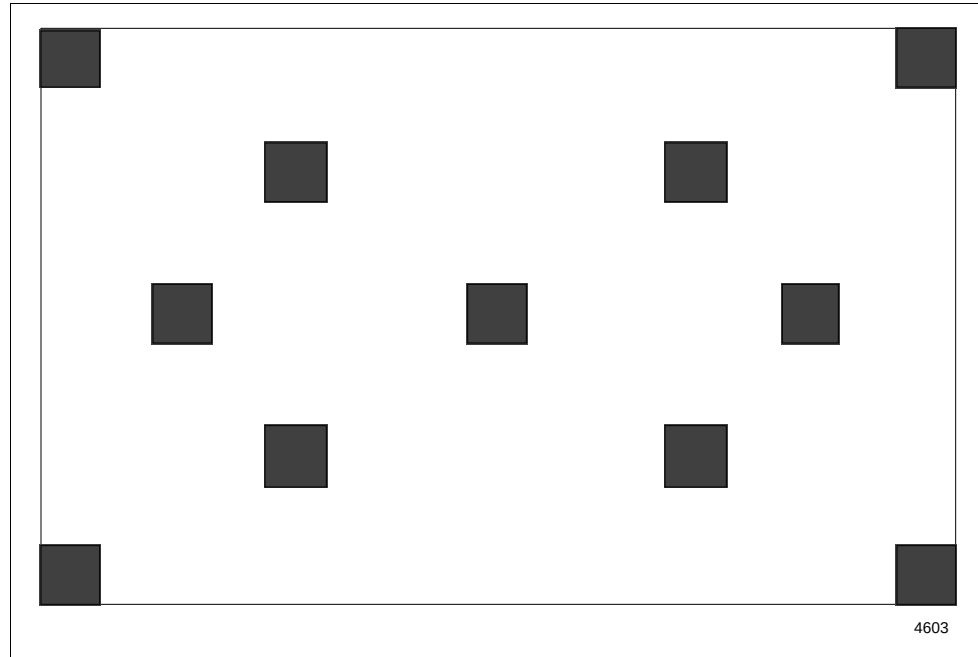
Step	Action
Horizontal Positioning	
1	Power-up or reset the Universal Station ^X . (Reset only after properly performing a shutdown per the instructions in subsection 1.3, Troubleshooting the Universal Station ^X .)
2	Using HVTs, run PDGI Test 15. After the pattern appears on the screen, abort PDGI.
3	Place your finger on the square located at the middle left edge of the pattern. Notice where the cursor appears.
4	Place your finger on the square at the middle right edge of the pattern.
5	After considering the results of the two tests, adjust the horizontal centering and horizontal width controls to make the cursor appear directly under your finger each time you touch the CRT surface.
6	Repeat this adjustment until the cursor is centered in the pattern square for each side.
Vertical Positioning	
7	Place your finger on the square located at the middle top edge of the pattern. Notice where the cursor appears.
8	Place your finger on the square at the middle bottom edge of the pattern.
9	After considering the results of the two tests, adjust the vertical centering and vertical width controls to make the cursor appear directly under your finger each time you touch the CRT surface.
10	Repeat this adjustment until the cursor is centered in the pattern square for each side. ATTENTION Note that the cursor will always be slightly below the finger when testing the top target. This is due to the need to have the touchscreen high in the bezel so that the bottom beam will not be partially blocked off by the side slot's bottom edge.

Continued on next page

4.4 Touchscreen Alignment Procedure, Continued

Illustration

Figure 4-1 HVTS PDGI Test 15 Test Pattern



4.5 Removing the Touchscreen Assembly

Scope

The touchscreen assembly is attached to the inside of the monitor bezel. The monitor must be removed from the station before the bezel assembly can be removed. See Section 3 for instructions for access and retraction of the monitor.

Removal Procedure

To remove a touchscreen assembly from the monitor, follow the procedure in Table 4-4. See Figure 4-1.

Table 4-4 Replacing the Touchscreen

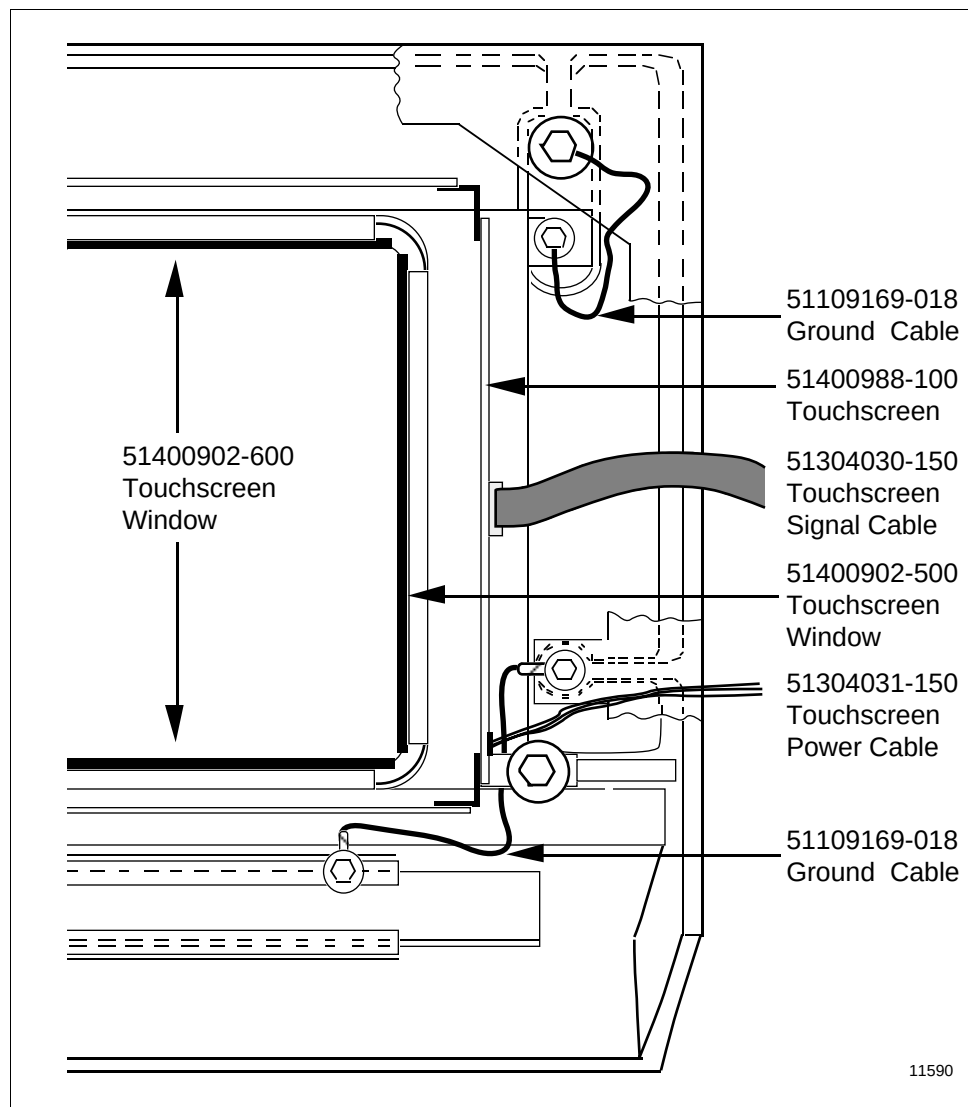
Step	Action
1	Place the breaker switch inside the station in the OFF position.
2	Disconnect all cables from the monitor and remove the monitor assembly from the station. If you do not remove it now, you'll have to later. After the monitor is removed, slide the monitor rails out of the way.
3	Disconnect the Interface cable from the touchscreen.
4	Disconnect the power cable from the touchscreen.
5	Disconnect the two ground wires (shown in Figure 4-1) from the touchscreen.
6	Remove the touchscreen after removing the four screws that fasten the four corners of the touchscreen to the bezel.
7	Slide the touchscreen out of the bezel.

Continued on next page

4.5 Removing Touchscreen Assembly, Continued

Illustration

Figure 4-2 Touchscreen Assembly Exploded View



4.6 Replacing the Touchscreen Assembly

Replacement Procedure

The touchscreen frame can be replaced by reversing the procedure in Table 4-1.

4.7 Spare Parts

Spare parts list

Table 4-5 Touchscreen Spare Parts

Part Number	Description
*51400988-100	Smartframe Touchscreen
*51304030-150	Touchscreen Interface Cable
*51304031-150	Smartframe Power Cable
51109399-100	Touchscreen Brush

* ORU = Optimum Replaceable Unit

Section 5 – Keyboard Service

5.1 Overview

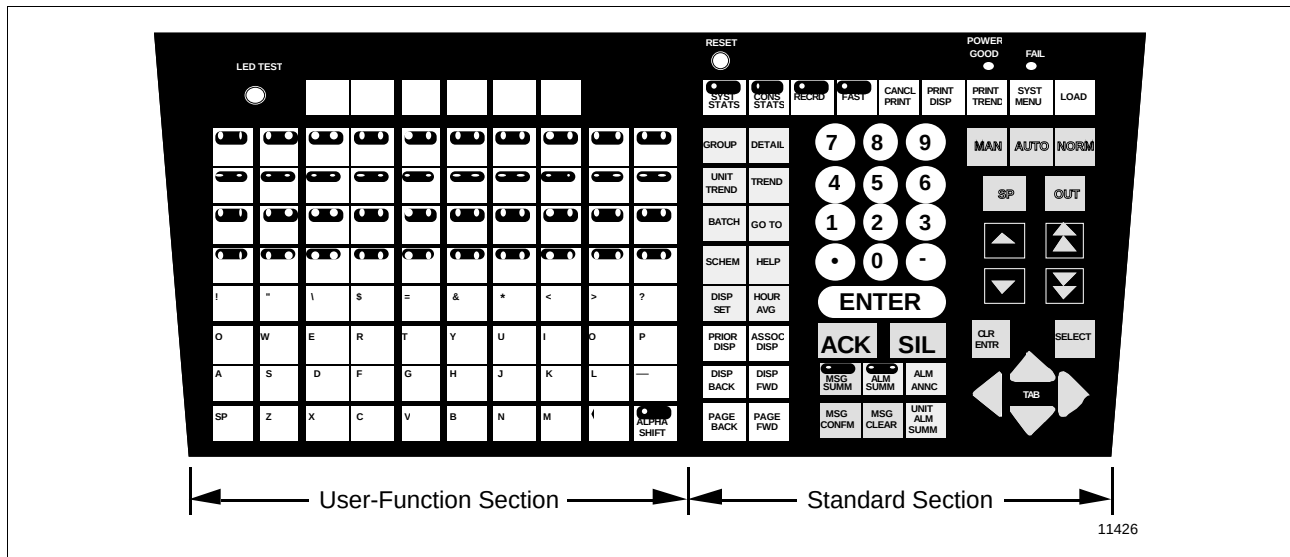
Section contents These are the topics covered in this section:

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5.5 Removing and Installing the U ^X S Auxiliary Keyboard.....	55
5.6 Test/Troubleshooting Keyboards.....	55
5.7 Spare Parts.....	56

Operator keyboard description The Operator's Keyboard has a QWERTY key configuration.

Illustration

Figure 5-1 QWERTY Operator's Keyboard



Continued on next page

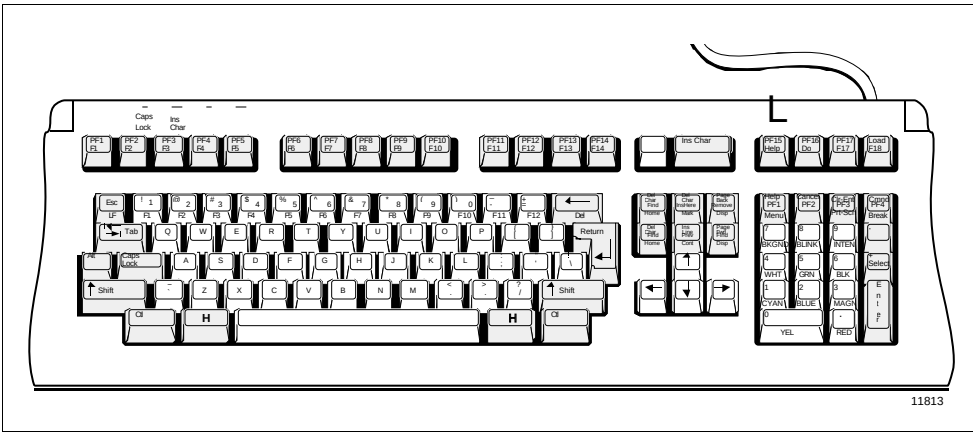
5.1 Overview, Continued

UXS Auxiliary keyboard description

The U^XS Auxiliary Keyboard is a QWERTY key configuration that can rest on any convenient work surface, or on the shelf (optional) that extends toward the operator from the Operator's Keyboard. This keyboard is referred to as the U^XS Auxiliary Keyboard and performs the engineer's keyboard function for Universal Station^X. The U^XS Auxiliary Keyboard is shown in Figure 5-2.

Illustration

Figure 5-2 U^XS Auxiliary Keyboard



5.2 Cleaning the Keyboards

Specification

Both keyboards can be cleaned with a soft cloth dampened, not saturated, with a light, dish washing soap solution.

Table 5-1 Cleaning the Keyboards

Step	Action
*1	Disconnect the keyboard from the station.
*2	Hold the keyboard in an upright position with the front edge of the keyboard resting on a stable surface.
3	Using a soft cloth dampened with a dish washing soap in water, wash the surface of the keyboard ensuring that no excess solution (there should not be any excess) runs into the keyboard assembly.
4	Rub the surfaces of the keyboard with a soft cloth to remove any remaining soap residue.
*5	Reconnect the keyboard to the station.

* Applies to U^XS Auxiliary Keyboard only.

5.3 Removing Operator's Keyboard

Procedure

To access the Operator's Keyboard, first remove the bezel.

Table 5-2 Removing the Operator's Keyboard

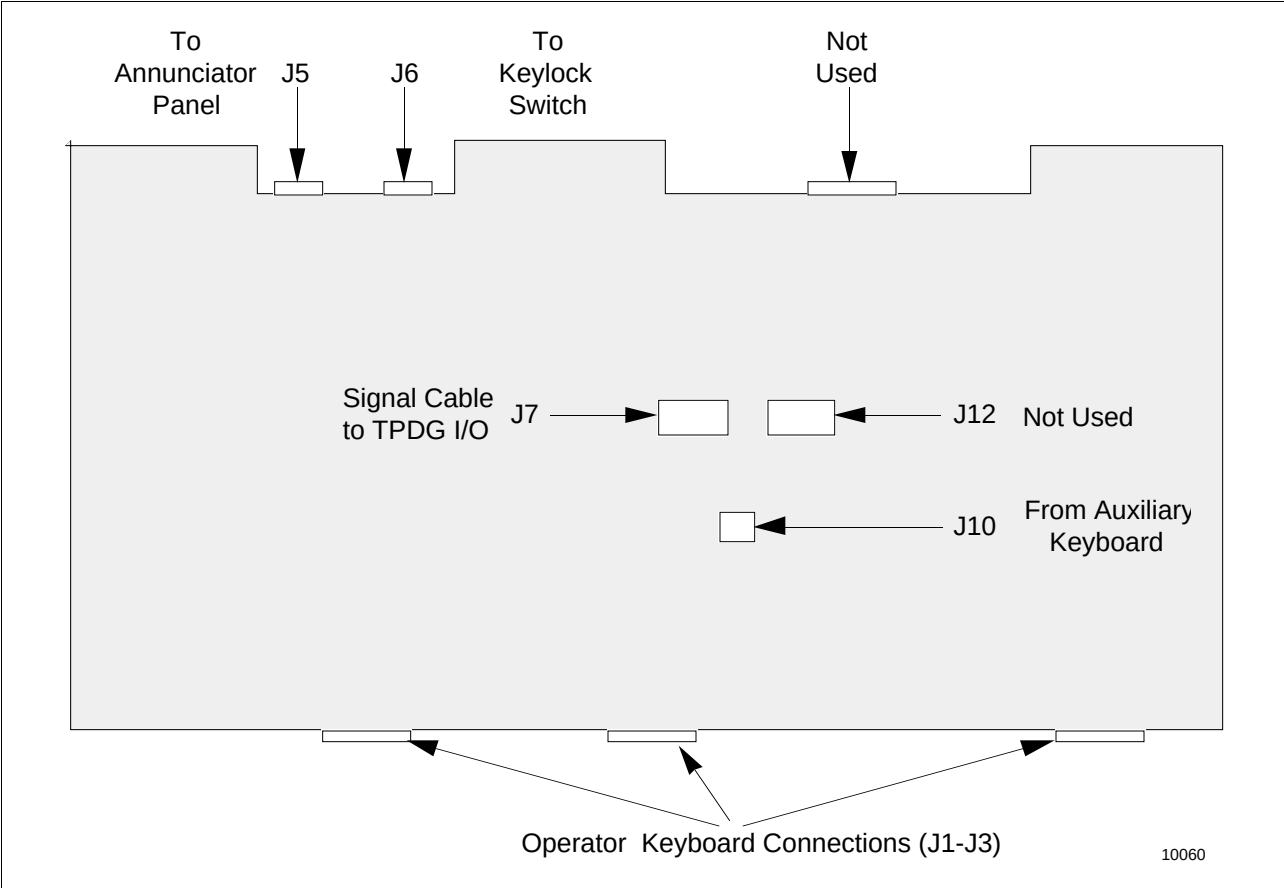
Step	Action
1	Remove the four M5 screws from under the keyboard tray. Use an 8 mm spin-tight or wrench.
2	Lift the bezel from the tray.
3	Push the RESET button cover towards the monitor.
4	Remove the four M5 screws from under the keyboard tray. Use an 8 mm spin-tight or wrench.
5	Lift and rotate the keyboard towards the monitor and disconnect the cables from J1, J2, and J3 on the near edge. See Figure 5-3.
6	Rotate the keyboard back to the tray and disconnect the rest of the cables.

5.4 Installing an Operator's Keyboard

Procedure Reverse the procedure in Table 5-2.

Illustration

Figure 5-3 Operator's Keyboard Cable Connections



5.5 Removing and Installing the U^XS Auxiliary Keyboard

Procedure

Removal or replacement of an U^XS Auxiliary Keyboard is accomplished by disconnecting or reconnecting the keyboard cable from the right-hand side of the gooseneck that supports the Operator keyboard.

5.6 Test/Troubleshooting Keyboards

Test

Refer to the *System Maintenance Guide*. In general, if one or a few keys do not work, the keyboard is defective. If no keys work, check the Five/Ten-Slot Module (see subsection 5.3). Replace the TPDG I/O (reinsert original board if it is not faulty). Check cabling between the module and the keyboard to ensure power is being supplied and signals are being passed. Keyboard cabling information is provided in Figure 1-4 and 1-5.

5.7 Spare Parts

Spare parts list

Table 5-3 Keyboard Spare Parts

Part Number	Description
51401560-100	QWERTY Keyboard
51401570-100	Keyboard Interface PWA
*51196712-100	U ^X S Auxiliary Keyboard (Engineer's Keyboard) Beige CE Compliant
*51196712-200	U ^X S Auxiliary Keyboard (Engineer's Keyboard) Gray CE Compliant
51304496-026	Engineer's Keyboard Interface Cable
51308058-100	Operator Keyboard Interface Cable
51308060-100	Engineer's Keyboard Interface Cable

* ORU = Optimum Replaceable Unit

Section 6 – Five-Slot Module Service

6.1 Overview

Section contents

These are the topics covered in this section:

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6.16	Hard Disk Drive Tray	75
6.17	525 MB Quantum Disk Drive (LPS525S).....	80
6.18	525 MB Seagate Drive (ST3600N).....	82
6.19	1.2 GB Digital Equipment Corp Drive (DSP3107L).....	86
6.20	1.2 GB Quantum Drive (LPS1080S).....	88
6.21	1.2 GB Quantum Drive (VP31110).....	90
6.22	1.2 GB Seagate Drive (ST31200).....	92
6.23	2 GB Quantum Drive (VP32210).....	96
6.24	2 GB Seagate Drive (ST32430).....	98
6.25	2 GB Hewlett-Packard Drive (C3325A).....	100
6.26	HDDT I/O Board.....	102
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Scope

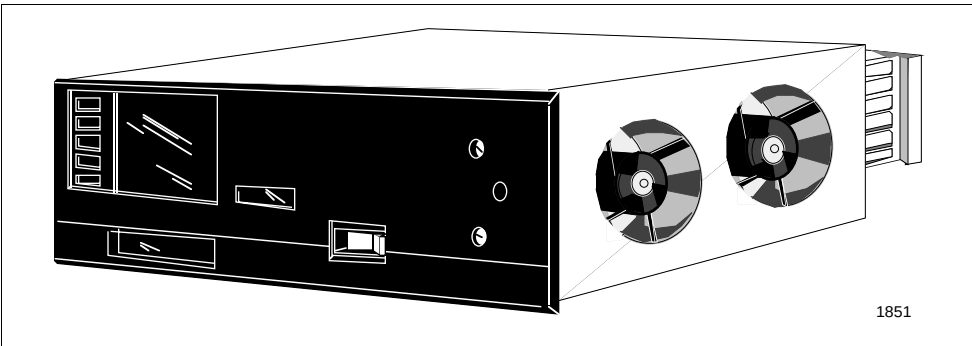
This section covers the module board-set unique to the Universal Station^X and the replacement of the module. The module service is covered in the *Five/Ten-Slot Module Service* manual.

Continued on next page

6.1 Overview, Continued

Illustration

Figure 6-1 Five-Slot Module



6.2 UXs Node Board Locations

Slot assignment

Table 6-1 Board Slot Assignments

Slot	Board Type	I/O Board Type
5	TPDGX	TPDGC (or TPDG I/O)
4	HDDT*	—
3	HDDT*	HDDT I/O
2	WSI2	WSI2 I/O or WSI2R I/O**
1	K2LCN-8/K4LCN-8	CLCN A/B (or LCN I/O)

* The HDDT board requires two slots.

** The WSI2R I/O is used in UXs that upgraded to Release 200.

6.3 K2LCN-8 Board Description

Features

A standard TPS module board. The board provides the processor (68020), memory (8 megabytes), and LCN interface to execute the TPS software personalities.

The kernel firmware (KFW) revision on the early production K2LCN-8 board (51401551-800) must be Rev. B or higher.

6.4 K4LCN-8 Board Description

Features

A standard TPS module board. The board provides the processor (68040), memory (8 megabytes), and LCN interface to execute the TPS software personalities.

6.5 CLCN A/B (or LCN I/O) Description

Features

This board provides the K2LCN/K4LCN board with the I/O connection to the LCN.

All aspects of both the K2LCN/K4LCN and CLCN A/B (or LCN I/O) boards are covered in the *Five/Ten-Slot Module Service* manual.

6.6 TPDGX Board Description

Features

The TPDGX (Turbo Peripheral Display Generator) is a high-speed dual ported graphics board supporting X Windows graphics from the WSI2 board and TPS graphics from the K2LCN/K4LCN board. The ROM-based firmware located on the TPDGX provides a real-time window, driven by the standard Universal Station software functions. This LCN window is given priority and special isolation from all other windows to provide a reliable “view to the valve.” The TPDGX is also directly connected to the coprocessor to provide a full X Window compliant window manager.

The TPDGX contains circuitry which provides a display resolution of 1280 X 1024 pixels. It is implemented as a superset of all EPDG features, including peripheral device control. See Figure 6-2.

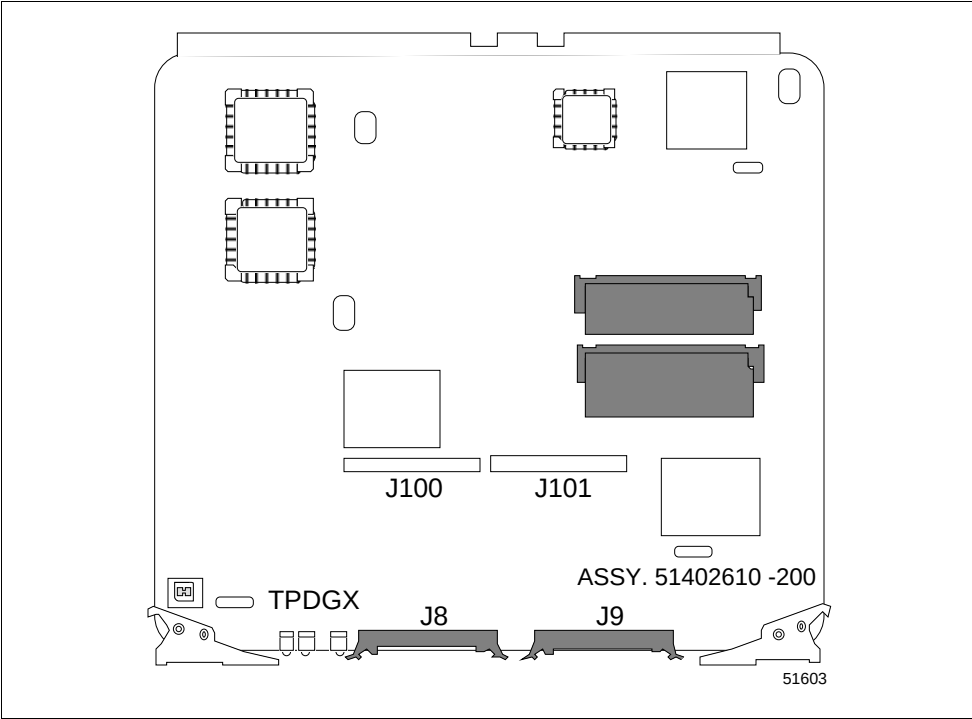
TPDGX is a faster video driver that replaced the TPDG at Release 200.

Continued on next page

6 . 6 TPDGX Board Description, Continued

Illustration

Figure 6-2 TPDGX Board



Indicators

Table 6-2 TPDG Indicators

Indicator	Action
RED—Self-Test Error	Self-test failed to complete due to error condition
GREEN—Pass Self-Test	Self-test completed
RED—Bus Transfer Error	An error occurred during a data bus transaction

6.7 TPDG I/O Board Description

Features

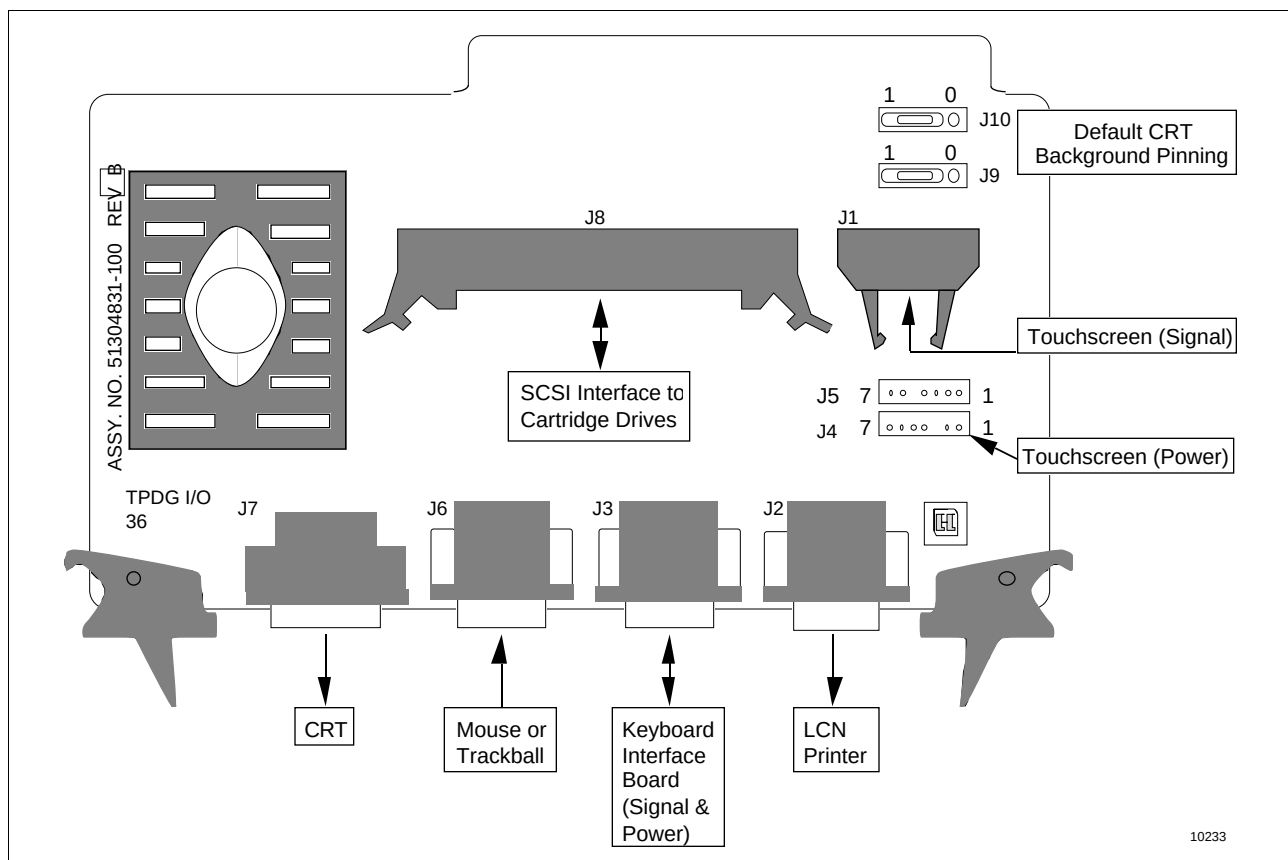
This I/O board provides these input/output connections for the TPDGX board:

- SCSI peripherals (cartridge drives)
- Touchscreen
- Mouse/Trackball
- LCN Printer
- Monitor
- Operator's keyboard
- Engineer's keyboard

This board has been replaced by the TPDGC board which is CE Compliant.

Illustration

Figure 6-3 TPDG I/O Board



6.8 TPDGC Board Description

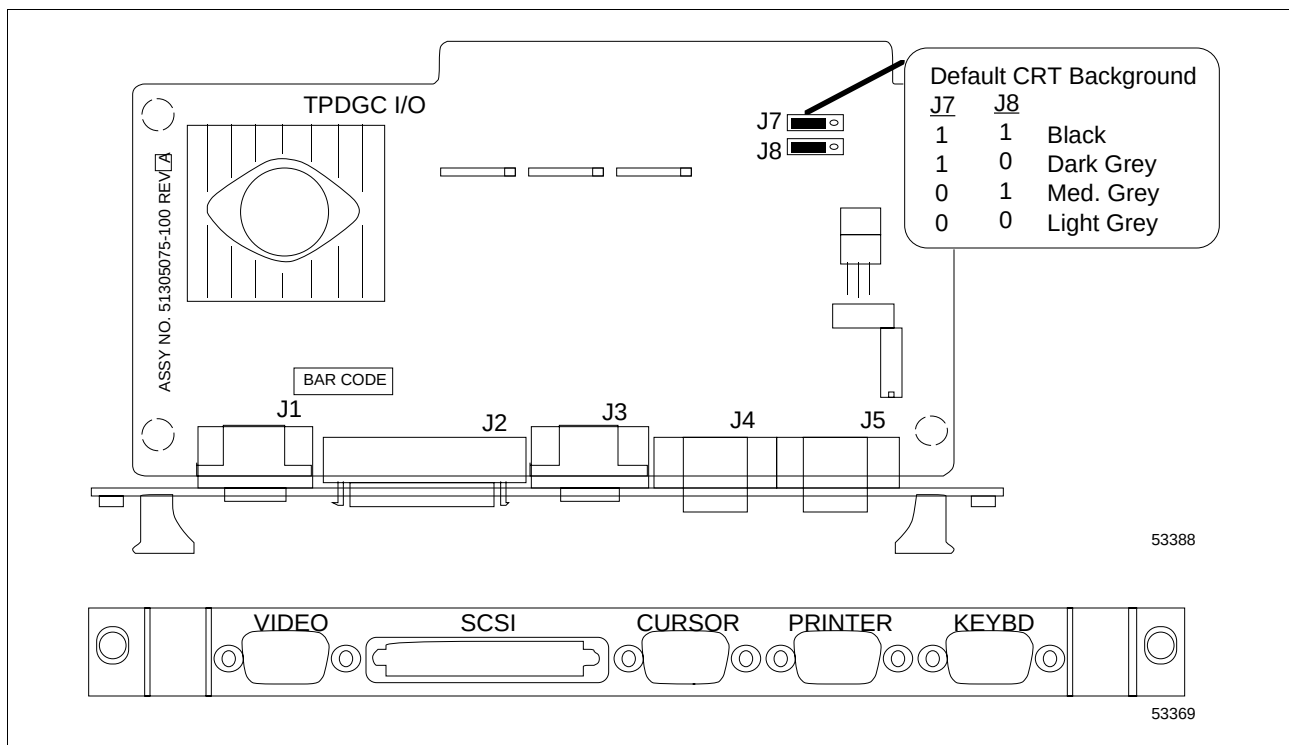
Features

This I/O board provides these input/output connections for the TPDGX board:

- SCSI peripherals (cartridge drives)
- Touchscreen
- Mouse/Trackball
- LCN Printer
- Monitor
- Operator's keyboard
- Engineer's keyboard

Illustration

Figure 6-4 TPDGC Board



6.9 WSI2 Board Description

Overview

The basic WSI2 board contains the coprocessor interface circuits required to adapt the HP 743 coprocessor into the U^XS node hardware. See the illustration that follows.

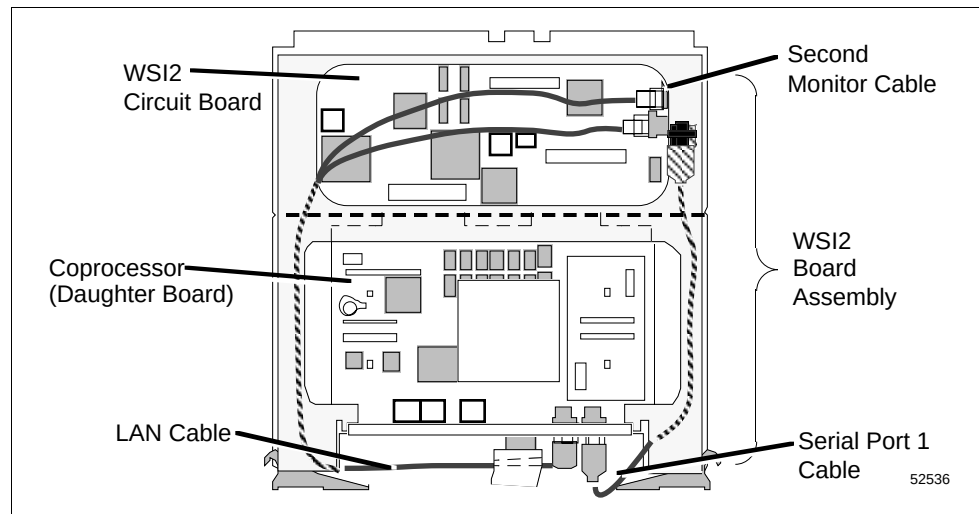
The HP 743 coprocessor board connects to the WSI2 board through a specially provided connector on the WSI2 board. There are additional small cables used to complete the interface to the WSI2 board. The total assembly of coprocessor (daughter board), WSI2 (mother board) and cables are referred to as the WSI2 board.

The complete WSI2 assembly is the Optimum Replaceable Unit (ORU). If a board is replaced the memory boards must be transferred from the old board to the new board.

The WSI2 board has separate boards that hold 32 MB, 64 MB or 128 MB of memory accessible to the coprocessor.

Illustration

Figure 6-5 WSI2 Board



Continued on next page

Continued

WSI indicators

There are two indicators on the WSI2 mother board as shown in Figure 6-6.

Figure 6-6 WSI2 Indicators and Switch

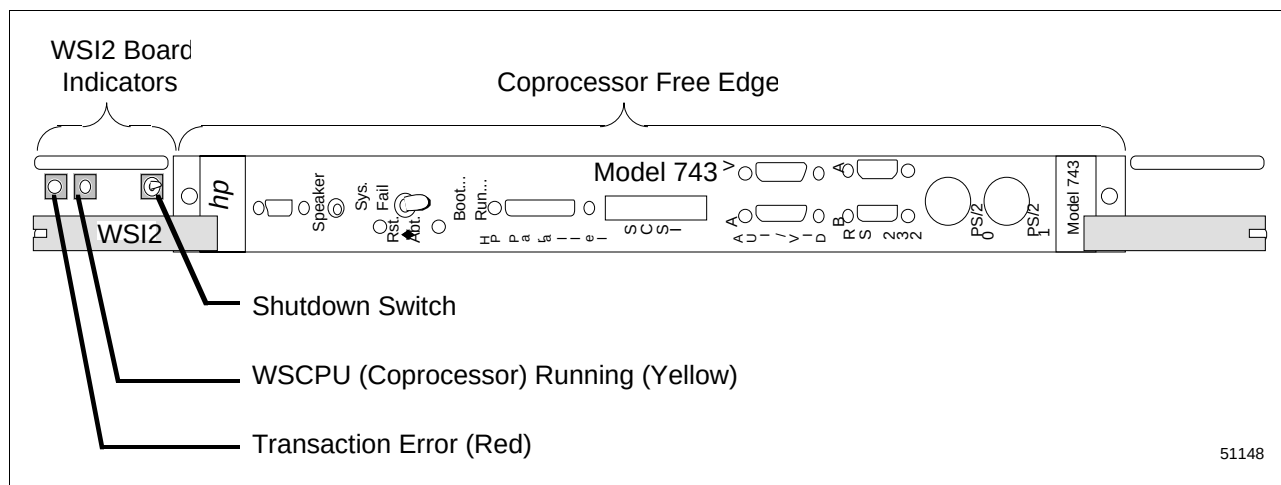


Table 6-3 WSI2 Board Indicator Description

LED	Description	Suspected Cause if Abnormal
Transaction Error	Indicates that a data parity error or bus error was encountered during normal operation on the module backpanel bus	WSI2 K2LCN K4LCN TPDGX
WSCPU Running	An ON condition indicates that the coprocessor software (HP-UX) is up and running	WSI2
Shutdown Switch	The switch can be used to force a processor shutdown. It is recommended that the keyboard method (as documented in the <i>Universal Station^X System Administration Manual</i>) be used whenever possible.	

Continued on next page

6.9 WSI2 Board Description, Continued

Illustration

In Release 100/110 units the SCSI Interface cable connects to J6 on the WSI2 board. The linkage to the hard drive and DAT drive is completed through the WSI2R I/O board.

In Release 200 equipment, the SCSI Interface is connected to the HDDT board. The HDDT I/O completed the linkage to the DAT drive.

Figure 6-7 WSI2 Coprocessor (HP 743) Indicators and Connectors

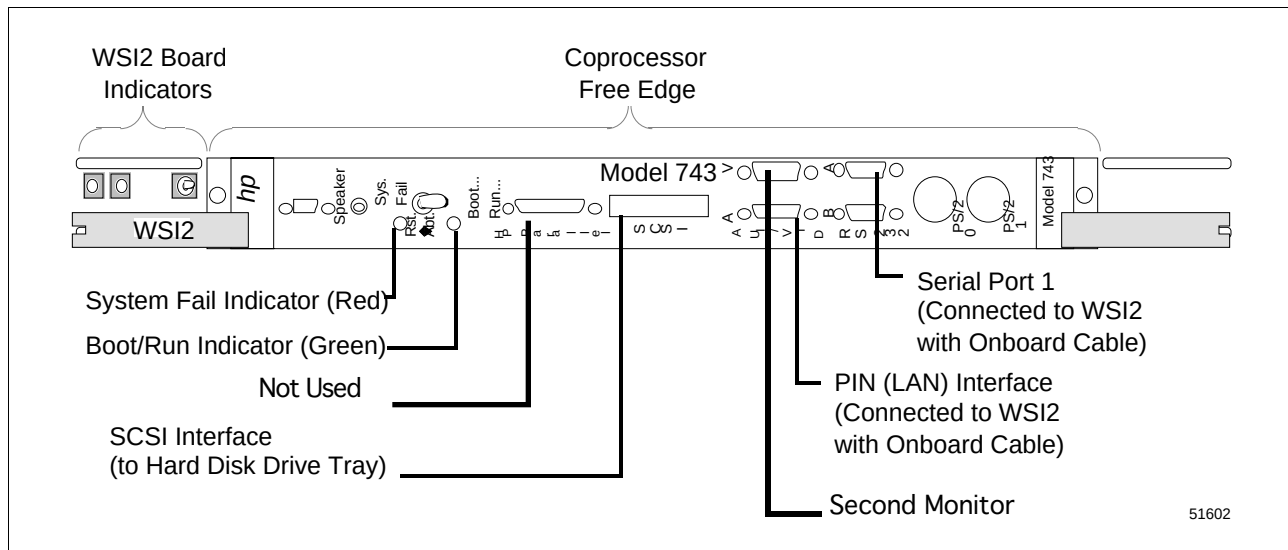


Table 6-4 HP 743 Coprocessor Indicators

LED	Description
Sys Fail (Red Indicator)	Comes on at power on and turns off after test routines complete successfully
Boot/Run (Green Indicator)	Blinks during the boot process and turns on (steady) when the boot is complete

Table 6-5 HP 743 Interface Connectors

Connector	Description
SCSI Interface	It is the interface used for connecting the coprocessor to the Hard Disk Drive Tray (HDDT) through a cable.
AUI/VID	Provides an interface to the Plant Information Network (PIN) through a Media Access Unit (MAU). The PIN is Honeywell's terminology for the Local Area Network (LAN).
RS-232-A	Commonly referred to as Serial Port 1. It provides the interface for the coprocessor console (terminal) or TAC access modem.

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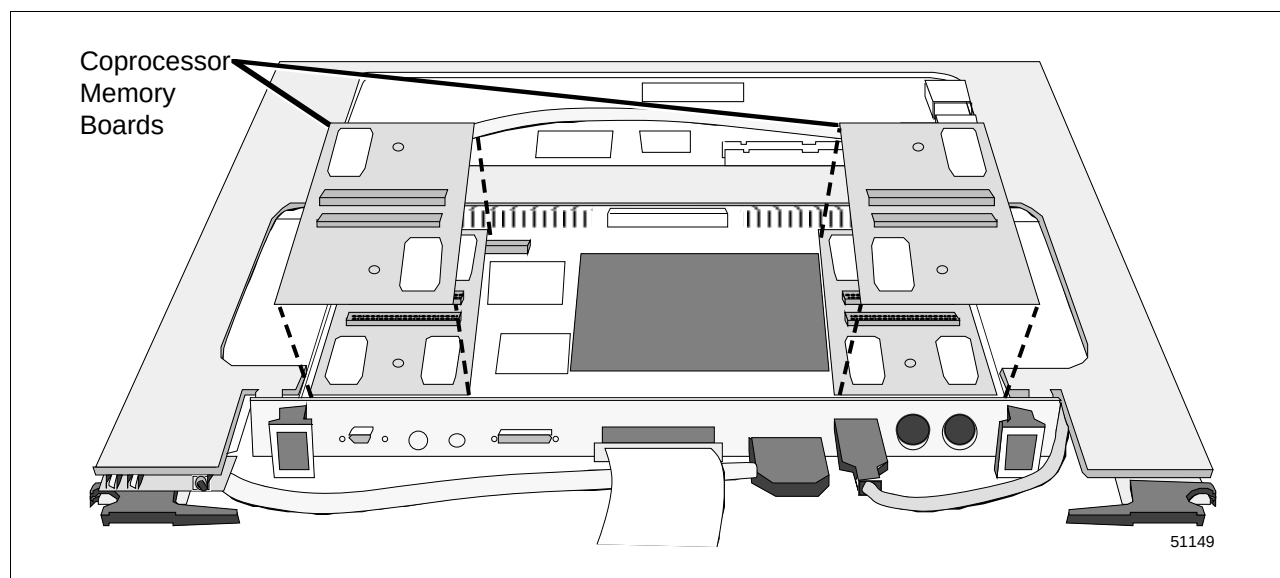
6.9 WSI2 Board Description, Continued

Coprocessor memory The coprocessor board holds 32 MB, 64 MB, or 128 MB of memory accessible to the coprocessor. This may be accomplished by using one or two memory boards. The table below details the memory configurations. The coprocessor memory can be upgraded by changing one or both of these boards.

Table 6-6 WSI Memory Configurations

Memory Size	Model Number	Number of Boards	Size of Board
32 MB	XL32MB-100	1	32 MB
	XL32MB-200	2	16 MB
64 MB	XL64MB-100	1	64 MB
	XL64MB-200	2	32 MB
128 MB	XL128M-200	2	64 MB

Figure 6-8 Coprocessor Memory



Board mounting

The coprocessor memory daughter boards are mounted with two Phillips-head screws threaded into standoffs on the coprocessor board. These screws and associated washers must be removed to change the memory boards.

ATTENTION

ATTENTION—Exercise care when pushing on the replacement memory board to ensure proper mating of the pins and sockets in the connectors.

Continued on next page

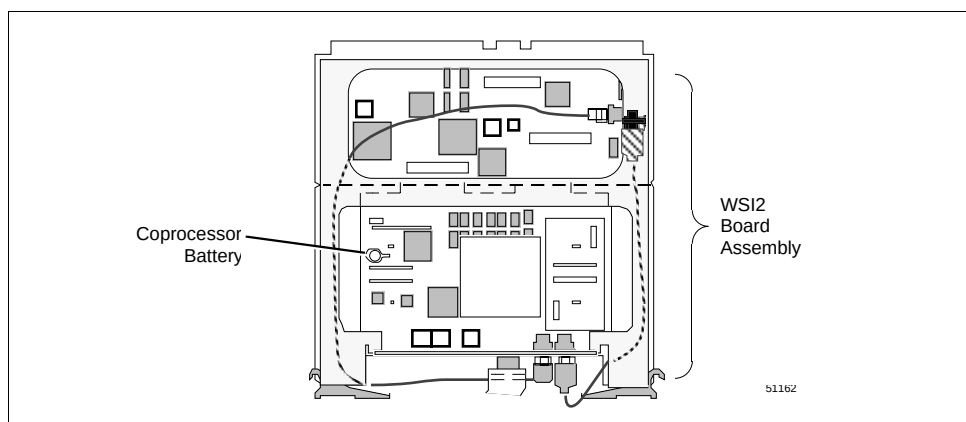
6.9 WSI2 Board Description, Continued

Coprocessor battery

The following figure illustrates the location of the coprocessor battery. It may not be clearly visible as shown here if an additional memory board is installed on your coprocessor. The battery will be covered by the additional memory board if it exists.

The WSI2 boards are shipped with a battery isolator in place to protect the battery life. When a WSI2 board is replaced the isolating paper under the battery end must be removed to activate the battery.

Figure 6-9 Location of Coprocessor Battery



6.10 WSI2 Board PIN Connection

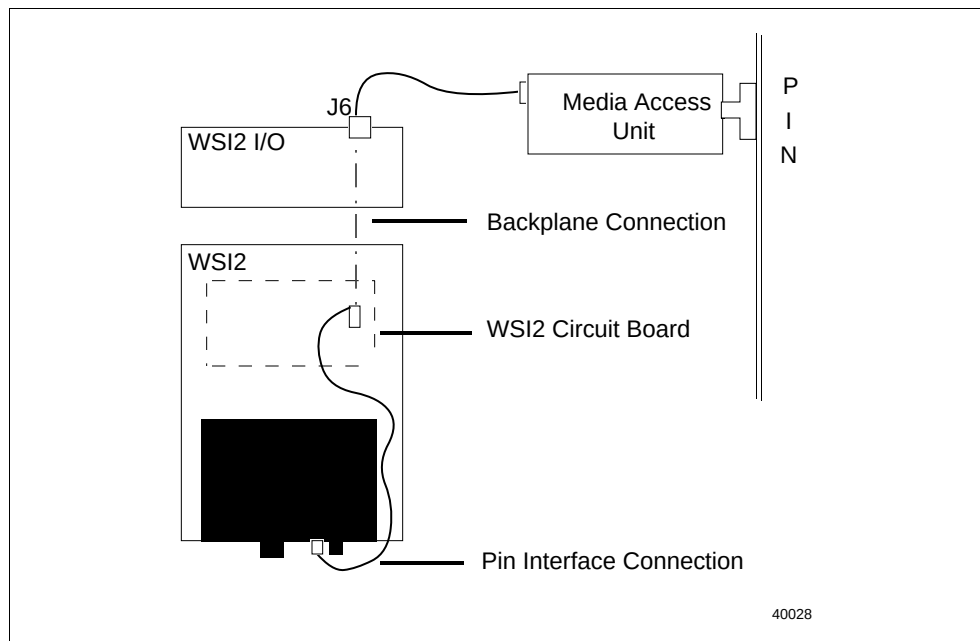
Description

The coprocessor is connected to the PIN network as shown below.

The PIN connection to the coprocessor starts at the front edge of the coprocessor AUI VID connector. See Figure 6-10. The PIN connection is routed from this point:

- through a cable to the WSI2 mother board,
- through the mother board, chassis backpanel, and on to the WSI2 I/O board,
- to a cable connector on the WSI2 I/O board,
- through a cable to the Media Access Unit (MAU),
- and the MAU connects to the PIN Network.

Figure 6-10 Coprocessor-PIN Connection



6.1.1 Coprocessor SCSI Interface to HDDT

Coprocessor SCSI interface details

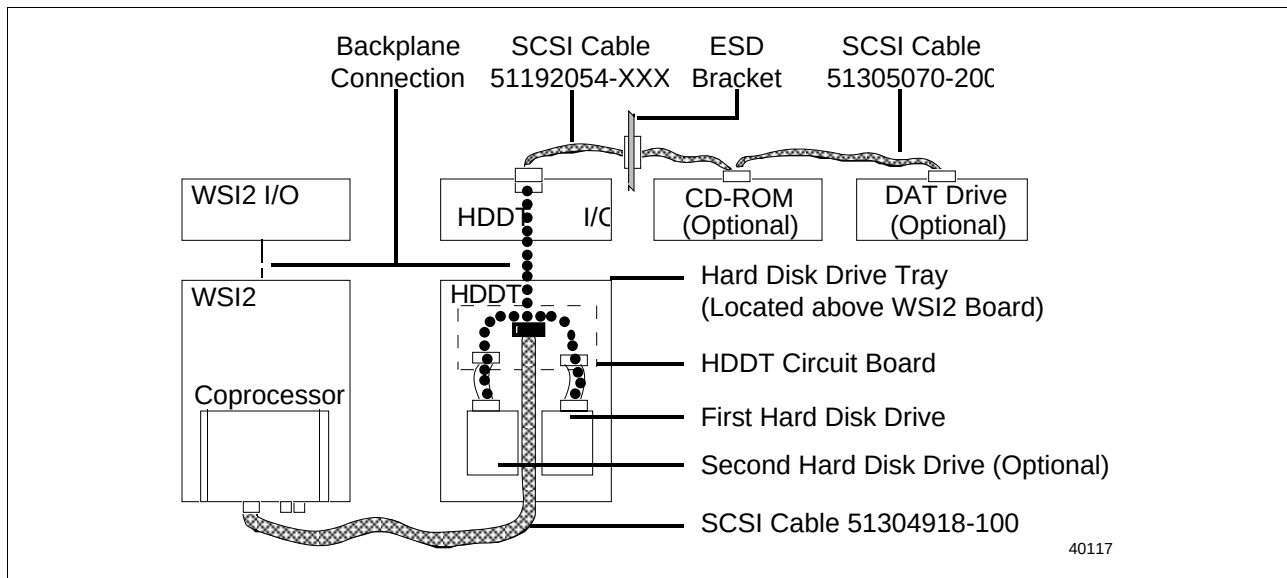
The coprocessor SCSI (hard disk drive) interface connection is made at the coprocessor front edge. See Figure 6-11. From there it is routed

- through a cable to the Hard Disk Drive Tray (HDDT),
- through the internal HDDT circuit board to the disk drive interface cables and also through the backpanel to the HDDT I/O board,
- through the HDDT I/O board to the cable connector.

The SCSI interface connector on the HDDT I/O board can

- connect to a Digital Audio Tape drive
 - A SCSI interface cable connects to the HDDT I/O board and to the DAT drive
- connect to a termination plug if there is no DAT drive connection.

Figure 6-11 WSI2 Coprocessor HDDT DAT Drive Connection

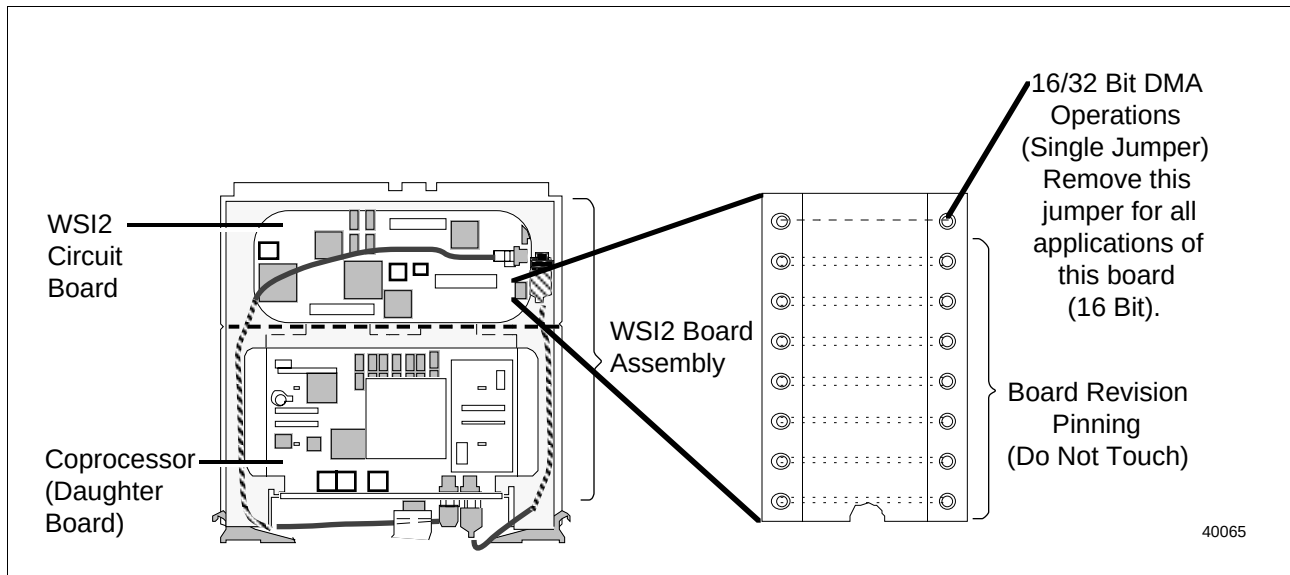


6.12 WSI2 Board Pinning

Description

The WSI2 board has a pinning requirement that is illustrated in Figure 6-12. Be sure to follow the requirement in the figure.

Figure 6-12 WSI2 Board Pinning



6.13 When Replacing a WSI2 Board

Coprocessor hardware identification number

Each coprocessor has a unique software visible hardware identification number manufactured into the board. This hardware identification number is tied directly to the Kinet software license. Not all users will have Kinet software installed on the coprocessor system.

Honeywell provides the appropriate Kinet address (correlated to the coprocessor hardware identification number) to the customer with the Kinet software at the time of shipment. **The customer must enter this provided address into the Kinet software configuration data.**

Replacement of a WSI2 board assembly due to failure (or any other reason) also changes the hardware identification number. This makes the existing Kinet address (originally provided) unusable. **A new Kinet address must be obtained when the WSI2 board is changed.**

The replacement Kinet address is obtained in the following manner.

Table 6-7 Procedure to Obtain a Replacement Kinet License

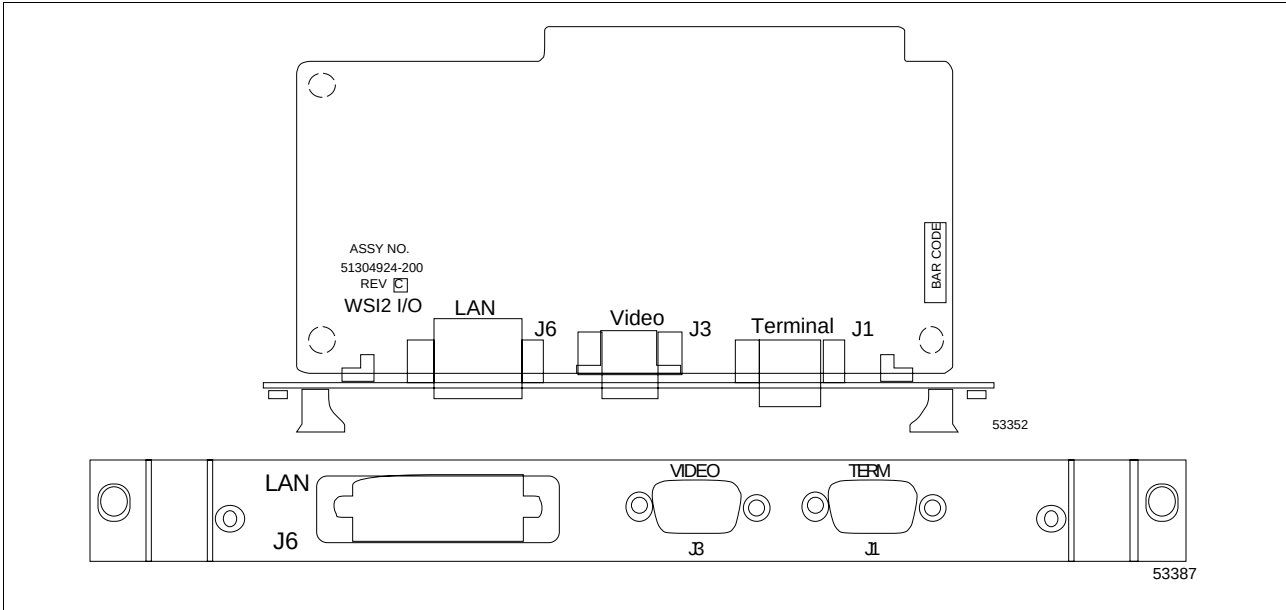
Step	Action
1	Install the new WSI2 board assembly as outlined in the replacement procedure provided in this section.
2	Place the node power supply switch in the On position and allow the coprocessor to complete its software boot.
3	Sign on as a user. This can be accomplished from any HP-UX compatible device (Workstation or Universal Station ^X) or the coprocessor console (terminal).
4	Enter the command "uname -a" The last 12 characters of the resultant output line is the new coprocessor hardware identification number.
5	Having obtained the new hardware identification number, call the Honeywell Technical Assistance Center (TAC) using the number listed on the Reader Comments sheet of this manual (last page.) Inform the TAC personnel that you have changed your Universal Station ^X WSI2 board and give them the new hardware identification number.
6	Honeywell will in turn, issue a new license and provide a new Kinet address which must be entered into the Kinet configuration data of your system. The procedure for doing this is given in the <i>Universal Station^X System Administration Manual</i> (section titled "Install and Update Application Module Software").

6.14 WSI2 I/O Board Description

WSI2 I/O board This TPS I/O module board provides these input/output connections for the WSI2 board.

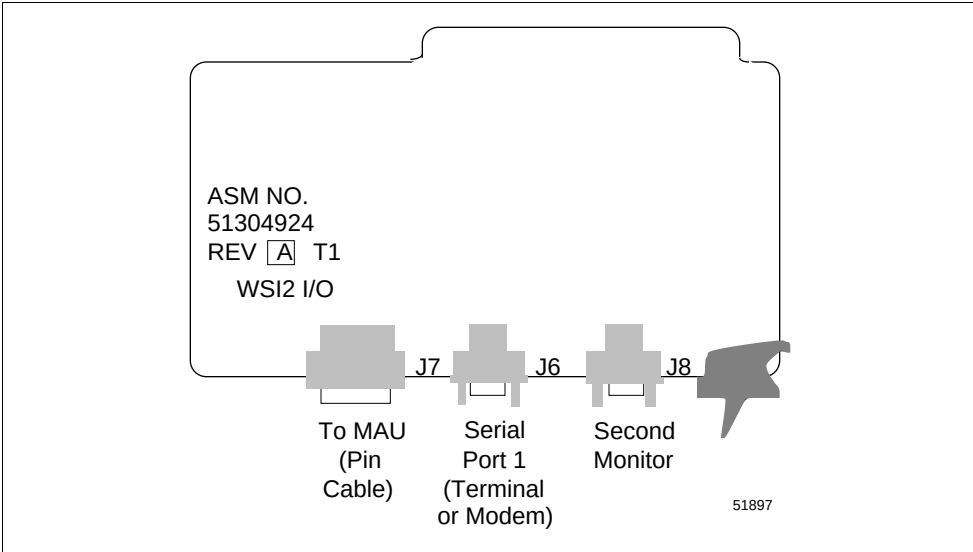
The figure below shows the different cable connectors on the CE Compliant WSI2 (I/O) board.

Figure 6-13 CE Compliant WSI2 (I/O) Board



Illustrations The original WSI2 I/O board is shown below.

Figure 6-14 WSI2 I/O Board



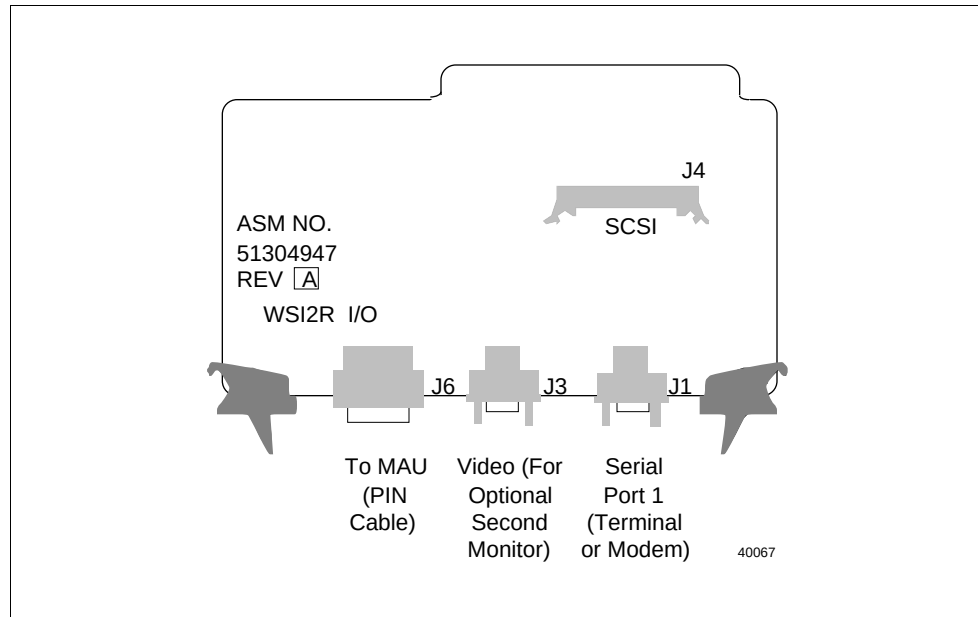
Continued on next page

6.14 WSI2 I/O Board Description, Continued

WSI2R I/O

Release 100/110 units upgraded to Release 200 use WSI2R I/O boards.

Figure 6-15 WSI2R I/O Board



6.15 Media Access Unit (MAU) Description

Purpose

The Media Access Unit (MAU) is a device that connects the U^XS to the Plant Information Network (PIN). It contains a transceiver device that adapts the LAN interface of the coprocessor to the Ethernet cable.

Source

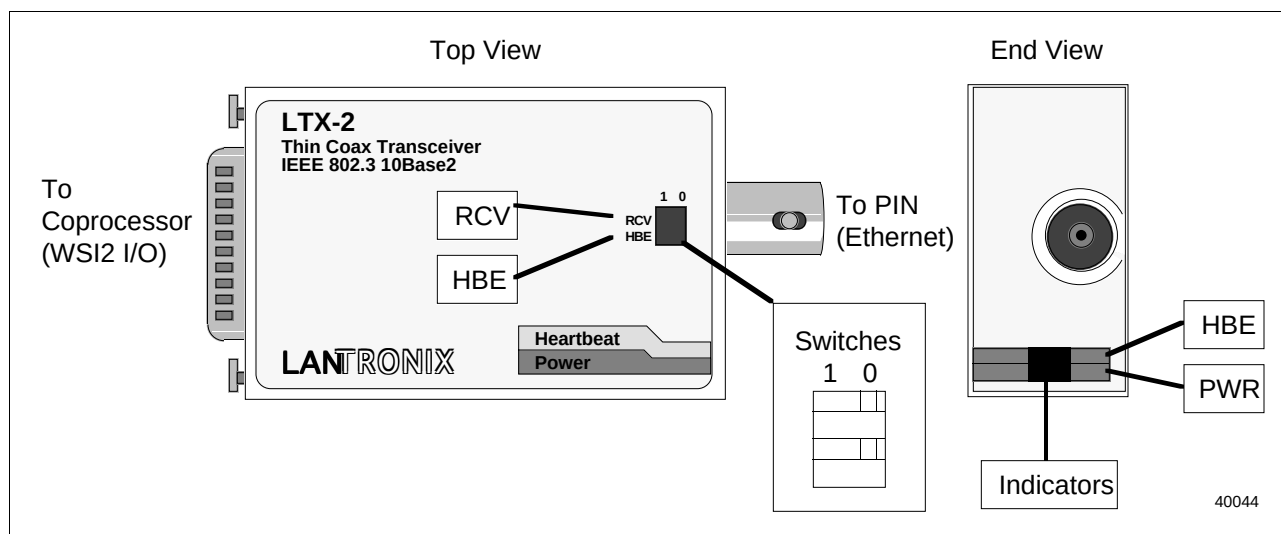
There are many vendors that produce this type of device. The one shown in the diagram is just one of many available.

Power source

Power for the MAU is supplied from the Universal Station^X through the PIN interface cable which is connected to the WSI2 I/O board.

Illustration

Figure 6-16 MAU Switches and Indicators



MAU indicators

Many of the MAU devices have indicators on them. The following indicator and switch explanation is specifically for the MAU shown in the previous diagram.

Table 6-8 LTX-2 Indicators and Switches

LED	Description	
PWR (green)	ON to indicate +12 Vdc power is applied from the WSI board through the WSI2 I/O and the interface cable. Normally ON.	
HBE (amber)	This feature (heartbeat) is not used with the WSI2 coprocessor. Normally OFF.	
Switch	Position Required	Comment
RCV	0	(Feature not used)
HBE	0	(Feature not used)

6.16 Hard Disk Drive Tray (HDDT)

Overview

The Hard Disk Drive Tray (HDDT) contains the coprocessor hard disk drive(s). A maximum of two drives can exist on a single HDDT.

The Primary (first drive) is always mounted on the right side. This is done for cooling reasons and must be adhered to in all cases. This drive will always have device address 6 (physical pinning).

An optional second drive is mounted on the left side. This drive will always have device address 5 (physical pinning).

A printed circuit board in the tray is used to provide a means of connecting the SCSI bus to each drive as well as routing it through the backpanel to the HDDT I/O board so it can be terminated or connected to the DAT drive. DC power from the backpanel is also routed through this board to the drives.

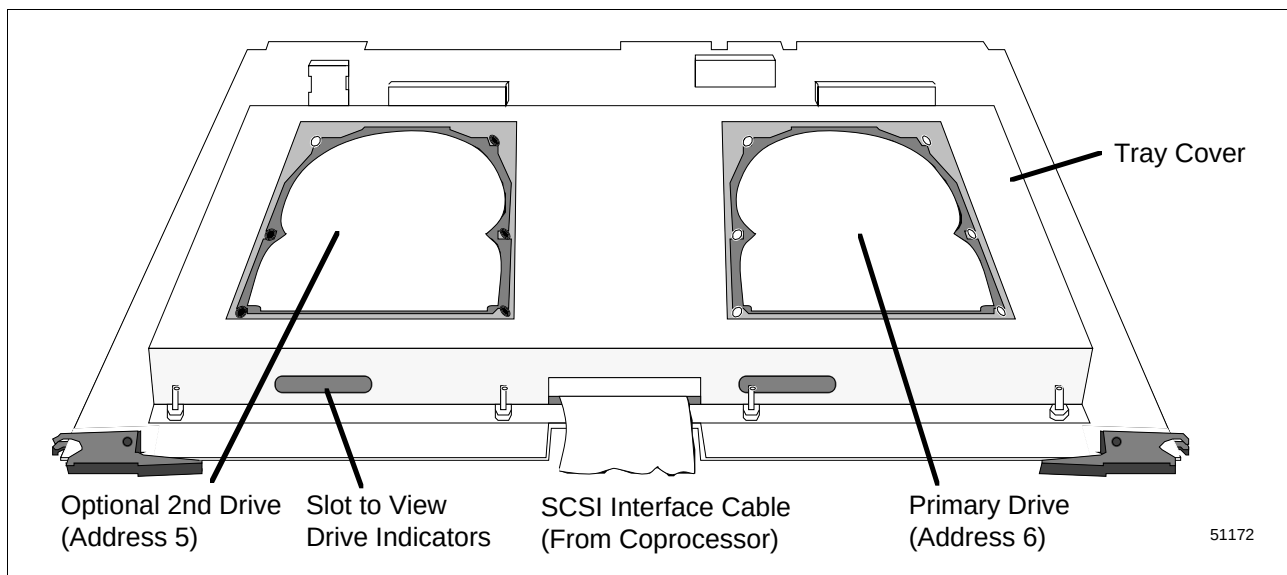
Disk drives are separately replaceable in the event of failure. The drive tray (including internal circuit board) is only replaceable as a single unit. When ordering a replacement drive tray, it will be supplied without drives. The old drive can be reused if it is still serviceable. Disk drives are ordered separately.

The spare parts section of this manual supports this strategy.

Hard disk drive tray (front view)

The following illustration shows the hard disk drive tray as you would see it after removing it from the node chassis. Notice the placement of the disk drives. The preliminary drive is always on the right side of the tray.

Figure 6-17 Front View of Hard Drive Disk Tray (HDDT Board)



Continued on next page

6.16 Hard Disk Drive Tray (HDDT), Continued

Disk drive physical placement and connections

Figure 6-18 shows the drive tray from the rear view. Notice the following.

- physical position of each drive
- drive tray internal circuit board
- SCSI interface ribbon cable for each drive

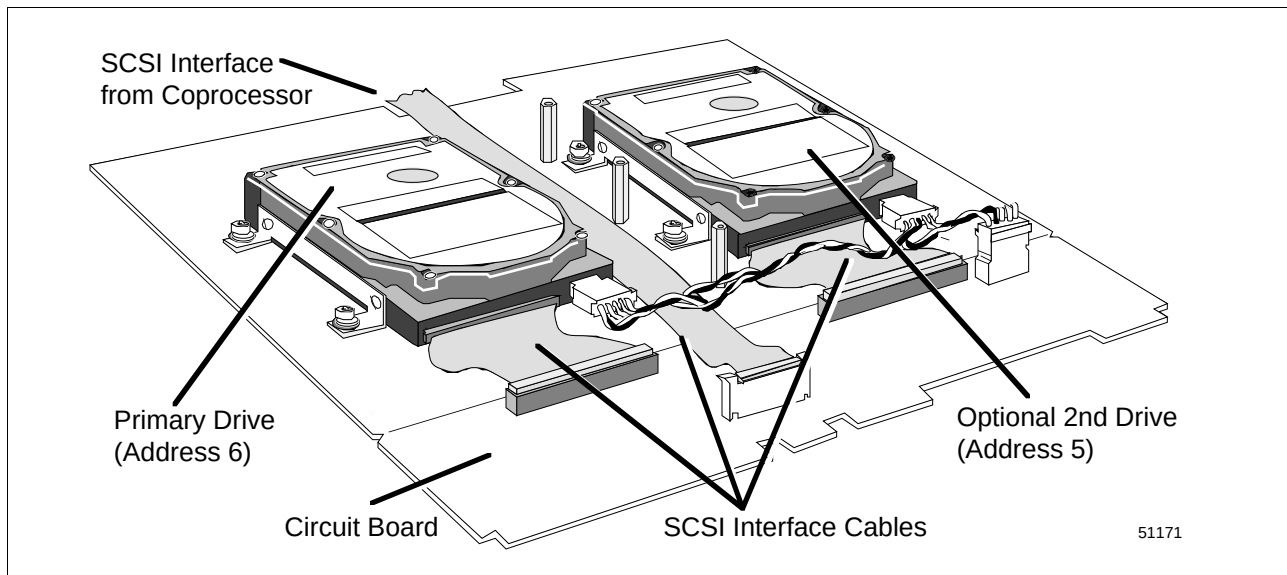
Disk drives currently originate from several different manufacturing sources. Their drive mounting and electrical connection properties are identical.

Each available disk is described later in this section.

Illustration

This is a view of the hard disk drive tray from the rear side of the board. Notice the SCSI interface cables. The SCSI interface originates at the front edge of the coprocessor board on the WSI2 assembly.

Figure 6-18 Rear View of Hard Disk Drive Tray (HDDT Board)



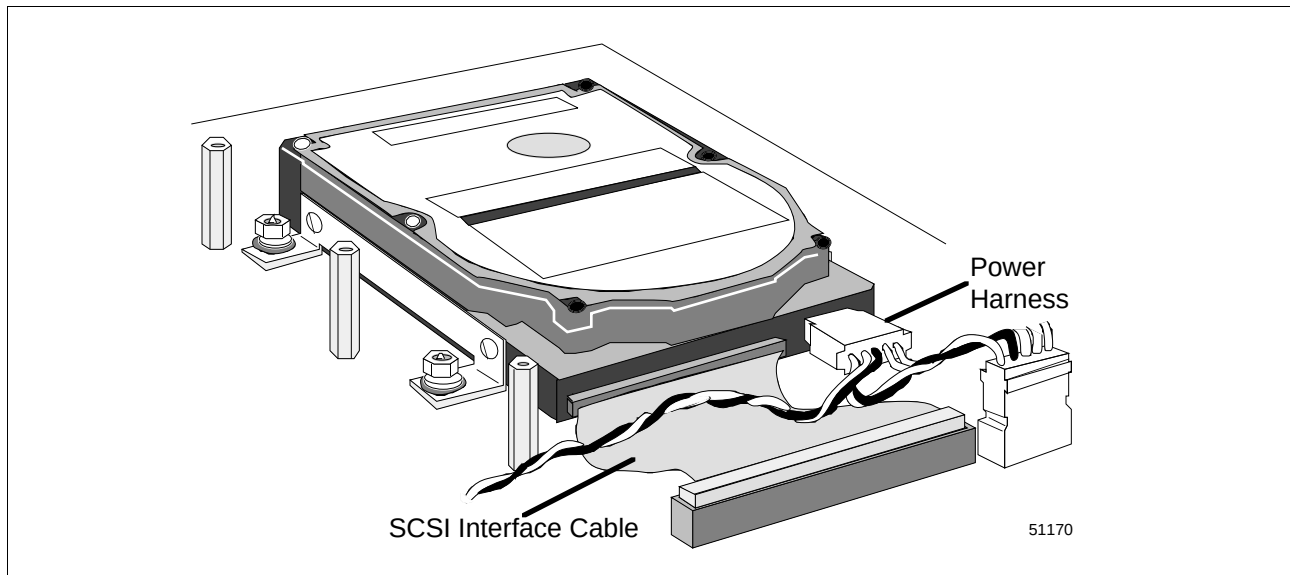
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6.16 Hard Disk Drive Tray (HDDT), Continued

Drive connection detail

Figure 6-19 shows greater detail of how the SCSI interface cable and power harness are connected to the hard drive.

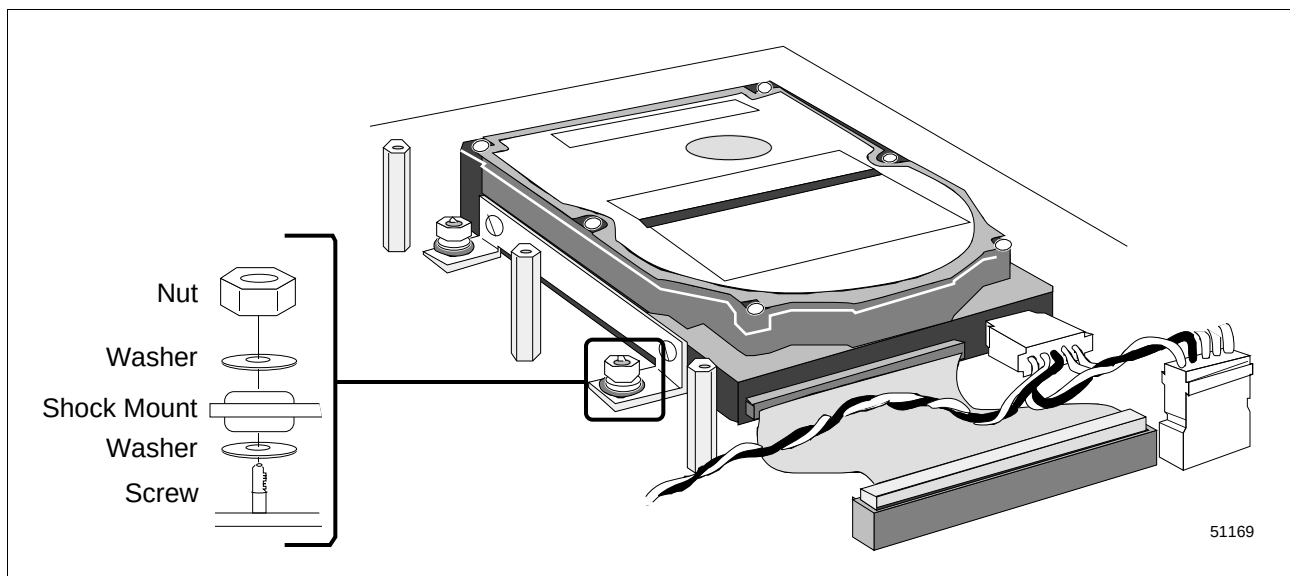
Figure 6-19 Hard Drive Cable Connections



Drive mounting detail

Figure 6-20 shows the detail of the drive mounting. All four mounting screws (one at each corner of the drive) are identical.

Figure 6-20 Hard Drive Mounting to HDDT



Continued on next page

6.16 Hard Disk Drive Tray (HDDT), Continued

Disk drives

Several types of disk drives are available for use in the drive tray. Table 6-9 identifies the drive types currently available. The drives of the same capacity (although from different manufacturers) are functionally the same. An order for a specific size spare drive will be filled with one from any of those listed in the table.

It is imperative that you can identify the different drive types by looking at the physical drive and the markings on it. Each drive type will have uniquely different physical pinning requirements associated with it. Drive pinning details are separately defined in the information following the table.

All of the disks are bought from Hewlett-Packard under a common purchase drawing list in the table.

Table 6-9 Disk Drives Currently Available

Capacity	Manufacturer	Identification Marking on Drive
525 MB	Quantum	Hewlett-Packard model A4280A Quantum model LPS525S
	Seagate	Hewlett-Packard model A4280A Seagate model ST3600N
1.2 GB	Digital	Hewlett-Packard model A2398A Digital model DSP 3107L
	Quantum	Hewlett-Packard model A2398A Quantum model LPS1080S
		Hewlett-Packard model A2398A Quantum model VP31110
	Seagate	Hewlett-Packard model A2398A Seagate model ST31200
2 GB	Quantum	Hewlett-Packard model no. S1164A Seagate Model no.VP32210
	Seagate	Hewlett-Packard model no. S1164A Seagate Model no.VP32210
	Hewlett-Packard	Hewlett-Packard model no. S1164A Seagate Model no.C3325A

Continued on next page

6.16 Hard Disk Drive Tray (HDDT), Continued

Considerations for ordering replacement drives

There are two things to consider when ordering a replacement disk drive. They are:

- drive capacity
- which drive (primary or optional second drive)
 - The primary drive (right side/address 6) is supplied with generic HP-UX software already loaded.

This is a requirement for booting the coprocessor after the primary drive is replaced. Upon a successful boot, using the generic software, access to the PIN must be configured and a complete software restore must be performed from a previously generated DAT drive "Backup Tape" (or other available source) over the PIN network. This will restore the coprocessor file system to reflect conditions that existed when the "Backup Tape" was generated.

- The optional second drive (left side/address 5) is supplied properly formatted, but without HP-UX software.

This drive can be put under system control as the second drive using the "mount " command without any preparation steps. A software restore must be performed from a previously generated DAT drive "Backup Tape" (or other available source) over the PIN network. This will restore the coprocessor file system to reflect conditions that existed when the "Backup Tape" was generated.

The parts list in this section of this publication makes the above distinctions and provides the appropriate number to order the appropriate part.

Replacement drives taken from another U^XS or A^XM

Care must be exercised when a replacement drive does not come directly from Honeywell spare parts (from hot spares or training system).

If the optional second drive is replaced with a drive containing files from another system, the existing files must be discarded by initializing the drive and then performing a software restore operation from DAT drive "Backup Tape" (or other available source) before returning the replaced drive under system control using the "mount" command.

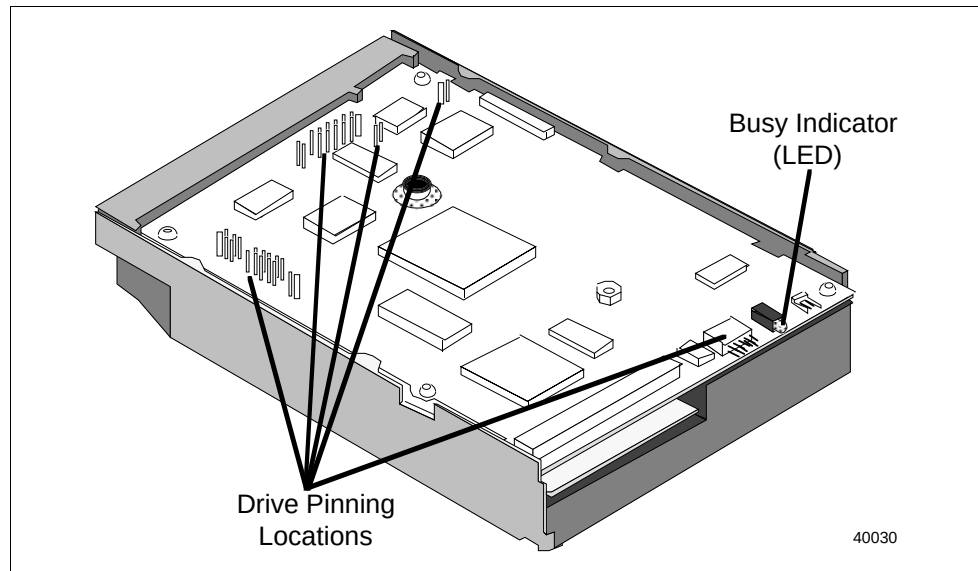
If the primary drive is replaced with a drive containing files from another system, it will have to have the proper files resident to allow a software boot operation (i.e., it must have been the primary drive in the previous system). The coprocessor will boot from this replacement device and assume the exact configuration that existed on the previous system. After making the appropriate configuration changes to access the PIN network, a complete software restore from DAT drive "Backup Tape" (or other available source) is required to get the coprocessor file structure to reflect conditions existing prior to the drive replacement.

6.17 525 MB Quantum Disk Drive (LPS525S)

Introduction

The figure below shows a 525 MB Quantum drive. Notice the parts identified.

Figure 6-21 525 Quantum Drive



Continued on next page

6.17 525 MB Quantum Disk Drive (LPS525S), Continued

525 MB Quantum drive pinning

The only pinning that should be required is the SCSI interface device address. The drive on the right of the tray is always address 6 and the optional drive on the left is always address 5. The remaining option jumpers should be correct from the factory. A quick reference is always advised.

Figure 6-22 525 Quantum Drive Pinning (Primary Drive)

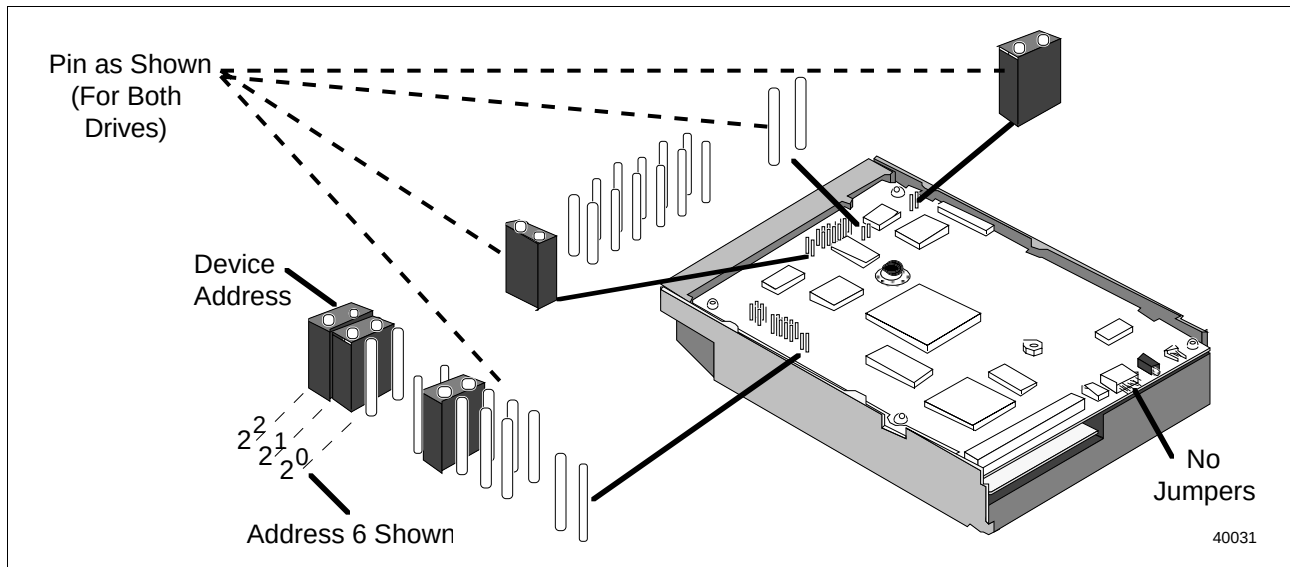
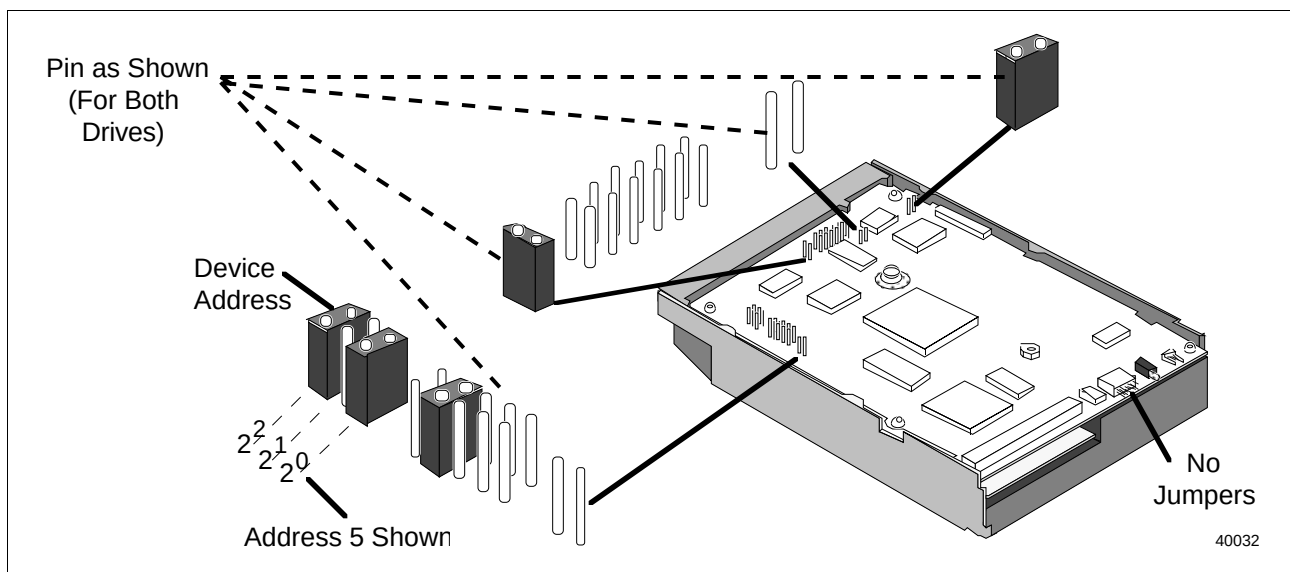


Figure 6-23 525 Quantum Drive Pinning (Optional Drive)

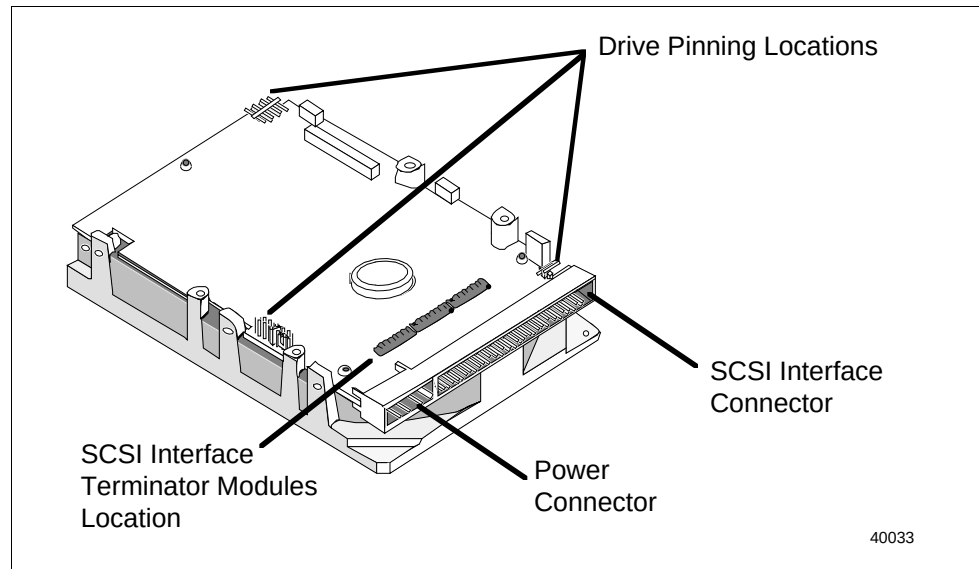


6.18 525 MB Seagate Drive (ST3600N)

Introduction

The figure below shows a 525 MB Seagate drive. Notice the identified parts.

Figure 6-24 525 MB Seagate Drive



Continued on next page

6.18 525 MB Seagate Drive (ST3600N), Continued

525 MB Seagate drive address pinning

The pinning to define the device address on the SCSI bus on the right of the tray is always pinned for address 6 and the optional drive on the left is always pinned for address 5.

Figure 6-25 illustrates the pinning for address 6. Notice that the drive is shown upside down. Figure 6-26 illustrates the pinning for the same drive for address 5.

Figure 6-25 525 MB Seagate Drive Pinning (Primary Drive)

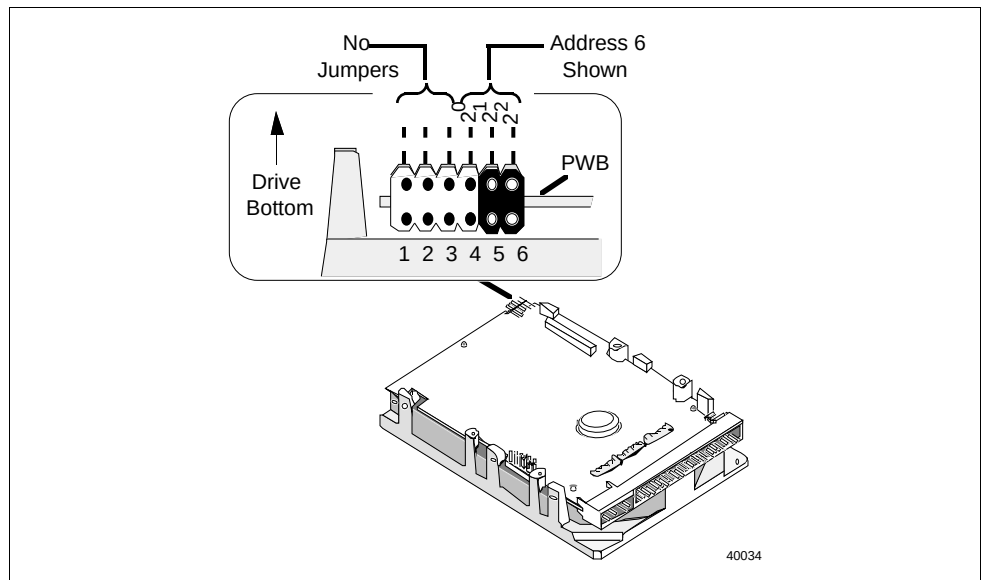
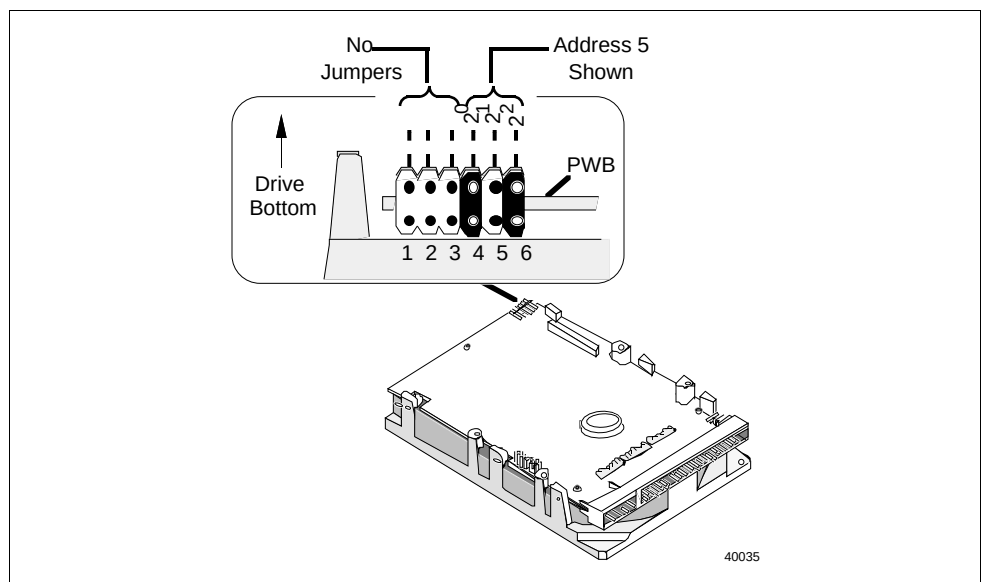


Figure 6-26 525 MB Seagate Drive Pinning (Optional Drive)



Continued on next page

6.18 525 MB Seagate Drive (ST3600N), Continued

525 MB Seagate drive SCSI interface bus termination

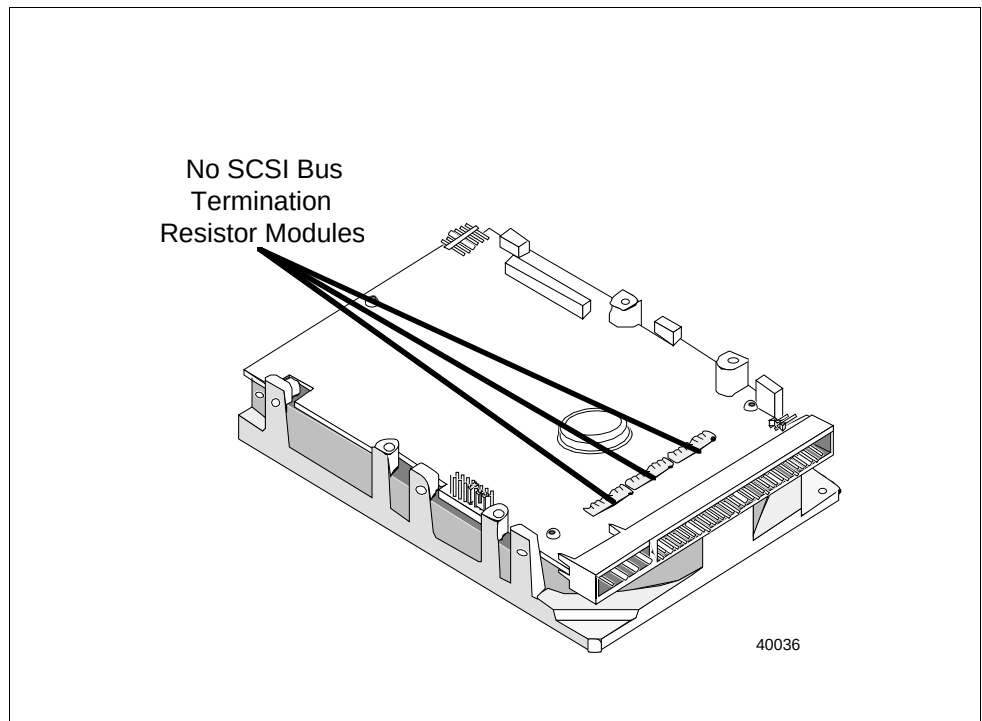
The Seagate drives have the capability to provide termination for the SCSI interface cable. This feature is not needed for the Universal Station^X applications

- in the absence of a DAT drive, a terminator is placed on the end of the bus at the connector of the HDDT I/O board.
- if a DAT drive is connected to the SCSI bus, a terminator is configured in the DAT drive.

The figure below illustrates the missing termination resistor modules on the drive. Replacement drives may have these resistor modules in place. Remove them if they are present. This applies to both drives in the disk tray.

Notice that the drive is upside down. You cannot see the terminator resistor module location once the drive is installed.

Figure 6-27 525 MB Seagate Drive Termination



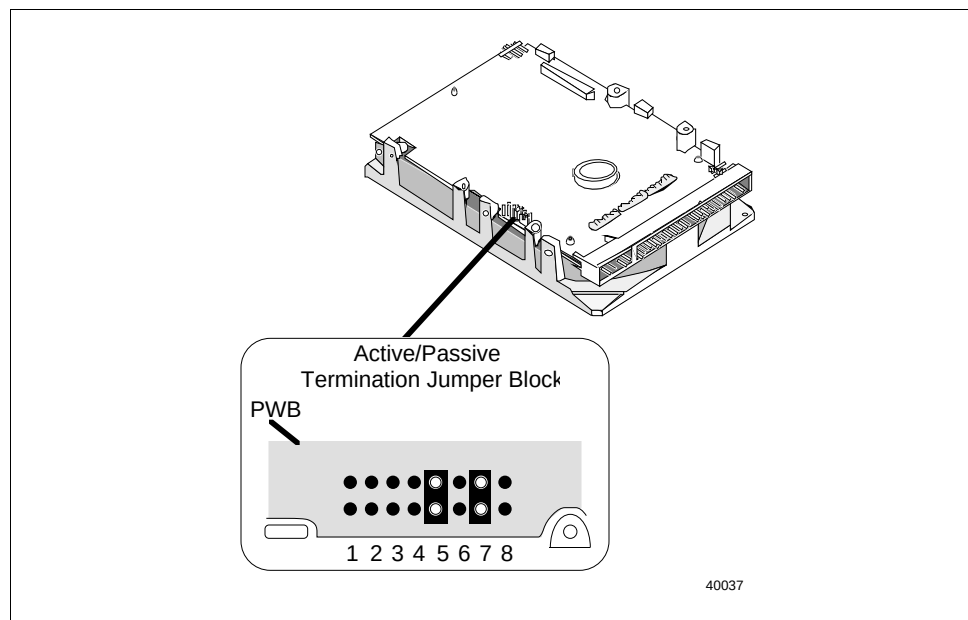
Continued on next page

6.18 525 MB Seagate Drive, (ST3600N) Continued

525 MB Seagate drive termination/parity pinning

Both drives 5 and 6 should have these options pinned as shown below.

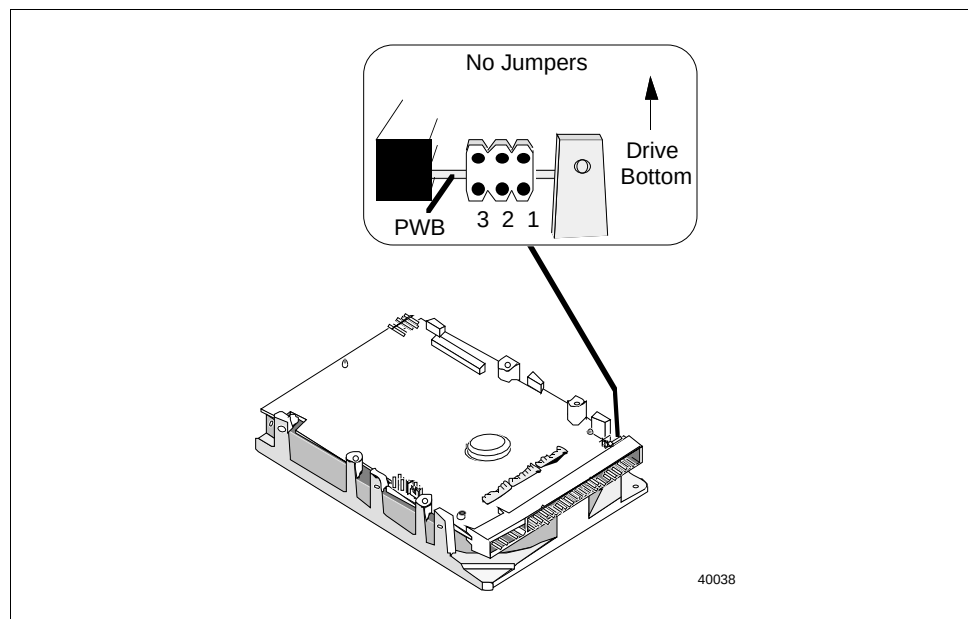
Figure 6-28 525 MB Seagate Drive Termination/Parity Pinning



525 MB Seagate drive unused pinning

There are no option jumpers required for this pinning area as shown below. This applies to both possible drives in the disk tray.

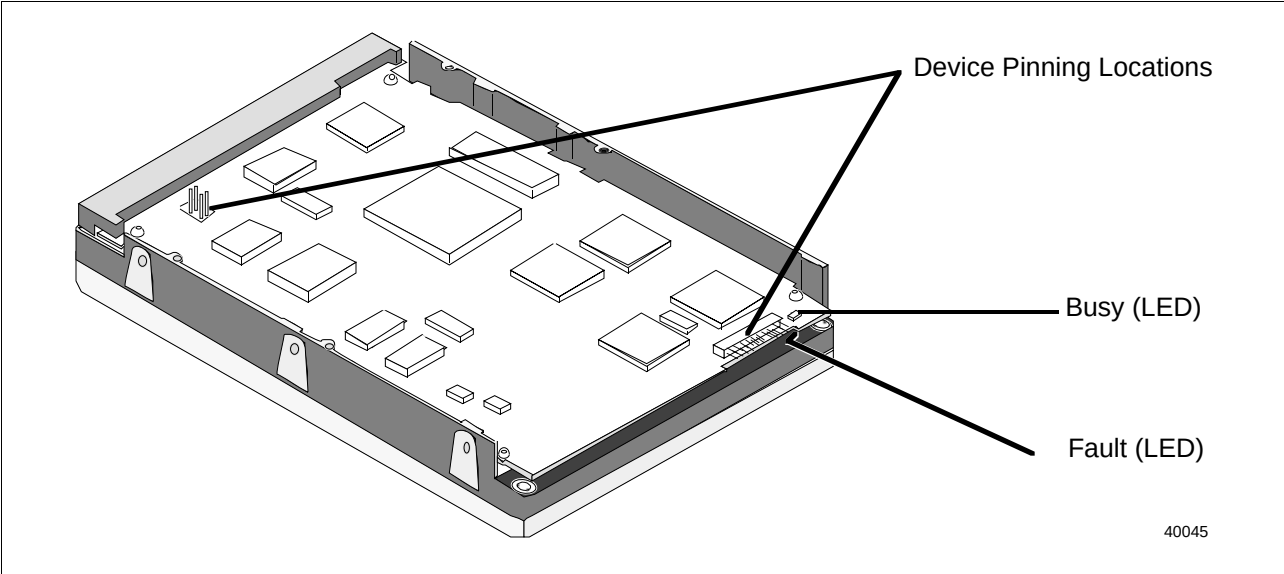
Figure 6-29 525 MB Seagate Drive Unused Pinning



6.19 1.2 GB Digital Equipment Corp Drive (DSP3107L)

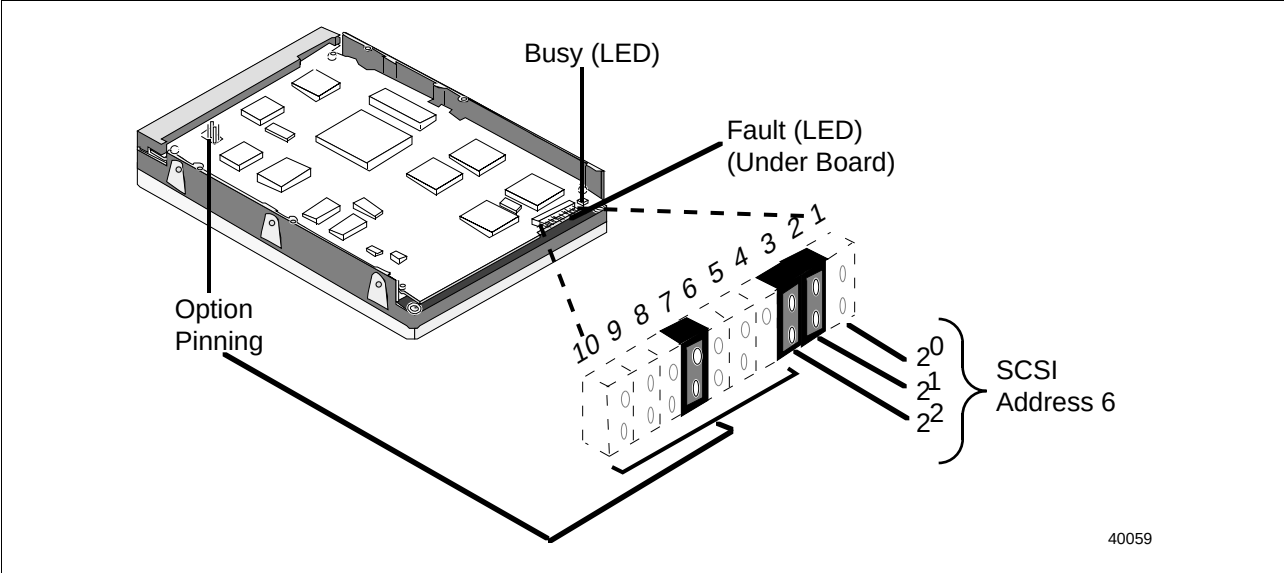
Introduction The figure below shows a 1.2 GB Digital Equipment Corp disk drive. The drive is upside down. Notice the identified parts.

Figure 6-30 1.2 GB Digital Equipment Corp Drive



1.2 GB DEC drive SCSI address pinning. The figure below shows the pinning for SCSI address 6.

Figure 6-31 1.2 GB DEC Drive SCSI Address Pinning (Address 6)



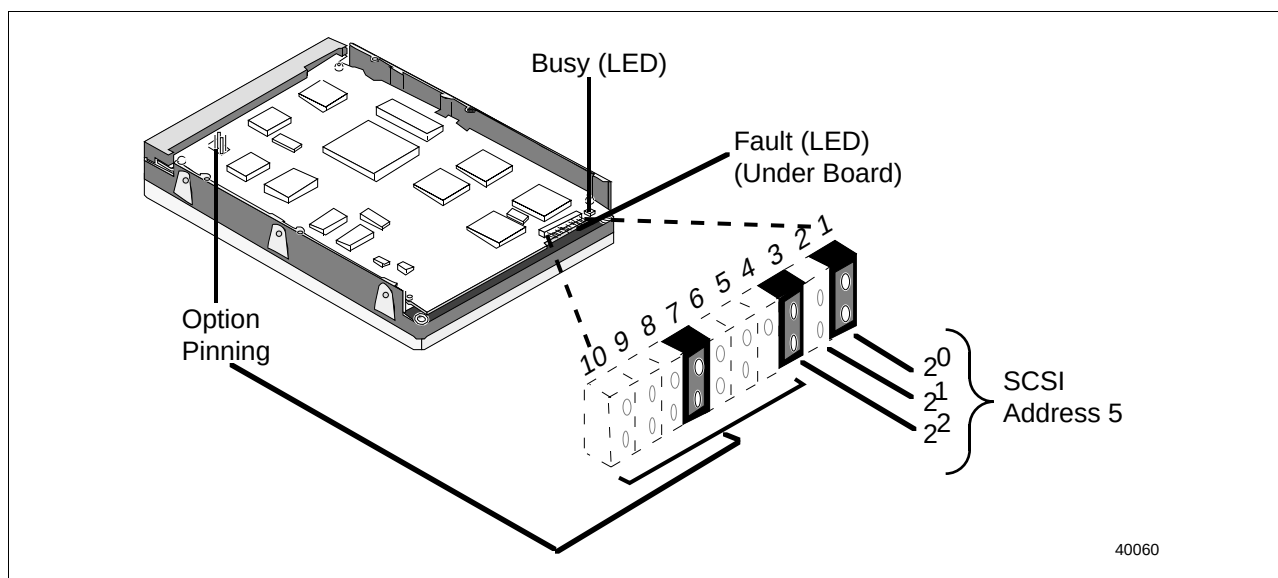
Continued on next page

6.19 1.2 GB Digital Equipment Corp Drive (DSP3107L), Continued

1.2 GB DEC drive SCSI address pinning

The figure below shows the SCSI address pinning for address 5.

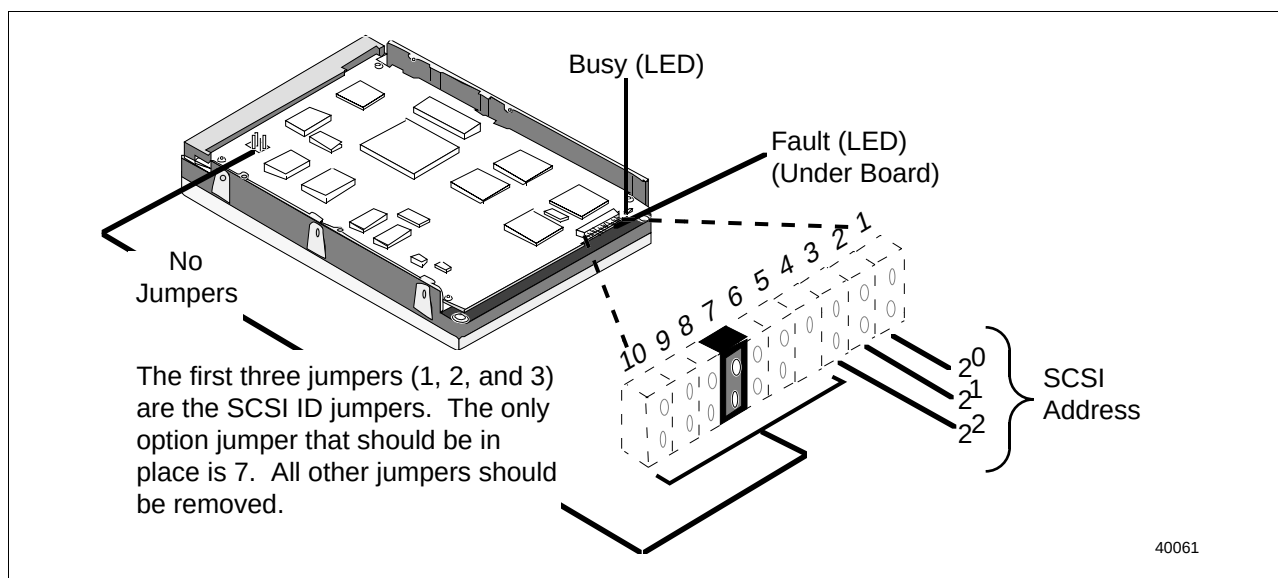
Figure 6-32 1.2 GB DEC Drive SCSI Address Pinning (Address 5)



1.2 GB DEC drive SCSI termination pinning

The SCSI termination pinning options provided on the drive are not used in Universal Station^X. They should all be removed if they are present. This applies to both drives in the disk tray.

Figure 6-33 1.2 GB DEC Drive SCSI Termination Pinning

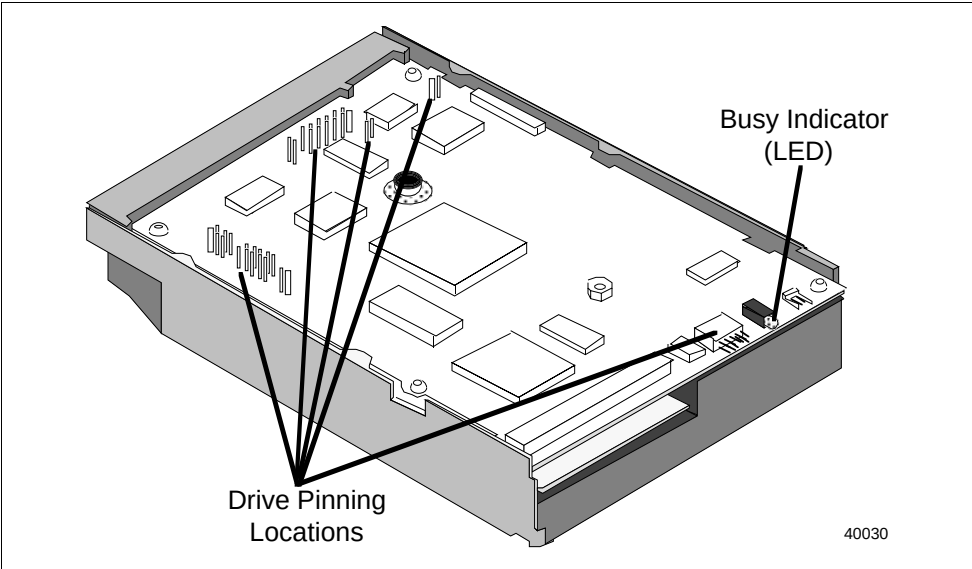


6.20 1.2 GB Quantum Drive (LPS1080S)

Introduction

The figure below illustrates the 1.2 GB Quantum disk drive. The drive is shown upside down. Notice the identified parts.

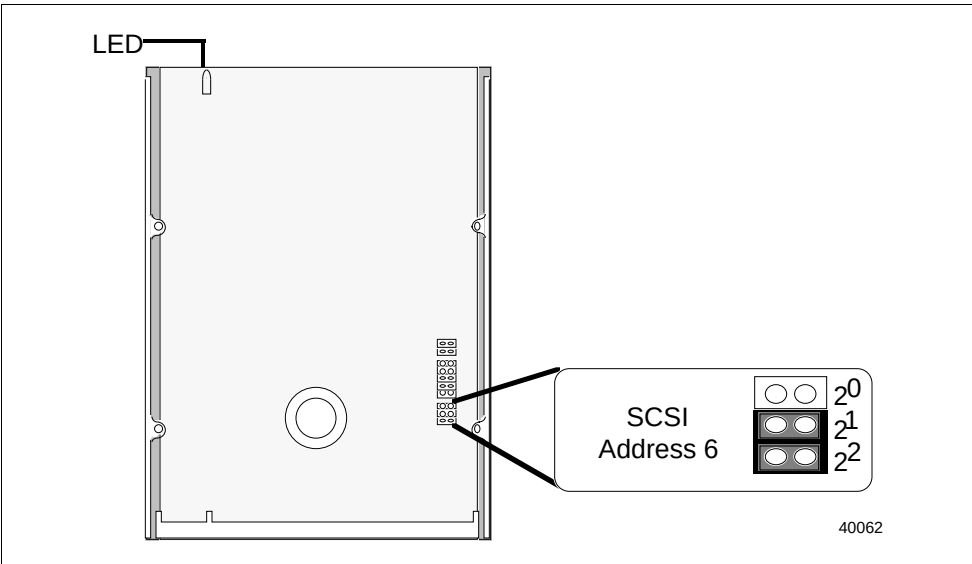
Figure 6-34 1.2 GB Quantum Disk Drive



1.2 GB Quantum drive SCSI address pinning

The figure below identifies the pinning required for address 6.

Figure 6-35 1.2 GB Quantum Disk Drive Address Pinning (Address 6)



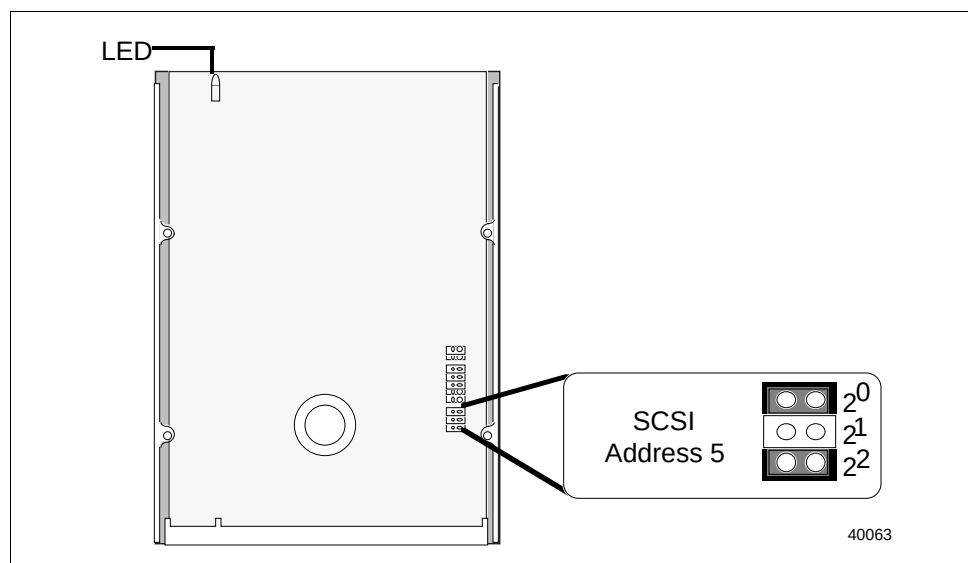
Continued on next page

6.20 1.2 GB Quantum Drive (LPS1080S), Continued

1.2 GB Quantum drive SCSI address pinning

The figure below identifies the address pinning for address 5.

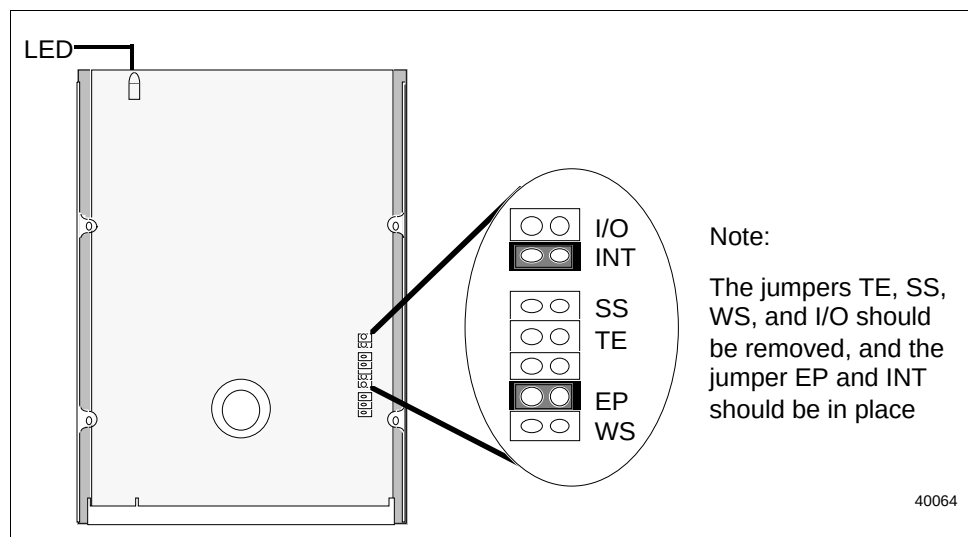
Figure 6-36 1.2 GB Quantum Drive SCSI Address Pinning (Address 5)



1.2 GB Quantum drive option pinning

The figure below illustrates the drive option pinning. Both drives must be pinned as shown.

Figure 6-37 1.2 GB Quantum Drive Option Pinning

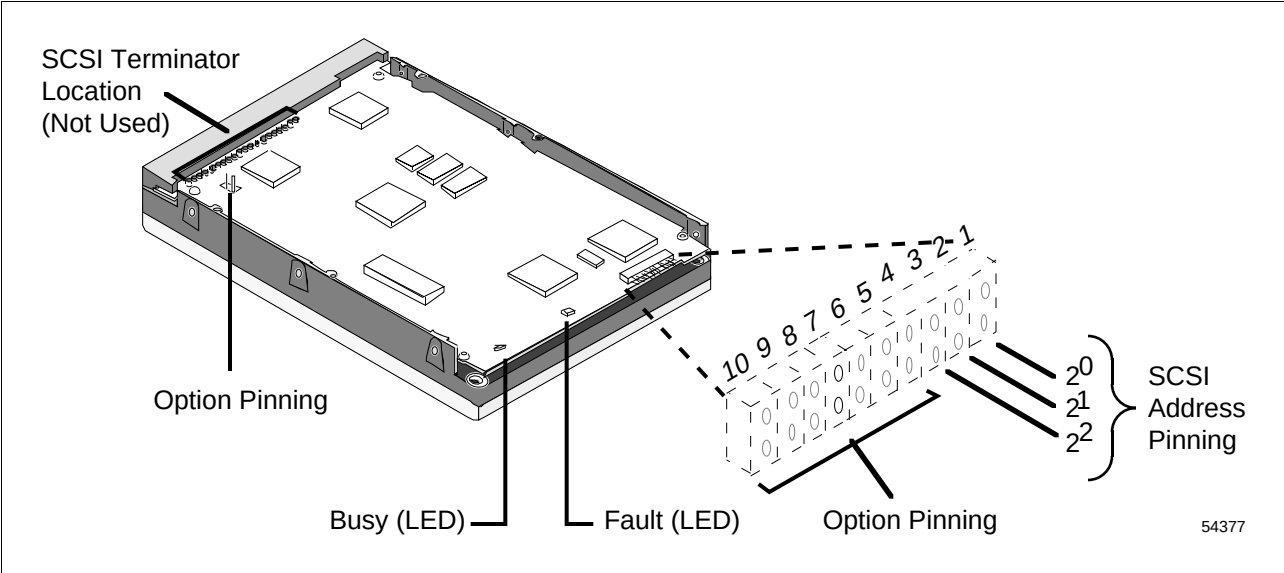


6.21 1.2 GB Quantum Drive (VP31110)

Introduction

The figure below illustrates the 1.2 GB Quantum (VP31110) disk drive. The drive is shown upside down. Notice the identified pinning location and “Busy” indicator.

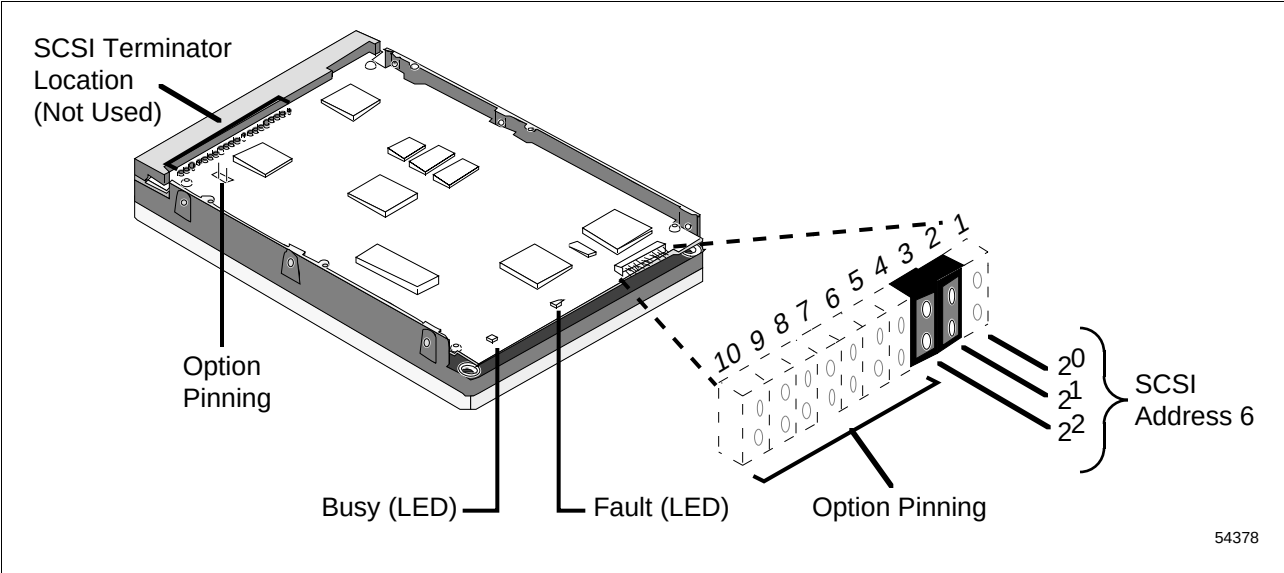
Figure 6-38 1.2 GB Quantum Disk Drive (VP-31110)



1.2 GB Quantum drive (VP31110) SCSI address pinning

The figure below identifies the pinning required for SCSI address 6.

Figure 6-39 1.2 GB Quantum Drive SCSI Address Pinning (Address 6)



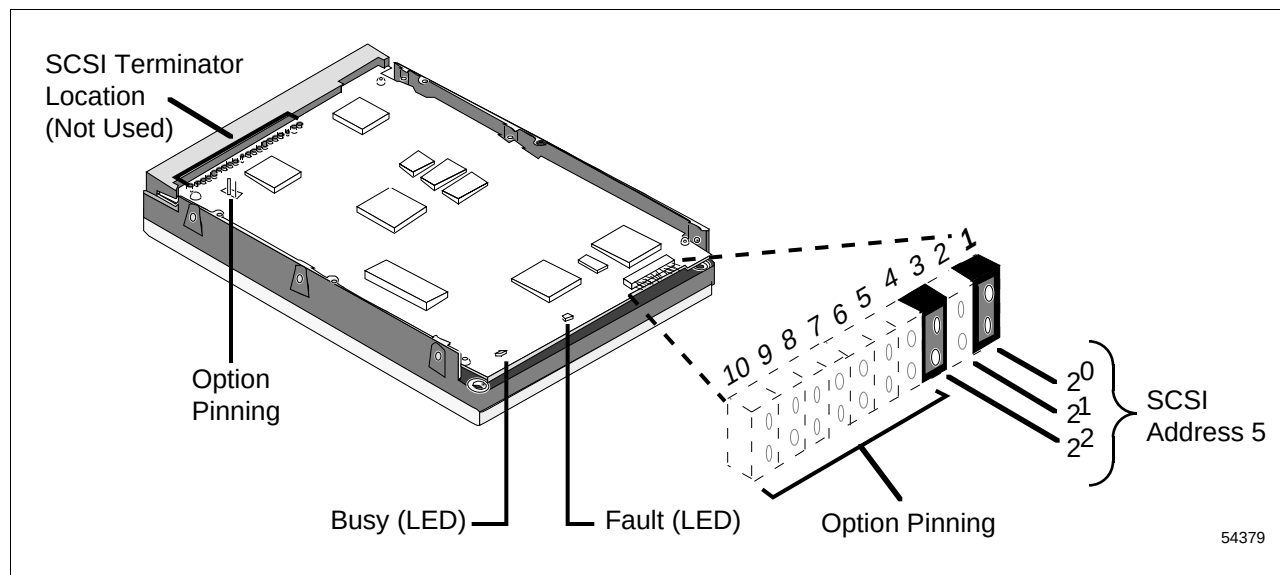
Continued on next page

6.2.1 1.2 GB Quantum Drive (VP31110), Continued

1.2 GB Quantum drive (VP31110) SCSI address pinning

The figure below is identical to the one in Figure 6-39 with the exception it is for SCSI address 5.

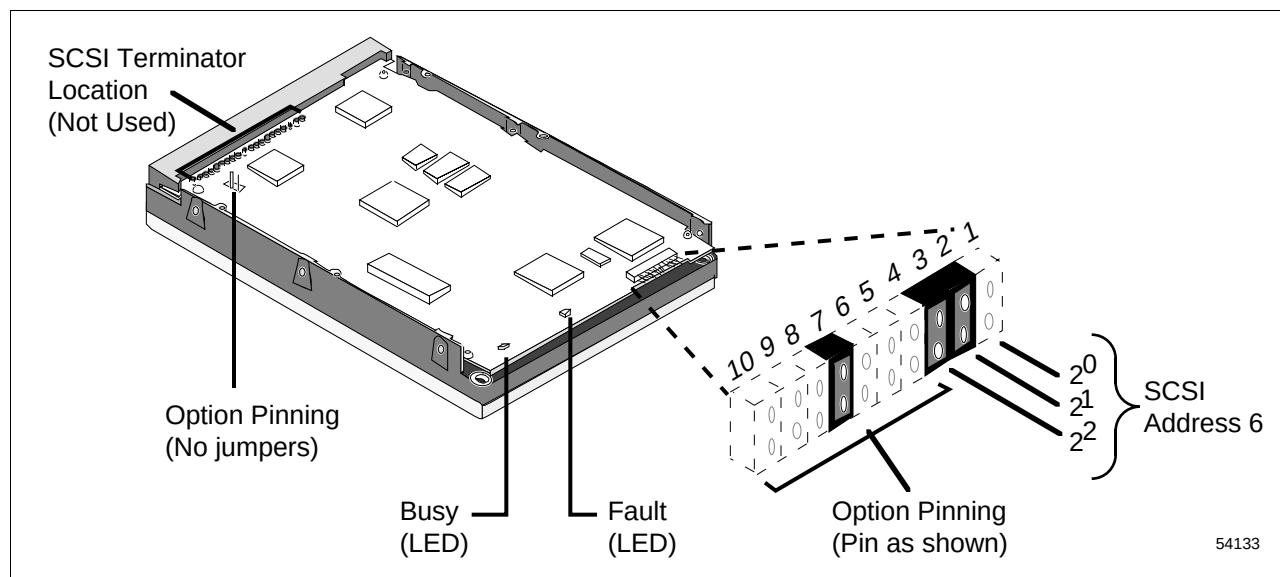
Figure 6-40 1.2 GB Quantum Drive SCSI Address Pinning (Address 5)



1.2 GB Quantum drive (VP31110) option pinning

The following figure illustrates the option pinning for use in the U^XS. Both drives must have the options pinned as shown.

Figure 6-41 1.2 GB Quantum Drive Option Pinning

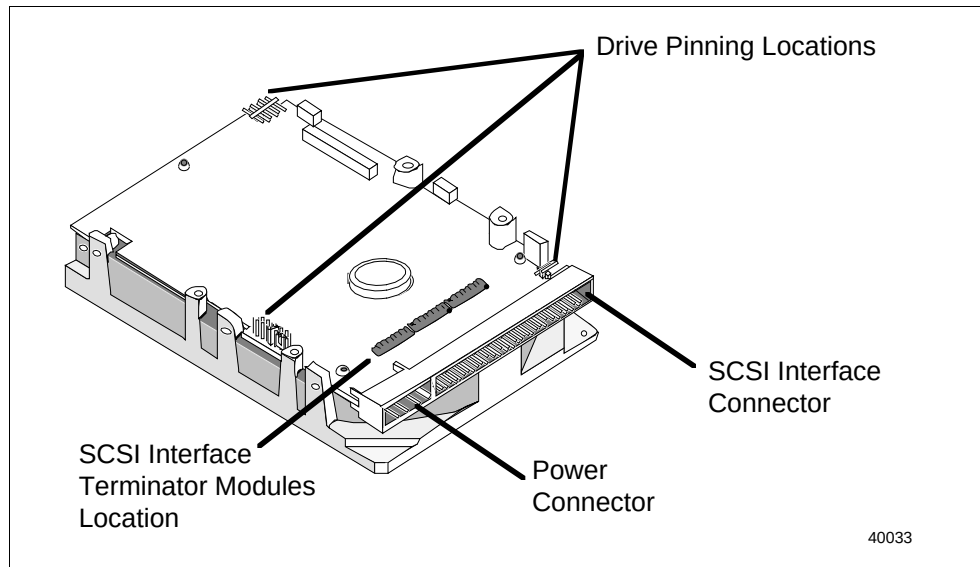


6.2.2 1.2 GB Seagate Drive (ST31200)

Introduction

The figure below illustrates the 1.2 GB Seagate disk drive. The drive is shown upside down. Notice the identified parts.

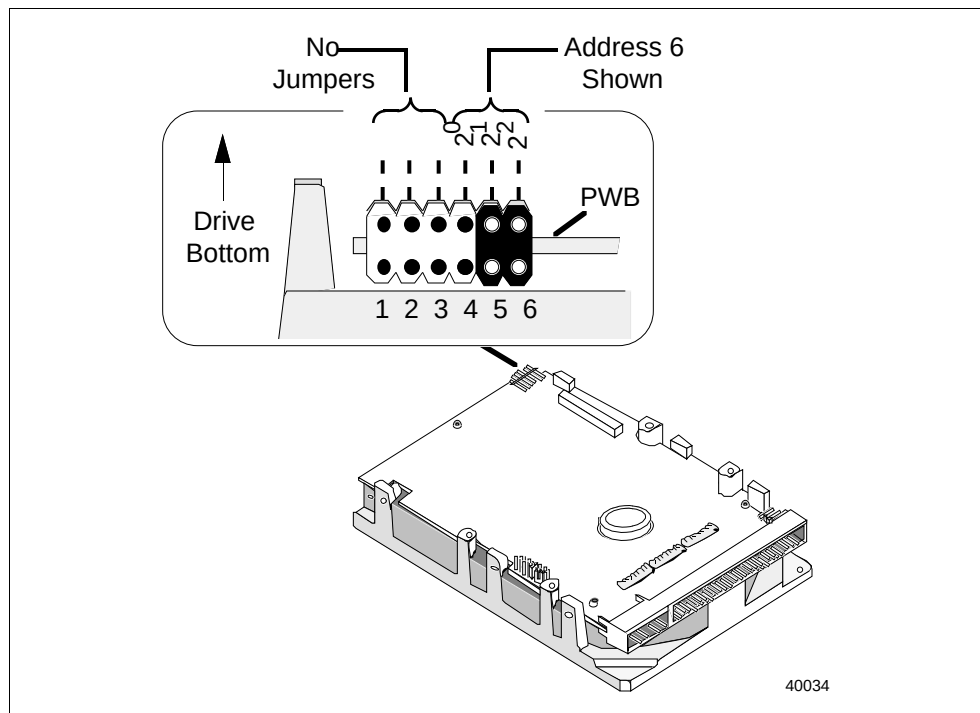
Figure 6-42 1.2 GB Seagate Disk Drive



1.2 GB Seagate drive SCSI address pinning

The figure below identifies the pinning required for address 6.

Figure 6-43 1.2 GB Seagate Drive SCSI Address Pinning (Address 6)



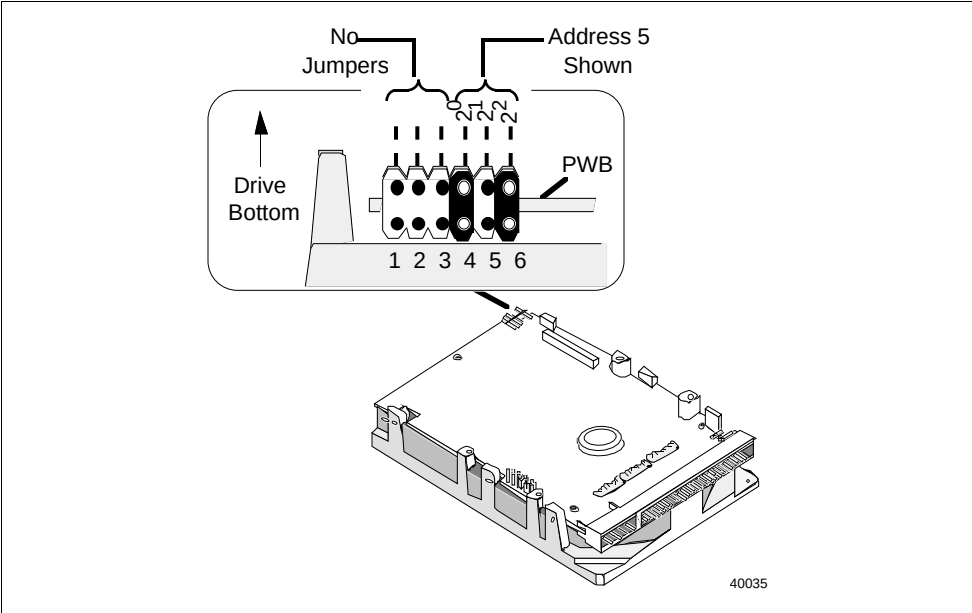
Continued on next page

6.2.2 1.2 GB Seagate Drive (ST31200), Continued

1.2 GB Seagate drive SCSI address pinning

The figure below identifies the pinning required for address 5.

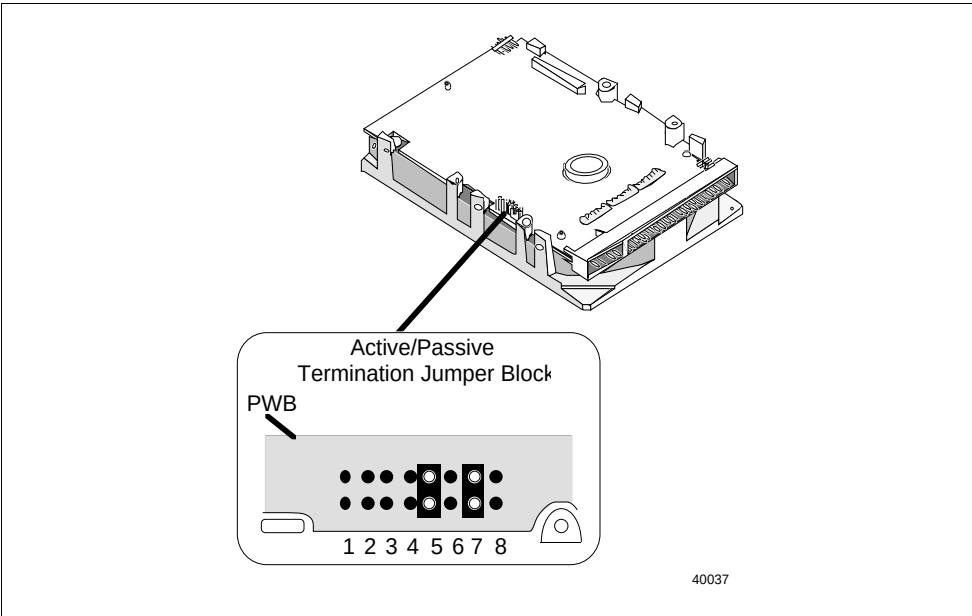
Figure 6-44 1.2 GB Seagate Drive SCSI Address Pinning (Address 5)



1.2 GB Seagate drive termination/parity pinning

The figure below shows the termination/parity pinning that used in both drives.

Figure 6-45 1.2 GB Seagate Drive Termination/Parity Pinning



Continued on next page

6.2.2 1.2 GB Seagate Drive (ST31200), Continued

1.2 GB Seagate drive interface termination

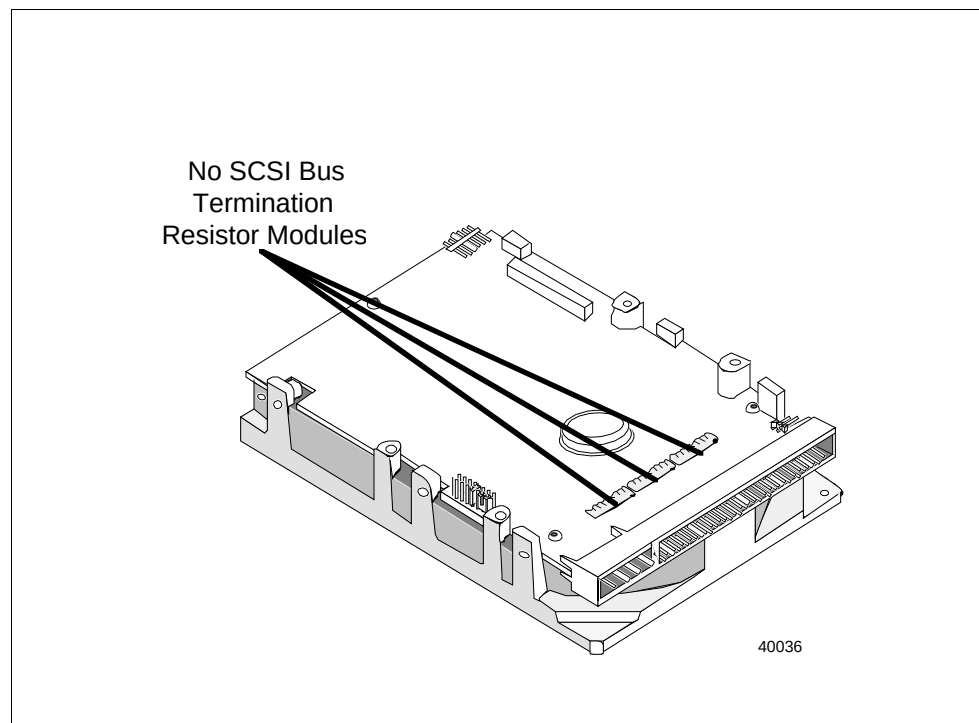
The Seagate drives have the capability to provide termination for the SCSI interface cable. This feature is not needed for the Universal Station^X applications since

- in the absence of a DAT drive, a terminator is placed on the end of the bus at the connector of the HDDT I/O board.
- if a DAT drive is connected to the SCSI bus, a terminator is configured in the DAT drive.

The figure below illustrates the missing termination resistor modules on the drive. Replacement drives may have these resistor modules in place. Remove them if they are present. This applies to both drives in the disk tray.

Notice that the drive is upside down. You cannot see the terminator resistor module location once the drive is installed.

Figure 6-46 1.2 GB Seagate Drive SCSI Interface Termination



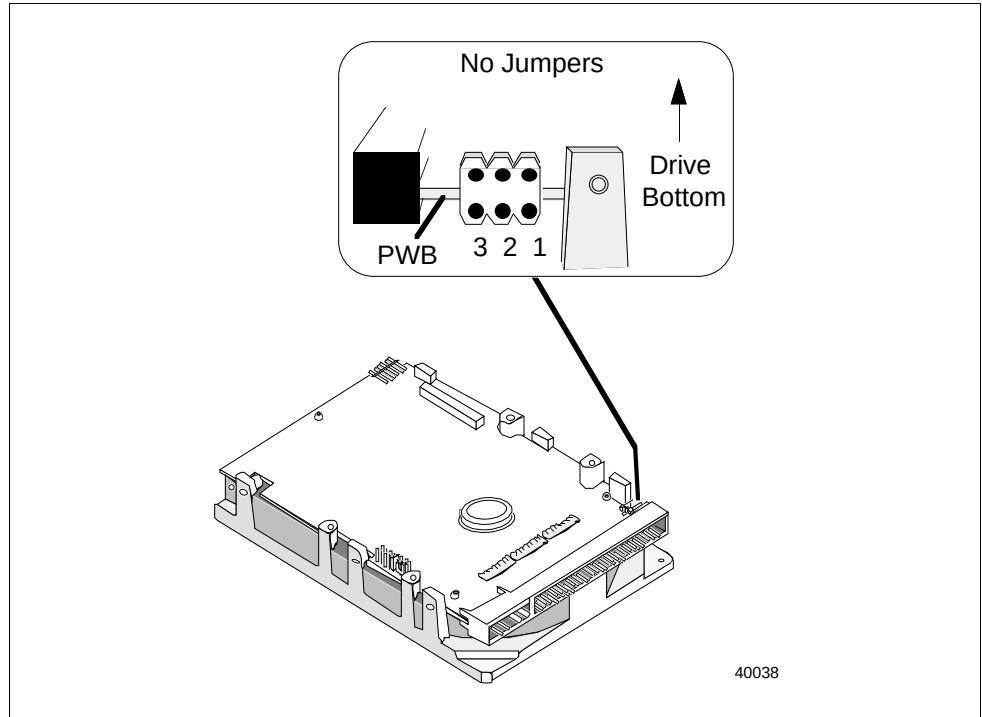
Continued on next page

6.2.2 1.2 GB Seagate Drive (ST31200), Continued

1.2 GB Seagate drive unused pinning

As shown below, there are to be no jumpers in this pinning area.

Figure 6-47 1.2 GB Seagate Drive Unused Pinning Area

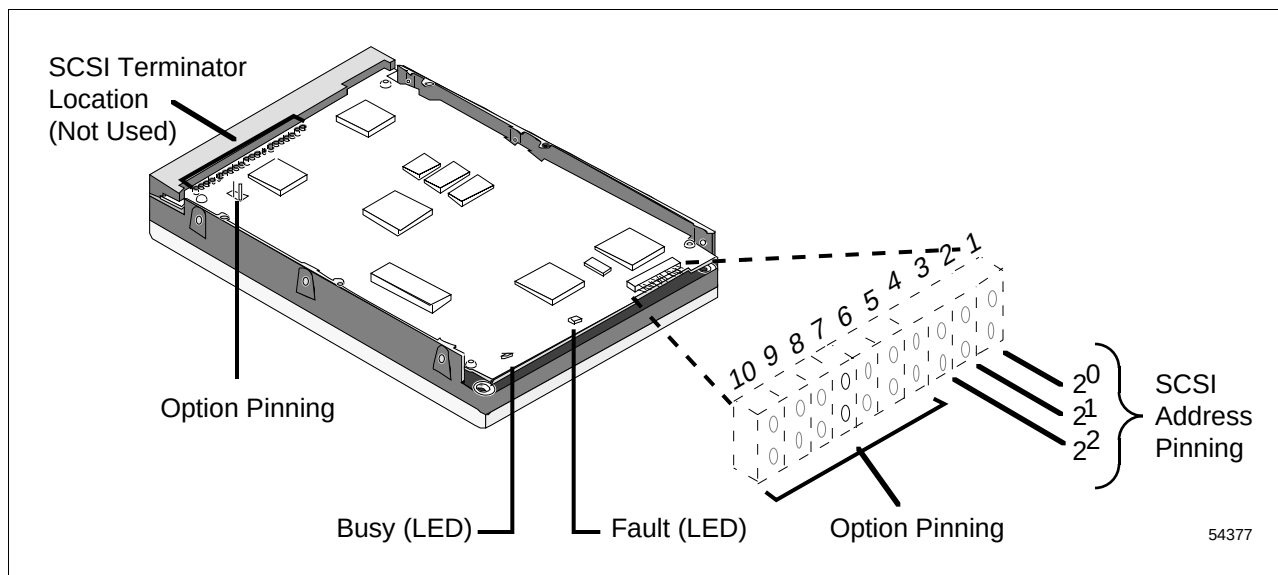


6.23 2 GB Quantum Drive (VP32210)

Introduction

The figure below illustrates the 2 GB Quantum disk drive (VP32210). The drive is shown upside down. Notice the identified pinning locations and the “Busy” indicator.

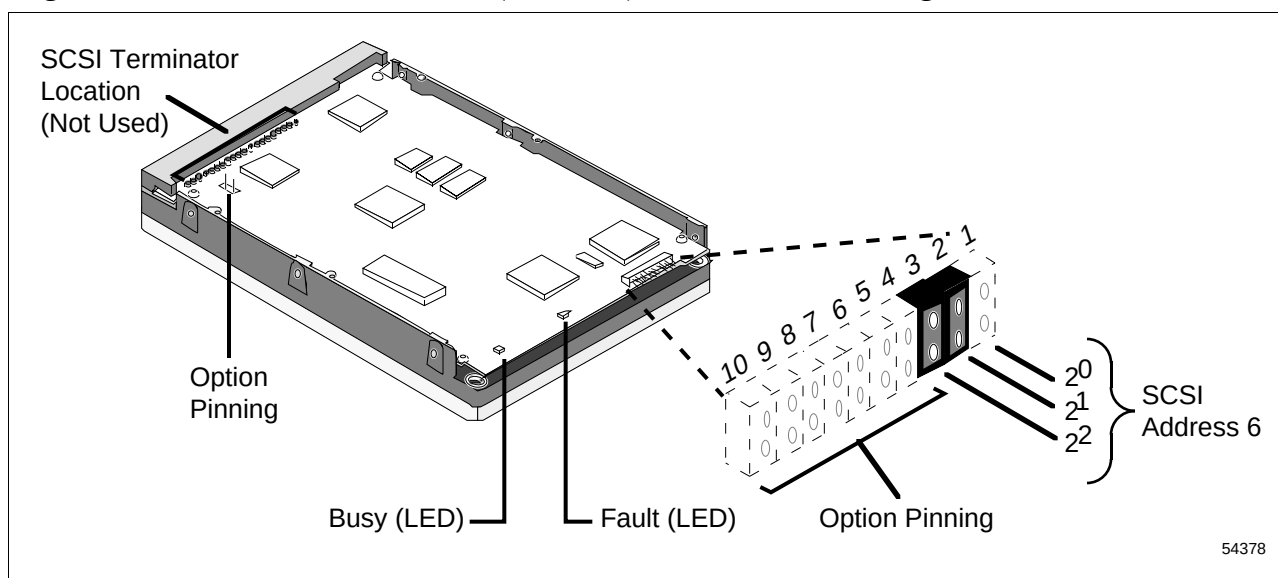
Figure 6-48 2 GB Quantum Drive (VP32210) Pinning Locations



2 GB Quantum drive (VP32210) SCSI address 6

The figure below identifies the pinning required for SCSI address 6.

Figure 6-49 2 GB Quantum Drive (VP32210) SCSI Address Pinning 6



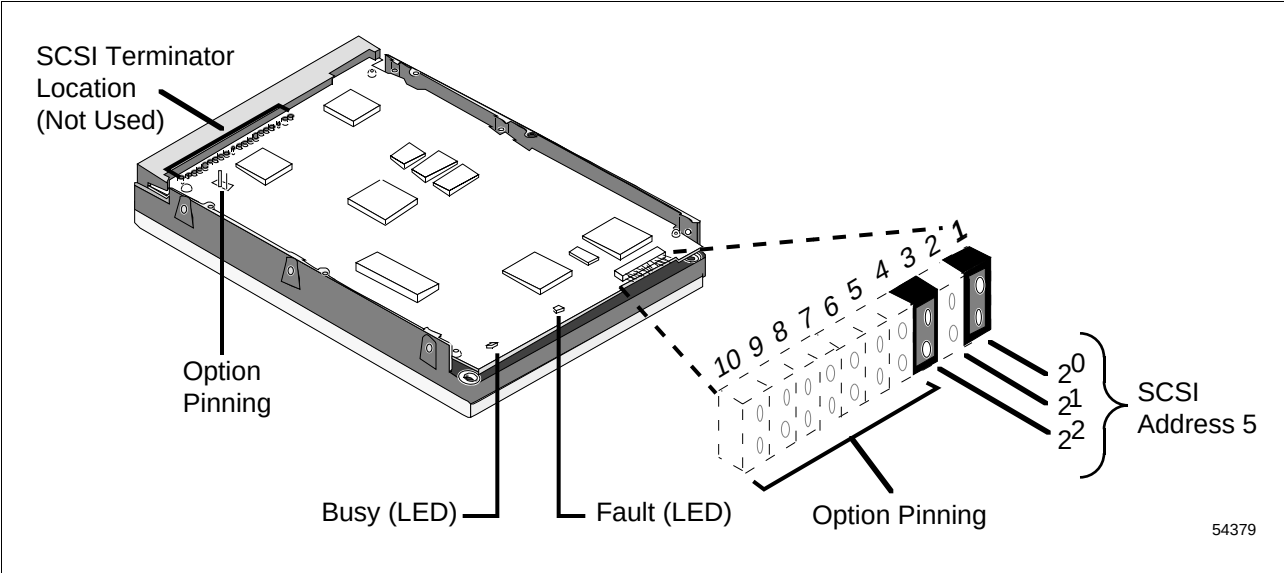
Continued on next page

6.23 2 GB Quantum Drive (VP32210), Continued

2 GB Quantum drive pinning for SCSI address 5

The figure below identifies the pinning required for SCSI address 5.

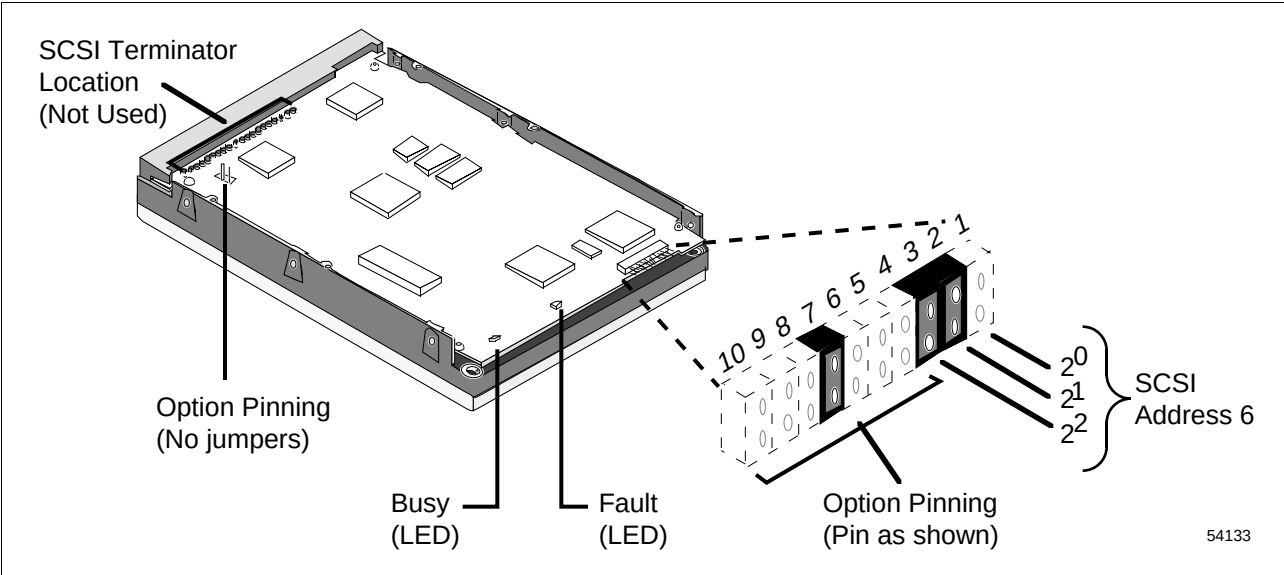
Figure 6-50 2 GB Quantum Drive (VP32210) Pinning for SCSI Address 5.



2 GB Quantum drive (VP32210) option pinning

The following figure illustrates the drive option pinning for use in U^XS. Both drives must have the options pinned as shown.

Figure 6-51 2 GB Quantum Drive (VP32210) Option Pinning

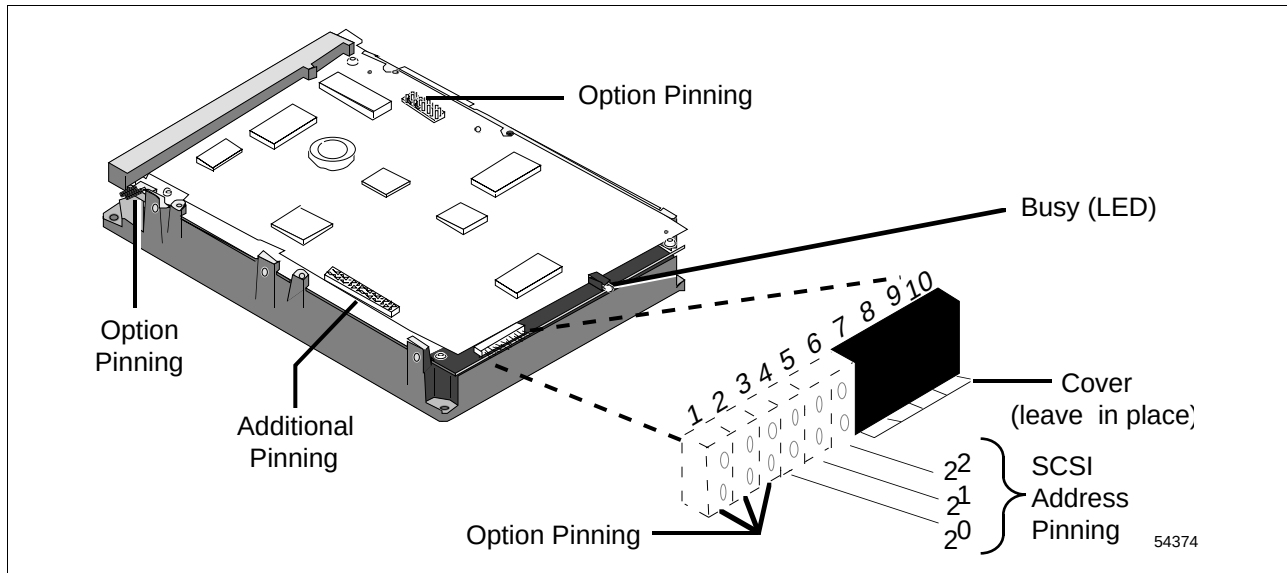


6.24 2 GB Seagate Drive (ST32430)

Introduction

The figure below illustrates the 2 GB Seagate disk drive (ST32430). The drive is shown upside down. Notice the identified pinning locations and the “Busy” indicator.

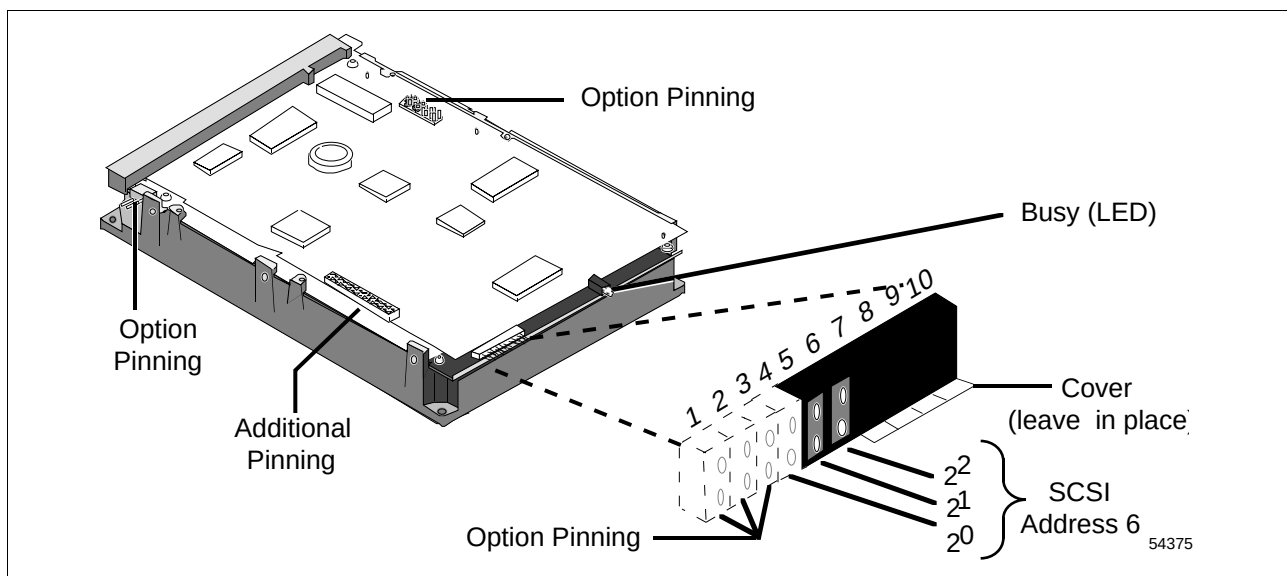
Figure 6-52 2 GB Seagate Drive (ST32430) Pinning Locations



2 GB Quantum drive (ST32430) SCSI address 6

The figure below identifies the pinning required for SCSI address 6.

Figure 6-53 2 GB Quantum Drive (VP32210) SCSI Address Pinning 6



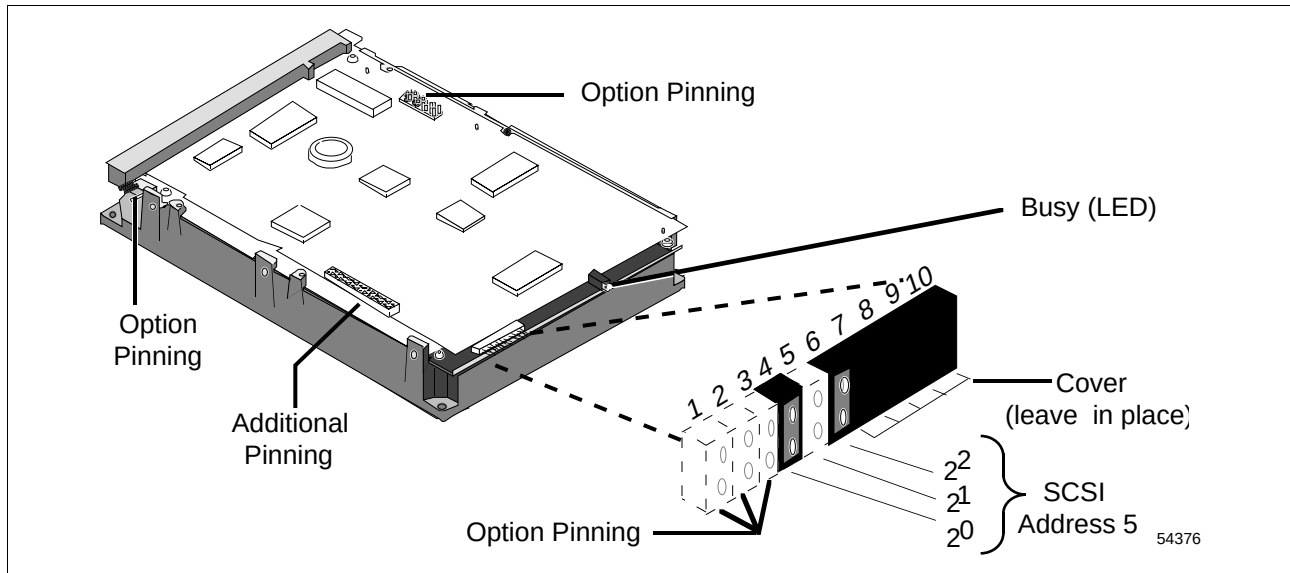
Continued on next page

6.24 2 GB Seagate Drive (ST32430), Continued

2 GB Seagate drive pinning for SCSI address 5

The figure below identifies the pinning required for SCSI address 5.

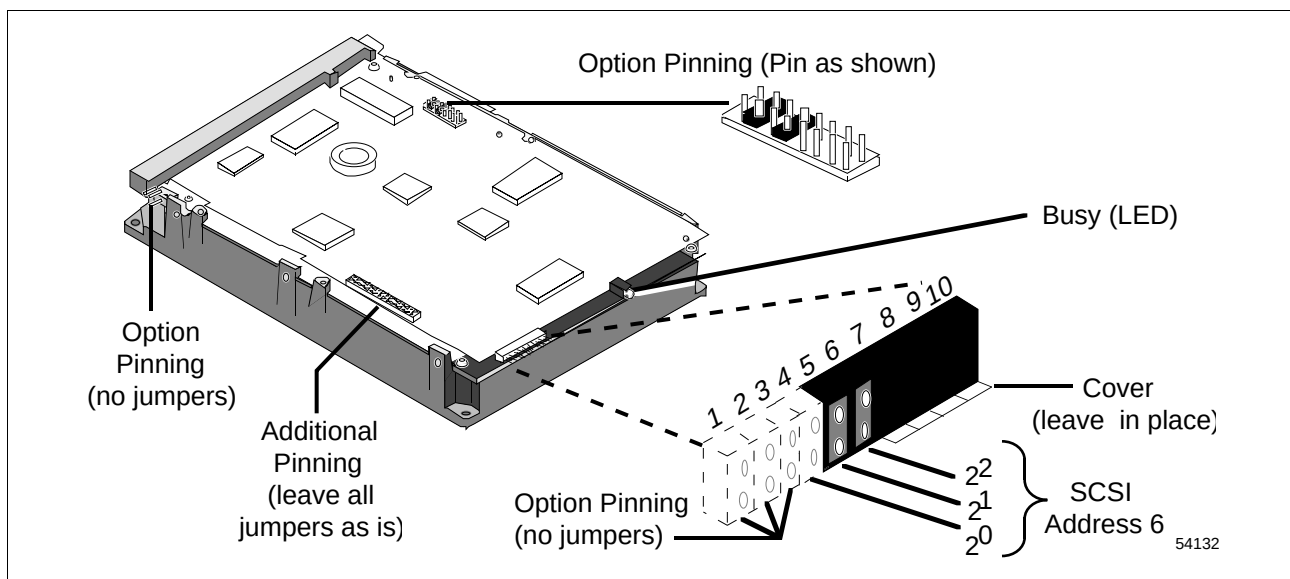
Figure 6-54 2 GB Seagate Drive (ST32430) Pinning for SCSI Address 5.



2 GB Seagate drive (ST32430) option pinning

The following figure illustrates the drive option pinning for use in U^XS. Both drives must have the options pinned as shown.

Figure 6-55 2 GB Seagate Drive (ST32430) Option Pinning

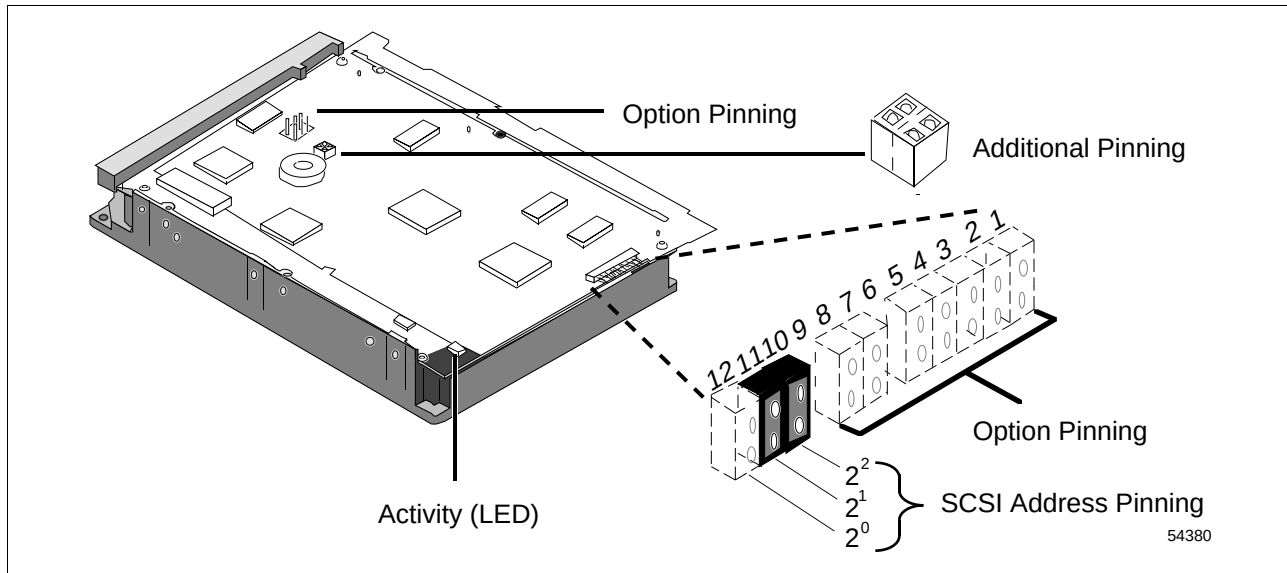


6.25 2 GB Hewlett-Packard Drive (C3325A)

Introduction

The figure below illustrates the 2 GB Hewlett-Packard disk drive (C3325A). The drive is shown upside down. Notice the identified pinning locations and the “Busy” indicator.

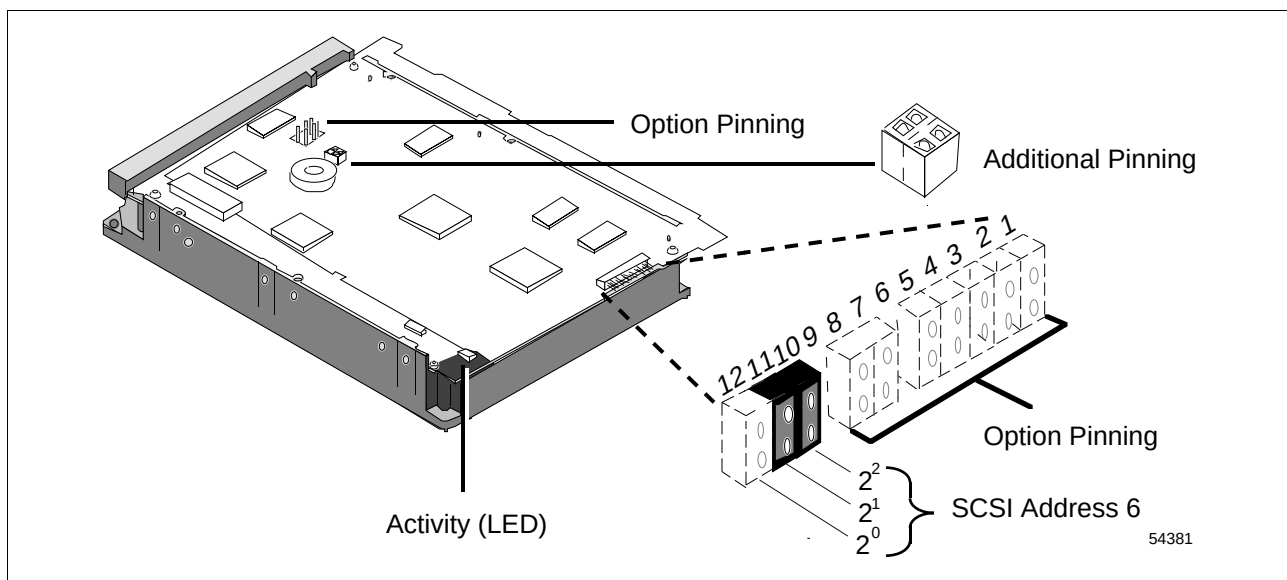
Figure 6-56 2 GB Hewlett-Packard Drive (C3325A) Pinning Locations



2 GB Hewlett-Packard drive (C3325A) SCSI address 6

The figure below identifies the pinning required for SCSI address 6.

Figure 6-57 2 GB Hewlett-Packard Drive (C3325A) SCSI Address Pinning 6

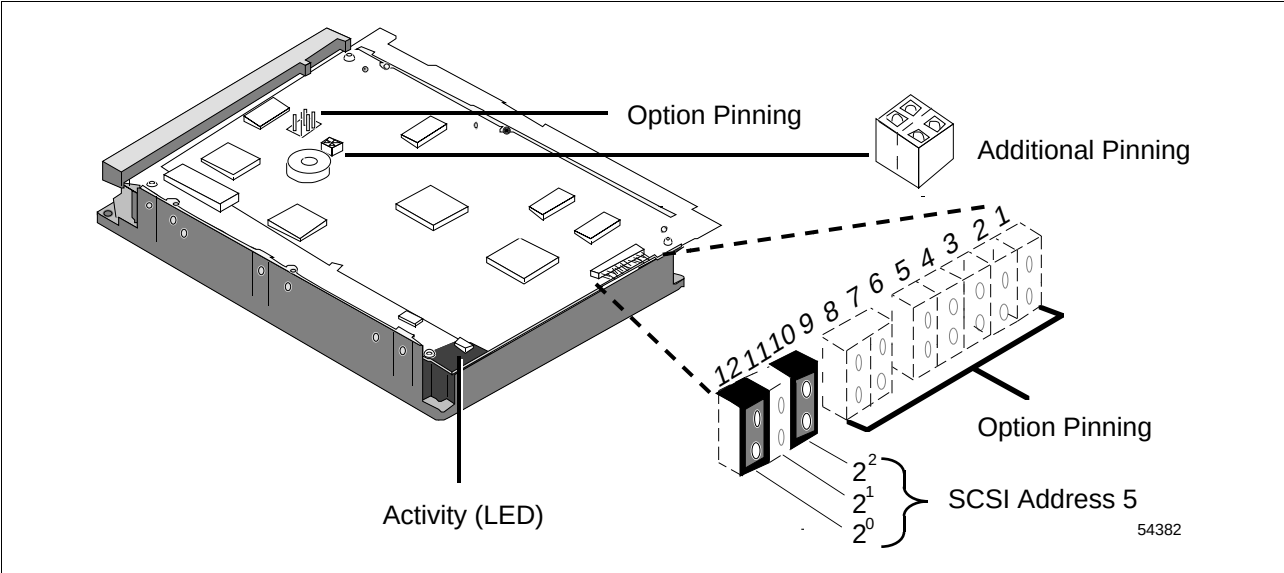


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6.25 2 GB Hewlett-Packard Drive (C3325A), Continued

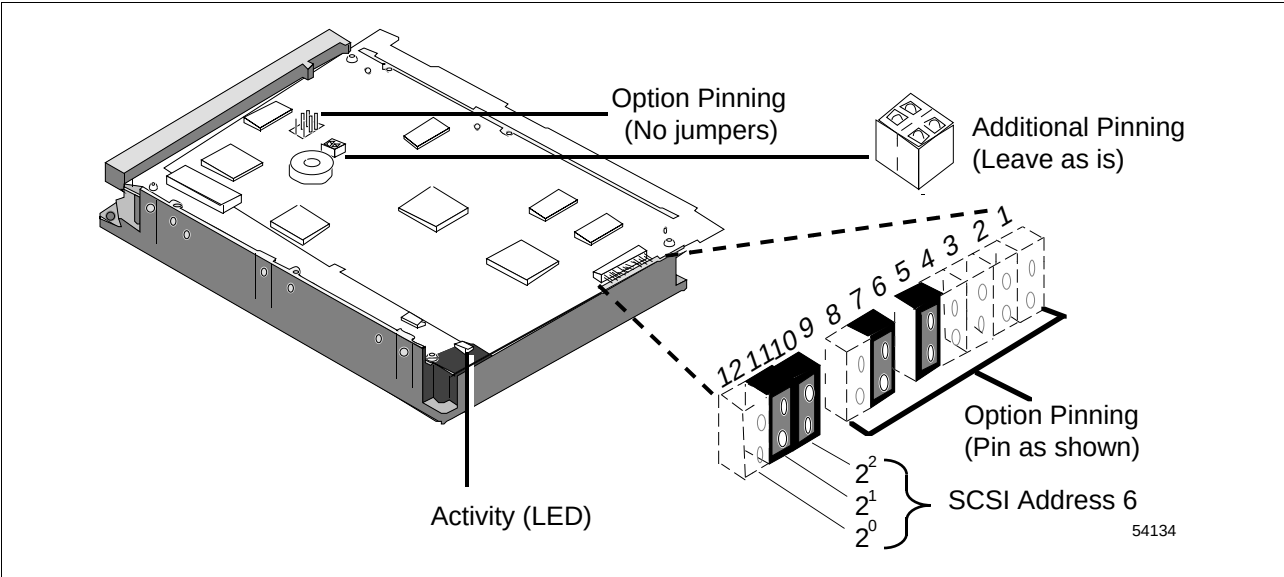
2 GB Hewlett-Packard drive (C3325A) pinning for SCSI address 5 The figure below identifies the pinning required for SCSI address 5.

Figure 6-58 2 GB Hewlett-Packard Drive (C3325A) Pinning for SCSI Address 5.



2 GB Hewlett-Packard drive (C3325A) option pinning The following figure illustrates the drive option pinning for use in U^XS. Both drives must have the options pinned as shown.

Figure 6-59 2 GB Hewlett-Packard Drive (C3325A) Option Pinning



6.2.6 HDDT I/O Board

Description

The HDDT I/O board interface provides a continuation of the SCSI bus coming from the HDDT board, through the module backpanel to the HDDT I/O board.

- If there is a DAT drive, the SCSI cable connects it to J1 on the HDDT I/O board.
- If no DAT drive is present, a terminator plug is connected to J1.

Figure 6-60 HDDT I/O and Terminator

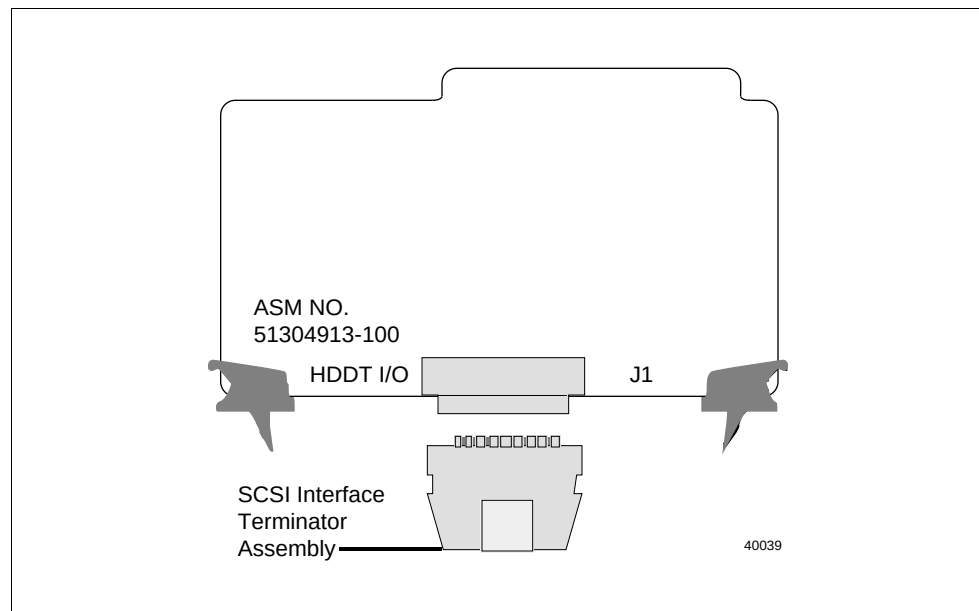
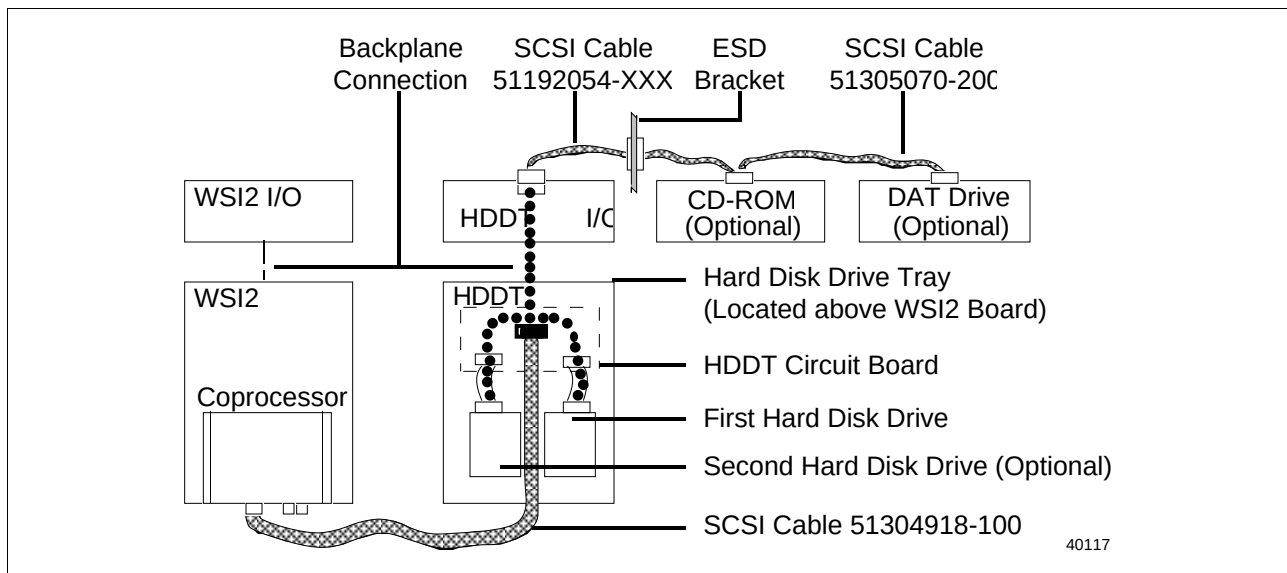


Figure 6-61 WSI2 Coprocessor HDDT DAT CD-ROM Connection



6.27 HDDT/Disk Drive Replacement

Overview

Node power must be turned off for the purpose of removing the Hard Disk Drive Tray (HDDT) from the module.

WARNING

WARNING—Damage can result—Circuit board damage can result if you do not practice proper ESD procedures or attempt to remove/insert a board with node power on.

CAUTION

CAUTION—Mandatory Steps—It is mandatory that appropriate software shutdown procedures are followed for the Universal Station^X coprocessor prior to removing power from the node.

ATTENTION

ATTENTION—It is highly recommended that you perform a node shutdown when preparing to turn the node power off. This provides for an orderly departure from the LCN.

Tray removal/replacement procedure

The HDDT is plugged into the Five-Slot Module and occupies two board slots. It must be removed if a disk drive is to be inspected or replaced.

Table 6-10 Disk Drive Replacement Procedure

Step	Action
1	If there are two disk drives determine if the primary or optional second disk is to be replaced. • Examine LED(s) • Refer to Universal Station^X Troubleshooting.
2	If it is the optional second disk that is to be replaced, use the appropriate command to dismount the drive. Refer to <i>Universal Station^X System Administration</i> .
3	Perform a coprocessor shutdown as detailed in subsection 1.7.
4	Perform a double shutdown procedure for this Universal Station ^X . Do a node shutdown from this or another station followed by a second node shutdown from another station. This prevents node isolation from occurring.
5	Place the node power switch in the OFF position.
6	Disconnect the SCSI cable from the free edge of the WSI2 board. This requires a Phillips-head screwdriver to loosen the two screws that secure the cable.
7	Remove the HDDT board from the module.

Continued on next page

6.27 HDDT/Disk Drive Replacement, Continued

Tray
removal/replacement
procedure, continued

Table 6-10 Disk Drive Replacement Procedure, Continued

Step	Action
8	Remove the tray cover by removing the three Phillips-head screws and eight lock nuts.
9	Disconnect the power connector from the drive to be replaced.
10	Disconnect the SCSI ribbon cable connector from the drive to be replaced.
11	Remove the bad disk drive from the HDDT by removing the four lock nuts from the drive mounting studs located under the drive at the four corners of the drive.
12	Ensure the drive pinning as detailed in the previous subsections.
13	Mount the new drive using the hardware previously removed.
14	Reconnect the drive power and SCSI cables.
15	Replace the drive tray cover using the hardware previously removed. Ensure that the cables are not pinched between the cover and tray at any point.
16	Insert the HDDT board into the module.
17	Reconnect the SCSI cable from the HDDT board to the SCSI connector on the coprocessor faceplate.
18	Place the node power supply switch in the ON position.
19	Observe the LED indicators on the boards in the module ensuring that the correct node number is displayed.
20	Refer to subsection 1.6 for the correct startup procedure.
21	If a second drive was replaced or added, refer to the <i>Universal Station^X System Administration</i> for procedures to activate and initialize the new drive.
22	If it was the primary drive that was replaced, the replacement will have the basic software loaded already. Any optional software should be loaded as detailed in the <i>Universal Station^X System Administration Manual</i> .

6.28 Module Replacement

Procedure

The Five-Slot Module replacement part is a bare chassis with backpanel and I/O card cage. The fan assembly and power supply must be removed and reused. **Remember, the front of the module faces the rear of the station.**

Refer to *Five-Slot Module Service* manual for the part number.

Table 6-11 Replacing a Module

Step	Action
1	Perform the node shutdown procedure found in Section 1
2	Remove the front and rear module covers.
3	Remove card file enclosure assembly by sliding it to the rear of the station and off the shelf. To do this, grasp the top and bottom edges of the enclosure, lift the edges away from the module chassis until the two tabs are clear of the module, and pull forward.
4	Extract the I/O cards from the back of the card file. The “T” may be disconnected from the board without disconnecting it from the cables. ATTENTION DO NOT DISCONNECT THE COAX FROM THE “T”—this would break the LCN cable continuity and isolate one part of the system from the other.
5	Disconnect the power cord from the power distribution panel.
6	Disconnect the ground wire from the back of the module.
7	Remove the boards from the front of the module and install them in the same slot in the new module.
8	Remove the fan assembly from the module after loosening the bail fasteners on the front of the assembly.
9	Remove the power supply by pulling the extractor levers located on each side of the power supply module.
10	Remove the two hex-head retaining screws located on the front of the module on each side.
11	Pull the module to the rear of the station until it clears the supporting shelf. WARNING Support the module so it does not fall.
12	Slide the new module onto the support shelf.

Continued on next page

6.28 Module Replacement, Continued

Procedure, continued

Table 6-11 Replacing a Module, Continued

Step	Action
13	Replace the four retaining screws.
14	Connect the ground wire to the chassis ground terminal.
15	Connect the module ac power cord to the power distribution panel.
16	Replace the power supply module, fan assembly, I/O cards, and card file enclosure assembly.
17	Replace the I/O cards in the rear of the new module.
18	Connect the "T" connectors according to the color-coded rings on the cable and card connectors.
19	Place the module power switch in the ON position.
20	The cursor should appear on the screen of the monitor after 60 seconds.
21	Load the station.

6.29 Filter Replacement

Procedure

The filter is located in the right side of the module enclosure. It is accessed by removing the rear door of the module.

Table 6-12 Replacing a Filter

Step	Action
1	Push on the front edge of the filter until it is pushed back approximately 1/4 inch.
2	Swing the front edge of the filter to the right one inch and slide forward.
3	To install a filter, slide the filter into the grooves in the right air chamber of the enclosure until some pressure is felt as the foam compresses against the stop.
4	Swing the front edge of the filter to the left and allow it to spring toward you until it catches in the detent groove.

6.30 When Servicing a Universal Station^X

WARNING

WARNING—Damage can result—Circuit board damage can result if you do not practice proper ESD procedures or attempt to remove/insert a circuit board with node power applied.

CAUTION

CAUTION—Mandatory Steps—It is mandatory that appropriate software shutdown procedures are followed for the Universal Station^X coprocessor prior to removing power from the Universal Station^X node.

Failure to do this can cause damage to the HP-UX file structure on the coprocessor hard disk.

ATTENTION

ATTENTION—It is highly recommended that you perform a node processor software shutdown when preparing to power off the Universal Station^X node. This provides for an orderly departure from the LCN network.

Locating shutdown procedure

See Section 1 in this publication, *Universal Station^X Shutdown*.

WARNING

WARNING—Damage can result—Observe ESD protection procedures when handling circuit boards that are outside the ESD protective bag and not plugged into the node. WEAR AN ESD STRAP CONNECTED TO CHASSIS GROUND.

6.31 WSI2 Board Replacement Procedure

Scope

Due to the complexity of the WSI2 board, follow the procedures when replacing it.

WARNING

WARNING—Damage can result—Circuit board damage can result if you do not practice proper ESD procedures or attempt to remove/insert a circuit board with node power applied.

CAUTION

CAUTION—Mandatory Steps—It is mandatory that appropriate software shutdown procedures are followed for the Universal Station^X coprocessor prior to removing power from the Universal Station^X node.

Failure to do this can cause damage to the HP-UX file structure on the coprocessor hard disk.

ATTENTION

ATTENTION—It is highly recommended that you perform a node processor software shutdown when preparing to power off the Universal Station^X node. This provides for an orderly departure from the LCN.

Locating shutdown procedure

See Section 1 in this publication, *Universal Station^X Shutdown*.

WARNING

WARNING—Damage can result—Observe ESD protection procedures when handling circuit boards that are outside the ESD protective bag and not plugged into the node. WEAR AN ESD STRAP CONNECTED TO CHASSIS GROUND.

Procedure

Table 6-13 WSI2 Board Replacement Procedure

Step	Action
1	Place the node power supply switch in the off position.
2	Disconnect the SCSI interface cable from the coprocessor free edge. It runs between the WSI2 and the HDDT above it.
3	Remove the second monitor cable from the coprocessor free edge. This cable is optional.
4	Ensure that the replacement WSI2 assembly pinning matches the existing board.

Continued on next page

6.31 WSI2 Board Replacement Procedure, Continued

Procedure, continued

Table 6-13 WSI2 Board Replacement Procedure, Continued

Step	Action
5	Remove the coprocessor battery isolator from the coprocessor battery clip. The isolator looks like a piece of cardboard.
6	Transfer the coprocessor memory boards (1 or 2) from the existing WSI2 board to the same position on the replacement board. These memory boards plug into sockets on the coprocessor board and are held in place with two screws each.
7	Ensure that the WSI2 board is pinned for a "K2LCN/K4LCN", see "WSI2 Board Pinning", in this section.
8	Ensure that all internal cables on the WSI2 board are tightly connected.
9	Insert the WSI2 board assembly into slot 2 of the Five-Slot Module.
10	Reconnect the SCSI cable removed in step 2.
11	Reconnect the second monitor cable removed in step 3, if it is present.
12	The node is ready for power to be applied. Place the power node power switch in the ON position.
13	Perform the node loading procedure, <i>Universal Station^X Startup</i> , found in Section 1.
14	If Kinet software is in use, perform the <i>Procedure to Obtain a Replacement Kinet License</i> found in this section.

6.32 UXs Node Spare Parts

Node parts and cables

Table 6-14 Node Spare Parts and Cables

Part Number	Description
*51108892-100	Five-Slot Card File Assembly
*51400646-100	Five-Slot Fan Assembly
51400918-100	Five-Slot Module Cover Assembly
*51201436-100	Logic Power Supply—P.S. Replacement Kit
*51402615-800	K2LCN Board with 8 Megabytes of Memory
*51401946-100	K4LCN Board without Daughter (Mezzanine) Memory Board
*51201645-800	8 Megabyte Daughter (Mezzanine) Memory Board
*51201759-160	16 Megabyte Daughter (Mezzanine) Memory Board
*51402755-800	K4LCN Board without DIMM Memory Board
*51201795-800	8 Megabyte DIMM Memory Board
*51201793-160	16 Megabyte DIMM Memory Board
*51402083-100	WSI2 Board, 64 MHz
*51402083-300	WSI2 Board, 64 MHz with Graphics (second Monitor support)
*51402610-200	TPDGX Board
*51402645-200	TPDGX2 Board
*51305072-100	CLCN A/B
*51107403-100	LCN I/O Board
*51304924-100	WSI2 I/O Board
*51305075-100	TPDGC Board
*51304831-100	TPDG I/O Board
See Table 6-16	Hard Disk Drive Tray (HDDT) Assembly
51304833-100	50 Pin SCSI Cable WSI J3 to WSCPU P5
51304835-100	50 Pin “Waterfall” Cable WSI to TPDGX

* ORU = Optimum Replaceable Unit

6.33 Module Spare Parts

Module parts

Table 6-15 Module Spare Parts

Part Number	Description
* 51403145-100	Five-Slot Card File Assembly w/o P.S. or Fan
51403177-100	Auxiliary Card File
51403089-200	Front Cover Assembly, Upper Module
51403090-200	Front Cover Assembly, Lower Module
51308086-201	Safety Barrier for Five-Slot Module
51308086-202	Safety Barrier for Five-Slot Module used for WDA History Module
51403152-200	Rear Cover Assembly (Complete for Five-Slot)
51403152-201	Rear Cover EMI Shield (with EMC strip for Five-Slot Module)
51197046-100	EMC Strip 3.35 inch
51197046-200	EMC Strip 4.46 inch
51403117-100	Card File Enclosure Assembly
51206056-100	Dust Filter with Gasket

* ORU = Optimum Replaceable Unit

6.34 HDDT Drive Spare Parts

Hard drive parts

The following table lists hard drive spare part numbers for hard drives of different capacities.

Table 6-16 HDDT Hard Drive Spare Parts

Part Number	Description
*51402176-100	Hard Disk Drive Tray (HDDT) Assembly
*51196433-100	525 MB Hard Disk Drive (applicable for both primary and secondary drive)
*51196432-100	1.2 GB Hard Disk Drive (applicable for both primary and secondary drive)
*51196479-100	2 GB Hard Disk Drive (applicable for both primary and secondary drive)
51304921-100	Disk Drive Power Cable Harness (inside HDDT)
51304918-100	SCSI Cable (between coprocessor on WSI2 board and HDDT)
*51304913-200	HDDT I/O (interfaces the rear of the HDDT) – CE Compliant
*51304913-100	HDDT I/O (interfaces the rear of the HDDT) – Non-CE Compliant
51191982-100	SCSI terminator (for SCSI termination on HDDT I/O board when there is not a DAT drive or CD-ROM drive)

* ORU = Optimum Replaceable Unit

Section 7 – DAT Drive Service

7.1 Overview

Section contents

This is a list of topics covered in this section:

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Description

The Digital Audio Tape (DAT) Drive has a Small Computer System Interface (SCSI) that is cabled, jointly with the hard drive, to the Work Station Interface (WSI I/O) board. The cartridge drive is on a separate SCSI bus connected to the TPDG I/O board.

- A single DAT drive can be mounted with a cartridge drive or by itself in a half-height upper pod or on a table top slide assembly.
 - The DAT drive is powered by the power supply in the Five-Slot Module.
-

7.2 Cleaning the DAT Drive Heads

Specification

The DAT drive playback heads should be cleaned:

- after every 28 hours of actual operation
 - when the Caution Signal is displayed (see subsection 7.3)
-

Procedure

To clean the DAT drive heads use a cleaning cassette, 51196257-200, (HP 92283K).

Table 7-1 Cleaning the DAT Drive Heads

Step	Action
1	Insert the cleaning cassette into the drive.
2	The drive will automatically load the cassette and clean the heads.
3	The drive will eject the cleaning tape when the end is reached.
4	Note the date on the label of the cleaning cassette to establish a record of the number of uses.
5	ATTENTION After 28 uses, discard the cleaning tape.

7.3 Troubleshooting the DAT Drive

Front panel display

The front panel has two bi-colored indicators (LEDs). Green is used to show normal operation, while amber is used to imply warning conditions. Blinking indicates drive activity. If the cassette light (left) shows steady amber, this indicates that the cassette is write-protected. If the Drive light (right) shows steady amber, this indicates a fault condition.

Table 7-2 DAT Drive LED Indicators

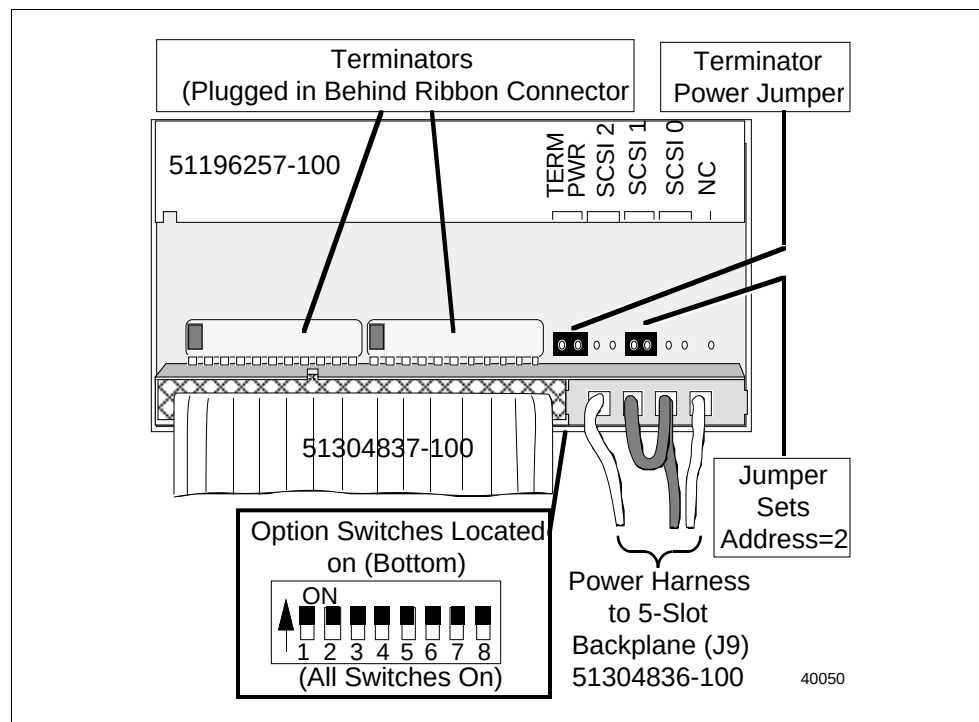
State	Cassette (Left)	Drive (Right)	Meaning
Read/Write	Green Blinking	Green Blinking	Cassette (un)Loading
	Green On	Green On	Cassette Loaded/Online
	Green On	Green Blinking	Cassette Loaded/Activity
	Green On	Off	Cassette Loaded/Offline
Write-Protect	Amber Blinking	Green On	Cassette (un)Loading
	Amber On	Green On	Cassette Loaded/Online
	Amber On	Green Blinking	Cassette Loaded/Activity
	Amber On	Off	Cassette Loaded/Offline
Error	Green On	Green/Amber Blinking	Media Wear (Caution Signal)
	Amber On	Amber On	High Humidity/ No Termination on SCSI Line
	Amber Blinking	Amber Blinking	Cassette Loaded/Activity
	Amber Blinking	Amber On	Self-Test (failure)

7.4 Pinning the DAT Drive

Instruction

The DAT drive address on the SCSI bus must be verified before installation of a replacement drive. The address is a binary “001” referred to as an octal “1.” See Figure 7-1. The Termination power is left unconnected.

Figure 7-1 Pinning the DAT Drive SCSI Address



Configuration

A set of eight option switches are located on the underside of the drive. They are to be configured according to Figure 7-1.

Table 7-3 DAT Drive Option Switches

Switch	Position	Function
1	ON	Used to configure data compression—a function this drive does not support. This switch is disabled.
2	ON	Used to configure data compression—a function this drive does not support. This switch is disabled.
3 - 8	ON	Specify customer functionality

7.5 Removing the DAT Drive

Procedure

Table 7-4 Removing the DAT Drive

Step	Action
1	Access the drive assembly either by - removing the upper half-height pod rear cover or - opening the rear door of the cabinet, releasing the slide assembly locks, and sliding the assembly out to its full extension.
2	Disconnect the four prong ac power cable from the rear of the DAT drive.
3	Disconnect the 50-pin ribbon cable from the rear of the DAT drive.
4	Remove the four hex-head machine screws holding the floppy mounting bracket to the slide assembly. See Figure 7-1.
5	Remove the four pan-head screws holding the drive to the floppy mounting bracket. See Figure 7-1.
6	Ensure the option switches are in the ON position.

7.6 Replacing the DAT Drive

Procedure

Table 7-5 Replacing the DAT Drive

Step	Action
1	Ensure the pinning per Figure 7-1 (DAT drives have a SCSI address of 1).
2	Ensure the option switches are in the ON position.
3	Replace the four pan-head screws holding the drive to the floppy mounting bracket.
4	Replace the four hex-head machine screws holding the floppy mounting bracket to the slide assembly.
5	Connect the four prong ac power cord to the rear of the DAT drive.
6	Connect the 50-pin ribbon cable to the rear of the DAT drive.

7.7 DAT Drive Cabling

Illustrations

The DAT drive is connected to an SCSI bus with the hard drive and optional CD-ROM as shown in Figure 7-2 and 7-3.

Figure 7-2 Dat Drive Cabling with CD-ROM Drive

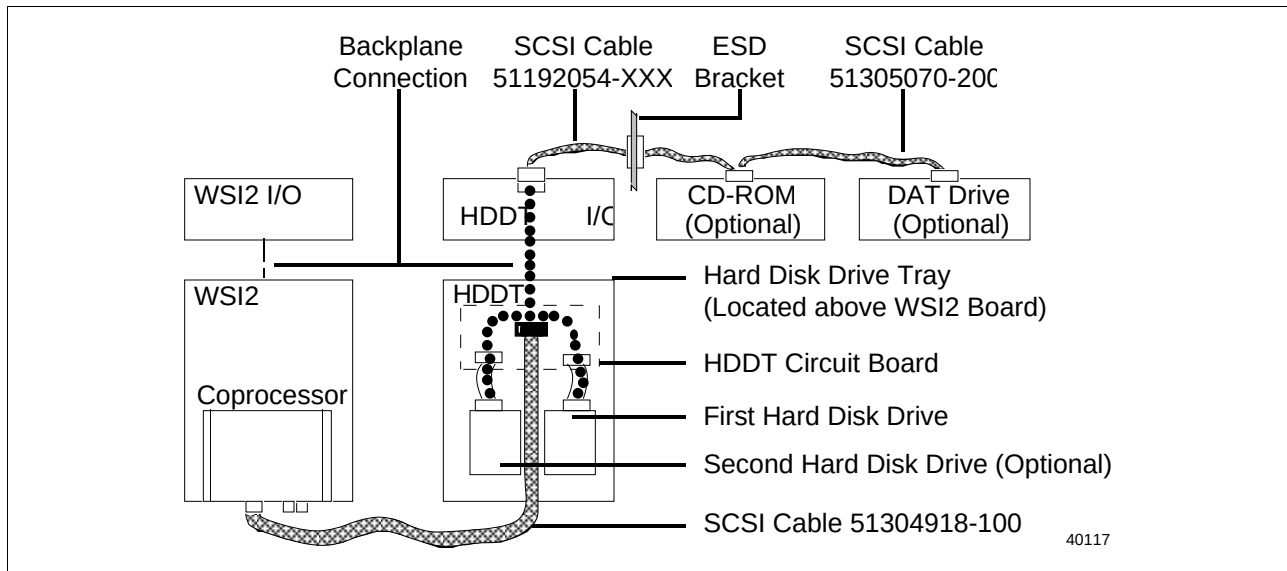
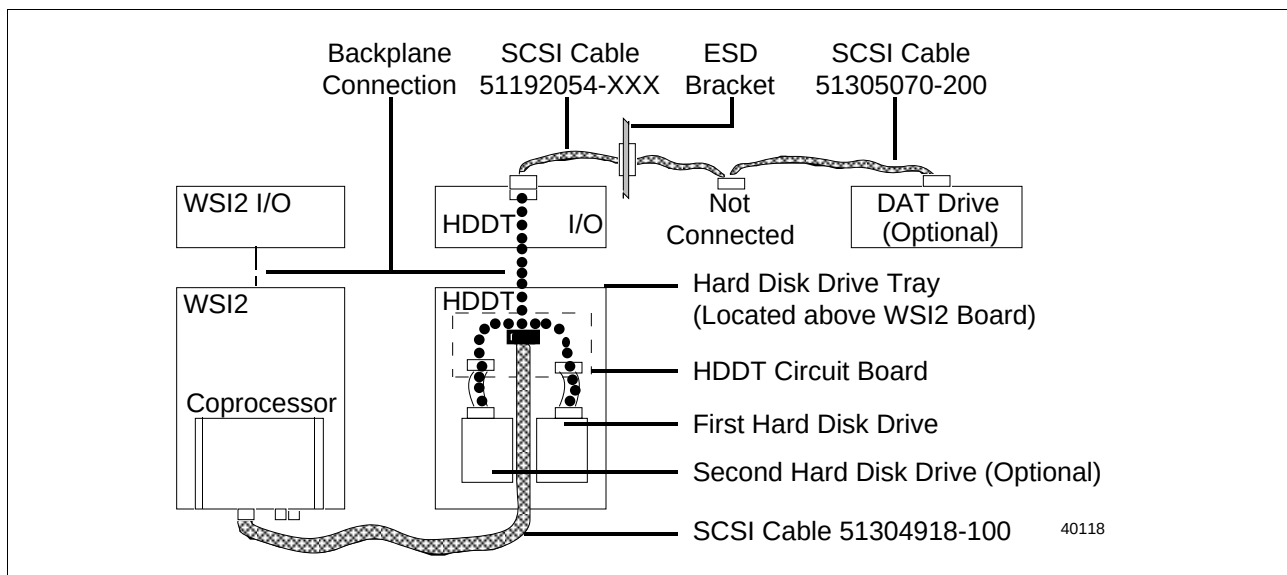


Figure 7-3 DAT Drive Cabling without CD-ROM Drive



7.8 Spare Parts

Spare parts list

Table 7-6 DAT Drive Spare Parts

Part Number	Description
*51196257-100	Digital Audio Tape (DAT) Drive (2 GB)
51196257-200	Head Cleaning Cassette
51195124-200	Fan Assembly
51304585-300	Fan DC Power Cable
51201201-500	Air Filter
51304836-100	DC Power Cable
51304837-100	SCSI Interface Cable

* ORU = Optimum Replaceable Unit

Section 8 – CE Compliant DAT Drive Service

8.1 Overview

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This is a list of topics covered in this section:

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Description

The Digital Audio Tape (DAT) Drive has a Small Computer System Interface (SCSI) that is cabled, jointly with the hard drive, to the Work Station Interface (WSI2 I/O or WSI2R I/O) board. The cartridge drive is on a separate SCSI bus connected to the TPDG I/O board.

- The single DAT and optional single cartridge drives are contained in an EMI enclosure mounted in the half upper tier or in a table top work surface cabinet.
- The DAT drive is powered by a periheral power supply in the enclosure.

8.2 Cleaning the DAT Drive Heads

Specification

The DAT drive playback heads should be cleaned:

- after every 28 hours of actual operation
 - when the Caution Signal is displayed (see subsection 8.3)
-

Procedure

To clean the DAT drive heads use a cleaning cassette, 51196257-200, (HP 92283K).

Table 8-1 Cleaning the DAT Drive Heads

Step	Action
1	Insert the cleaning cassette into the drive.
2	The drive will automatically load the cassette and clean the heads.
3	The drive will eject the cleaning tape when the end is reached.
4	Note the date on the label of the cleaning cassette to establish a record of the number of uses.
5	ATTENTION After 28 uses, discard the cleaning tape.

8.3 Troubleshooting the DAT Drive

Front panel display

The front panel has two bi-colored indicators (LEDs). Green is used to show normal operation, while amber is used to imply warning conditions. Blinking indicates drive activity. If the cassette light (left) shows steady amber, this indicates that the cassette is write-protected. If the drive light (right) shows steady amber, this indicates a fault condition.

Table 8-2 DAT Drive LED Indicators

State	Cassette (Left)	Drive (Right)	Meaning
Read/Write	Green Blinking	Green Blinking	Cassette (un)Loading
	Green On	Green On	Cassette Loaded/Online
	Green On	Green Blinking	Cassette Loaded/Activity
	Green On	Off	Cassette Loaded/Offline
Write-Protect	Amber Blinking	Green On	Cassette (un)Loading
	Amber On	Green On	Cassette Loaded/Online
	Amber On	Green Blinking	Cassette Loaded/Activity
	Amber On	Off	Cassette Loaded/Offline
Error	Green On	Green/Amber Blinking	Media Wear (Caution Signal)
	Amber On	Amber On	High Humidity/ No Termination on SCSI Line
	Amber Blinking	Amber Blinking	Cassette Loaded/Activity
	Amber Blinking	Amber On	Self-Test (failure)

8.4 Removing the DAT Drive

Procedure

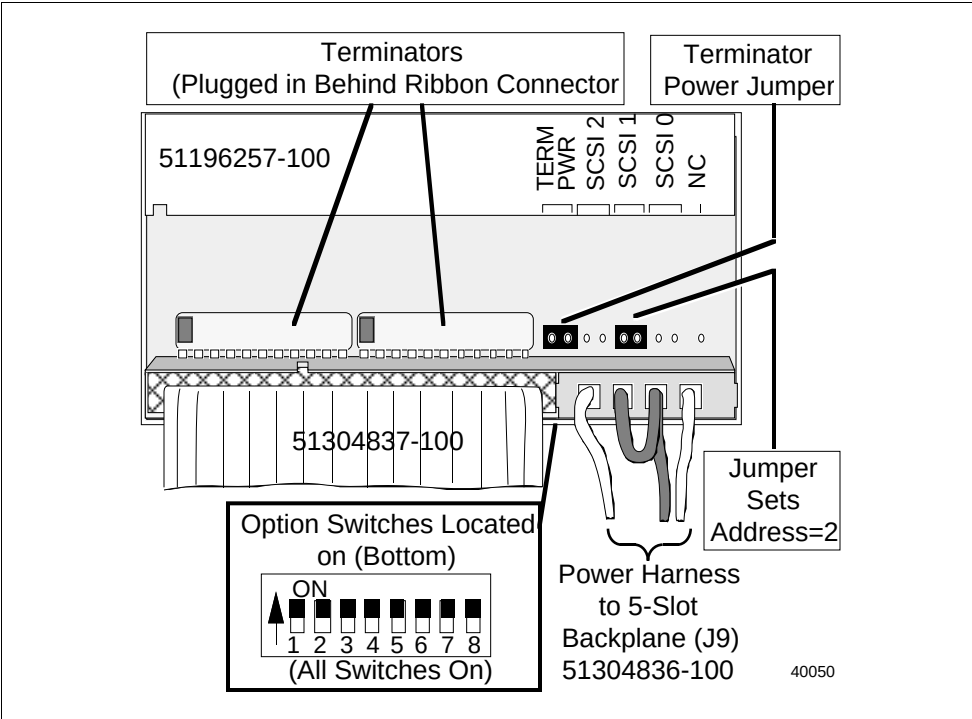
Table 8-3 Removing the DAT Drive

Step	Action
1	Remove the rear cover from the upper pod after removing the four retaining screws.
2	Disconnect the cables from the back of the drive enclosure.
3	Remove the two hex-head screws (from each side) that mount the enclosure to the mounting brackets.
4	Lift the enclosure off the support tray and place on a stable work surface.
5	Remove the 16 screws from the top cover of the enclosure.
6	Remove the top cover and set it aside.
7	Disconnect the four prong dc power cable from the rear of the DAT drive.
8	Disconnect the 50-pin ribbon cable from the rear of the DAT drive.
9	Remove the four hex-head machine screws holding the DAT drive to the removable media drive disk tray assembly.
10	Pin the new drive according to the instructions in the previous section. CAUTION Observe ESD precautions while handling the cartridge drive.

8.5 Pinning the DAT Drive

Instruction The DAT drive address on the SCSI bus must be verified before installation of a replacement drive. The address is a binary “001” referred to as an octal “1.” See Figure 8-1. The Termination power is left unconnected.

Figure 8-1 Pinning the DAT Drive SCSI Address



Configuration A set of eight option switches are located on the underside of the drive. Configure them according to Figure 8.1.

Table 8-4 DAT Drive Option Switches

Switch	Position	Function
1	ON	Used to configure data compression—a function this drive does not support. This switch is disabled.
2	ON	Used to configure data compression—a function this drive does not support. This switch is disabled.
3 - 8	ON	Specify customer functionality.

8.6 Replacing the DAT Drive

Procedure

Table 8-5 Replacing the DAT Drive

Step	Action
1	Ensure the pinning per Figure 8.1 (DAT drives have a SCSI address of 1).
2	Ensure the option switches are in the ON position.
3	Replace the four pan-head screws holding the drive to the floppy mounting bracket.
4	Replace the four hex-head machine screws holding the floppy mounting bracket to the slide assembly.
5	Connect the 4-prong ac power cord to the rear of the DAT drive
6	Connect the 50-pin ribbon cable to the rear of the DAT drive.

8.7 DAT Drive Cabling

Illustrations

The DAT drive is connected to an SCSI bus with the hard drive and optional CD-ROM as shown in Figure 8.2 and 8.3.

Figure 8-2 DAT Drive Cabling with Cartridge Drive

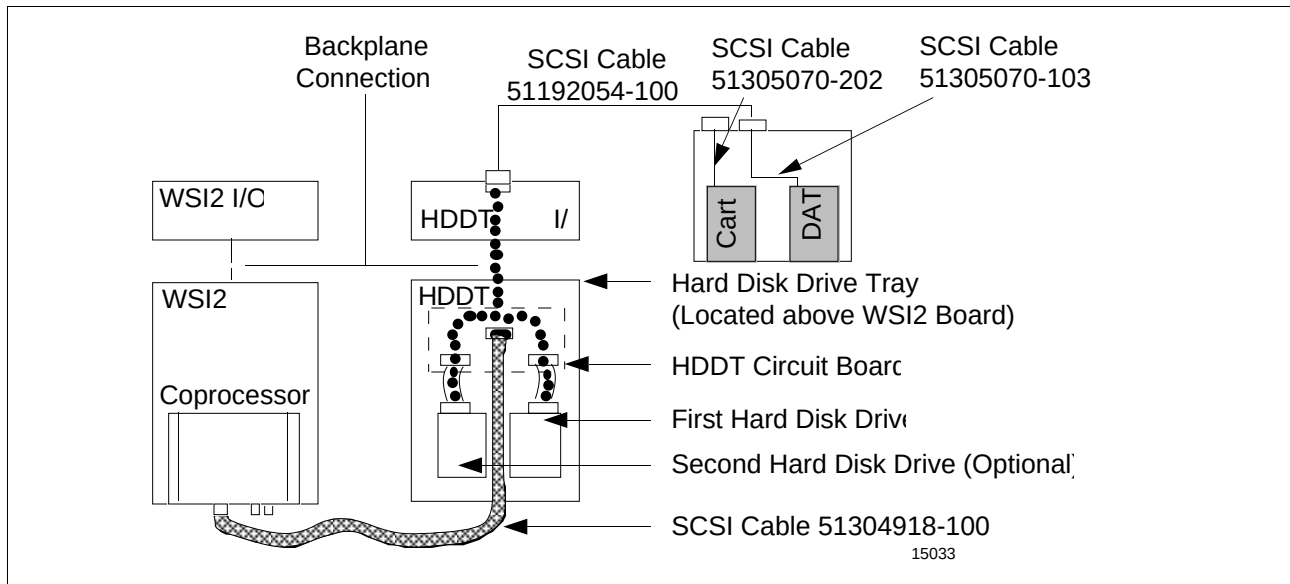
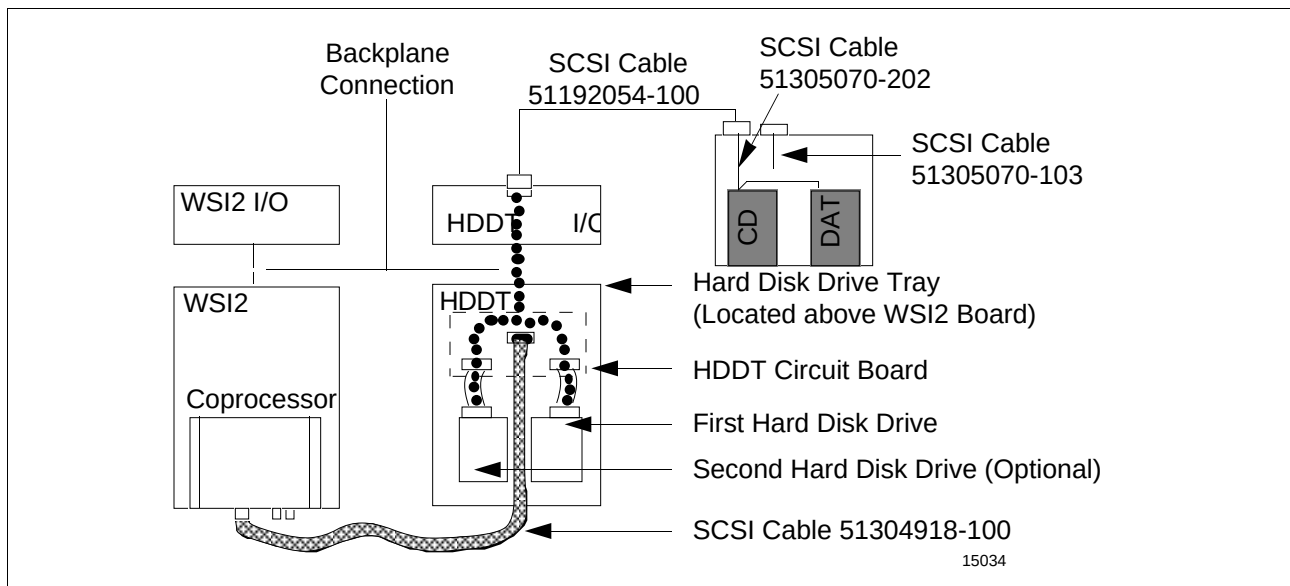


Figure 8-3 DAT Drive Cabling with CD-ROM Drive



8.8 Spare Parts

Spare parts list

Table 8-6 DAT Drive Spare Parts

Part Number	Description
*51196257-100	Digital Audio Tape (DAT) Drive (2 GB)
51196257-200	Head Cleaning Cassette
51195124-200	Fan Assembly
51304585-300	Fan DC Power Cable
51201201-500	Air Filter
51305341-100	DC Power Cable
51305070-103	Cable SCSI (1 drive connector)
51305070-202	Cable SCSI (2 drive connectors)
*51197004-300	DC Power Supply (Skynet)

* ORU = Optimum Replaceable Unit

Section 9 – Cartridge Drive Service

9.1 Overview

Section contents

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Description

Cartridge drives exist as:

- Dual drives (if there is no DAT drive, CD-ROM drive, or Zip drive) in a half upper pod or in a table top slide tray.
- Single drive mounted with a DAT drive in a half upper pod or in a table top slide tray.

The Universal Station^X node is in a Five-Slot Module, the cartridge drive is powered by the Acme power supply in the Five-Slot Module.

Drive size

There are two different drives in use:

- 20 MB Iomega Bernoulli drive in production 1987 to 1994
- 150 MB Iomega Bernoulli drive in production late 1994
 - Multidisk capability for 35, 65, 105, 150 MB disks

The 150 MB drive is a direct replacement for the 20 MB drive but it cannot read or write to 20 MB cartridges. Some users may want to replace one of a pair of 20 MB drives with a 150 MB drive for a more flexible transition.

ATTENTION

ATTENTION—THE CARTRIDGE SHOULD NOT REMAIN IN THE CARTRIDGE DRIVE FOR EXTENDED PERIODS OF TIME WHEN IT IS NOT IN USE. The drive automatically retracts the heads after a period of time without use, but the media is still spinning. This shortens the life of the cartridge and could result in lost data.

The 150 MB cartridge drive has an "auto sleep" feature that spins down the drive after approximately 30 minutes. The drive will spin up again when accessed.

9.2 Loading/Unloading

Procedure

Follow the steps in Table 9-1 to load a cartridge in the drive

Table 9-1 Loading a Cartridge

Step	Action
1	Insert the cartridge into the drive in a smooth motion until it clicks (with about 3 cm [1 inch] still sticking out) and starts to spin.
2	The green LED will start flashing, then will remain steadily on when the drive is ready. • If the green LED starts to flash before the cartridge is in all the way, stop and wait till the LED stops flashing. • If the LED does not stop flashing, press the STOP button. • When the LED stops flashing, remove the cartridge.

Follow the steps in Table 9-2 to unload a cartridge from the drive.

Table 9-2 Unloading a Cartridge

Step	Action
1	Press the front panel button and release it once. The LED indicator will flash on and off. WARNING Do not pull on the cartridge until the LED indicator stops flashing.
2	When the LED indicator stops flashing, remove the cartridge by firmly pulling on the rear center of the cartridge.

9.3 Cleaning the Cartridge Drive

Periodic cleaning

Periodic cleaning of the read/write head is important for prolonging head and cartridge life. The head should be cleaned:

- every 50 hours of use (more frequently in dusty environments)
- whenever frequent errors occurs

ATTENTION

Each drive should have its own cleaning cartridge.

Cleaning cartridge

A cleaning cartridge is available from Honeywell. See Table 9-7 for the part number. The kit includes:

- a head-cleaning cartridge
- wiper pads
- an approved head-cleaner solution

The head-cleaning cartridge inserts into the drive as does a standard data cartridge. Instructions are included with the head-cleaning kit.

Procedure

The cleaning cartridge has a pad that is mounted on the pad-arm assembly. Change the pad after five uses.

Table 9-3 Cleaning a Cartridge Drive

Step	Action
1	Remove any cartridge that is in the drive.
2	Press the front panel button on the disk drive and hold until the green LED starts flashing (about 2 seconds). • This places the drive in the mode to accept the cleaning cartridge.
3	CAUTION Follow instructions on the container that holds the disk drive cleaner solution to avoid vapor inhalation or skin exposure.
4	Apply the cleaner by squeezing the solution from the container onto the pad through the cartridge window and center slot. ATTENTION The cleaner should be applied to the pad just before inserting the cartridge into the drive. The pad should be dampened, but not saturated with the solution.
5	Place the cleaning cartridge in the drive. The LED will glow steadily.
6	The wiper lever moves back and forth for 30 cleaning cycles.
7	After the 30 cleaning cycles are complete, press the front panel button.
8	When the green light goes out, remove the cartridge.

9	Record the date of cleaning on the label provided with the cleaning cartridge. Place the label on the outside of the storage sleeve.
----------	--

9.4 Troubleshooting an Inoperative Cartridge Drives

Failure modes

In general, if only one disk drive fails, the disk drive is most likely at fault. If both disk drives are inoperative, check the power source.

- If the node is in a Five-Slot Module, the module's power supply provides the power.
 - If the node is in a Dual node Module, a peripheral power supply provides the power.
-

Isolation procedure

Table 9-4 Troubleshooting a Cartridge Problem

Step	Action
1	If the node is in a Five-Slot Module, check the indicators on the module power supply—refer to the <i>Five/Ten Slot Module Service</i> manual.
2	Check the TPDGX board error LED (red).
3	Check the power cable by disconnecting it from the cartridge drive. • Pins 1 & 2 = 12 Vdc $\pm$.1 V • Pins 3 & 4 = 5.0 Vdc $\pm$.25 V

9.5 Removing the Cartridge Drive

Procedure

Table 9-5 Removing a Cartridge Drive

Step	Action
1	Remove the rear cover from the upper pod after removing the four retaining screws.
2	Place the upper pod power supply switch in the OFF position. If there is not an upper pod power supply the power is provide by the Five-Slot Module. Place the Five slot module power supply switch in the OFF position.
3	Remove the four hex-head retaining screws from each side of the slide tray the cartridge drive(s) are mounted on.
4	Slide the slide tray back to access the cartridge drive. Be careful not to slide it too far back on the support tray. Hint: temporarily replace one of the slide tray retaining screws to hold the slide tray in place while replacing the cartridge drive.
5	Disconnect the four prong ac power cable from the rear of the cartridge drive.
6	Disconnect the 50-pin ribbon cable from the rear of the cartridge drive.
7	Remove the four hex-head machine screws holding the cartridge drive to the removable media drive disk tray assembly.

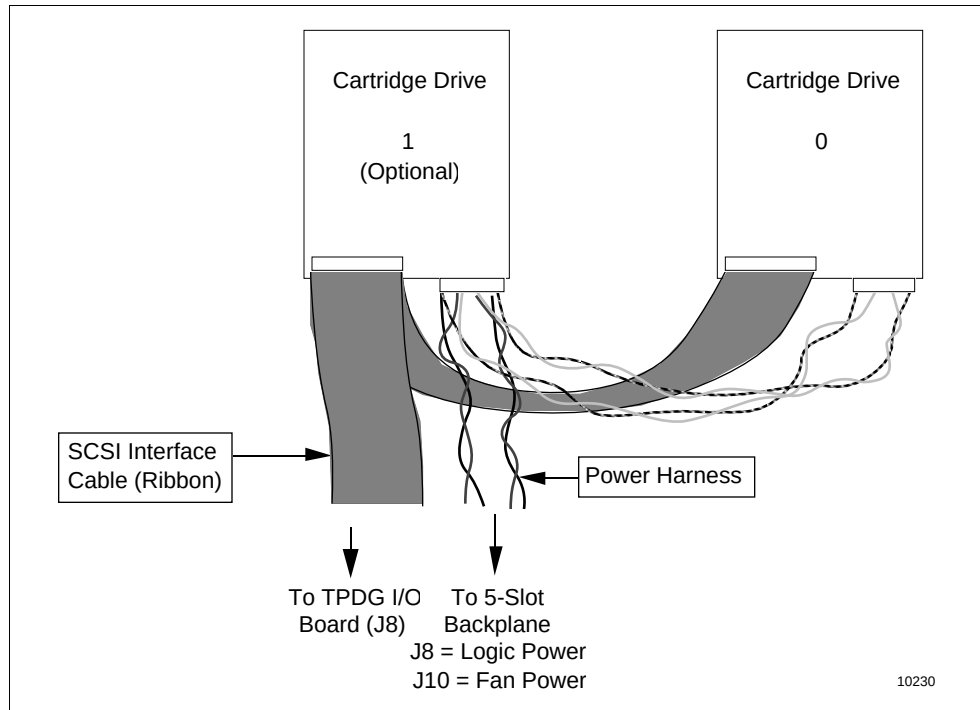
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9.5 Removing the Cartridge Drive, Continued

Illustration

The ribbon cable **plugs on the top connector** on a 20 MB cartridge drive.

Figure 9-1 20 MB Cartridge Drive Cabling



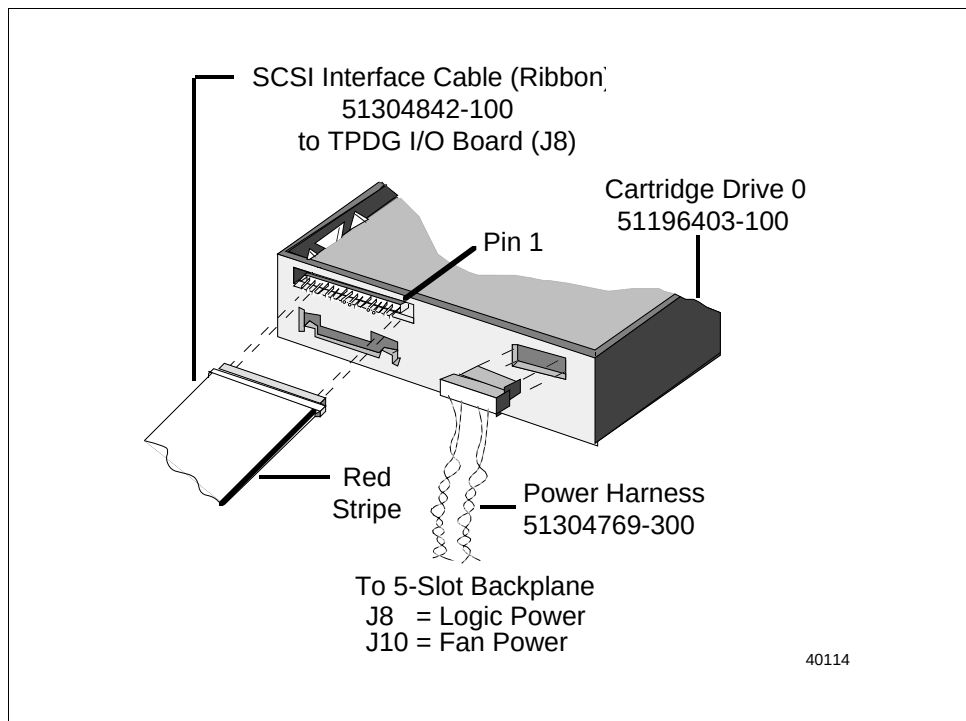
Continued on next page

9.5 Removing the Cartridge Drive, Continued

Illustration

The ribbon cable **plugs on the back connector (the only one)** on a 150 MB cartridge drive.

Figure 9-2 150 MB Cartridge Drive Cabling

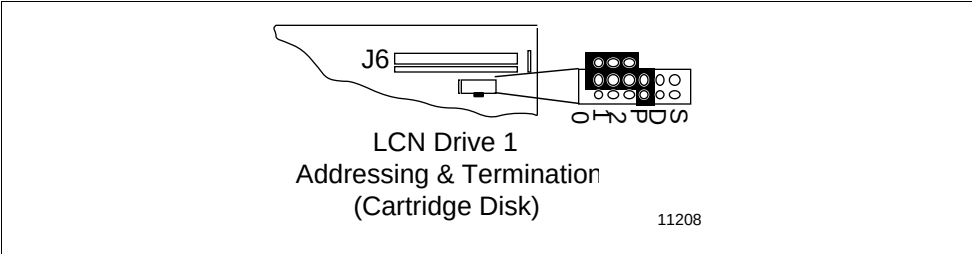


9.6 Pinning the Cartridge Drive

20 MB drive pinning illustration

Pin the replacement 20 MB cartridge drive as shown in Figure 9-3

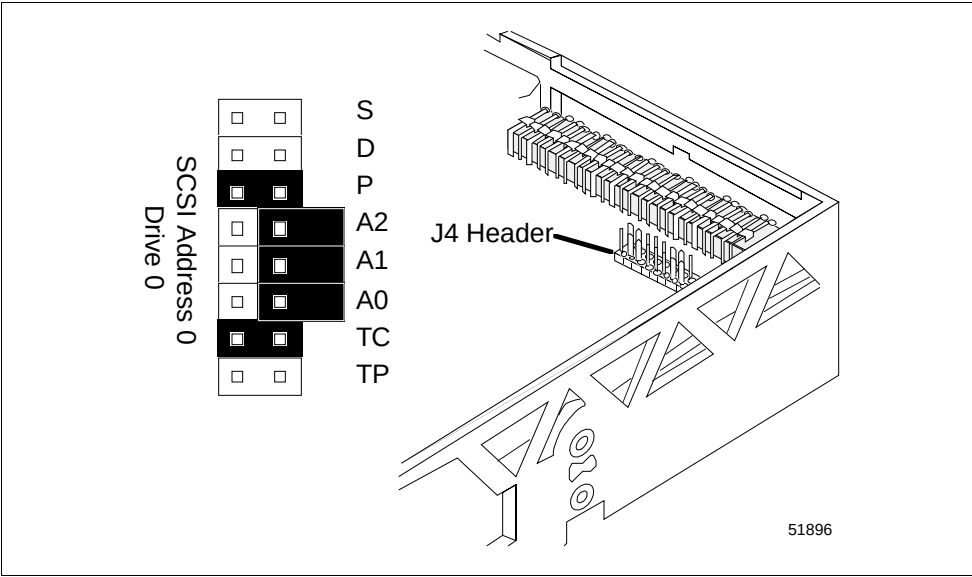
Figure 9-3 20 MB Iomega Bernoulli Cartridge Drive Pinning



150 MB drive pinning illustration

The Universal Station^X can only have one cartridge drive, Drive 0, as shown in Figure 9-4.

Figure 9-4 150 MB Iomega Bernoulli Cartridge Drive Pinning



9.7 Replacing a Cartridge Drive

Procedure

Table 9-6 Replacing a Cartridge Drive

Step	Action
1	CAUTION Observe ESD precautions while handling the cartridge drive.
2	Verify that the new drive is correctly pinned. See Figures 9-3 and 9-4.
3	Place the hard drive on the removable media drive tray and fasten it in place with the four mounting screws.
4	Reconnect the dc power cable at the rear of the drive.
5	Reconnect the SCSI bus 50-pin ribbon cable at the rear of the drive.
6	Move the slide tray fully forward to the front of the station. Check to see that the face plate is properly in contact with the bezel.
8	Install the four retaining screws at each side of the slide tray. These screws can be used to adjust the position of the faceplate/bezel fit.
9	Fold the SCSI bus ribbon cable as necessary. Fasten it in place with the cable camp.
10	Place the upper pod power supply or Five-Slot Module power supply switch in the ON position. Check the indicators for proper operation.
11	Replace the upper pod rear cover assembly.
12	Replace the four retaining screws in the upper pod rear cover.

9.8 Spare Parts

20 MB drive spare parts list

Table 9-7 20 MB Cartridge Drive Spare Parts

Part Number	Description
*51195156-100	Beta 0 (Iomega Cartridge Drive)
51304842-200	SCSI Interface Cable (Single Drive)
51304199-300	SCSI Interface Cable (Dual Drive)
51308055-100	Drive Cable DC Power
51195156-101	Cartridge Drive Cleaning Kit

* ORU = Optimum Replaceable Unit.

150 MB drive spare parts list

Table 9-8 150 MB Cartridge Drive Spare Parts

Part Number	Description
*51196403-100	Beta 0 (Iomega Cartridge Drive)
51304842-200	SCSI Interface Cable (Single Drive)
51304199-300	SCSI Interface Cable (Dual Drive)
51308055-100	Drive Cable DC Power
51196485-100	35 MB Multidisk 150 MB compatible 5.25" cartridge
51196485-200	65 MB Multidisk 150 MB compatible 5.25" cartridge
51196485-300	105 MB Multidisk 150 MB compatible 5.25" cartridge
51196485-400	150 MB Multidisk 150 MB compatible 5.25" cartridge
51196485-500	Cartridge Drive Cleaning Kit—includes • cleaning cartridge • three 0.25 oz bottles of cleaning solution • two extra cleaning pads

51196485-600	Cartridge Drive Cleaning Solution—refill kit • three 0.25 oz bottles of cleaning solution
51196485-700	Wiper Arm Pads (quantity of 2)

* ORU = Optimum Replaceable Unit.

Section 10 – CE Compliant Cartridge Drive Service

10.1 Overview

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Description

Cartridge drives exist as:

- Dual drives (if there is no DAT drive, CD-ROM drive, or Zip drive) in a half upper pod or in a table top slide tray.
- Single drive mounted with a DAT drive in a half upper pod or in a table top slide tray.

The cartridge drive is located in an EMI isolation enclosure and is powered by the a peripheral power supply in the enclosure.

Drive size

The cartridge drive is a 150 MB Iomega Bernoulli drive that has multidisk capability for 35, 65, 105, 150 MB disks.

ATTENTION

ATTENTION—THE CARTRIDGE SHOULD NOT REMAIN IN THE CARTRIDGE DRIVE FOR EXTENDED PERIODS OF TIME WHEN IT IS NOT IN USE. The drive automatically retracts the heads after a period of time without use, but the media is still spinning. This shortens the life of the cartridge and could result in lost data.

The 150 MB cartridge drive has an "auto sleep" feature that spins down the drive after approximately 30 minutes. The drive will spin up again when accessed.

10.2 Loading/Unloading

Procedure

Follow the steps in Table 10-1 to load a cartridge in the drive.

Table 10-1 Loading a Cartridge

Step	Action
1	Insert the cartridge into the drive with a smooth motion until it clicks into place (with about 3 cm [1 inch] still sticking out) and starts to spin.
2	The green LED will start flashing, then will remain steadily on when the drive is ready. • If the green LED starts to flash before the cartridge is in all the way, stop and wait until the LED stops flashing. • If the LED does not stop flashing, press the STOP button. • When the LED stops flashing, remove the cartridge.

Follow the steps in Table 10-2 to unload a cartridge from the drive.

Table 10-2 Unloading a Cartridge

Step	Action
1	Press the front panel button and release it once. The LED indicator should flash on and off. WARNING Do not pull on the cartridge until the LED indicator stops flashing.
2	When the LED indicator stops flashing, remove the cartridge by firmly pulling on the rear center of the cartridge.

10.3 Cleaning the Cartridge Drive

Periodic cleaning

Periodic cleaning of the read/write head is important for prolonging head and cartridge life. The head should be cleaned:

- every 50 hours of use (more frequently in dusty environments).
- whenever frequent errors occur.

ATTENTION

ATTENTION—Each drive should have its own cleaning cartridge.

Cleaning cartridge

The cleaning cartridges are available from Honeywell. See Table 10-8 for part numbers. The kits include:

- a head-cleaning cartridge.
- wiper pads.
- an approved head-cleaner solution.

The head-cleaning cartridge inserts into the drive the same as a standard data cartridge. Instructions are included with the head-cleaning kit.

Procedure

The cleaning cartridge has a pad that is mounted on the pad-arm assembly. Change the pad after five uses.

Table 10-3 Cleaning a Cartridge Drive

Step	Action
1	Remove any cartridge that is in the drive.
2	Press the front panel button on the disk drive and hold it until the green LED starts flashing (about two seconds). This places the drive in the a mode to accept the cleaning cartridge.
3	CAUTION Follow the instructions on the container that holds the disk drive cleaner solution to avoid vapor inhalation or skin exposure.
4	Apply the cleaner by squeezing the solution from the container onto the pad through the cartridge window and center slot. ATTENTION The cleaner should be applied to the pad just before inserting the cartridge into the drive. The pad should be dampened but not saturated with the solution.
5	Place the cleaning cartridge in the drive. The LED will glow steadily.
6	Allow the wiper lever to move back and forth for 30 cleaning cycles.

7	After the 30 cleaning cycles are complete, press the button on the front panel.
----------	---

Continued on next page

10.3 Cleaning the Cartridge Drive, Continued

Procedure, continued

Table 10-3 Cleaning a Cartridge Drive, Continued

Step	Action
8	When the green light extinguishes, remove the cartridge.
9	Record the date of cleaning on the label provided with the cleaning cartridge. Place the label on the outside of the storage sleeve.

10.4 Troubleshooting an Inoperative Cartridge Drive

Failure modes

In general, if only one drive fails, the drive is most likely at fault. If both drives are inoperative, check the power source. The Five-Slot Module's power supply provides the drive power.

Isolation procedure

Table 10-4 Troubleshooting a Cartridge Drive Problem

Step	Action
1	Check the indicators on the Five-Slot Module's power supply. Refer to the <i>Five/Ten Slot Module Service</i> manual for guidance.
2	Check the error LED (red) on the TPDGX board
3	Check the voltages in the power cable by disconnecting it from the cartridge drive. Pins 1 and 2 = $12 \pm .1$ Vdc Pins 3 and 4 = $5.0 \pm .25$ Vdc

10.5 Removing the Cartridge Drive

Procedure

Table 10-5 Removing a Cartridge Drive

Step	Action
1	Remove the rear cover from the upper pod after removing the four retaining screws.
2	Disconnect the cables from the back of the drive enclosure.
3	Remove the two hex-head screws (from each side) that attach the enclosure to the mounting brackets.
4	Lift the enclosure off the support tray and place it on an ESD-safe work surface.
5	Remove the 16 screws from the top cover of the enclosure.
6	Remove the top cover and set it aside.
7	Disconnect the 4-prong dc power cable from the rear of the cartridge drive.
8	Disconnect the 50-pin ribbon cable from the rear of the cartridge drive.
9	Remove the four hex-head machine screws that hold the cartridge drive to the removable media drive disk tray assembly.
10	Pin the new drive according to the instructions in subsection 10.6. CAUTION Observe ESD precautions while handling the cartridge drive.

Continued on next page

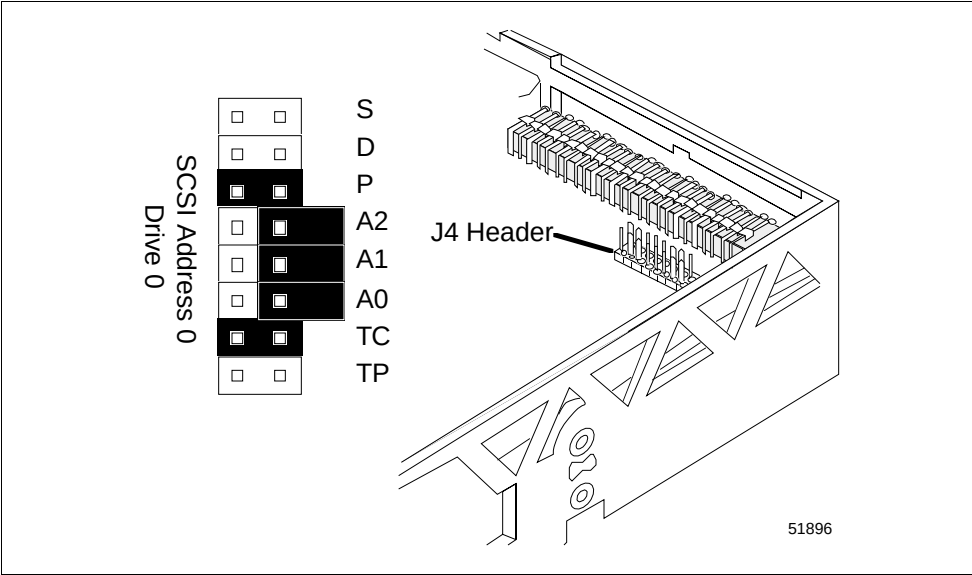
10.6 Pinning the Cartridge Drive

Scope

A Universal Station^X can only have one cartridge drive, Drive 0, as shown in Figure 10-1.

If a RULA is installed, see the RULA section of this publication for special drive pinning.

Figure 10-1 150 MB Iomega Bernoulli Cartridge Drive Pinning



10.7 Replacing a Cartridge Drive

Procedure

Table 10-6 Replacing a Cartridge Drive

Step	Action
1	Place the hard drive on the removable media drive tray and fasten it in place with the screws.
2	Reconnect the dc power cable to the rear of the drive.
3	Reconnect the SCSI 50-pin ribbon cable to the rear of the drive.
4	Place top cover on the enclose and replace the 16 screws to ensure a good ground.
5	Place the drive enclosure on the mounting tray and replace the four hex-head screws that attach the enclosure.
6	Reconnect the power and interface cables to the enclosure.
7	Replace the half upper tier rear cover or return the slide tray fully forward to the front of the station. In either case, check to see if the faceplate is in proper contact with the bezel.

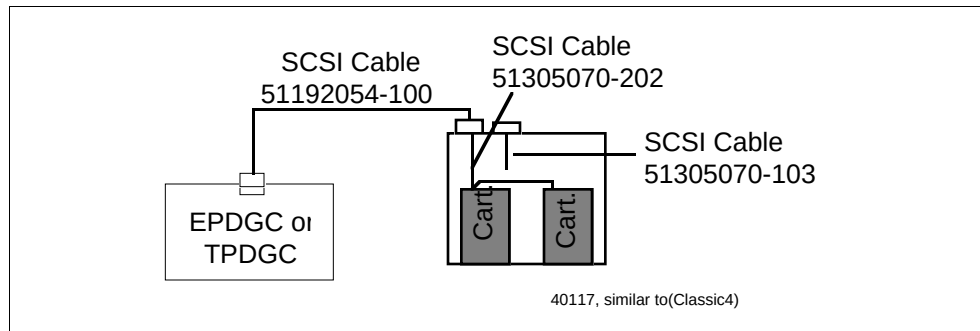
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10.6 Replacing a Cartridge Drive, Continued

Illustration

The ribbon cable **connects to the back connector (the only one)** on a 150 MB cartridge drive.

Figure 10-2 150 MB Cartridge Drive Cabling



10.8 Spare Parts

Spare parts list

Table 10-7 150 MB Cartridge Drive Spare Parts

Part Number	Description
*51196403-100	Beta 0 (Iomega Cartridge Drive)
51192054-102	SCSI Interface Cable TPDGC I/O Board to Drive Enclosure
51305070-202	SCSI Interface Cable for Drive Enclosure (3 connector)
51305070-103	SCSI Interface Cable for Drive Enclosure (2 connector)
51305341-100	Drive Cable DC Power
51196485-100	35 MB Multidisk 150 MB compatible 5.25" cartridge
51196485-200	65 MB Multidisk 150 MB compatible 5.25" cartridge
51196485-300	105 MB Multidisk 150 MB compatible 5.25" cartridge
51196485-400	150 MB Multidisk 150 MB compatible 5.25" cartridge
51196485-500	Cartridge Drive Cleaning Kit—includes: • cleaning cartridge • three 0.25 oz bottles of cleaning solution • two extra cleaning pads
51196485-600	Cartridge Drive Cleaning Solution—refill kit includes: • three 0.25 oz bottles of cleaning solution
51196485-700	Wiper Arm Pads (quantity of 2)

* ORU = Optimum Replaceable Unit.

Section 11 – CD-ROM Drive Service

11.1 Overview

Section contents These are the topics covered in this section:

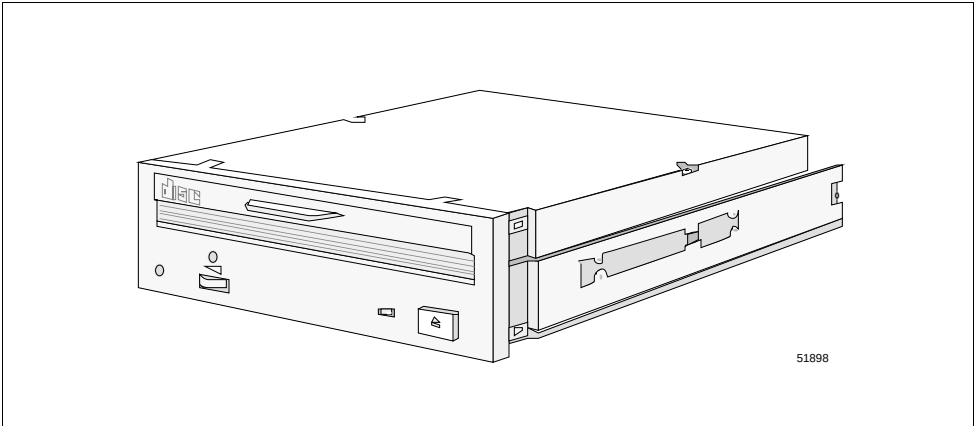
Topic		See Page
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11.3	Spare Parts.....	155

Description

The CD-ROM drive has the following attributes:

- small Computer System Interface (SCSI) bus
- capable of removing disk with power off
- CD-ROM drive is powered by the Five-Slot Module power supply.
- CD-ROM is present for Universal Station^X nodes that have a DAT drive, cartridge drive, or neither DAT or cartridge drives.
- CD-ROM drive is located on the removable media drive tray and is accessible through a panel above the keyboard tray.
- The Audio output earphone and volume control do work because there is no software support for it.

Figure 11-1 CD-ROM Drive



Continued on next page

11.1 Overview, Continued

Location of drive

The CD-ROM may be located on either side of the upper pod or table top slide tray (as you face the front of the station).

Peripherals	Cable Type	CD-ROM SCSI Address
Cartridge on left, CD-ROM on right	Single-ended at CD-ROM end	1
DAT on right, CD-ROM on left	Double-ended at CD-ROM end	1
Blank on left, CD-ROM on right	Double-ended at CD-ROM end	1

If a CD-ROM is installed by itself, the second connector is there to provide for adding a DAT drive later.

- When a DAT drive is added, the CD-ROM must be moved to the left side.

Indicator

The LED on the front indicates activity is present.

Operation

Operation is as follows:

- Load the CD by inserting the cartridge in the reader.
- Press the button to unload the CD.

Continued on next page

11.1 Overview, Continued

SCSI cabling

The CD-ROM drive is connected to a SCSI bus with the hard drive and optional DAT drive as shown in Figures 11-2 and 11-3.

- The DAT drive is always at the end of the cable when present.
- SCSI bus termination is always located on the device at the cable's end.
- The ESD bracket provides the grounding for the cable shield.
- If the DAT drive is not present, connect the CD-ROM to the end of the cable and use termination as shown Figure 11-6.

Figure 11-2 CD-ROM SCSI Cabling with DAT Drive

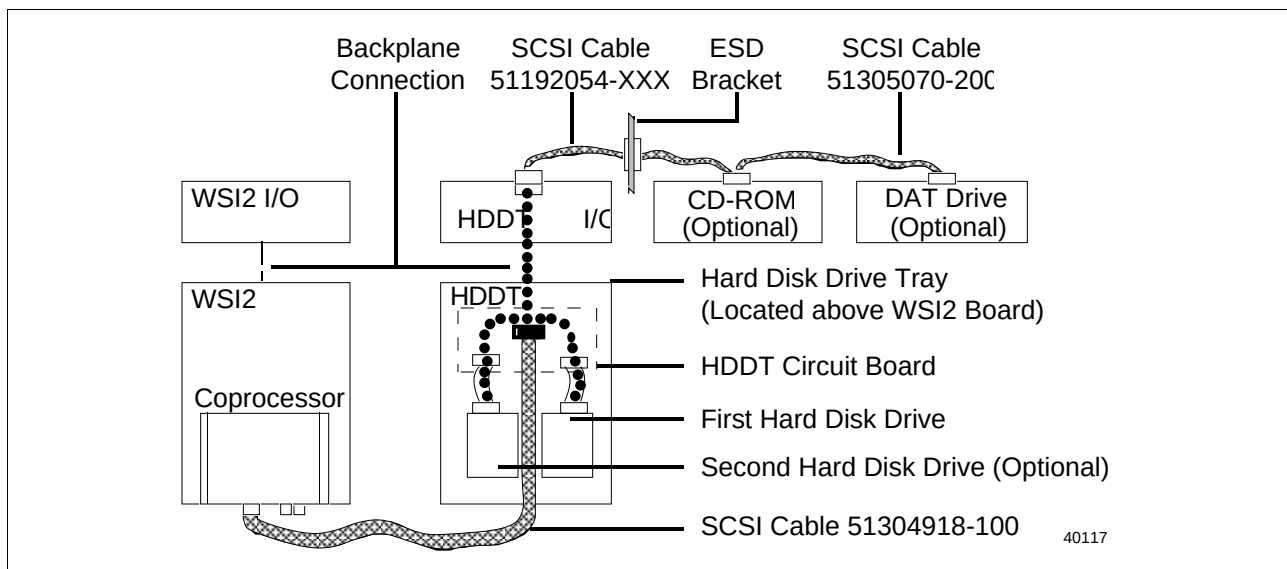
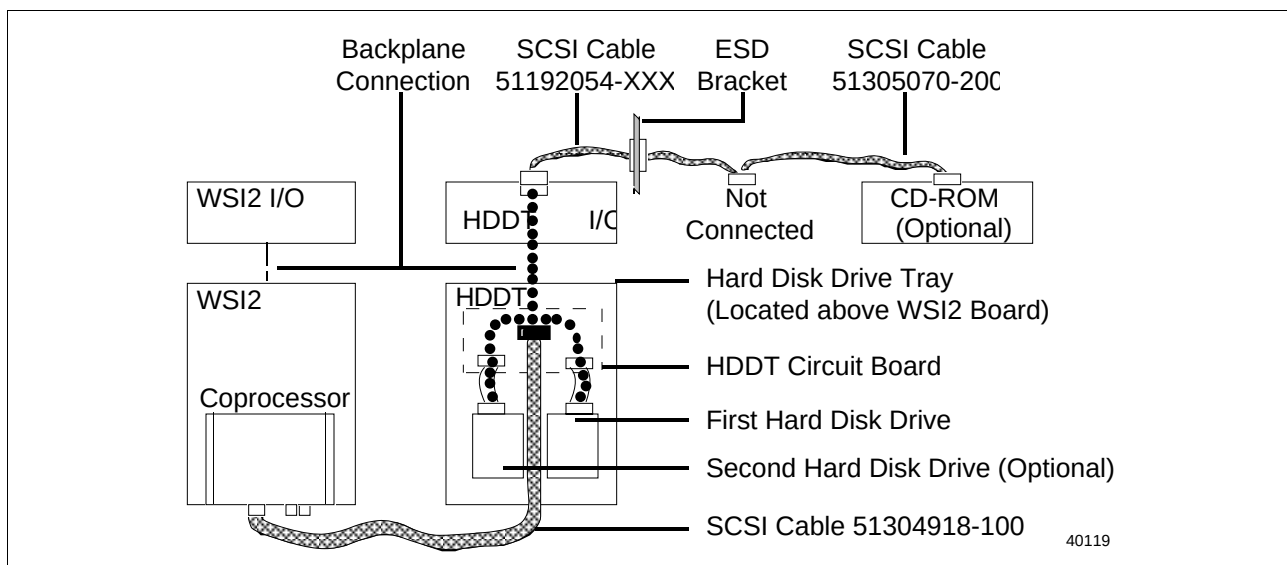


Figure 11-3 CD-ROM SCSI Cabling without DAT Drive



11.2 Replacing the CD-ROM Drive

Procedure

The CD-ROM drive is located in either the Half upper pod or table top slide tray area.

Table 11-1 Replacing a CD-ROM Drive in the Upper Pod

Step	Action
1	Remove the rear cover from the upper pod after removing the four retaining screws.
2	Place the upper pod power supply switch in the OFF position. If there is not an upper pod power supply the power is provide by the Five-Slot Module. Place the Five -Slot Module power supply switch in the OFF position.
3	Remove the four retaining screws from each side of the slide tray the CD-ROM is mounted on.
4	Slide the slide tray back to access the CD-ROM. Be careful not to slide it too far back on the support tray. Hint: temporarily replace one of the slide tray retaining screws to hold the slide tray still while replacing the CD-ROM.
5	Disconnect the four-prong ac power cable from the rear of the CD-ROM drive.
6	Disconnect the 50-pin ribbon cable from the rear of the drive.
7	Remove the four hex-head machine screws holding the CD-ROM drive to the removable media tray assembly. See Figure 11-4.
8	Ensure that the CD-ROM drive is pinned for a SCSI address of 1 and terminated properly for the cases with/without a DAT Drive. See Figures 11-5 and 11-6.
9	Replace the hard drive on the removable media drive tray and fasten it in place with the four screws.
10	Reconnect the dc power cable to the rear of the drive.
11	Reconnect the SCSI 50-pin ribbon cable to the rear of the drive.
12	Return the slide tray fully forward to the front of the station. Check to see if the faceplate is properly in contact with the bezel.
13	Replace the four retaining screws to each side of the slide tray. These screws can be used to adjust the position of the faceplate/ Bezel fit.
14	Fold the SCSI ribbon cable as it had been before and fasten it in place with the cable camp.
15	Place the upper pod power supply and Five-Slot Module power supply switches in the ON position. Check indicators for proper operation.
16	Replace the upper pod rear cover assembly.
17	Replace the four retaining screws in the upper pod rear cover.

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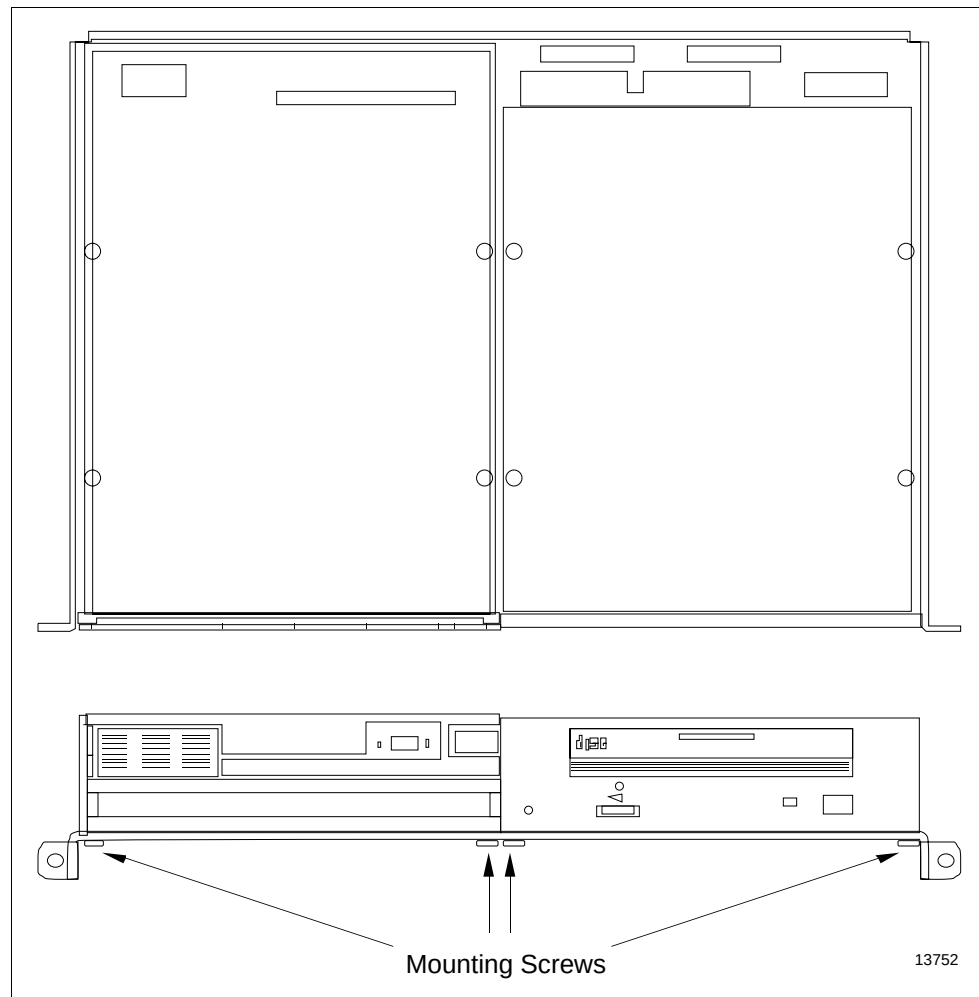
11.2 Replacing the CD-ROM Drive, Continued

Illustration

The CD-ROM drive can be mounted to either side of the removable media drive tray assembly.

- to the right of the cartridge drive
- to the left of the DAT drive
- on either side if there are not any cartridge or DAT drives

Figure 11-4 CD-ROM Drive Mounting



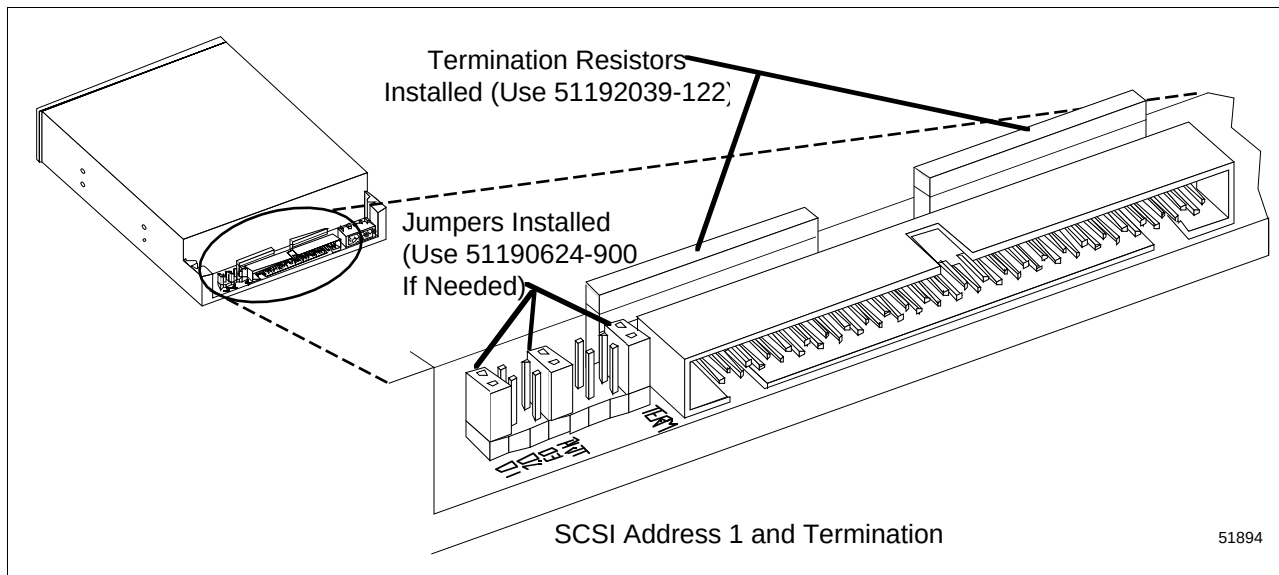
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11.2 Replacing the CD-ROM, Continued

Pinning

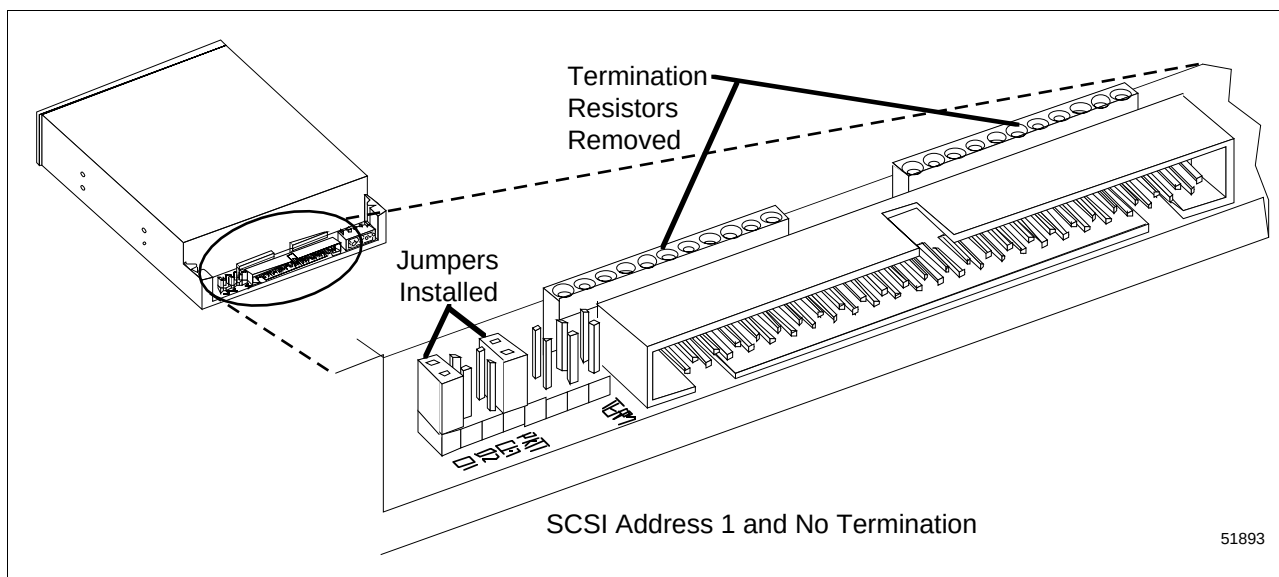
The SCSI bus must be terminated by placing resistors at the drive end of the cable, farthest from the HDDT I/O board. This is done on the DAT when it is present. The CD-ROM should be pinned as in Figure 11-5.

Figure 11-5 CD-ROM Drive Pinning and Termination—DAT Drive Present



The SCSI bus must be terminated by placing resistors at the drive end of the cable, farthest from the HDDT I/O board. When there is not a DAT drive, the termination must be on the CD-ROM drive and the pinning jumper must be in place as shown in Figure 11-6.

Figure 11-6 CD-ROM Drive Pinning and Termination—No DAT Drive Present



11.3 Spare Parts

Spare parts list

Table 11-2 CD-ROM Drive Spare Parts

Part Number	Description
*51196480-100	CD-ROM Drive
51192039-122	Termination Resistors (need 2)
51190624-900	Jumper for Termination
51308080-200	Cartridge/DAT Drive DC Power Cable
51305070-103	Cable SCSI (1 drive connector)
51305070-202	Cable SCSI (2 drive connectors)

* ORU = Optimum Replaceable Unit

Section 12 – CE Compliant CD-ROM Drive Service

12.1 Overview

Section contents

These are the topics covered in this section:

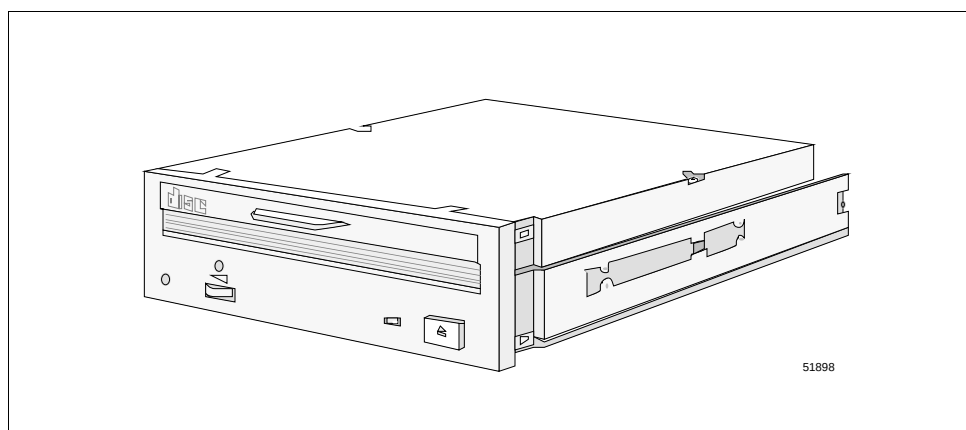
Topic		See Page
SECTION 12 – CE COMPLIANT CD-ROM DRIVE		157
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12.2	Replacing the CD-ROM Drive	160
12.3	Spare Parts.....	163

Description

The CD-ROM drive has the following attributes:

- small Computer System Interface (SCSI) bus
- capable of removing disk with power off
- CD-ROM drive is powered by the Five-Slot Module power supply.
- CD-ROM is present for Universal Station^X nodes that have a DAT drive, cartridge drive, or neither DAT or cartridge drives.
- CD-ROM drive is located on the removable media drive tray and is accessible through a panel above the keyboard tray.
- The Audio output earphone and volume control do not work because there is no software support for it.

Figure 12-1 CD-ROM Drive



Continued on next page

12.1 Overview, Continued

Location of drive

The CD-ROM may be located on either side of the upper pod or table top slide tray. As you face the front of the station

Peripherals	Cable Type	CD-ROM SCSI Address
Cartridge on left, CD-ROM on right	Single ended at CD-ROM end	1
DAT on right, CD-ROM on left	Double-ended at CD-ROM end	1
Blank on left, CD-ROM on right	Double-ended at CD-ROM end	1

If a CD-ROM is installed by itself, the second connector is there to provide for adding a second DAT drive at a later time.

When a DAT drive is added, the CD-ROM must be moved to the left side.

Indicator

The LED on the front indicates activity is present.

Operation

Operation is as follows:

- Load the CD by inserting the cartridge in the reader.
- Press the button to unload the CD.

Continued on next page

12.1 Overview, Continued

SCSI cabling

The CD-ROM drive is connected to a SCSI bus with the hard drive and optional DAT drive as shown in Figures 12-2 and 12-3.

- The DAT drive is always at the end of the cable when present.
- SCSI bus termination is always located on the device at the cable's end.
- The ESD bracket provides the grounding for the cable shield.
- If the DAT drive is not present, connect the CD-ROM to the end of the cable and use termination as shown Figure 12-6.

Figure 12-2 CD-ROM SCSI Cabling with DAT Drive

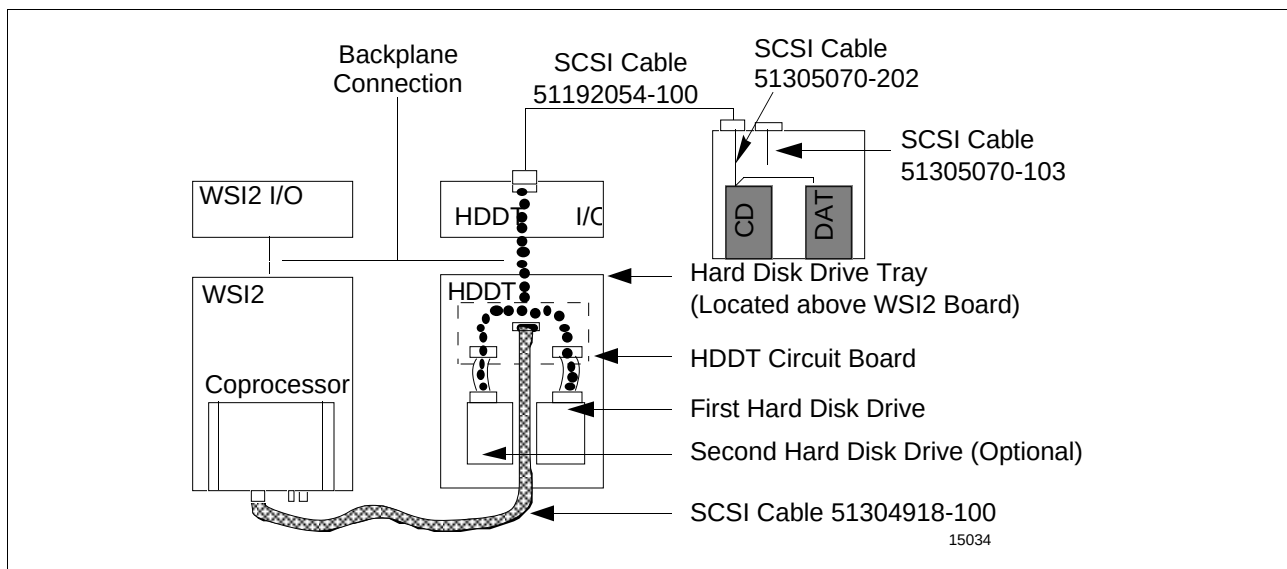
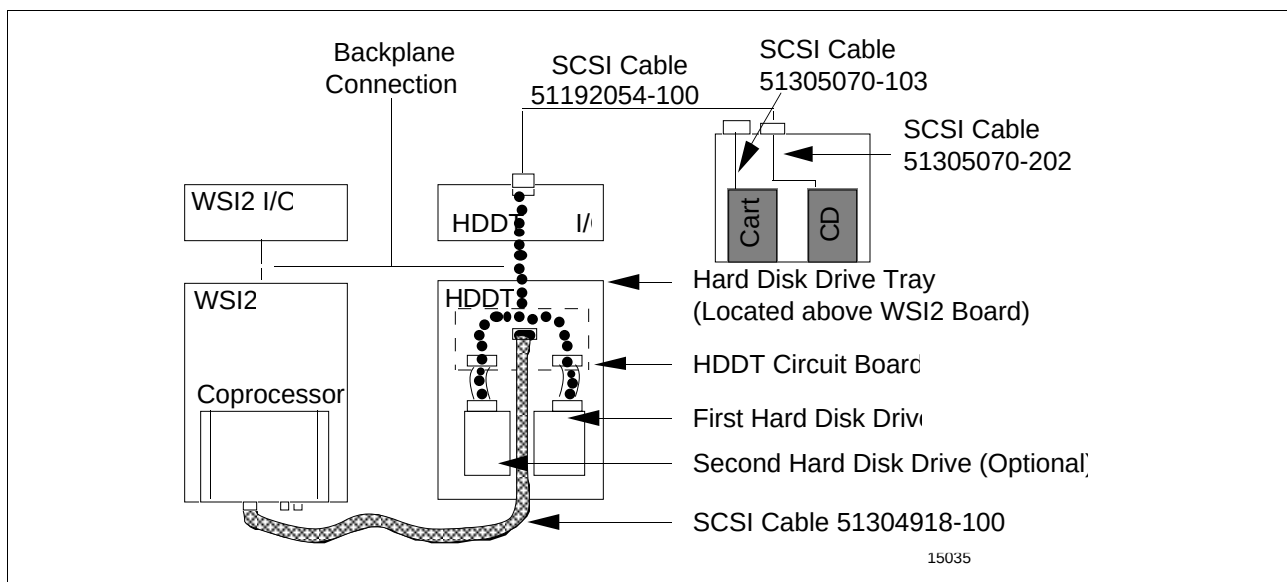


Figure 12-3 CD-ROM SCSI Cabling with Cartridge Drive



12.2 Replacing the CD-ROM Drive

Procedure

The CD-ROM drive is located in the half upper pod or in the table top slide tray area.

Table 12-1 Replacing a CD-ROM Drive in the Upper Pod

Step	Action
1	Remove the rear cover from the upper pod after removing the four retaining screws.
2	Place the upper pod power supply switch in the OFF position. If there is not an upper pod power supply the power is provide by the Five-Slot Module. Place the Five-Slot Module power supply switch in the OFF position.
3	Remove the four retaining screws from each side of the slide tray the CD-ROM is mounted on.
4	Slide the slide tray back to access the CD-ROM. Be careful not to slide it too far back on the support tray. Hint: temporarily replace one of the slide tray retaining screws to hold the slide tray still while replacing the CD-ROM.
5	Disconnect the four-prong ac power cable from the rear of the drive.
6	Disconnect the 50-pin ribbon cable from the rear of the drive.
7	Remove the four hex-head machine screws holding the CD-ROM drive to the removable media tray assembly. See Figure 12-2.
8	Ensure that the CD-ROM drive is pinned for a SCSI address of 1 and terminated properly for the cases with/without a DAT Drive. See Figures 12-3 and 12-4.
9	Replace the hard drive on the removable media drive tray and fasten it in place with the four screws.
10	Reconnect the dc power cable to the rear of the drive.
11	Reconnect the SCSI 50-pin ribbon cable to the rear of the drive.
12	Return the slide tray fully forward to the front of the station. Check to see if the face plate is properly in contact with the bezel.
13	Replace the four retaining screws to each side of the slide tray. These screws can be used to adjust the position of the faceplate/ Bezel fit.
14	Fold the SCSI ribbon cable as it had been before and fasten it in place with the cable camp.
15	Place the upper pod power supply and Five-Slot Module power supply switches in the ON position. Check indicators for proper operation.
16	Replace the upper pod rear cover assembly.
17	Replace the four retaining screws in the upper pod rear cover.

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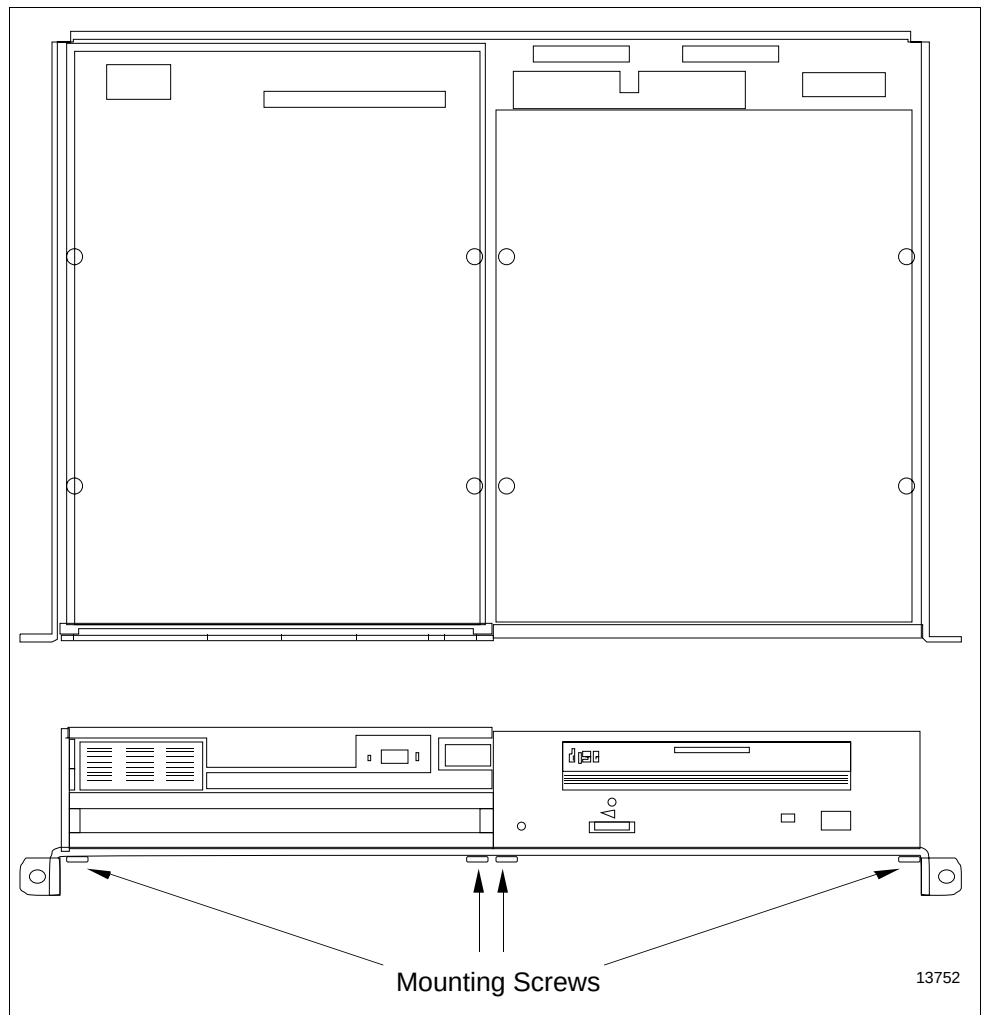
12.2 Replacing the CD-ROM Drive, Continued

Illustration

The CD-ROM drive can be mounted to either side of the removable media drive tray assembly.

- to the right of the cartridge drive
- to the left of the DAT drive
- on either side if there are not any cartridge or DAT drives

Figure 12-4 CD-ROM Drive Mounting



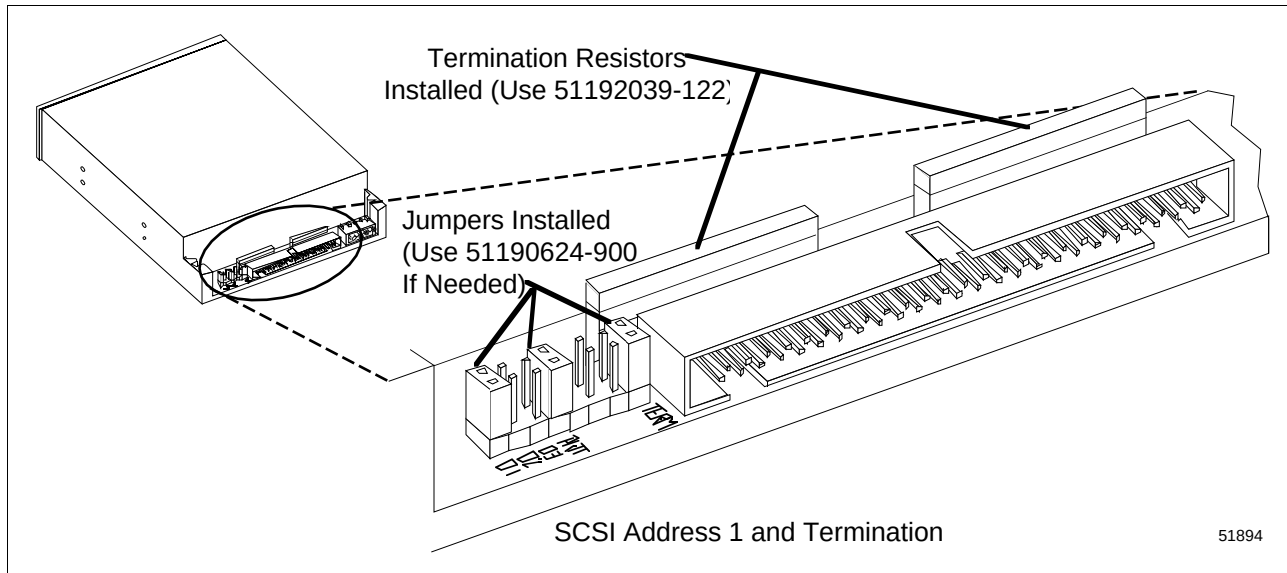
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12.2 Replacing the CD-ROM, Continued

Pinning

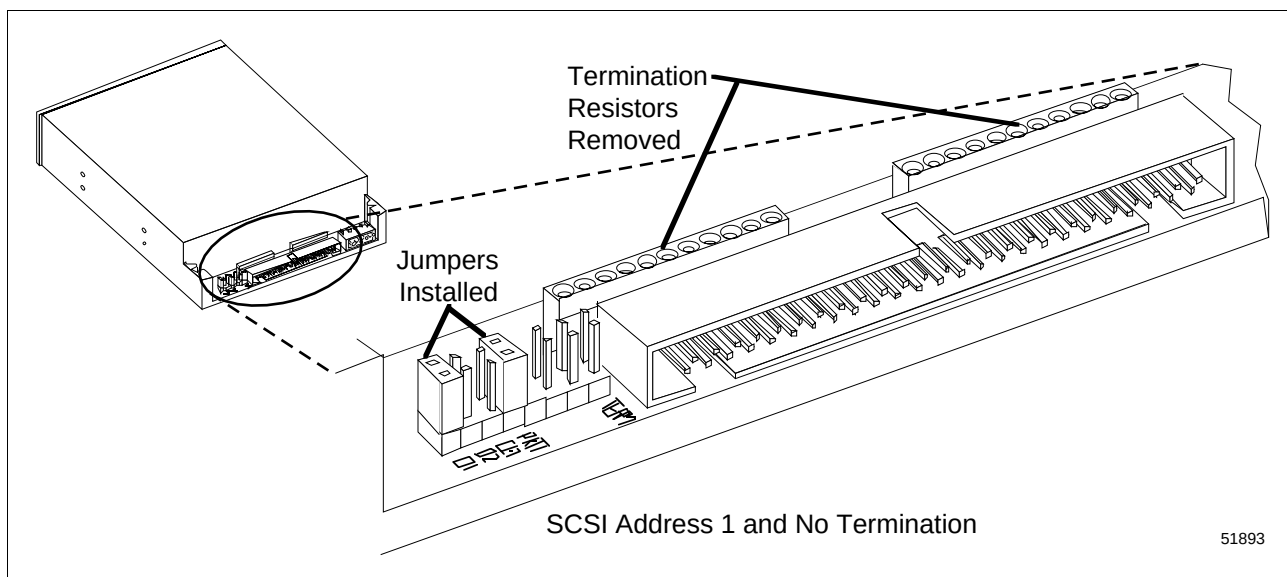
The SCSI bus must be terminated by placing resistors at the drive end of the cable, farthest from the HDDT I/O board. This is done on the DAT when it is present. The CD-ROM should be pinned as in Figure 12-5.

Figure 12-5 CD-ROM Drive Pinning and Termination—DAT Drive Present



The SCSI bus must be terminated by placing resistors at the drive end of the cable, farthest from the HDDT I/O board. When there is not a DAT drive, the termination must be on the CD-ROM drive and the pinning jumper must be in place as shown in Figure 12-6.

Figure 12-6 CD-ROM Drive Pinning and Termination—No DAT Drive Present



12.3 Spare Parts

Spare parts list

Table 12-2 CD-ROM Drive Spare Parts

Part Number	Description
*51196480-100	CD-ROM Drive
51192039-122	Termination Resistors (need 2)
51190624-900	Jumper for Termination
51308080-200	Cartridge/DAT Drive DC Power Cable
51305070-103	Cable SCSI (1 drive connector)
51305070-202	Cable SCSI (2 drive connectors)

* ORU = Optimum Replaceable Unit

Section 13 – Zip Drive Service

13.1 Overview

Section contents

These are the topics covered in this section:

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13.5	Zip Drive SCSI Bus Address Pinning Procedure	170
13.6	Zip Drive Installation Procedure.....	171
13.7	Spare Parts.....	172

Introduction

The Iomega Zip disk drive uses a 100-megabyte (MB) disk for loading or backing up data.

Zip drive configurations can consist of:

- dual drives (if there is no Bernoulli cartridge drive, CD-ROM drive, or DAT drive) in a half upper pod or on a table top slide-tray.
- a single drive that coexists with a Bernoulli cartridge drive, CD-ROM drive, or DAT drive in a half upper pod or on a table top slide-tray.

The Universal Station^X node is resident in a Five-Slot Module and the Zip drive(s) is powered by the Acme power supply in the Five-Slot Module.

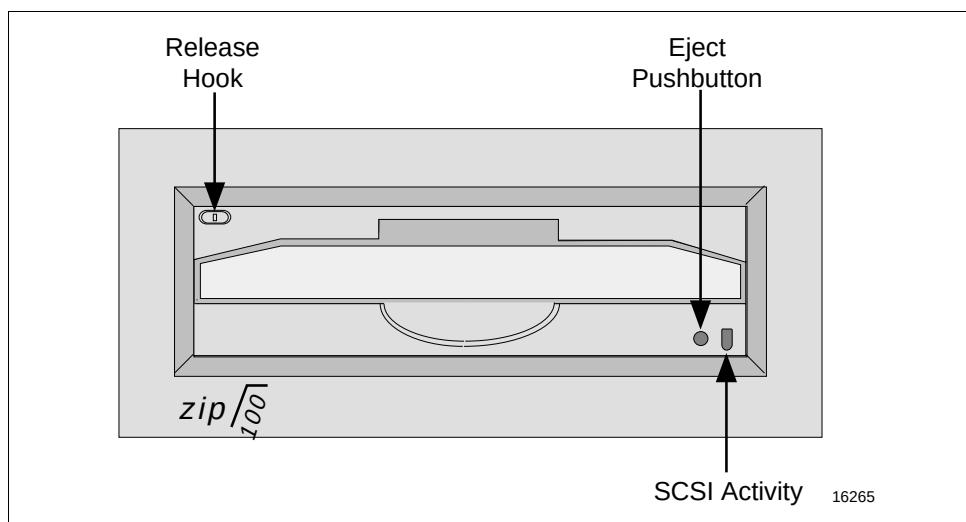
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13.1 Overview, Continued

Front panel of Zip drive

Figure 13-1 illustrates the front panel of the Zip disk drive and identifies its features. Depressing the eject pushbutton normally ejects the disk. If the disk cannot be ejected by this method because of an electronic failure, the release hook that is located in the upper left corner of the front panel can be used to mechanically eject the disk. Use a small paper clip or similar tool to release the disk by pulling forward on the hook.

Figure 13-1 Iomega Zip Disk Drive Front Panel



13.2 Loading/Unloading the Drive

ATTENTION

ATTENTION—Avoid leaving the disk in the drive when it is not being used to reduce wear. The drive automatically retracts its heads after a period of time when not in use, but the disk still rotates. Drive rotation stops completely if the disk is not used during a 15-minute time period.

Disk loading procedure

Use the procedure in Table 13-1 to load a disk into the Zip drive.

Table 13-1 Zip Drive Disk Loading Procedure

Step	Action
1	Insert the disk into the drive with a smooth motion until a click is felt. The disk should start to rotate and the activity LED indicator should flash and then extinguish. The edge of the disk should be flush with the front surface of the drive.
2	The LED indicator is essentially a disk activity monitor. The intensity of the indicator represents the rate of data transfer on the SCSI bus.

Disk unloading procedure

Use the procedure in Table 13-2 to unload a disk from the Zip drive.

Table 13-2 Zip Drive Disk Unloading Procedure

Step	Action
1	Press the eject pushbutton once and release it. The activity LED indicator should flash briefly. The disk will then be ejected.
2	Remove the disk by grasping the center of the disk.

13.3 Zip Drive Troubleshooting an Inoperative Zip Drive

Failure modes

In general, if two drives are installed and only one drive fails, the drive is most likely defective. If both drives are inoperative, check the power supply in the Five-Slot Module.

Troubleshooting procedure

Use the troubleshooting procedure in Table 13-3 to isolate a drive problem.

Table 13-3 Troubleshooting Procedure

Step	Action
1	Check the error (red) LED indicator on the TPDG I/O or TPDGX I/O board.
2	Check the indicators on the Five-Slot Module's power supply.
3	If two drives are installed and both fail, disconnect the power cable from the module's power supply and check for the presence of +5 Vdc and +12 Vdc at the power supply connector. Refer to the <i>Five/Ten-Slot Module Service</i> manual as needed. If the voltages are present, reconnect the power cable and go to Step 4. If the voltages are not present, check the module's power supply.
4	Disconnect the power cable at the drive and check for the presence of +5 Vdc and +12 Vdc in the cable. • Pins 1 and 2 = 12 Vdc $\pm$.1 V • Pins 3 and 4 = 5.0 Vdc $\pm$.25 V If present, reconnect the power cable and go to Step 5.
5	If the power cable connection(s) is secure, replace the drive.
6	If the problem still exists, check the SCSI bus cable connections.

13.4 Zip Drive Removal Procedure

Zip drive removal procedure

Use the procedure in Table 13-4 to remove a Zip drive.

Table 13-4 Zip Drive Removal Procedure

Step	Action
1	If the Zip drive is located in an upper pod, remove the pod's rear cover after removing the four screws that secure the cover. If the Zip drive is located in a desk top cabinet, open the cabinet's rear door, loosen the tray's fasteners, and extend the tray out of the cabinet.
2	Place the upper pod peripheral power supply's power switch in the OFF (0) position. If there is not an upper pod peripheral power supply, the power is provided by the Five-Slot Module's power supply. Place the Five-Slot Module's power supply's power switch in the OFF (0) position.
3	Remove the four hex-head retaining screws from each side of the slide-tray on which the Zip drive(s) is mounted.
4	Move the slide-tray backward enough to provide access to the Zip drive. Be careful not to move the slide-tray too far back on the support tray. Hint: Temporarily replace one of the slide-tray's attachment screws to hold the slide-tray in place while replacing the Zip drive.
5	Disconnect the dc power cable from the rear of the Zip drive. An adapter cable is used between the dc power harness connector and the drive.
6	Disconnect the 50-pin SCSI bus interface ribbon cable from the rear of the Zip drive.
7	Remove the four pan-head machine screws that attach the Zip drive assembly to the the vertical brackets on the slide-tray.

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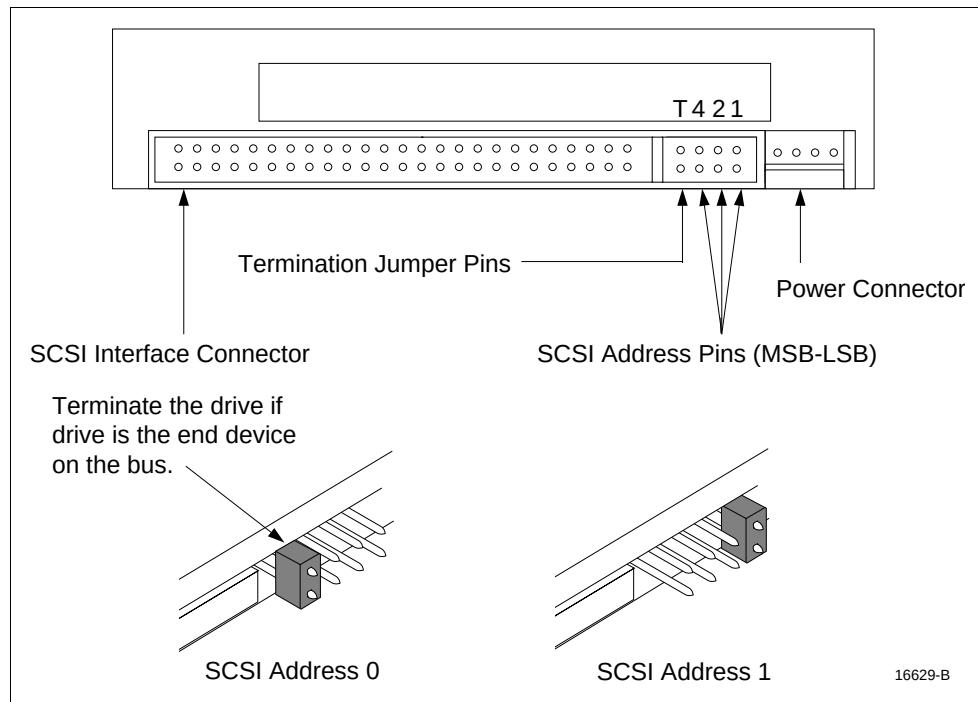
13.5 Zip Drive SCSI Bus Address Pinning Procedure

Drive SCSI bus pinning

Figure 13-2 illustrates the method of pinning the Zip drive. Pinning for SCSI bus addresses of one (1) and zero (0) are illustrated in the figure. By normal convention, the drive that is pinned for a SCSI bus address of zero (0) is the end device on the bus, and the termination jumper must be in place. Duplicate the pinning of the drive that is being replaced.

If a RULA is installed, see the RULA section of this publication for special drive pinning instructions.

Figure 13-2 Zip Drive SCSI Bus Pinning



13.6 Zip Drive Installation Procedure

CAUTION

CAUTION—Observe ESD precautions while handling the Zip drive.

Zip drive installation procedure

Use the procedure in Table 13-5 to install a replacement Zip drive.

Table 13-5 Zip Drive Installation Procedure

Step	Action
1	Verify that the replacement drive is correctly pinned for its application. See Figure 13-2.
2	Attach the drive assembly to the vertical brackets on the slide-tray and secure it in place with the four screws that were previously removed.
3	Reconnect the dc power adapter cable at the rear of the drive.
4	Reconnect the 50-pin SCSI bus interface ribbon cable at the rear of the drive.
5	Move the slide-tray forward to the front of the station. Check to see that the faceplate is properly in contact with the bezel.
6	Install the four retaining screws at each side of the slide-tray. The screw slots in the slide-tray are used to align the tray with the faceplate/bezel.
7	Fold the SCSI bus interface ribbon cable as necessary. Fasten it in place with the cable clamp(s).
8	Place the peripheral power supply or Five-Slot Module power supply switch in the ON (1) position. Check the indicators for proper operation.
9	If the Zip drive is located in an upper pod, replace the pod's rear cover and secure the cover with the four screws that were previously removed. If the Zip drive is located in a desk top cabinet, return the tray to its normal position, secure it with its fasteners, and close the cabinet's rear door.

13.7 Spare Parts

Zip drive parts list

Table 13-6 is a list of Zip drive related parts.

Table 13-6 Zip Drive Replacement Parts

Part Number	Description
*51196933-100	Iomega 100-Megabyte Zip Drive Assembly
51304842-200	SCSI Bus Interface Cable (single drive)
51304199-300	SCSI Bus Interface Cable (dual drives)
51308055-100	DC Power Cable

* ORU = Optimum Replaceable Unit.

Section 14 – CE Compliant ZIP Drive Service

14.1 Overview

Section contents

These are the topics covered in this section:

Topic		See Page
SECTION 14 – CE COMPLIANT ZIP DRIVE SERVICE.....		173
14.1	Overview.....	173
14.2	Loading/Unloading a Zip Drive	175
14.3	Troubleshooting an Inoperative Zip Drive	176
14.4	Zip Drive Removal Procedure	177
14.5	Zip Drive SCSI Bus Address Pinning Procedure	178
14.6	Zip Drive Installation Procedure.....	179
14.7	Spare Parts.....	180

Introduction

The Iomega Zip disk drive uses a 100-megabyte (MB) disk for loading or backing up data.

Zip drive configurations can consist of:

- dual drives if there is no Bernoulli cartridge drive, CD-ROM drive, or DAT drive in the drive enclosure.
- a single drive that coexists with a Bernoulli cartridge drive, CD-ROM drive, or DAT drive in the drive enclosure.

The Universal Station^X node is resident in a Five-Slot Module and the Zip drive(s) is powered by the peripheral power supply in the drive enclosure.

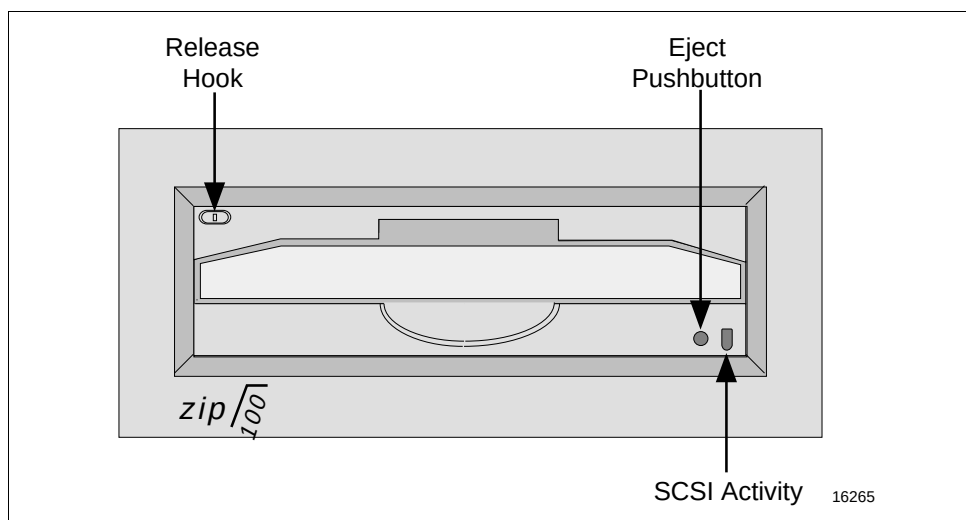
Continued on next page

14.1 Overview, Continued

Front panel of Zip drive

Figure 14-1 illustrates the front panel of the Zip disk drive and identifies its features. Depressing the eject pushbutton normally ejects the disk. If the disk cannot be ejected by this method because of an electronic failure, the release hook that is located in the upper left corner of the front panel can be used to mechanically eject the disk. Use a small paper clip or similar tool to release the disk by pulling forward on the hook.

Figure 14-1 Iomega Zip Disk Drive Front Panel



14.2 Loading/Unloading a Zip Drive

ATTENTION

ATTENTION—Avoid leaving the disk in the drive when it is not being used to reduce wear. The drive automatically retracts its heads after a period of time when not in use, but the disk still rotates. Drive rotation stops completely if the disk is not used during a 15-minute time period.

Disk loading procedure

Use the procedure in Table 14-1 to load a disk into the Zip drive.

Table 14-1 Zip Drive Disk Loading Procedure

Step	Action
1	Insert the disk into the drive with a smooth motion until a click is felt. The disk should start to rotate and the activity LED indicator should flash and then extinguish. The edge of the disk should be flush with the front surface of the drive.
2	The LED indicator is essentially a disk activity monitor. The intensity of the indicator represents the rate of data transfer on the SCSI bus.

Disk unloading procedure

Use the procedure in Table 14-2 to unload a disk from the Zip drive.

Table 14-2 Zip Drive Disk Unloading Procedure

Step	Action
1	Press the eject pushbutton once and release it. The LED indicator should flash on and off. Do not pull on the disk until the LED indicator stops flashing.
2	When the LED indicator stops flashing, remove the disk by firmly pulling on the center of the disk.

14.3 Troubleshooting an Inoperative Zip Drive

Failure modes

In general, if only one drive fails, the drive is most likely at fault. If both drives are inoperative, check the drive enclosure's peripheral power supply.

Troubleshooting procedure

Use the procedure in Table 14-3 to isolate the Zip drive problem. Refer to the *Five/Ten-Slot Module Service* manual as needed.

Table 14-3 Zip Drive Troubleshooting Procedure

Step	Action
1	Check the error (red) LED indicator on the TPDG I/O or TPDGX I/O board.
2	Remove the drive enclosure's top cover.
3	Check to be sure that the power cable connection(s) inside the drive enclosure is proper.
4	Disconnect the adapter power cable from the Zip drive and check for the presence of +5 Vdc and +12 Vdc in the cable. • Pins 1 and 2 = 12 Vdc $\pm$.1 V • Pins 3 and 4 = 5.0 Vdc $\pm$.25 V If present, replace the drive. If not present, replace the peripheral power supply.
5	If the problem still exists, check the SCSI bus interface cable connections.

14.4 Zip Drive Removal Procedure

CAUTION

CAUTION—Observe ESD precautions while handling the Zip drive.

Zip drive removal procedure

Use the procedure in Table 14-4 to remove a Zip drive.

Table 14-4 Zip Drive Removal Procedure

Step	Action
1	If the Zip drive is located in an upper pod, remove the rear cover from the pod after removing the four screws that secure it in place. If the Zip drive is located in a table top cabinet, open the rear door, release the latches at each side of the slide-tray, and fully extend the tray.
2	If the Zip drive is located in an upper pod, disconnect the power and interface cables from the back of the drive enclosure. Label them, if necessary, to ensure that they will be properly reconnected.
3	If the Zip drive is located in an upper pod, remove the eight hex-head screws that attach the drive tray to the cabinet structure. Move the tray backward enough to provide access to the drive enclosure's top cover.
4	Remove the 16 screws that attach the drive enclosure's top cover.
5	Remove the top cover.
6	Disconnect the 4-pin dc power adapter cable from the rear of the Zip drive.
7	Disconnect the 50-pin SCSI bus interface ribbon cable from the rear of the Zip drive.
8	Remove the four machine screws that attach the Zip drive assembly to the vertical mounting brackets.
9	Remove the Zip drive assembly from the drive enclosure.

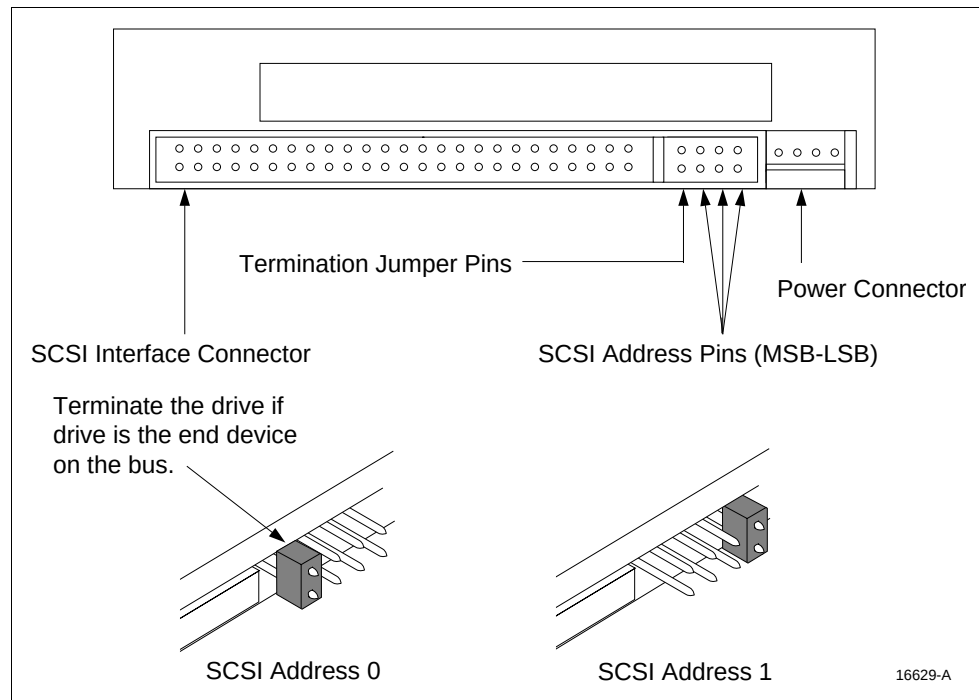
14.5 Zip Drive SCSI Bus Address Pinning Procedure

Drive SCSI bus pinning

Figure 14-2 illustrates the method of pinning the Zip drive. Pinning for SCSI bus addresses of one (1) and zero (0) are illustrated in the figure. By normal convention, the drive that is pinned for a SCSI bus address of zero (0) is the end device on the bus, and the termination jumper must be in place. Duplicate the pinning of the drive that is being replaced.

If a RULA is installed, see the RULA section of this publication for special drive pinning instructions.

Figure 14-2 Zip Drive SCSI Bus Pinning



14.6 Zip Drive Installation Procedure

CAUTION

CAUTION—Observe ESD precautions while handling the Zip drive.

Zip drive installation procedure

Use the procedure in Table 14-5 to install a replacement Zip drive.

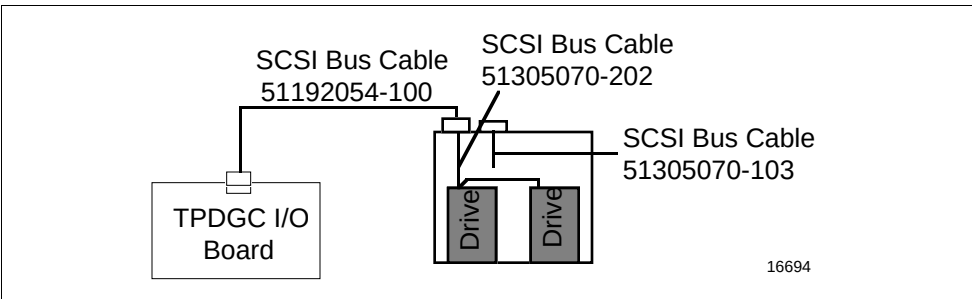
Table 14-5 Zip Drive Installation Procedure

Step	Action
1	Be sure the replacement drive is pinned correctly. Attach the drive assembly to the vertical mounting brackets in the drive enclosure with the four screws that were previously removed.
2	Reconnect the dc power adapter cable to the rear of the drive.
3	Reconnect the 50-pin SCSI bus interface ribbon cable to the rear of the drive.
4	Place the cover on the drive enclosure and secure it with the sixteen screws that were previously removed. All screws must be in place for EMI containment.
5	If the Zip drive is located in an upper pod, move the slide-tray forward against the front bezel. Attach the drive-tray to the cabinet structure with the eight hex-head screws that were previously removed.
6	Reconnect the power and SCSI bus interface cables to the drive enclosure.
7	If the Zip drive is located in an upper pod, replace the rear cover on the pod and secure it with the four screws that were previously removed. If the Zip drive is located in a table top cabinet, release the slide-tray stops and slide it fully into the cabinet. Secure it in place with its latches. Close the rear door.

Drive enclosure internal and external SCSI bus cabling

Figure 14-3 illustrates the drive enclosure’s internal and external SCSI bus cabling.

Figure 14-3 Drive Enclosure Internal/External SCSI Bus Cabling



14.7 Spare Parts

Zip drive parts list

Table 14-6 is a list of Zip drive related parts.

Table 14-6 Zip Drive Replacement Parts

Part Number	Description
*51196933-100	Iomega 100-Megabyte Zip Drive Assembly
51305070-202	Drive Enclosure Interior SCSI Bus Interface Cable (3-connector)
51305070-103	Drive Enclosure Interior SCSI Bus Interface Cable (2-connector)
51192054-102	TPDGC I/O Board to Drive Enclosure SCSI Bus Cable

* ORU = Optimum Replaceable Unit

Section 15 – Mouse Service

15.1 Overview

Section contents

These are the topics covered in this section:

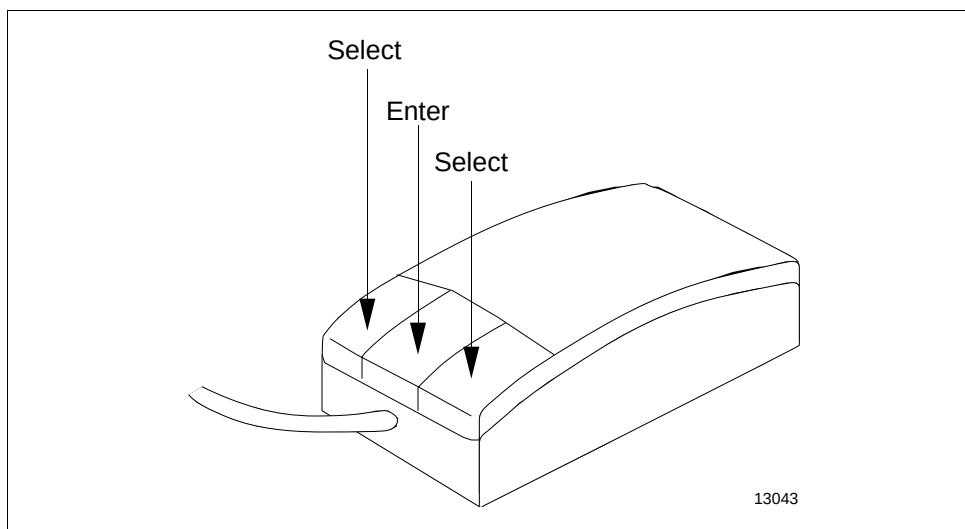
Topic		See Page
SECTION 15 – MOUSE SERVICE		181
15.1	Overview.....	181
15.2	Cleaning the Mouse.....	182
15.3	Troubleshooting the Mouse	183
15.4	Mouse Spare Parts.....	183

Description

The mouse has three select switches:

Switch Location	Function
Left	Select
Center	Not used
Right	Enter

Figure 15-1 Universal Station^X Mouse



15.2 Cleaning the Mouse

Procedure

Table 15-1 Cleaning the Mouse

Step	Action
1	Turn the mouse upside down.
2	Unscrew the circular ring around the ball.
3	Turn the mouse over into the palm of your other hand—the ball and ring will fall out of the mouse.
4	Wipe the surface of the ball with a slightly damp cloth that has been dipped in a mild soap solution.
5	Polish the ball with a soft, dry, lint-free cloth.
6	Place the ball in the mouse.
7	Place the ring over the ball.
8	Press the ring into the mouse and twist the ring until it locks.

15.3 Troubleshooting the Mouse

Procedure

Table 15-2 Troubleshooting the Mouse

Step	Action
1	If the cursor on the monitor does not move when the mouse is scrolled, check the cable connection at J6 on the TPDG I/O board. See Figure 6-2.
2	Ensure that the mouse cable is connected to J6 on the TPDG I/O board.
3	If checking the cable connection does not correct the problem, install each of the following until the problem is corrected: • Mouse, 51195083-200 • TPDG I/O Board • TPDG Board

15.4 Mouse Spare Parts

Spare parts list

Table 15-3 Mouse Spare Parts

Part Number	Description
*51195083-200	Mouse, Universal Station ^X

* ORU = Optimum Replaceable Unit

Section 16 – Trackball Service

16.1 Overview

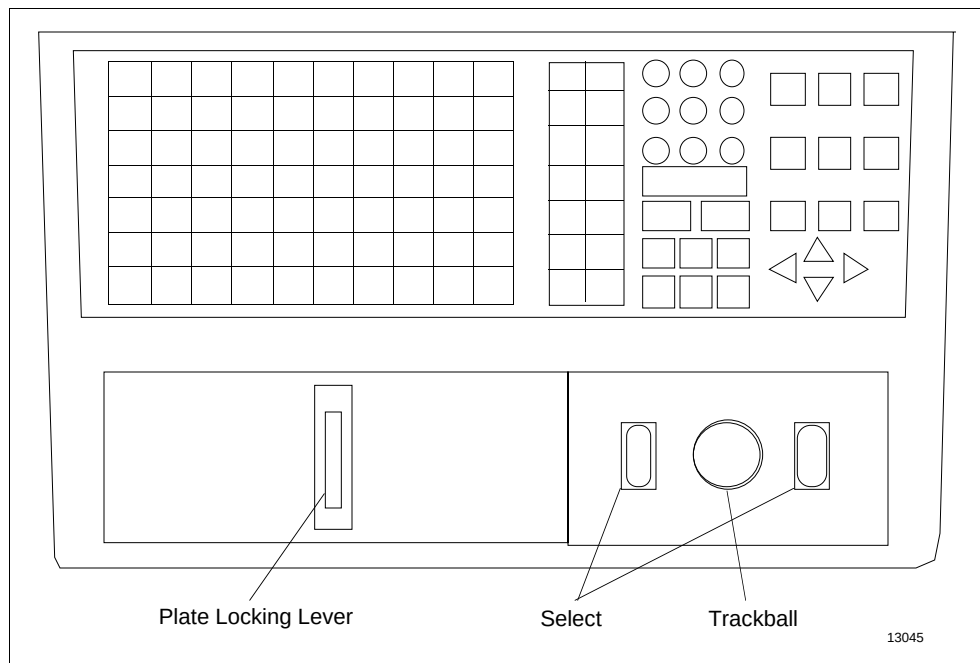
Section contents These are the topics covered in this section:

Topic		See Page
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16.1	Overview	185
16.2	Operating the Trackball	186
16.3	Cleaning the Trackball	187
16.4	Troubleshooting the Trackball	187
16.5	Replacement of the Trackball Interface Board	188
16.6	Trackball Spare Parts	189

Description

The Trackball is designed to operate with either the right hand or left hand. The plate located beside the Trackball fits on either side of the Trackball.

Figure 16-1 Universal Station^X Trackball



16.2 Operating the Trackball

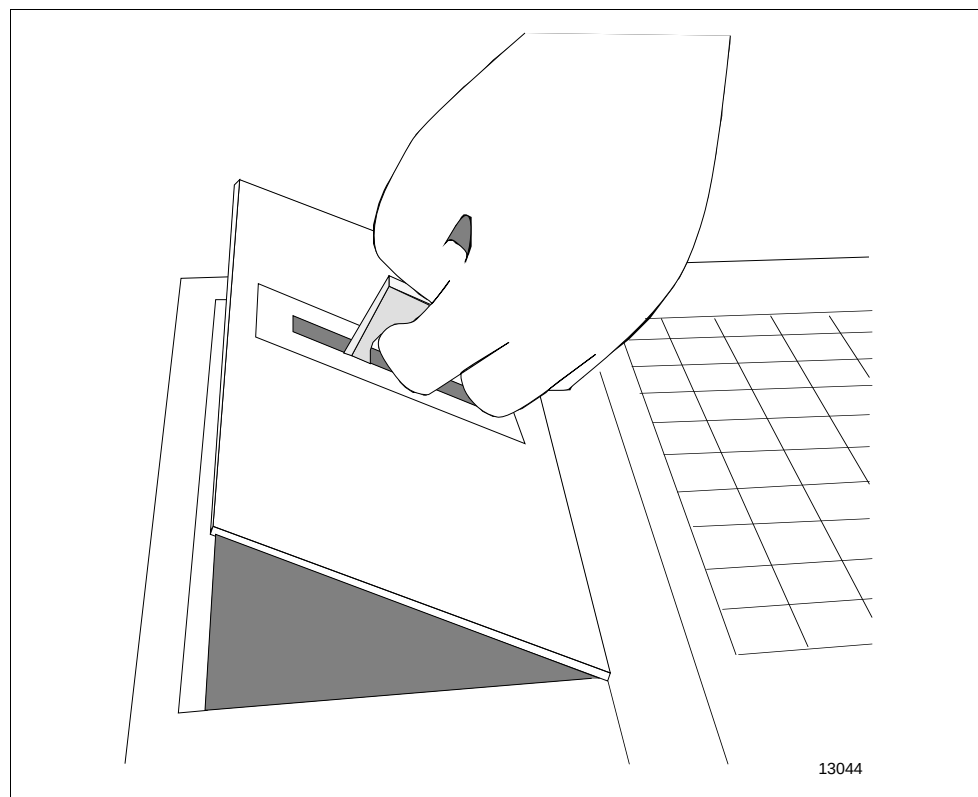
Switching right-to-left

Table 16-1 Switching the Trackball

Step	Action
1	Remove the filler plate. • Press down on the front edge of the narrow rectangle located at the center of the filler plate until a lever with a finger hole rises up in the air. • Hook a finger through the finger hole and raise/rotate the plate in the air, removing it from the tray.
2	Slide the Trackball to the other side, ensuring that the cable, visible to the right of the Trackball, is moving freely.
3	Replace the filler plate by reversing Step 1.

Illustration

Figure 16-2 Removing Filler Plate



16.3 Cleaning the Trackball

Procedure

Table 16-2 Cleaning the Trackball

Step	Action
1	Wipe the surface of the ball with a slightly damp cloth that has been dipped in a mild soap solution.
2	Polish the ball with a soft, dry, lint-free cloth.

16.4 Troubleshooting the Trackball

Procedure

Table 16-3 Troubleshooting the Trackball

Step	Action
1	If the cursor on the monitor does not move when the Trackball is scrolled, check the cable connection at J6 on the TPDG I/O board. See Figure 6-2.
2	If no problem is corrected at the TPDG I/O board, install a spare board for each of the following until the problem is corrected: • Trackball Interface Board, 51108312-200, located under the Operator's Keyboard. See subsection 16.4 • TPDG I/O Board • TPDG Board

16.5 Replacement of the Trackball Interface Board

Procedure

Table 16-4 Replacing the Trackball Interface Board

Step	Action
1	Open the rear door of the cabinet and turn off the Universal Station ^X node power switch.
2	Remove the two Phillips-head screws located under the keyboard tray, 2.5 inches from the front, on each side.
3	Remove four hex-head screws located on the two sides near the edge of the bottom of the keyboard tray.
4	Remove the two Phillips-head screws located on the top, at the rear of the Operator's Keyboard.
5	Pivot the keyboard bezel/trackball assembly, raising it toward yourself.
6	Tilt the operator towards the CRT.
7	Disconnect the trackball cable at the nine-pin connector attached to the cable connecting to the Trackball Interface Card.
8	Remove the cable connecting the Trackball Interface Card to the EPDGP I/O card (through the gooseneck). Release the tie-downs as necessary.
9	Replace the Trackball Interface Card after removing the two slotted -head screws.
10	Reconnect the Trackball Interface Cable to the TPDG I/O card connector J6.
11	Retie the tie-downs previously released in Step 8.
12	Reconnect the Trackball cable to the nine-pin connector that is attached to the Trackball Interface Card.
13	Pivot the Trackball/bezel assembly up toward yourself.
14	Place the Operator's Keyboard in the keyboard tray.
15	Place the Trackball/bezel assembly on the keyboard tray.
16	Replace the four hex-head screws removed in Step 4.
17	Replace the two Phillips-head screws removed in Step 3.

16.6 Trackball Spare Parts

Spare parts list

Table 16-5 Trackball Spare Parts

Part Number	Description
*51108312-100	Trackball Assembly
*51108312-200	Interface Card and Attached Interface Cables
51304673-200	Switch, Trackball, Right
51304673-100	Switch, Trackball, Left

* ORU = Optimum Replaceable Unit

Section 17 – LCN Printers Service

17.1 Overview

Section contents These are the topics covered in this section:

Topic		See Page
SECTION 17 – LCN PRINTERS SERVICE		191
17.1	Overview	191
17.2	LCN Printer ORUs	192

TPS and PIN Printers All TPS (LCN) printing requirements are supported by ASPI-32, ASPI-41, ASPI-46, or Signum 2043 printers. These printers are covered in the *Universal Station Service* manual.

ATTENTION

ATTENTION—Transmittal and printing of bit-mapped graphics on the laser printer requires large memory and appears to take more time than normal (up to 5 minutes). Do not assume a failure has occurred if you are waiting for a print job to complete.

17.2 LCN Printer ORUs

Spare parts list

ASPI-32 Printers are no longer available. Use the ASPI-41, ASPI-46, or Signum 2043 printer.

Table 17-1 Printer ORUs

Part Number	Description
*51198305-100	ASPI-41 Printer
*51195091-100	ASPI-46 Printer
*51196690-100	Signum 2043 – 120 Vac
*51196690-200	Signum 2043 – 240 Vac

* ORU = Optimum Replaceable Unit

Section 18 – Laser Printer Service

18.1 Overview

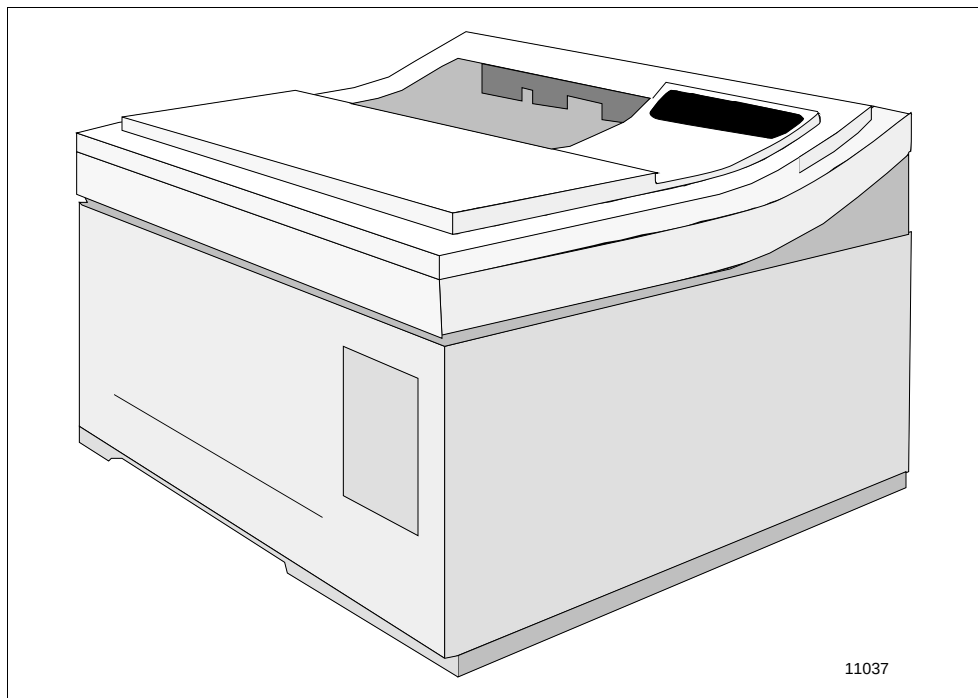
Section contents These are the topics covered in this section:

Topic		See Page
SECTION 18 – LASER PRINTER SERVICE.....		193
18.1	Overview.....	193
18.2	Operating Instruction and Self-Test Procedure.....	194
18.3	Spare Parts.....	194

Description

The Hewlett-Packard LaserJet4 printer is the printer for the coprocessor system. The interface to the Laser Jet Printer is accomplished using the serial interface port on the printer (the bottom connector on the rear of the printer).

Figure 18-1 HP LaserJet4 Printer



18.2 Operating Instruction and Self-Test Procedure

Reference

The LaserJet4 HP LaserJet4 and 4M Printer User's Manual, part number C2001-90912, (shipped with the printer) should be used to support the operation of the printer. Because the printer is the Optimum Replaceable Unit (ORU), Honeywell does not supply the service manual. It can be ordered from Hewlett-Packard.

18.3 Spare Parts

Spare parts list

Table 18-1 LaserJet4 Spare Parts

Part Number	Description
*51196314-100	LaserJet4 Printer
*51196314-300	LaserJet4 Printer Signal Cable (115 Vac)

* ORU = Optimum Replaceable Unit

Appendix A – Release 100/110 Hardware

A.1 Overview

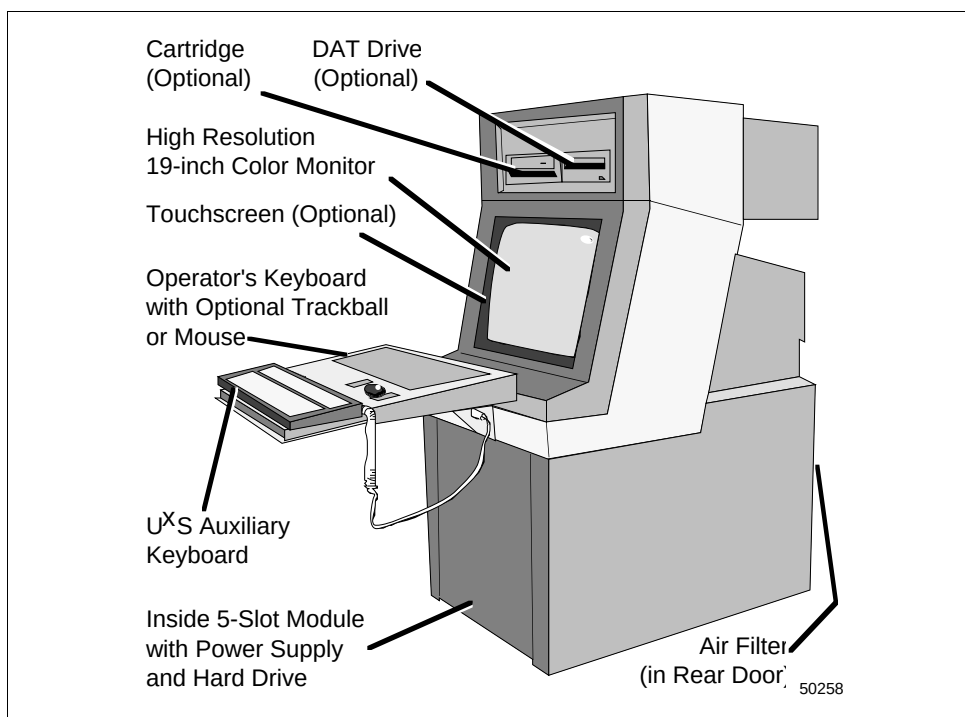
Section contents

These are the topics covered in this section:

Topic		See Page
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A.2	WSI Board.....	197
A.3	WSI I/O Board Description.....	199
A.4	WSI Daughter Board Description.....	200
A.5	400 Hard Drive.....	201
A.6	Replacing the Hard Drive.....	202
A.7	R100 Universal StationX Cabling.....	204
A.8	Universal StationX Shutdown.....	206
A.9	Universal StationX Node Troubleshooting	209
A.10	Spare Parts.....	214

Illustration

Figure A-1 Universal Station^X ORU Locations



Continued on next page

A.1 Overview, Continued

Scope

This appendix contains support information for the original Release 100 hardware. The Release 110 hardware is identical to Release 100.

Only the portions of the U^XS that are different from the Release 200 U^XS are supported in this appendix.

Differences

There are two primary differences:

- WSI and WSI I/O boards
- 400 MB hard drive

Release 100/110 Features of the Station

A.2 WSI Board Description

Features

A board that provides connections to the coprocessor daughter board. The WSI board holds 16 MB (optionally 32 MB) of memory accessible to the coprocessor. See Figure A-2.

Connectors J8 and J9, on the “free edge” of the WSI opposite from the backpanel connector, are connected by two ribbon cables to the two matching connectors on the “free edge” of the TPDG board.

WSI Indicators

Table A-1 WSI Board Indicator Description

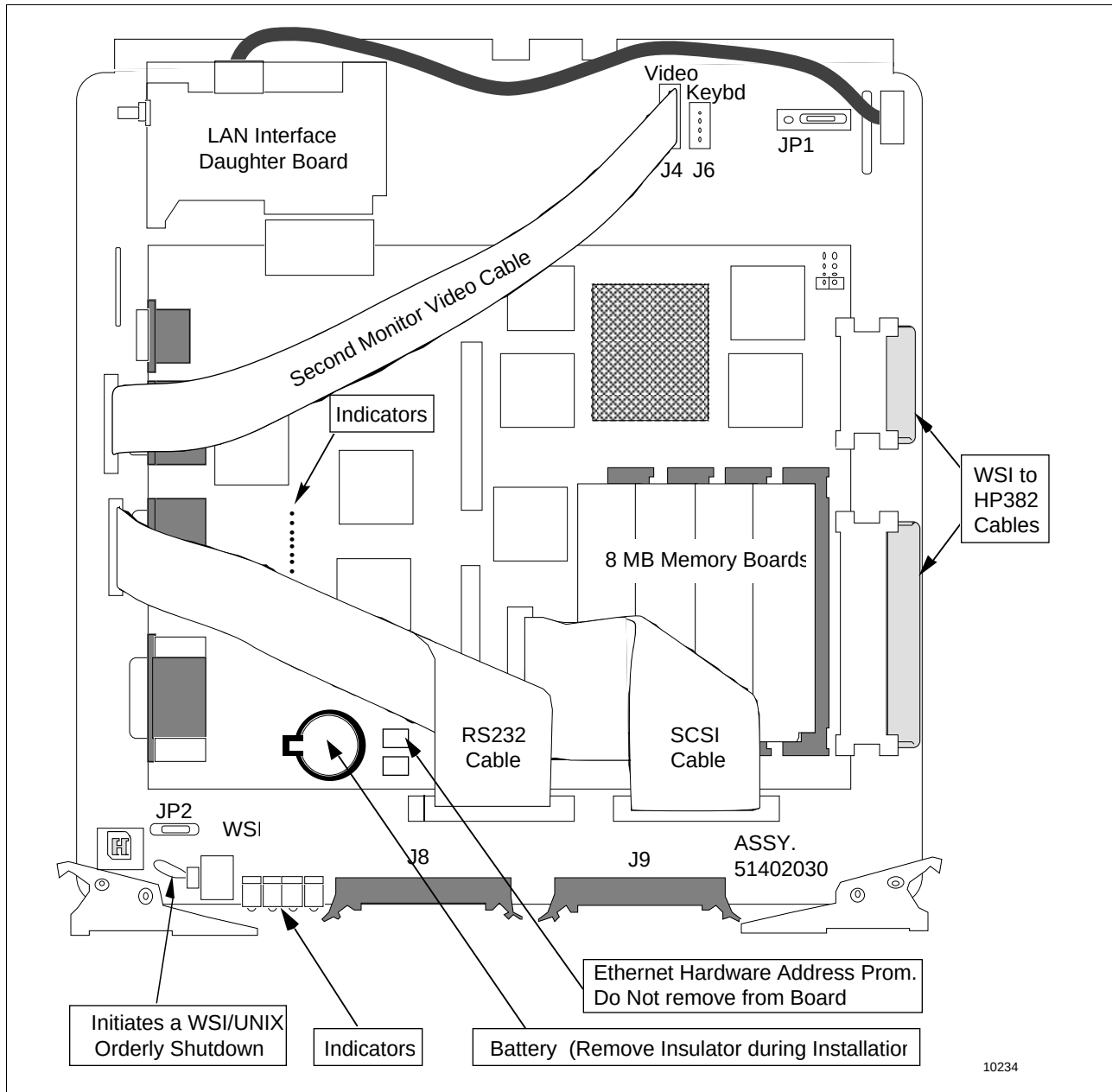
LED	Description	Suspected Cause If Abnormal
SELF-TEST ERROR	On to indicate a data transfer acknowledge failure. Normally off.	WSI board or WSI I/O board
PASS MOD TEST	On to indicate the board has passed self-tests. Normally on.	WSI board or WSI I/O board
BUS TRANS ERROR	On to indicate a detected backpanel bus parity error. Normally off.	WSI board or other boards in chassis or backpanel
UNIX	On to indicate that UNIX is operational. Normally on.	WSI board or WSI I/O board or coprocessor hard disk

Continued on next page

A.2 WSI Board Description, Continued

Illustration

Figure A-2 WSI Board



A.3 WSI I/O Board Description

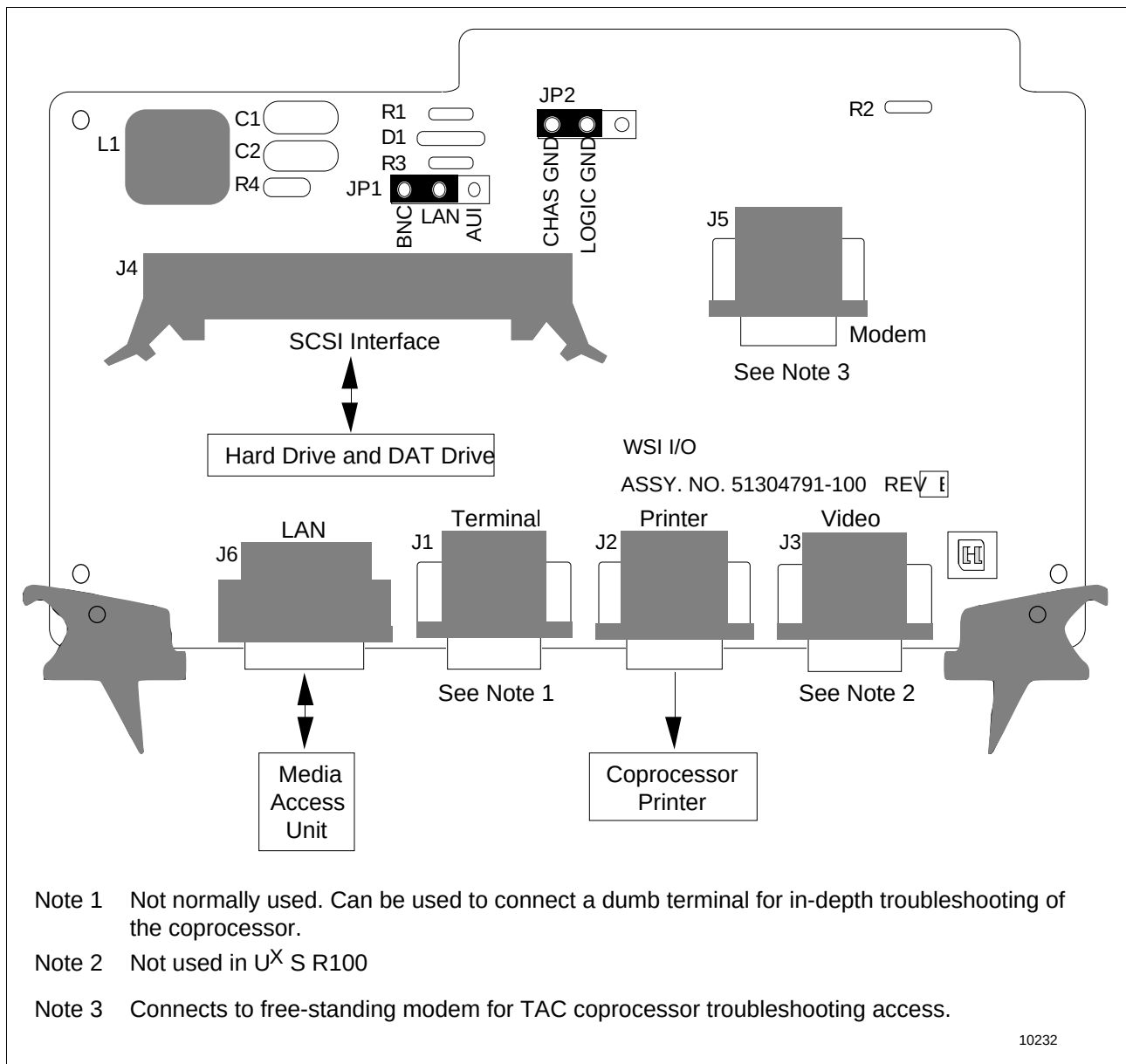
WSI I/O board

This TPS I/O module board provides these input/output connections for the WSI board:

- SCSI peripherals (hard drive, DAT tape drive, etc.)
- LAN connection (Ethernet)
- Serial Port (Term0 connection for maintenance)

See Figure A-3.

Figure A-3 WSI I/O Board



A.4 WSI Daughter Board Description

HP9000-382 board

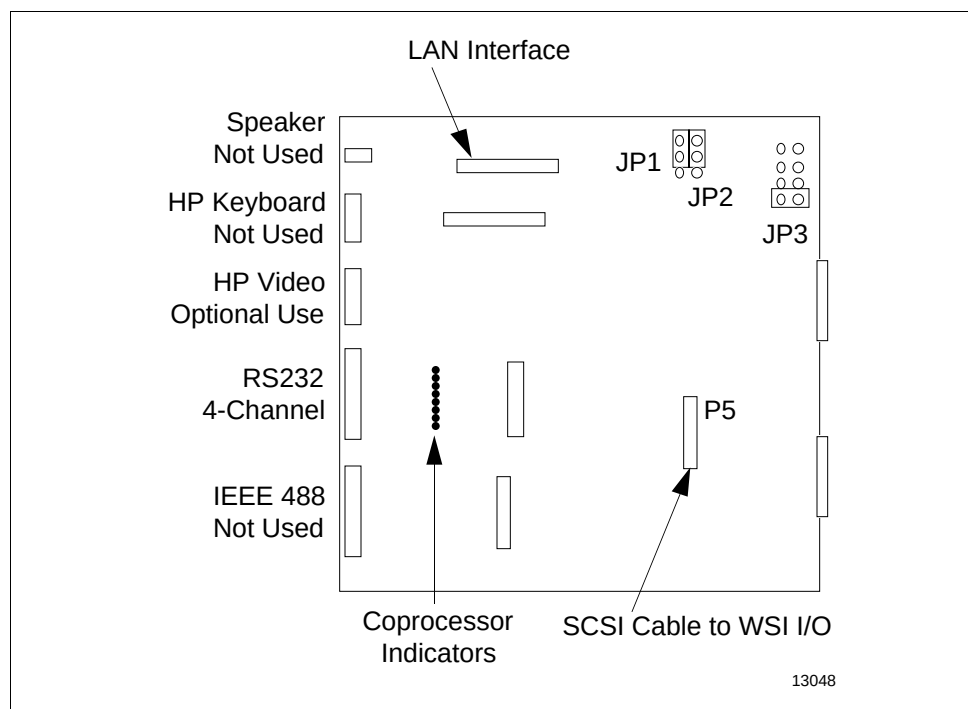
This is the UNIX coprocessor and a daughter board to the WSI board. The board is an HP9000-382 processor (68040) with 16 MB of RAM, interfaced to the WSI module board utilizing the HP DIO-II bus.

The workstation processor executes HP-UX version 8.0 operating system.

HP9000-382 board pinning

The HP processor board should be pinned as shown below.

Figure A-4 WSI Daughter Board



A.5 400 Hard Drive

Scope

Release 100 of the U^XS uses a 400 MB hard drive (for the coprocessor WSI board) operating system storage.

Description

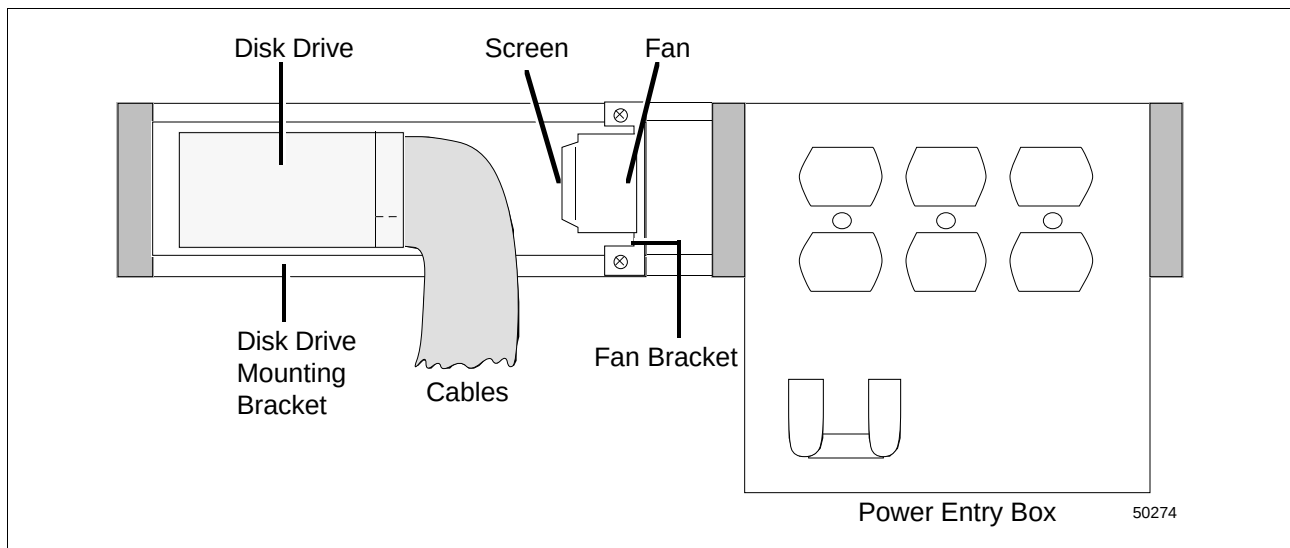
The hard drive has the following attributes:

- Small Computer System Interface (SCSI) bus (also connected to the DAT drive—both feed the WSI I/O board)
- 400 MB size
- Contents of drive can be updated from DAT drive
- Hard drive is powered by the Five-Slot Module power supply.
- Hard drive is present only for U^XS nodes (in standard US node, this space is filled by the peripheral power supply).
- Hard drive is located on a panel over the Five-Slot Module power supply.

Refer to Figure A-7 for cabling.

Illustration

Figure A-5 400 MB Hard Drive Mounted Next to Breaker Box



A.6 Replacing the Hard Drive

Procedure

The hard drive is located in the power supply enclosure located behind the media drive (cartridge/floppy) and is accessible from the rear.

Table A-2 Replacing the 400 MB Hard Drive

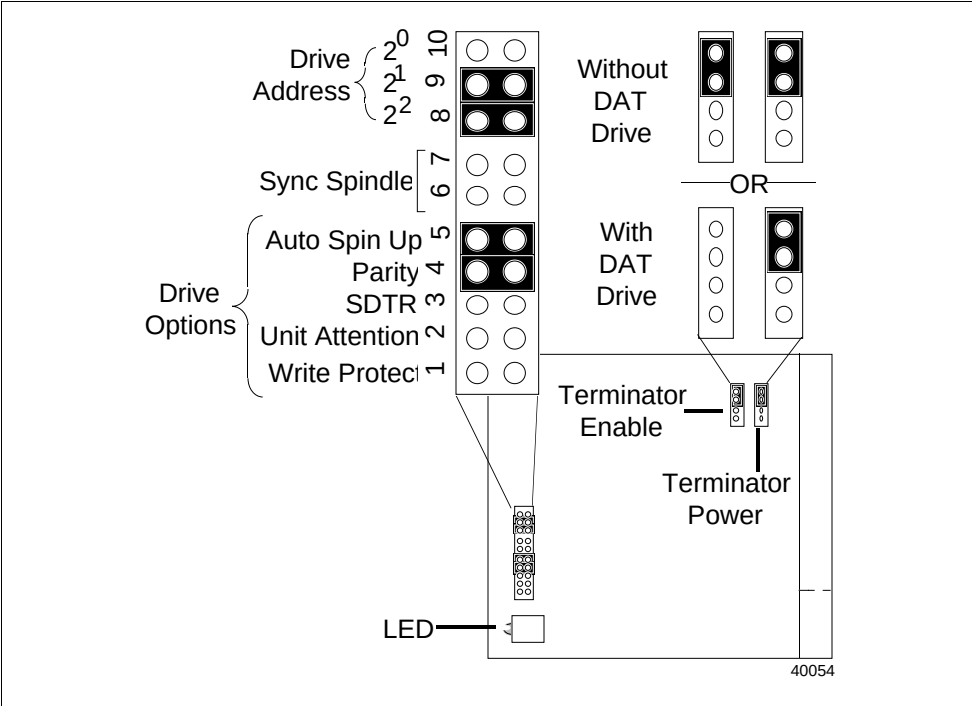
Step	Action
1	Loosen the two retaining screws from the power supply enclosure.
2	Slide the enclosure out of the station.
3	Disconnect the power cable and the 50-pin signal cable located on the right-hand side of the assembly.
4	Disconnect the fan power cable from dc fan power cable (this connection is just outside the enclosure on the left-hand side).
5	Remove the two screws from each end of the enclosure cover and pull the cover off the enclosure.
6	Remove the four mounting screws that fasten the drive mounting bracket to the enclosure.
7	Remove the four screws holding the drive to the mounting bracket.
8	Ensure that the hard drive is pinned for a SCSI address of 6 and terminated properly for the cases with/without a DAT Drive. See Figure A-6.
9	Replace the hard drive and fasten it in place with the four screws.
10	Mount the mounting bracket and drive to the enclosure with the four screws.
11	Reconnect the fan power cable from dc fan (this connection is just outside the enclosure on the left-hand side.)
12	Reconnect the power cable and 50-pin signal cable to the right side of the assembly.
13	Return the enclosure to the station.
14	Replace the two retaining screws.

Continued on next page

A.6 Replacing the Hard Drive, Continued

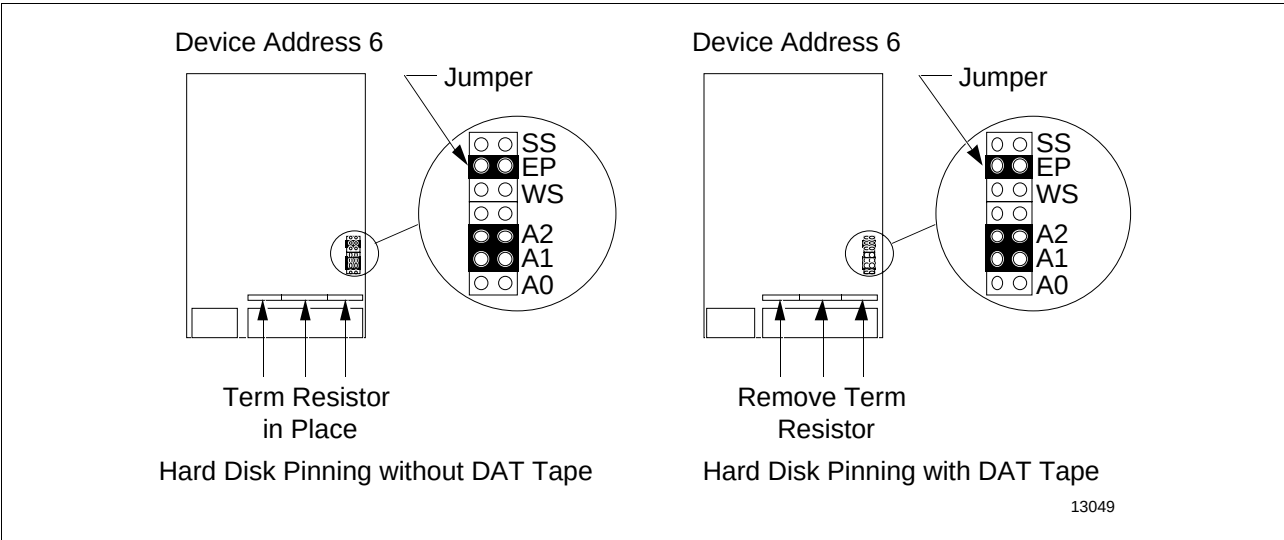
Illustration Ensure the pinning of the new hard drive as shown in Figure A-6.

Figure A-6 Hard Drive Pinning and Terminations



Illustration

Figure A-7 Hard Drive Pinning and Terminations-Quantum Model 425S



A.7 R100 Universal Station^X Cabling

Cable connections Figures A-8 and A-9 are connection diagrams for the TPDG I/O and WSI I/O boards.

Figure A-8 TPDG I/O Board Cabling

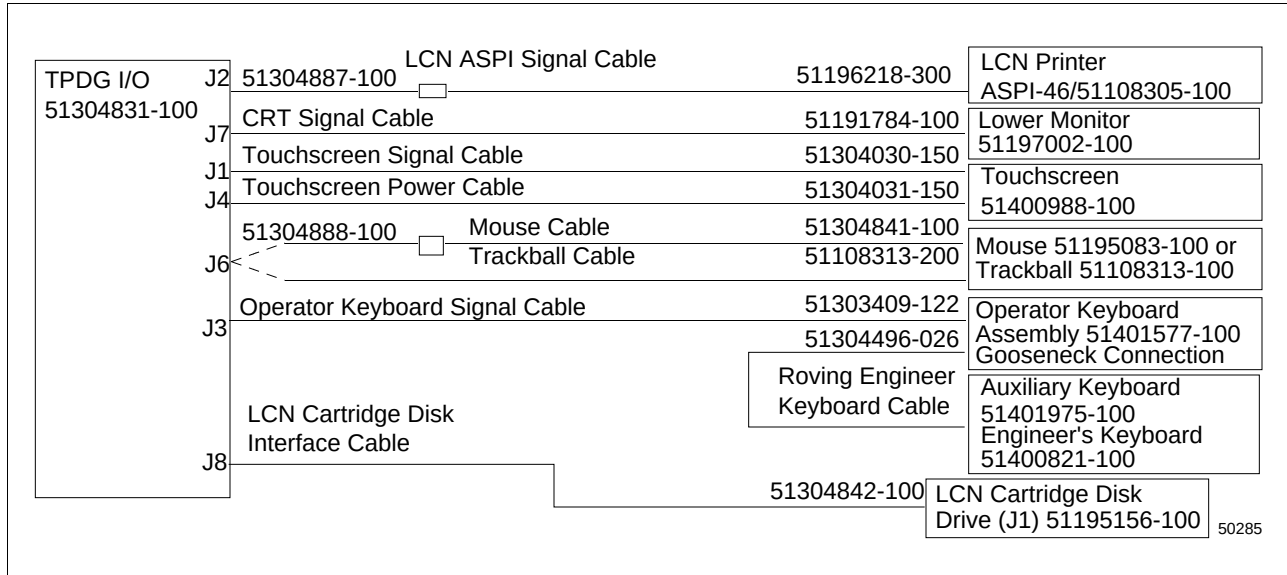
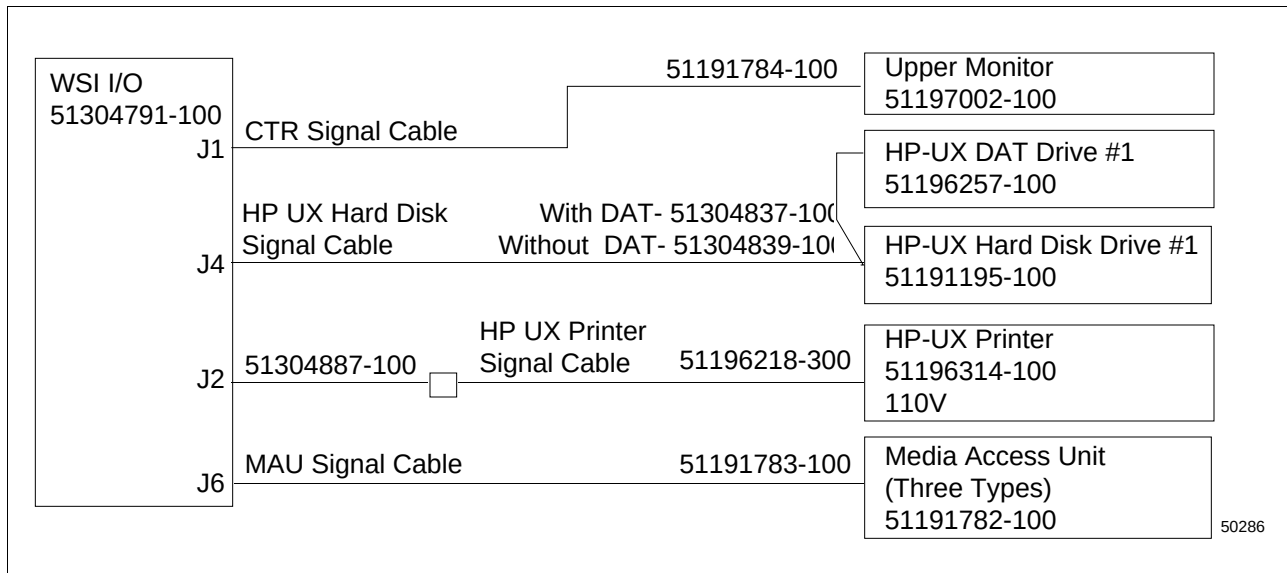


Figure A-9 WSI I/O Board Cabling

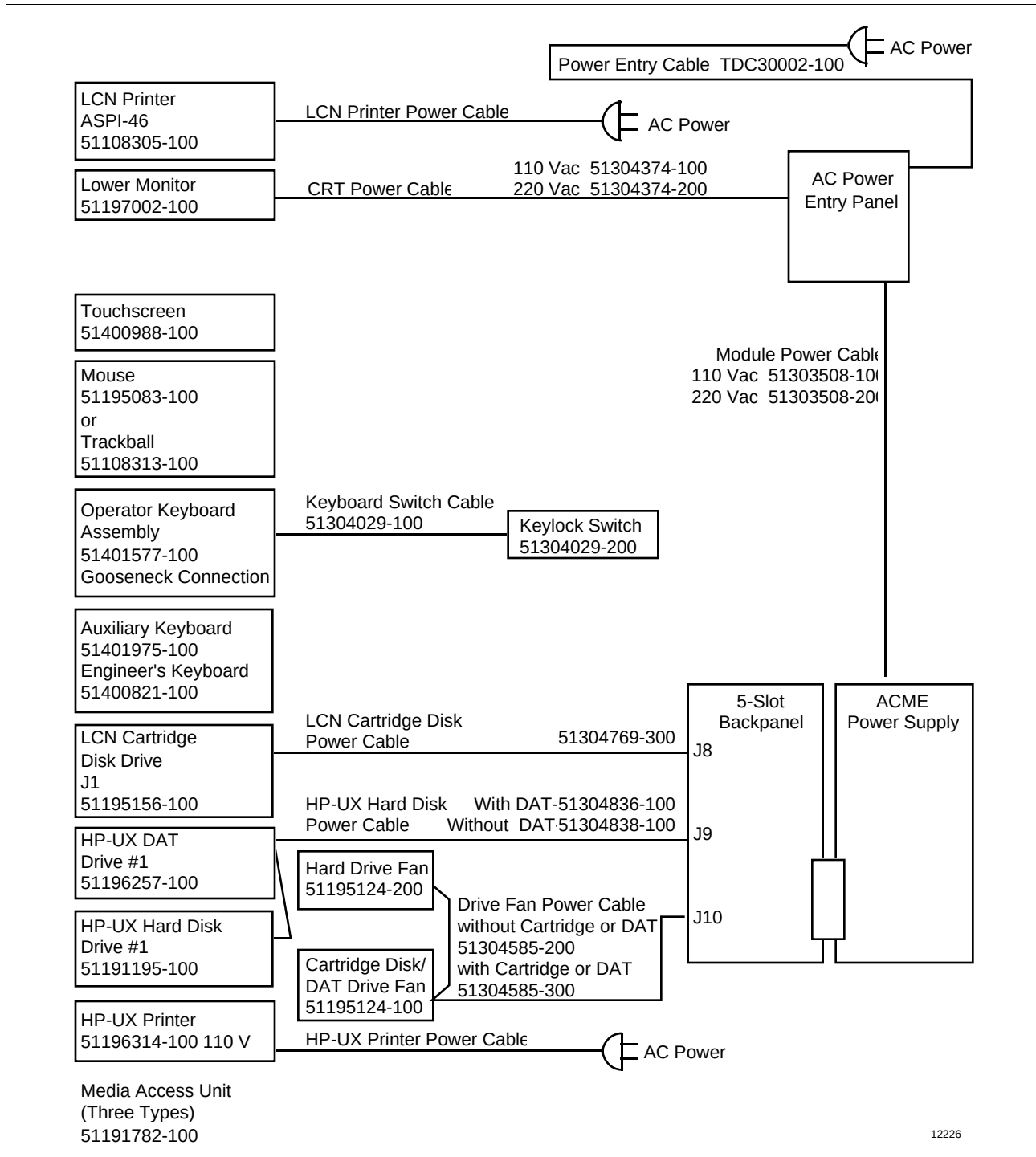


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A.7 Universal Station^X Hardware Features, Continued

Cable part numbers Figure A-10 is a Universal Station^X cabling diagram that illustrates cable part numbers.

Figure A-10 Universal Station^X Cabling



A.8 Universal Station^X Shutdown

WARNING

WARNING—It is imperative that both the LCN node processor and UNIX coprocessor operations be terminated prior to powering off the Universal Station^X.

Failure to do so can cause damage to the coprocessor file system. It also causes a disruption to the orderly token passing scheme on the LCN. (Node goes isolated.)

Following the shutdown procedure ensures the integrity of the process environment and the coprocessor file system.

ATTENTION

ATTENTION—The user logout steps are not needed if there is no user logged onto the coprocessor at the time. This situation can be recognized by observing if a login/shutdown prompt is present across the bottom of the display.

Procedure Overview

The shutdown procedures include these operations:

- User logout
- Coprocessor shutdown
- LCN node processor shutdown
- Power off

Follow the steps in Table A-3 to properly power down the station.

User logout

Steps 1-5 are required if a user is currently logged onto the coprocessor. Begin with Step 6 if the user has already logged out.

Table A-3 Shutdown Procedure—Coprocessor Logout

Step	Action	Remarks
1	Move the display cursor to a blank portion of the display screen using the mouse (or trackball).	The Work Space menu can only be accessed from a blank portion of the screen.
2	Press the left mouse button (or select button on trackball assembly).	This displays the Work Space menu.
3	Select the X-MENU	X-MENU appears.
4	Select the LOGOUT target.	Quit Mwm ? prompt appears.
5	Select the OK target.	Logout process commences. The login/shutdown window appears after several seconds.

Continued on next page

A.8 Universal Station^X Shutdown, Continued

ATTENTION

ATTENTION—The user must be authorized to shutdown the coprocessor—see IWSM in the *Universal Station^X Administration Manual*.

Table A-3 Shutdown Procedure—Coprocessor Logout, Continued

Step	Action	Remarks
6	Enter your user name and password.	Do not press the carriage return or enter key after entering the password.
7	Select the SHUTDOWN WSI target.	This will initiate the coprocessor shutdown operation. A message indicating the coprocessor is in the idle state appears after several seconds. Note: An optional method of coprocessor shutdown can be forced by using the shutdown switch on the WSI board free edge (see Figure 6-3) and waiting for the yellow indicator to turn off. This performs a proper shutdown of the coprocessor file system, but it does not perform an orderly departure from the LAN. Other LAN nodes will see it as a failure of this node. This technique should be avoided. If you do use this procedure, return the switch to the Normal position.

Continued on next page

A.8 Universal Station^X Shutdown, Continued

LCN node processor shutdown

Table A-3 Shutdown Procedure—Coprocessor Logout, Continued

Step	Action	Remarks
8	Press the CONSOLE STATUS key on the operator keyboard.	The Console Status display appears.
9	Select your Universal Station ^X .	The selected node number turns white.
10	Select the SHUTDOWN target.	“Press enter to execute” prompt appears at the top of the display.
11	Press ENTER.	The shutdown of the LCN node processor in the Universal Station ^X commences. The Console Status display disappears from the screen.
12	Option: Do a second SHUTDOWN from another US.	If you do a second SHUTDOWN sequence, the node will go to FAILED instead of ISOLATED and you avoid several error messages in the journal.

Power off

Table A-3 Shutdown Procedure—Coprocessor Logout, Continued

Step	Action	Remarks
13	Turn OFF the power switch on the 5-slot chassis power supply.	The front of the power supply is accessible through the rear door of the station furniture. This only removes power from the five-slot chassis.
14	Turn OFF the circuit breaker located on the AC power distribution panel.	It is located inside the door under the operator keyboard (top right). This step removes power from all components in the Universal Station ^X furniture. Free-standing printers can be separately powered.

A.9 Universal Station^X Node Troubleshooting

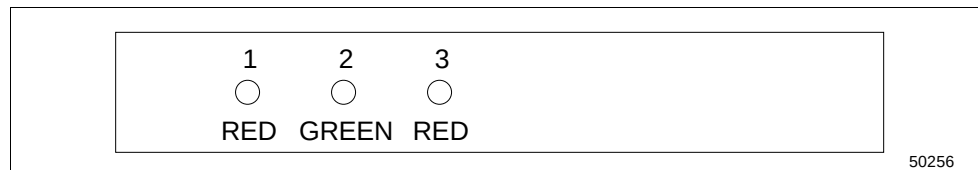
Overview

Use standard hardware practices common to the Five-Slot Module for troubleshooting/service procedures. The added UNIX system components have status indicators similar to those found in existing USs.

TPDG board status

The TPDG board has three status indicator LEDs. They are found on the left side along the front edge of the board.

Figure A-11 TPDG Indicators



Use the table below to interpret TPDG LED indicators:

Table A-4 TPDG Indicators

LED	Indication	Action
1	Board fails self-test	Replace module board
2	Board passes self-test	Good indication—hardware OK
3	Bus transaction error	Investigate further—possible backpanel problems

Continued on next page

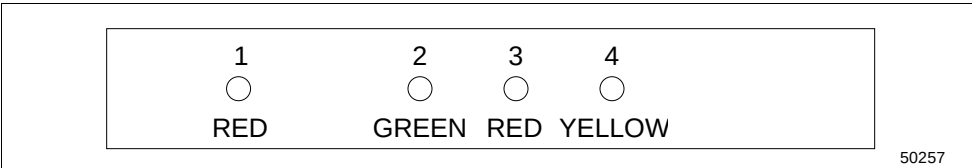
A . 9 Universal Station^X Node Troubleshooting,

Continued

WSI status

The WSI board has four status indicator LEDs. They are found on the left side along the front edge of the board (as viewed from power supply side of 5-slot module).

Figure A-12 WSI Indicators



Use the table below to interpret WSI LED indicators:

Table A-5 WSI Indicators

LED	Indication	Action
1	Board fails self-test	Replace WSI board
2	Board passes self-test	Good indication—hardware OK
3	Bus transaction error	Investigate further—possible backpanel problems
4	Coprocessor alive	Good indication—UNIX OK

Continued on next page

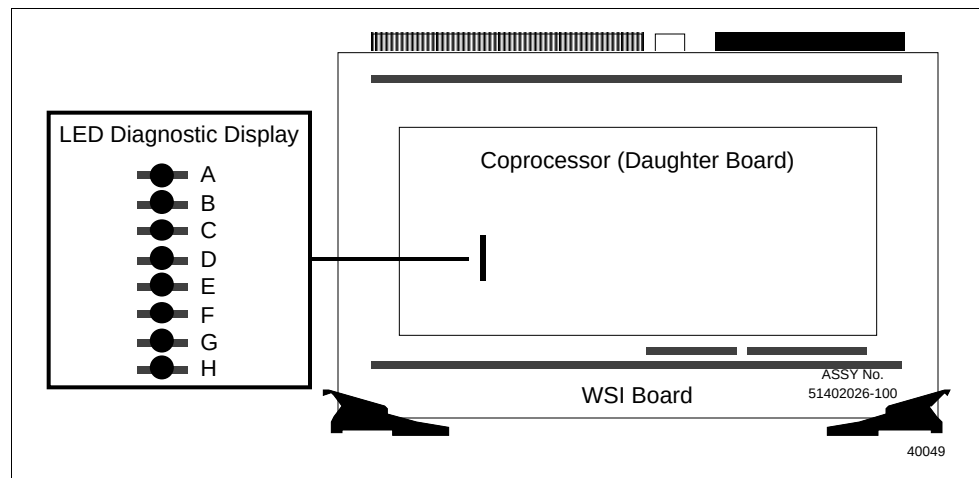
A.9 Universal Station^X Node Troubleshooting,

Continued

HP 382 Coprocessor status

The WSI coprocessor daughter board has eight indicator LEDs that are difficult to see (but not impossible) when the board is in place. A small mirror (with a nonconductive frame) may be needed to examine these indicators. Figure A-13 gives the orientation of these indicators. Table A-6 gives the description of the indicators.

Figure A-13 WSI Coprocessor Indicators



LED D is the Operating System Running indicator. It is known as the "heartbeat" LED because its blinking indicates that the UNIX software system is running. If the Universal Station^X fails to display a Login banner within 5 minutes of Power ON, check the UNIX alive LED (Yellow) on the WSI board front edge. If either indicator is on, set the Shutdown switch (located on the left front edge of the WSI board) to the SHUTDOWN position and wait 1 minute before turning the Five-Slot Module power switch to the OFF position.

Continued on next page

A.9 Universal Station^X Node Troubleshooting,

Continued

Coprocessor LEDs

Table A-6 Coprocessor Indicator Descriptions

Display LEDs A B C D E F G H	Description	Remarks
X X X X 0 0 0 0 Note: X = on or blinking 0 = off	Normal operation	LEDs A, B, C, and D all blink at different rates
Indicator notes: A = LAN transmit in progress B = LAN receive in progress C = Hard disk access in progress D = Operating system running (Heartbeat)		
Any indication other than described above identifies a significant failure. To interpret the indicators, see <i>Universal Station^X Troubleshooting</i>.		

Continued on next page

A.9 Universal Station^X Node Troubleshooting,

Continued

Coprocessor does
not boot

Table A-7 Coprocessor Troubleshooting Procedure

Step	Action
1	DO NOT TURN POWER OFF! Booting the coprocessor can take up to 5 minutes. If, after 6 minutes the banner does not appear at the bottom of the Monitor screen, proceed to Step 2; otherwise go the Step 3.
2	Ensure the boards (TPDG, TPDGC (or TPDG I/O), WSI, WSI I/O, K2LCN/K4LCN, CLCN A/B (or LCN I/O)) are properly seated in the correct slot positions with all cabling correctly installed and secure.
3	At Power ON: - The HP 382 searches its SCSI (pronounced "scuzzy") bus for a bootable device. - Finding the internal hard drive, it loads the Unix file system. - The coprocessor performs FSCK, which automatically repairs inconsistencies in the file system. ATTENTION This repair adds time to the coprocessor boot process (less than 5 minutes total boot time for 400 MB hard drives. DO NOT GET IMPATIENT AND CYCLE POWER DURING FSCK REPAIR. Unit will recover from an improper shutdown, but may fail if the FSCK operation is interrupted during repairs. This may leave the file system in a nonrepairable state and require the drive to be reformatted and the Unix/Honeywell software to be reloaded. However, if the coprocessor completes the FSCK process, the Honeywell X-Window software is initialized and the coprocessor LOGIN banner is displayed.
4	If the LOGIN banner does not appear within 10 minutes: - Place the Shutdown switch on the WSI board to the shutdown position. This will disable the coprocessor. - Wait 1 minute before turning the Power switch to the OFF position. - Return the Shutdown switch to the Normal position. - Contact your local service representative or call the Honeywell Technical Assistance Center to resolve the problem.

ATTENTION

ATTENTION—If the coprocessor displays the LOGIN banner, refer to the *Universal Station^X User's Guide* for complete booting procedures.

Continued on next page

A.9 Universal Station^X Node Troubleshooting,

Continued

Troubleshooting procedure

Table A-8 Coprocessor Troubleshooting Procedure Using a VGA

Step	Action
1	To recover from a failure which prevents the coprocessor from booting, it is necessary to connect a Dec VT-100 compatible monitor to J3 of the WSI I/O board, and an HP compatible keyboard to J1 of the WSI I/O board.

A.10 Spare Parts

WSI and WSI I/O boards

Refer to *Five/Ten-Slot Module Service* for spare parts information.

Hard drive parts

Table A-9 Hard Drive Spare Parts

Part Number	Description
*51191195-100	400 MB Hard Disk (Already Loaded with Software)
51304838-100	WDD Power Cable (Installations with Hard Drive Only)
51304836-100	WDD Power Cable (Hard Drive and DAT Drive Installations)
51304839-100	Hard Disk Signal Cable (Hard Disk Installations Only)
51304837-100	Hard Disk Signal Cable (Hard Disk and DAT Drive Installations)
51195124-200	Fan Assembly
51304585-200	Fan DC Power Cable (Installations with Hard Drive Only)
51304585-300	Fan DC Power Cable (Hard Drive and DAT Drive Installations)

Appendix B – Other Reference Materials

Listings

Table B-1 Other Reference Materials

Publication Title	Publication Number
<i>Installing and Administering Network Services</i>	B1012-90001
<i>Installing and Administering NFS Services</i>	B1013-90001
<i>Installing and Administering OSI Transport Services</i>	
<i>Installing and Administering X.400 Administrators Guide</i>	32034-90005
<i>Installing and Configuring the X Window System</i>	98794-90004
<i>Installing and Maintaining NS-ARPA Services</i>	B1014-90007
<i>Installing Peripherals HP 9000 Series 300</i>	97005-90002
<i>Introducing UNIX System V</i>	98597-90620
<i>LAN Node Managers Handbook</i>	
<i>LAN/X.25 Reference Pages</i>	98194-90014
<i>Laserjet III Printer User's Manual</i>	
<i>Model 382 and R/382 Owner's Guide</i>	
<i>NetIPC Programmer's Guide</i>	98194-60527
<i>Network Services Reference Pages</i>	B1012-90002
<i>Networking Overview</i>	B1012-90003
<i>NFS Services Reference Pages</i>	B1013-90003
<i>NS Cross-System NFT Reference Manual</i>	

Continued on next page

Appendix B – Other Reference Materials, Continued

Listings, continued

Table B-1 Other Reference Materials, Continued

Publication Title	Publication Number
<i>Patriot Monitor (Aydin) Operation and Maintenance Manual</i>	
<i>Programming and Protocols for NFS Services</i>	B1013-90002
<i>Quantum 210 MB Disk Drive Manual</i>	
<i>Quantum 423 MB Disk Drive Manual</i>	
<i>Quick Reference Card for vi Editor</i>	98597-90000
<i>Troubleshooting HP-UX Systems Error Diagnosis and Recovery</i>	92453-90026
<i>Using ARPA Services</i>	B1014-90000
<i>Using Network Services</i>	B1012-90000
<i>Using NFS Services</i>	B1013-90000
<i>Using the X11 Window System</i>	98794-90001
<i>X.25 Programmer's Guide</i>	
<i>X.25: The PSN Connection</i>	

Appendix C – Coprocessor Console

C.1 Overview

Requirements for coprocessor console

A console terminal connection to the coprocessor Serial Port 1 is required to initially configure the coprocessor software in PIN connected Universal Station^X. This (terminal) will serve as a “console” to the coprocessor HP-UX software environment.

The fundamental coprocessor software configuration task is to make the Universal Station^X conversant with appropriate other nodes on the PIN (LAN). Once the communications over the PIN are established, further configuration and implementation tasks can be directed from a work station or U^XS connected to the PIN.

When a console is needed

A “console” terminal connection is normally not required after the initial PIN (LAN) configuration has been performed and a suitable work station or Universal Station^X is available on the PIN.

Troubleshooting some coprocessor related problems will require direct access with a “console” terminal for the purpose of problem isolation and recovery.

Some Universal Station^Xs will be used on systems where the Universal Station^X is in a stand-alone mode. **This means the Universal Station^X has a connection to the LCN, but no connection to the PIN.** In this case, the coprocessor “console” terminal is needed only for troubleshooting and recovery activities. It is not envisioned that users of stand-alone A^XM^s will be developing solutions.

Suitable consoles

A System Administration and Development workstation (HP 712/60) or U^XS can be used as a console terminal.

A personal computer running terminal emulator software that is compatible with the Digital Equipment Corporation (DEC) VT-100 protocol can also be used as a console terminal.

Continued on next page

C.1 Overview, Continued

Overview of activities to be performed

There are several distinct tasks required to connect a console terminal to the coprocessor as a console.

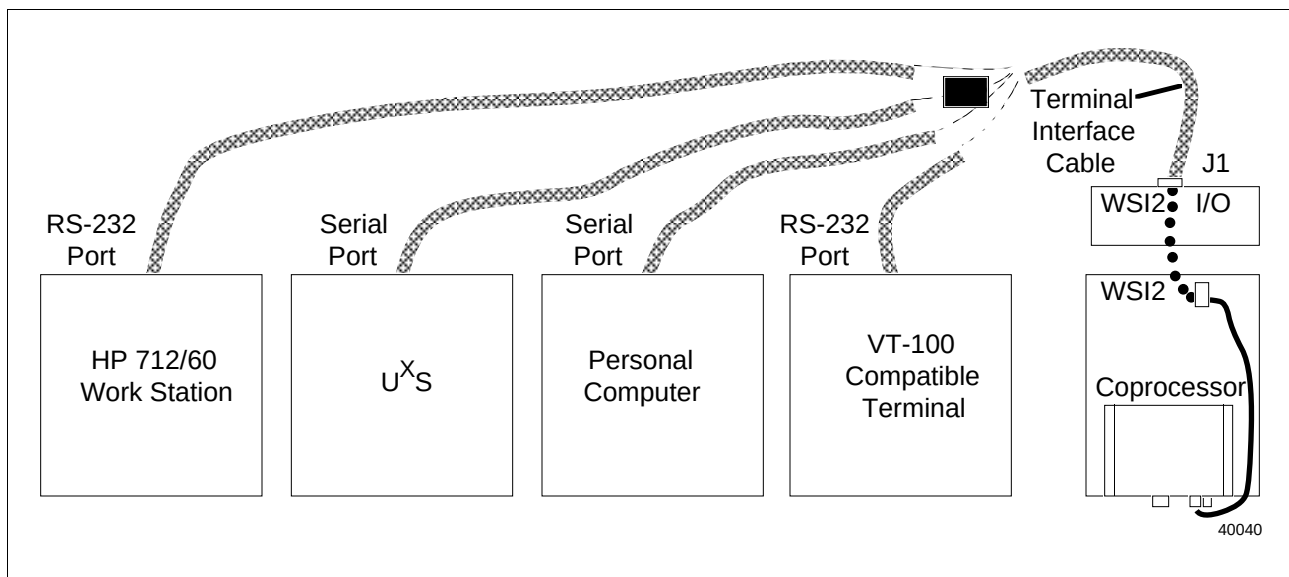
- Select the proper terminal options.
- Connect the terminal to the WSI2 I/O board.
- The coprocessor software defaults to the proper configuration of Serial Port 1 and no coprocessor configuration actions are required.
- Apply power to the terminal.
- Ensure the terminal emulation software is activated (if using a personal computer or HP 712/60 PA-RISC workstation or U^XS) to perform the coprocessor console operations through serial port 1 of the A^XM coprocessor.
- Initiate communications using the keyboard on the terminal

The usage of this console terminal is defined in the procedures provided in the *Universal Station^X System Administration Manual*.

Illustration

An illustration of a coprocessor console connection is shown below.

Figure C-1 Example of Various Console to Coprocessor Connections



C.2 Coprocessor Terminal Description and Configuration

Recommended terminal

Honeywell recommends the Digital Equipment Corporation (DEC) VT-100 terminal (or any simple terminal with equivalent functionality) for use as a “console” terminal connected to the coprocessor. This terminal is needed for initial configuration of PIN connected Universal Station^Xs and for troubleshooting coprocessor related problems.

A U^XS, or System Administration and Development workstation (HP 712/60), can be used as a coprocessor console. The serial port of these devices must be connected to the serial port of the Universal Station^X coprocessor using the provided terminal cable to support this functionality.

You may also use a PC running suitable terminal emulation software (such as Microsoft Windows Terminal) to provide compatible terminal functionality.

Required terminal Options

The terminal must be set to provide the following functionality to successfully communicate with the coprocessor serial port 1:

- Asynchronous operation
 - 9600 bits per second
 - 8 bits/character
 - No parity
-

C.3 Coprocessor Console Terminal Connection

Overview

A direct connect interface cable must be connected between the coprocessor console terminal device and the WSI2 I/O board (Serial Port 1).

Several different devices can be used to provide this coprocessor console connection as shown in Figure C-1.

Coprocessor console terminal interface cable schematics

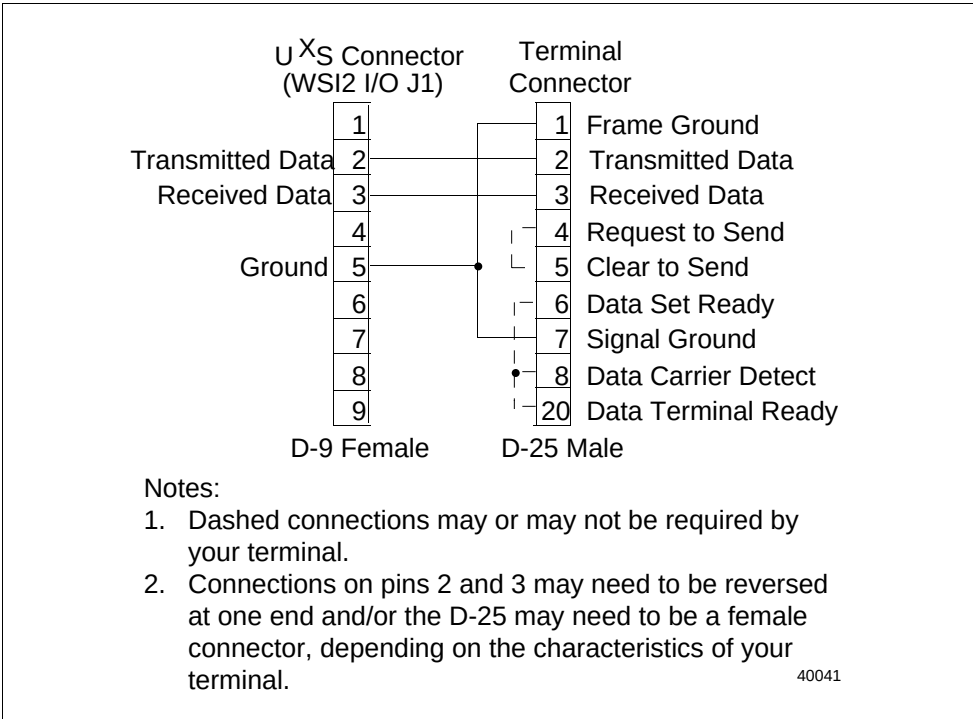
Examples of the direct connect terminal interface cable wiring details are shown in the following diagrams. Examples of the direct connect terminal interface cable wiring details are shown in the following diagrams.

Notice that two cable types are described. One is a 9 to 25 pin direct connect cable and the other is a 9 to 9 pin direct connect cable.

The 9 to 25 pin cable (in the first diagram) is available from Honeywell logistics using the part number 51196218-300. This is an optional cable type. It is not supplied with the Universal Station^X.

The 9 to 9 pin cable (in the second diagram) is available from Honeywell logistics using the part number MP-XLLPLK. It is the cable type provided with the Universal Station^X 9-pin to 25-pin cable.

Figure C-2 25-Pin Coprocessor Console Cable Pin-Out

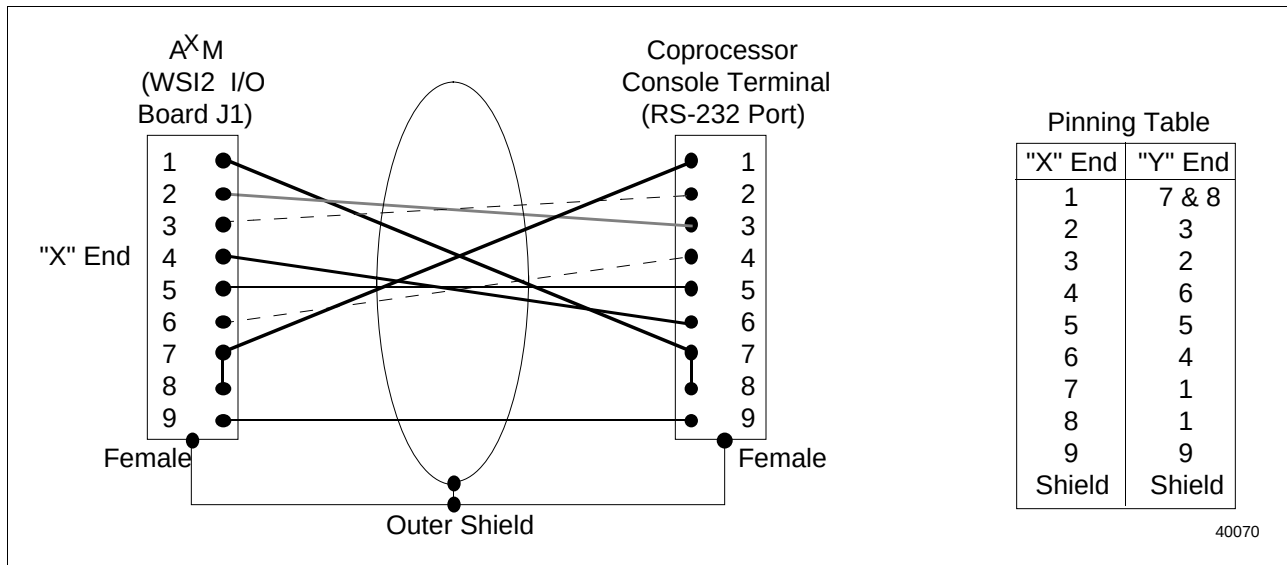


Continued on next page

C.3 Coprocessor Console Terminal Connection, Continued

Coprocessor console terminal interface cable schematics, continued

Figure C-3 9-Pin to 9-Pin Coprocessor Console Cable Pin-Out



ATTENTION

ATTENTION—The dashed lines shown in the 9 to 9 cable diagram above are required connections. The dashed notation for some of the connections is used to make the specific wiring easier to read on screen.

Coprocessor console terminal connection procedure

The following describes the connection details for a console terminal. The usage of this terminal is defined in procedures provided in the *Universal Station^X System Administration Manual* and *Application Module^X Troubleshooting manual*.

CAUTION

CAUTION—The terminal can be connected while power is applied to the AXM and the terminal. Proper care should be exercised to ensure the cable ends do not come in contact with other electrically charged components.

Continued on next page

C.3 Coprocessor Console Terminal Connection,

Continued

Procedure

Table C-1 Terminal Connection procedure

Step	Action
1	Make sure the terminal is properly configured (switches/jumpers/etc.)
2	Connect the direct connect terminal interface cable to the terminal. See the diagram following this table for the proper connection (depends on which device you are using as the console terminal).
3	Connect the terminal interface cable to the WSI2 I/O board connector for Serial Port 1 (connector J1). You must disconnect the modem cable from J1 if a modem is currently connected to the Universal Station ^X coprocessor.
4	Make sure the terminal is powered on.
5	If you are using a PC as the coprocessor console, invoke the emulation software and configure it for the following characteristics: • 9600 bits/second • 8 bits/character • no parity
6	The procedure for using a VT-100 compatible terminal (simple terminal, Personal Computer, HP 712/60 workstation, or Universal Station ^X) as a coprocessor console is provided in the <i>Universal Station^X System Administration Manual</i> .
7	Establish communications using the terminal keyboard.
8	The activities that require the coprocessor console terminal are documented in the <i>Universal Station^X System Administration Manual</i> .

C.3 Coprocessor Console Terminal Connection,

Continued

Connection of console to R100/110 U^XS

The cable connection to the U^XS is different depending on if it is Release 100 or 200 hardware. The U^XS-100 connection is to the WSI I/O board J2 (normally used for printer). The U^XS-200 connection is to the WSI2 I/O board J1 (normally used for U^XS coprocessor console terminal).

The following table defines the different cable types available for connecting coprocessor terminals.

Table C-2 Console Cable Selection

Applicable Terminal Devices	Terminal Connector	Cable Part Number
HP 712/60 Workstation	RS-232	51305069-100 9 pin(F) to 9 pin(F)
U ^X S-100	WSI I/O board (J2)	51305069-100 9 pin(F) to 9 pin(F)
U ^X S-200	WSI2 I/O board (J1)	51305069-100 9 pin(F) to 9 pin(F)
Other VT-100 compatible devices with 9 pin RS-232 connector.	Serial port	51305069-100 9 pin(F) to 9 pin(F)
Other VT-100 compatible devices with 25 pin RS-232 connector.	Serial port	51196218-300 9 pin(F) to 25 pin(M)

* This cable is not supplied with the Universal Station^X. It is available by separate order.

Appendix D – TAC Access To Coprocessor

D.1 Overview

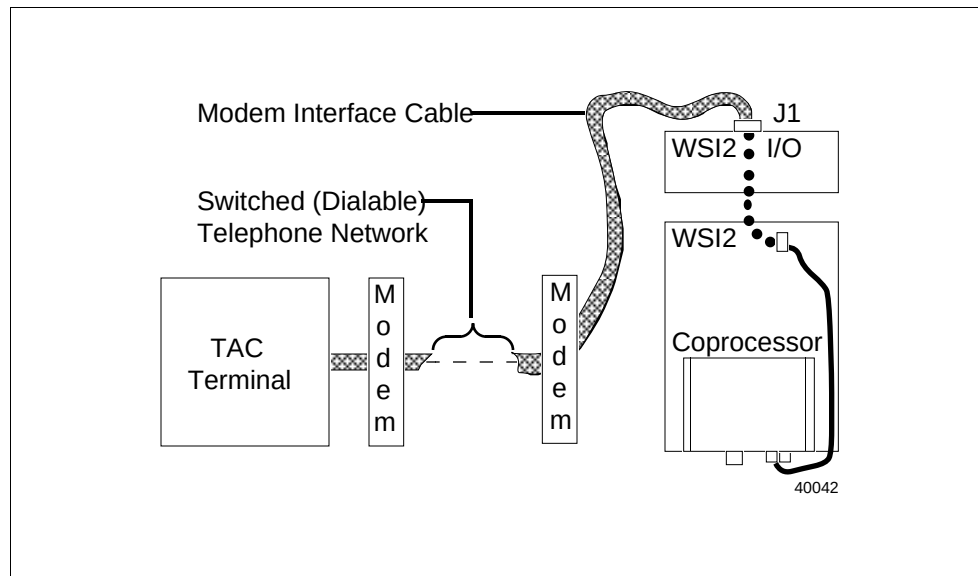
TAC support for coprocessor problems

The Technical Assistance Center (TAC) has the capability of providing assistance with troubleshooting coprocessor related problems.

Troubleshooting some problems may require direct access to the coprocessor. A telephone communications link must be connected to the coprocessor to allow direct access from TAC.

The diagram on the right illustrates this communications link.

Figure D-1 TAC Connection to Universal Station^X by Modem



Overview of activities to enable TAC access

There are several distinct tasks required to establish the TAC communications link to the coprocessor.

- Configure the modem options using switches and internal registers.
- Connect the modem to the WSI2 I/O board.
- Connect the modem to a switched (disable) telephone line.
- The coprocessor software defaults to the proper configuration of Serial Port 1 and no coprocessor configuration actions are required.
- Apply power to the modem.
- Establish the telephone connection with TAC under direction of the TAC engineer.

D.2 Communications Modem Description and Configuration

Recommended modem

Honeywell recommends the following modem for the TAC access communications link.

U.S. Robotics Sportster 9600

This is a commercially available device and is not supplied by Honeywell. The configuration switches and registers for this modem are given later in this section.

Other modems that are fully compatible with the U. S. Robotics Sportster 9600 and its configuration may work.

The proper functionality of the communications link to TAC must be verified at the time of Universal Station^X hardware installation. Failing to do so can drastically effect the availability/efficiency of TAC support when it is needed.

Modem configuration switches

The following table identifies the required setting of the modem switches to make it function as a TAC access modem.

Table D-1 U.S. Robotics 9600 Modem Configuration

Switch	Position	Description
1	Down	DTR override
2	Down	Verbal result codes
3	Up	Suppress result codes
4	Down	No echo, offline commands
5	Up	Auto answer on ring
6	Down	Carrier detect override
7	Up	Load NVRAM defaults
8	Down	Smart mode

Continued on next page

D.2 Communications Modem Description and Configuration, Continued

Modem configuration registers

The internal configuration registers of the U.S. Robotics Sportster 9600 modem must be properly loaded to finish the modem configuration task. This is done by connecting a terminal to the modem and then entering the required configuration data into the registers.

The configuration data for the U.S. Robotics 9600 modem is provided in a table following in this section.

Once the registers are configured, the modem registers will retain the information. This information is retained even if power is turned off and on again.

Terminal requirements for setting modem configuration registers

The terminal used for setting the internal configuration registers of the U.S. Robotics 9600 modem must have the following configuration characteristics:

- Asynchronous operation
- 9600 baud rate
- 8 bits/no parity character format

Continued on next page

D.2 Communications Modem Description and Configuration, Continued

Required modem configuration register data

The following table contains the required configuration register data to make it support the TAC access connection.

Table D-2 U.S. Robotics 9600 Configuration Register Data

Register	Description
B0	Needed to answer above 4800
E0	Local echo off
F1	Local echo of transmitted data off (full duplex)
M1	Speaker on until carrier established
Q1	Result codes suppressed
V0	Numeric result codes
X0	Result code messages (min)
&A0	Disable additional ARQ result codes
&B1	Serial port remains fixed at last AT command (during configuration of the modem)
&C0	CD override
&D2	Coprocessor must set DTR before modem will accept commands
&G0	No guard tone
&H1	Hardware flow control
&I0	Flow control disabled
&K1	Auto data compression enabled
&M0	Normal mode, error control disabled
&N0	Variable link operations
&P0	Pulse dialing US/Canada
&R1	Ignore RTS
&S0	DSR override
&T5	Prohibit RDL
&Y1	Break handling, destructive expedited

CAUTION

CAUTION—Be certain to perform a save/write (AT&W) to NVRAM registers after entering the register data.

Continued on next page

D.2 Communications Modem Description and Configuration, Continued

Modem register configuration procedure

The procedure for configuring the U.S. Robotics Sportster 9600 registers is defined in the product manual supplied from the manufacturer with the modem.

D.3 Modem Connection

Overview

An interface cable must be connected between the optional modem and the WSI2 I/O board J1 (Serial Port 1). Locate J1 in the following diagram.

Figure D-2 WSI2 I/O Board Connections

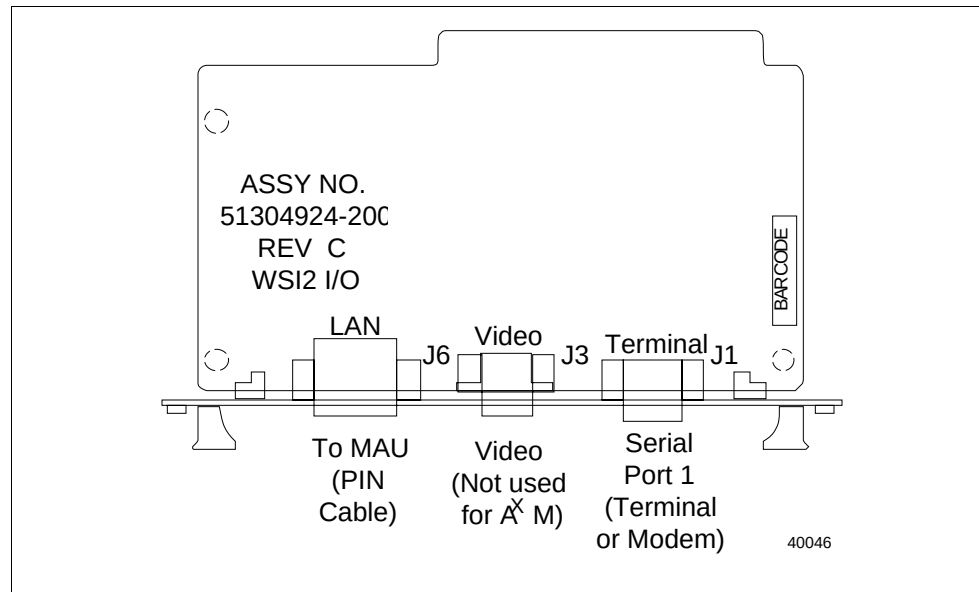
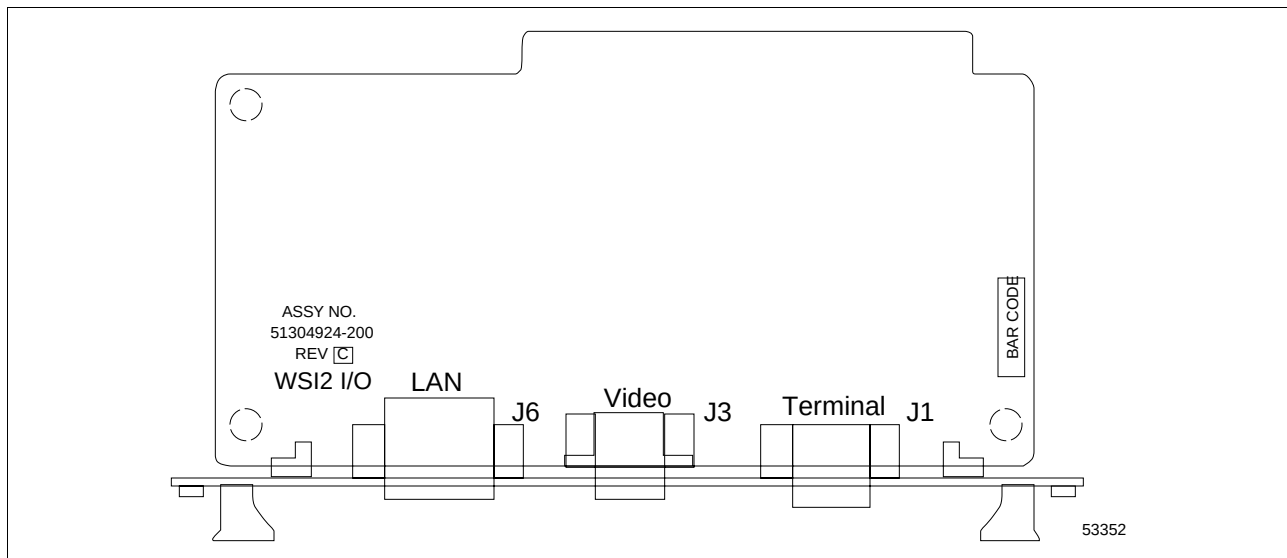


Figure D-3 CE Compliant WSI2 Board



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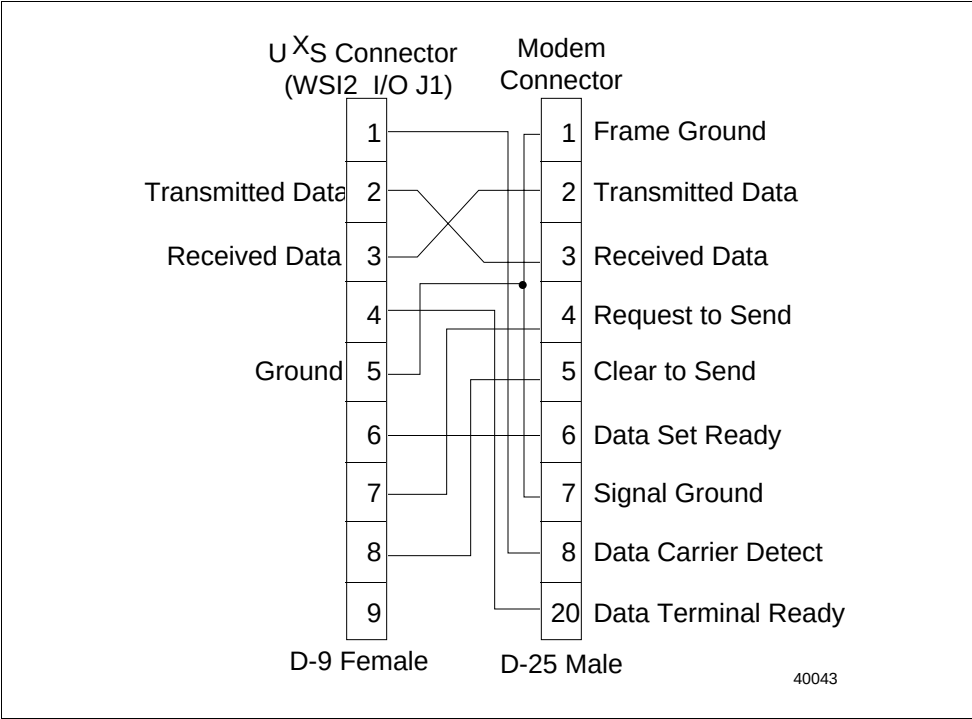
D.3 Modem Connection, Continued

**Modem Interface
cable Schematic**

An example of the modem interface cable internal wiring details are shown in the diagram.

This cable is available from Honeywell logistics with the part number 51305073-100. It must be ordered separately.

Figure D-4 Universal Station^X Modem Cable Schematic



D.4 Modem Cable Connection

CAUTION

CAUTION—The modem can be connected while power is applied to the Universal Station^X and the modem. Proper care should be exercised to ensure the cable ends do not come in contact with other electrically charged components.

Modem Connection procedure for U.S.Robotics Sportster 9600

Table D-3 Modem Connection Procedure

Step	Action
1	Make sure the modem is properly configured (switches and internal configuration registers).
2	Connect one end of the modem interface cable to the modem.
3	Connect the other end of the modem interface cable to the WSI2 I/O board for serial Serial Port 1 (connector J1). You will have to disconnect the coprocessor console terminal cable from this jack if the terminal is currently connected.
4	Make sure the telephone connection to the modem is properly made.
5	Ensure modem power is on.
6	The communications link hardware is ready for making the telephone connection and subsequent sign-on by TAC.

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